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M.SC ZOOLOGY

STA100 - General Mathematics and Biostatistics



HANDOUTS TOPIC NO 1 TO 45

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Lecture 1-3

Sets and Numbering Systems Topic 1

The study of Mathematics begins with a study of sets and the development of the numbering systems. Every mathematical system can be represented as a "set"; therefore, it is important for us to understand the definitions, notations and properties of "sets"

- Definition: A **set** is an *unordered* collection of *distinct* objects. Objects in the collection are called **elements** of the set.
- Examples:
  - The collection of persons living in Lahore is a set.
    - Each person living in Lahore is an element of the set.
  - The collection of all towns in the Punjab province is a set.
    - Each town in Punjab is an element of the set.
  - The collection of all quadrupeds is a set.
    - Each quadruped is an element of the set.
  - The collection of all four-legged dogs is a set.
    - Each four-legged dog is an element of the set.
  - The collection of counting numbers is a set.
    - Each counting number is an element of the set.
  - The collection of pencils in your bag is a set.
    - Each pencil in your bag is an element of the set.
- Notation: Sets are usually designated with capital letters. Elements of a set are usually designated with lower case letters.
  - *D* is the set of all four legged dogs.
  - An individual dog might then be designated by *d*.
- The **roster method** of specifying a set consists of surrounding the collection of elements with braces. For example the set of counting numbers from 1 to 5 would be written as {1, 2, 3, 4, 5}.
- **Set builder** notation has the general form {variable | descriptive statement }.

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The vertical bar (in set builder notation) is always read as "such that".

Set builder notation is frequently used when the roster method is either inappropriate or inadequate.

For example, {x | x < 6 and x is a counting number} is the set of all counting numbers less than 6.
Note this is the same set as {1,2,3,4,5}.

- Other Notation: If *x is an element of the set A,* we write this as x ∈ A. x ∉ A means *x is not an element of A.*

  If A = {3, 17, 2 } then 3 ∈ A, 17 ∈ A, 2 ∈ A and 5 ∉ A.
  If A = { x | x is a prime number } then 5 ∈ A, and 6 ∉ A.

- Definition: The set with no elements is called the **empty set** or the **null set** and is designated with the symbol ∅.

- Definition: The **universal set** is the set of all things pertinent to a given discussion and is designated by the symbol *U*

  For example, when dealing with all the students enrolled at the Virtual University, the Universal set would be
  U = {all students at the Virtual University}
  Some sets living in this universal set are:
  A = {all Computer Technology students}
  B = {first year students}
  C = {second year students}

- Definition: The set A is a **subset** of the set B, denoted A ⊆ B, if every element of A is an element of B.
  If A is a subset of B and B contains elements which are not in A, then A is a **proper subset** of B. It is denoted by A ⊂ B.

  If A is not a subset of B we write A ⊄ B to designate that relationship.

- Definition: Two sets A and B are **equal** if A ⊆ B and B ⊆ A. If two sets A and B are equal we write A = B to designate that relationship.
  In other words, two sets, A and B, are equal if they contain the same elements

- Definition: The **intersection** of two sets A and B is the set containing those elements which are elements of A **and** elements of B. We write A ∩ B to denote A Intersection B.

- Example: If A = {3, 4, 6, 8} and B = { 1, 2, 3, 5, 6} then A ∩ B = {3, 6}

- Definition: The **union** of two sets A and B is the set containing those elements which are elements of A **or** elements of B. We write A ∪ B to denote A Union B.

- Example: If A = {3, 4, 6} and B = { 1, 2, 3, 5, 6} then A ∪ B = {1, 2, 3, 4, 5, 6}.

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- Algebraic Properties of Sets:
  - Union and intersection are **commutative** operations. In other words, **A ∪ B = B ∪ A** and **A ∩ B = B ∩ A**
  - Union and intersection are **associative** operations. In other words, **(A ∪ B) ∪ C = A ∪ (B ∪ C)** and **(A ∩ B) ∩ C = B ∩ (A ∩ C)**
    Union and Intersection are **distributive** with respect to each other. In other words **A ∩ (B ∪ C)= (A ∩ B) ∪ (A ∩ C)** and **A ∪ (B ∩ C)= (A ∪ B) ∩ (A ∪ C)**

- A few other elementary properties of intersection and union.
  A ∪ ∅ =A, A ∩ ∅ = ∅, A ∪ A = A, A ∩ A = A.

- Numbering Systems:

  - Counting numbers are called **Natural numbers** and the set of Natural numbers is denoted by **N = {**1,2,3…}
  - **Integers** are Natural numbers, their opposites and zero. The set of integers is denoted by **Z = {**…-3, -2, -1, 0,1,2,3…}
  - **Rational numbers** such as $^{2}/_{3}$, $-3^{1}/_{2}$ 0.3333, are numbers that can be written as a ratio of two integers. The set of rational numbers is denoted by **Q.** This set includes
    - Repeating decimals, terminating decimals and fractions
    - Integers are also rational numbers since every integer *a* can be written as a fraction *a/1*
  - **Irrational numbers** are numbers that can't be written as fractions.
    - 3. 45455455545555… has a pattern but doesn't repeat. It isn't rational. It can't be written like a fraction.
    - Square root of 2, π (Pi) and *e* are also irrational.
  - The Union of the set of rational numbers and the set of irrational numbers is the set of **Real numbers,** denoted by **R**
  - **N ⊂ Z ⊂ Q ⊂ R**

  - Definition: **Cardinality** refers to the number of elements in a set
    - A *finite* set has a countable number of elements
    - An *infinite* set has at least as many elements as the set of *natural numbers*

  - Notation: |A| represents the *cardinality* of Set *A*

  - History: Initially numbers were used for counting and the natural numbers did that job well. However there were no solutions for equations of the form *x + 4 = 0*.

    To resolve this, the natural numbers were extended by inventing the negative integers. This was done by attaching a symbol "-" (which we now call the minus sign) to each natural number and calling the new number the "negative" of the original number. This was further extended to all real numbers.

    Now people had solutions for equations of the form *x + 4 = 0*, but equations of the form $x^2 + 4 = 0$ still had no solutions. There is no real number whose square is -4.

    The numbering system had to be extended once again to accommodate for square roots of negative numbers. A symbol, $= \sqrt{-1}$, was invented and it was called the "imaginary unit". The real numbers were extended by attaching this imaginary unit to each number and calling it the "imaginary copy" of the real numbers.

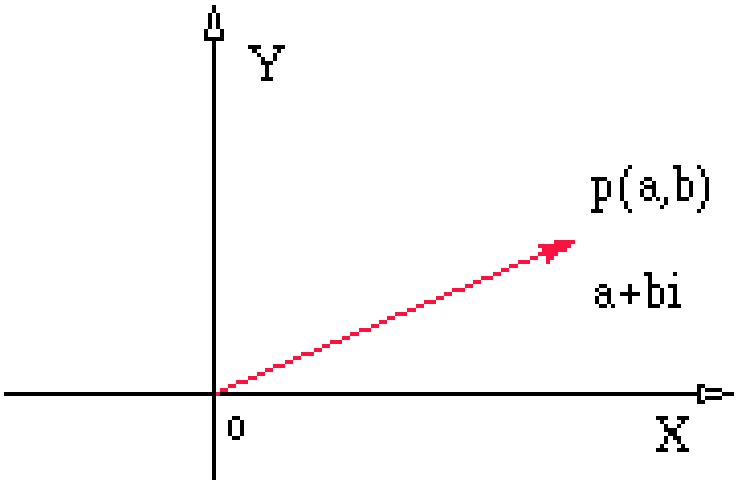
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- Definition: Numbers of the form $a + bi$ are called **complex numbers.**
  - a is the **real** part
  - b is the **imaginary** part

  The set of complex numbers is denoted by **C**

- Examples : 2 - 4i , -3 + 5i and -5 + $^3/_4$i are all complex numbers

- Graphical Representation: Recall that real numbers are represented on a **line.** A complex number has a representation in a **plane**. Simply take the x-axis as the real numbers and y-axis as the imaginary numbers. Thus, giving the complex number $a + bi$ the representation as point P with coordinates (a,b) as the following diagram shows:



- Properties of Complex Numbers:
  - Addition and Subtraction: For complex numbers $a + bi$ and $c + di$,

    $$(a+bi)+(c+di)=(a+c)+(b+d)i$$

    $$(a+bi)-(c+di)=(a-c)+(b-d)i$$
  - Examples
    - $(4 - 6i) + (-3 + 7i) = [4 + (-3)] + [-6 + 7]i = 1 + i$
    - $(10 - 4i) - (5 - 2i) = (10 - 5) + [-4 - (-2)]i = 5 - 2i$
  - Multiplication of Complex Numbers: For complex numbers $a + bi$ and $c + di$,

    $$(a+bi)(c+di)=(ac-bd)+(ad+bc)i.$$
  - The product of two complex numbers is found by multiplying as if the numbers were binomials and using the fact that $i^2 = -1$.
  - Example: $(2 - 4i)(3 + 5i)$

    $$=2(3)+2(5i)-4i(3)-4i(5i)$$

    $$=6+10i-12i-20i^2$$

    $$=6-2i-20(-1)$$

    $$=26-2i$$

- Definition: Given a complex number $z = a + ib$, its **conjugate** is defined as $z^* = a - ib$

- Properties of conjugates:
  - $(z^*)^* = z$
  - $z.z^* = a^2 + b^2$ (real)
  - $z + z^* = 2a$ (real)
  - $z - z^* = 2ib$ (imaginary)

- Division of Complex Numbers: For complex numbers $a + bi$ and $c + di$,

$$\frac{(a+bi)}{(c+di)} = \frac{(a+bi)}{(c+di)} x \frac{(c-di)}{(c-di)}$$

$$= \frac{(ac+bd)}{c^2+d^2} + i\frac{(bc-ad)}{c^2+d^2}$$

The quotient of two complex numbers is found by multiplying and dividing by the conjugate of the denominator.

- $Example: \ \frac{3+11i}{-1-2i} * \frac{-1+2i}{-1+2i} = \frac{(3+11i)(-1+2i)}{(-1-2i)(-1+2i)} = \frac{-3+6i-11i+22i^2}{1-2i+2i-4i^2} = \frac{-3-5i+22(-1)}{1-4(-1)}$

$$= \frac{-3-5i-22}{1+4} = \frac{-25-5i}{5} = \frac{-25}{5} - \frac{5i}{5} = -5-i$$

- Definition: The **absolute value** or **modulus** of a complex number is the distance the complex number is from the origin on the complex plane.

- If you have a complex number $z = (a+bi)$ the absolute value can be found using $|z| = \sqrt{a^2 + b^2}$
- Example: $|-2+5i| = \sqrt{(-2)^2 + (5)^2} = \sqrt{4+25} = \sqrt{29}$

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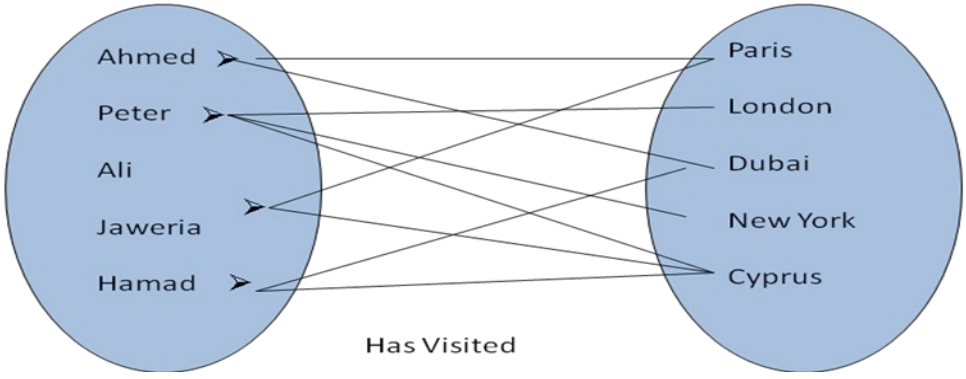
Lecture 4-7

Functions and Quadratics Topic 2

- Definition: A **mapping** between two sets A and B is simply a rule for relating elements of one set to the other. A mapping is also called a **relation**.
- Types of Relations:
  - **One-One Relations** are mappings where each member of the pre-image is mapped to exactly one member of the image.

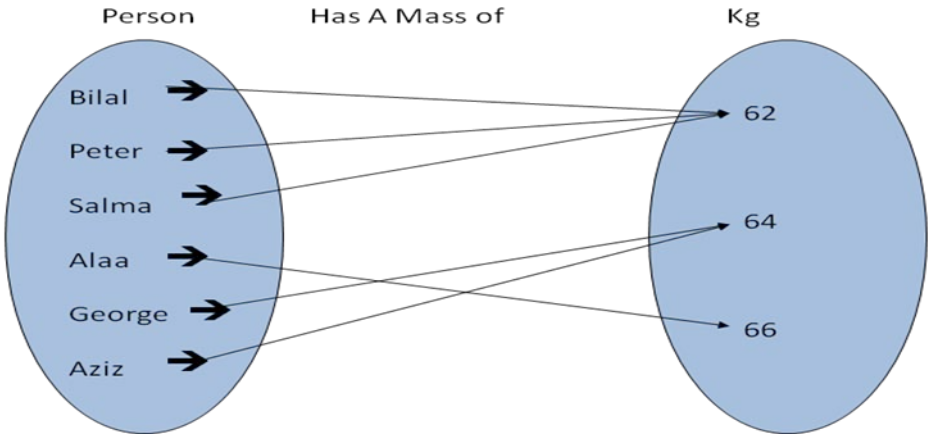


  - **Many – Many Relations** are the mappings where many members of the image are images of more than one member of the pre-image, and members of the pre-image are mapped to more than one image.

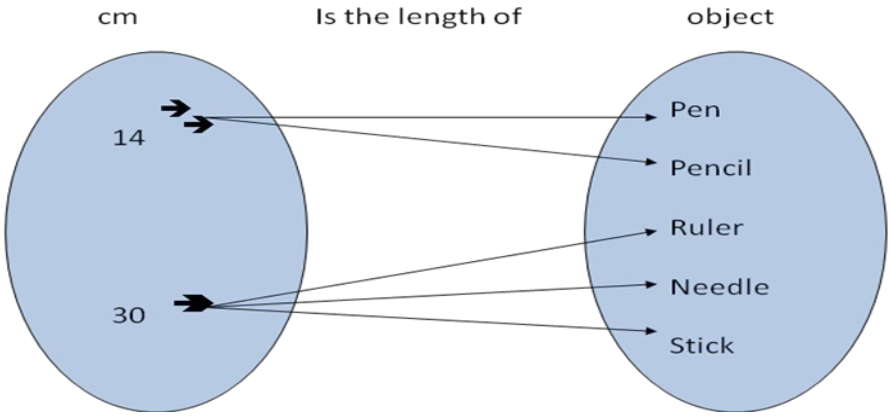


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- **Many - One Relations** are the mappings where two or more members of the pre-image are mapped to exactly one member of the image.



- **One-Many Relations** are mappings where one member of the pre-image is mapped to two or more members of the image.

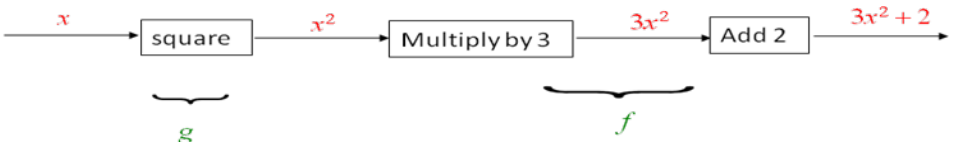


- Definition: Many-One and One-One relationships are called **functions**.
- Definition: The set consisting of members of the pre-image or inputs of a function is called its **domain.** For a given domain the set of possible outcomes or images of a function is called its **range**.

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- It is important to note that to define a function we need two things: One, the formula for the function, and two, the domain.

- Examples:
  - $f(x) = 2x + 1$; Domain: x = R, Range: $f(x)$ = R
  - $g(x) = \frac{1}{x-2}$; Domain: x = R – {2}, Range: $g(x)$ = R
  - $h(x) = \sqrt{x-3}$; Domain: $\{x \in R \mid x \geq 3\}$, Range: $h(x) \geq 0$
  - $q(x) = (x+1)^2 + 2$; Domain: x = R, Range: $q(x) \geq 2$
- Definition: A function is called an **even function** if its graph is symmetric with respect to the vertical axis, and it is called an **odd function** if its graph is symmetric with respect to the origin.
- **Theorem:**
  - If $f(-x) = f(x)$, then f is an even function
  - If $f(-x) = -f(x)$, then f is an odd function
- Example: $f(x) = x^2$ is an even function
- Example: $f(x) = x^3$ is an odd function
- Definition: The **sum**, **difference**, **product** and **quotient** of the functions f and g are the functions defined by
  - $(f + g)(x) = f(x) + g(x)$
  - $(f - g)(x) = f(x) - g(x)$
  - $(fg)(x) = f(x)g(x)$
  - $(f/g)(x) = f(x)/g(x)$, provided $g(x) \neq 0$
- Definition: Given functions f and g, then the function fog is a **composite function**, where g is performed first and then f is performed on the result of g.
- Example: Consider 2 function s $f(x) = 3x + 2$ and $g(x) = x^2$. The function fog may be found using a flow diagram as follows:



Thus $f \circ g = 3x^2 + 2$

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- Remember, the domain of *fog* is the set of all real numbers *x* in the domain of *g* where *g(x)* is in the domain of *f*.

The domain of *fog* cannot always be determined simply by examining the final form of *(fog)(x).* Any numbers that are excluded from the domain of *g* must also be excluded from the domain of *fog*.

- Example: Given $g(x) = \sqrt{3-x}$ and $f(x) = \sqrt{4-x^2}$ find $(f \circ g)(x)$ and its domain.

Solution: Now Domain of *f: -2* $\leq x \leq$ 2 and Domain of *g: x* $\leq$ *3*.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{3-x})$$

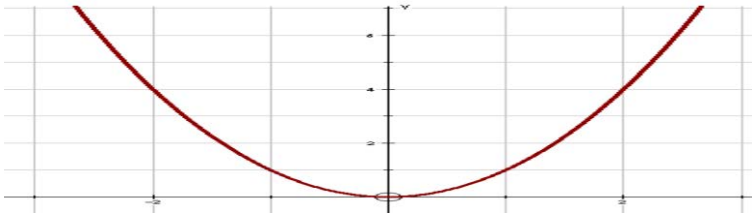
$$= \sqrt{4-(\sqrt{3-x})^2} = \sqrt{4-(3-x)} = \sqrt{1+x}$$

Even though $\sqrt{1+x}$ is defined for all $x \geq -1$, we must restrict the domain of *fog* to those values that are also in the domain of *g*. Thus, Domain *fog*: $-1 \leq x \leq 3$

- Definition: If *f* is a one-one function, then the **inverse** of *f*, denoted by f^{-1}, is the function obtained by reversing the order of *f*. In other words, if *f(a) = b* then $f^{-1}(b) = a$.

- If a function is to have an inverse which is also a function then it must be **one-one**. This means that a horizontal line will never cut the graph more than once; i.e. we cannot have *f(a) = f(b)* if $a \neq b$. Two different inputs (*x* values) are not allowed to give the same output (*y* value).

- Example**:** $f(x) = x^2$ with domain $x \in \Re$ is not one to one. So, for example, the inverse of 4 would have two possibilities: -2 or 2. This means that the inverse is not a function. We say that the inverse function of *f* does not exist. However, if the Domain is restricted to $x \geq 0$, then the function would be one to one and its inverse would be $f^{-1}(x) = \sqrt{x}$, $x \geq 0$



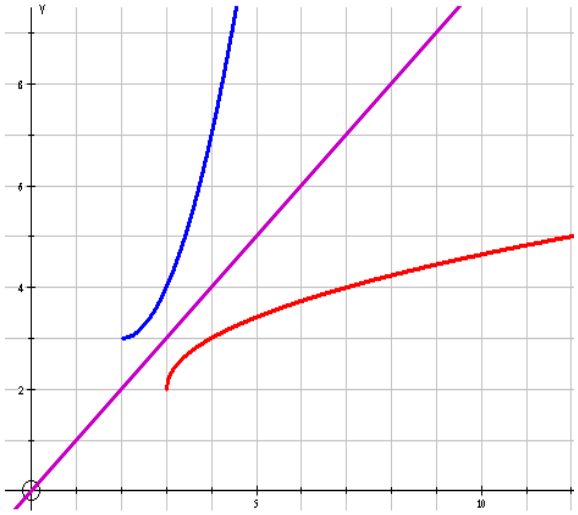
- Properties of Inverse Functions:
  - Domain of the inverse is equal to the Range of *f*.
  - Range of inverse is equal to the Domain of *f*.

- Steps for finding inverse of a function f:
  - Find the domain of *f* and verify that f is one-to-one. If *f* is not one-to-one, then stop as the inverse does not exist.
  - Solve the equation *y = f(x)* for *x*. i.e. make *x* the subject.
  - Interchange *x* and *y* in step two. This will give the inverse function in terms of *x*.
  - Find the domain of the inverse function.
  - Check that the inverse function is correct.

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- Example**:** Find the inverse of the function $y = f(x) = (x-2)^2 + 3$, $x \geq 2$. Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same axes showing the relationship between them.
  Solution
  - **Step 1:** In order for the function to be one to one, we must restrict its domain to $x \geq 2$. The Range of f is $y \geq 3$ and so the domain of f^{-1} will be $x \geq 3$.
  - **Step 2:** Make x the subject. $y - 3 = (x-2)^2$ ➔ $\sqrt{(y-3)} = x-2$ ➔ $x = 2 + \sqrt{(y-3)}$
  - **Step 3:** Interchange x and y in the above equation to get $y = 2 + \sqrt{(x-3)}$. So Final Answer is: $f^{-1}(x) = 2 + \sqrt{(x-3)}$, $x \geq 3$
  - **Step 4:** Verification: $f[f^{-1}(x)] = f[2 + \sqrt{(x-3)}] = \{[2 + \sqrt{(x-3)}] - 2\}^2 + 3 = [\sqrt{(x-3)}]^2 + 3 = (x-3) + 3 = x$. And $f^{-1}[f(x)] = f^{-1}[(x-2)^2 + 3] = 2 + \sqrt{([(x-2)^2 + 3] - 3)} = 2 + \sqrt{(x-2)^2} = 2 + (x-2) = x$

  Graph: Reflect in $y = x$ to get the graph of the inverse function.



- Definition: A function of the type $y = ax^2 + bx + c$ where a, b, and c are called the coefficients, is called a **quadratic function**.
  The graph of a quadratic function will form a **parabola**. Each graph will have either a **maximum** or **minimum** point. There is a **line of symmetry** which will divide the graph into two halves.

- What happens if we change the value of a and c?

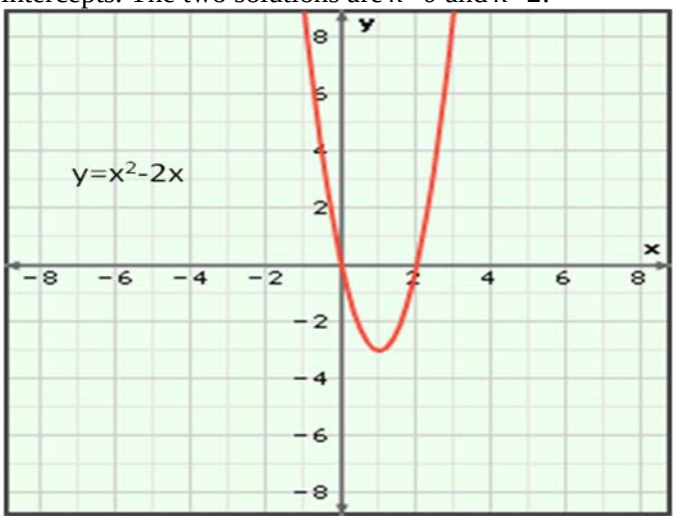
  - When a is positive, the graph concaves downward.
  - When a is negative, the graph concaves upward.
  - When c is positive, the graph moves c units up.

    When c is negative, the graph moves c units down

- Solving Quadratic Equations: Since $y = ax^2 + bx + c$, by setting $y=0$ we set up a quadratic equation. To find the solutions means we need to find the x-intercept. Since the graph is a parabola, there will be at most two solutions.

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- Graphing Method: In this method, we use a scientific calculator and graph the equation. Then we read the *x*-intercepts from the graph.
-
    - Example: $x^2 - 2x = 0$
    - To solve the equation, write $y = x^2 - 2x$ into your graphing calculator. Find the *x*-intercepts. The two solutions are $x=0$ and $x=2$.



- Factorization Method: To solve a quadratic equation we get it in the standard form $y = ax^2 + bx + c$ and see if it will factorize.
    - Example: $x^2 = 5x - 6$ ➔ $x^2 - 5x + 6 = 0$ ➔ $(x-3)(x-2) = 0$
      ➔ $x - 3 = 0$ or $x - 2 = 0$
      ➔ $x = 3$ and $x = 2$

- Completing the Square Method: For this method we need the coefficient of x^2 to be 1. We then divide the take the coefficient of *x* and add and subtract the square of half of the coefficient of *x* from the equation to form a perfect square on one side of the equation.
    - Example: $x^2 + 6x + 3 = 0$. This does not factorize. So we will use the completing the square method here.

      $x^2 + 6x = -3$ The coefficient of *x* is 6. So, the square of 6/2 is 9.
      $x^2 + 6x + \underline{9} = -3 + \underline{9}$. We added 9 to both sides of the equation

      $x^2 + 6x + 9 = 6$. The left side becomes a perfect square.

      $(x+3)^2 = 6$ ➔ $x + 3 = \pm\sqrt{6}$ ➔ $x = -3 \pm \sqrt{6}$

      $x = -3 + \sqrt{6} \approx -0.55$ And $x = -3 - \sqrt{6} \approx -5.45$

- Formula Method: Using the completing the square method, we get the general quadratic formula. Given $ax^2 + bx + c = 0$, the quadratic formula is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

    - Example: $x^2 + 6x + 3 = 0$
      Here, $a = 1$, $b = 6$ and $c = 3$. So, using the formula, we get:

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-6 \pm \sqrt{6^2 - 4(1)(3)}}{2(1)} = \frac{-6 \pm \sqrt{24}}{2} = -3 \pm \sqrt{6}$$

- Note that the expression $b^2 - 4ac$ under the square root – called the **discriminant** - determines how many solutions (if any) the quadratic equation will have.
    - If $b^2 - 4ac > 0$. there will be two distinct real solutions.
    - If $b^2 - 4ac = 0$ there will be exactly one real solution.
    - If $b^2 - 4ac < 0$ there will be no real solutions.
- Examples**:** In each of the following cases determine if the equations has one, two or zero real solutions.
    - $2x^2 + 7x + 4 = 0$
      $b^2 - 4ac = 7^2 - 4(2)(4) = 49 - 32 = 17 > 0$
      Therefore, there are two distinct real solutions
    - $2x^2 + 4x + 2 = 0$
      $b^2 - 4ac = 4^2 - 4(2)(2) = 16 - 16 = 0$
      Therefore, there is exactly one real solution
    - $3x^2 + 4x + 2 = 0$
      $b^2 - 4ac = 4^2 - 4(3)(2) = 16 - 24 = -8 < 0$
      Therefore, there are no real solutions.

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Lecture 8-9

Matrices and Determinants Topic 3

- Definition: A **matrix** is a rectangular arrangement of numbers in rows and columns. The **order** of a matrix is the number of the rows and columns. The **entries** are the numbers in the matrix.

- Notation: A matrix A is written as follows:

$$A=\begin{bmatrix} a_{11}\ a_{12}\ \ldots a_{1n} \\ a_{21}\ a_{22}\ \ldots a_{2n} \\ \ldots\ldots\ldots\ldots \\ a_{m1}\ a_{m2}\ \ldots a_{mn} \end{bmatrix}=\left(a_{ij}\right)$$

where a_{rs} represents the entry in the rth row and sth column. A matrix with m rows and n columns is said to have order m x n.

- Operations on Matrices:

  - Addition of Matrices: To add two matrices, they must have the same order. To add, we simply add corresponding entries.
  - Example:

$$\begin{bmatrix} 5 & -3 \\ -3 & 4 \\ 0 & 7 \end{bmatrix}+\begin{bmatrix} -2 & 1 \\ 3 & 0 \\ 4 & -3 \end{bmatrix}=\begin{bmatrix} 5+(-2) & -3+1 \\ -3+3 & 4+0 \\ 0+4 & 7+(-3) \end{bmatrix}=\begin{bmatrix} 3 & -2 \\ 0 & 4 \\ 4 & 4 \end{bmatrix}$$

  - Subtraction of Matrices: To subtract two matrices, they must have the same order. We simply subtract corresponding entries.
  - Example:

$$\begin{bmatrix} 9 & -2 & 4 \\ 5 & 0 & 6 \\ 1 & 3 & 8 \end{bmatrix}-\begin{bmatrix} 4 & 0 & 7 \\ 1 & 5 & -4 \\ -2 & 3 & 2 \end{bmatrix}=\begin{bmatrix} 9-4 & -2-0 & 4-7 \\ 5-1 & 0-5 & 6-(-4) \\ 1-(-2) & 3-3 & 8-2 \end{bmatrix}$$

$$=\begin{bmatrix} 5 & -2 & -3 \\ 4 & -5 & 10 \\ 3 & 0 & 6 \end{bmatrix}$$

  - Multiplication by a Scalar: In matrix algebra, a real number is often called a **scalar**. To multiply a matrix by a scalar, we multiply each entry in the matrix by that scalar.
  - Example:

$$4\begin{bmatrix} -2 & 0 \\ 4 & -1 \end{bmatrix}=\begin{bmatrix} 4(-2) & 4(0) \\ 4(4) & 4(-1) \end{bmatrix}=\begin{bmatrix} -8 & 0 \\ 16 & -4 \end{bmatrix}$$

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- Matrix Multiplication: Given an *r* x *c* matrix *A* and an *s* x *d* matrix *B*, we can multiply *A* with *B* to form the matrix *AB* if *c = s.* The resulting matrix will have dimension *r* x *d.*
  To multiply two matrices, we multiply each row in the first matrix by each column in the second matrix. An example is shown below:

$$A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad B=\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

  , ,

  We can multiply to form the matrix *C = AB* since the number of columns of *A* is equal to the number of rows of *B.* The multiplication is carried out as follows:

$$C=AB=\begin{bmatrix} a_{11}b_{11}+a_{12}b_{21} & a_{11}b_{12}+a_{12}b_{22} & a_{11}b_{13}+a_{12}b_{23} \\ a_{21}b_{11}+a_{22}b_{21} & a_{21}b_{12}+a_{22}b_{22} & a_{21}b_{13}+a_{22}b_{23} \end{bmatrix}$$

- Example:

$$A=\begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \text{ and } B=\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$C=AB=\begin{bmatrix} (2)(1)+(3)(1)=5 & (2)(1)+(3)(0)=2 & (2)(1)+(3)(2)=8 \\ (1)(1)+(1)(1)=2 & (1)(1)+(1)(0)=1 & (1)(1)+(1)(2)=3 \\ (1)(1)+(0)(1)=1 & (1)(1)+(0)(0)=1 & (1)(1)+(0)(2)=1 \end{bmatrix}=\begin{pmatrix} 5 & 2 & 8 \\ 2 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix}$$

- Definition: A Square matrix with ones on the diagonal and zeros elsewhere is called an **identity** matrix. It is denoted by *I.*

- Example: The *4* x *4* identity matrix is:

$$I=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Definition: Given a square matrix *B,* if there exists a matrix *D* such that *BD = DB = I,* then *D* is called the **inverse** of *B,* and is denoted by $D = B^{-1}$.

- Given a system of equations $\sum_{j=1}^{n} a_{ij}x_j = b_i$, it can be written as a matrix equation $A\underline{x} = \underline{b}$,

  where *A* is the matrix of the coefficients, $\underline{x}$ is the column matrix of the variables and $\underline{b}$ is the column matrix of the constants. i.e.

$$A=\begin{bmatrix} a_{11}\ a_{12} \ldots a_{1n} \\ a_{21}\ a_{22} \ldots a_{2n} \\ \ldots\ldots\ldots\ldots \\ a_{m1}\ a_{m2} \ldots a_{mn} \end{bmatrix}$$

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$$\underline{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \qquad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

,

If A^{-1} exists then the system has a unique solution given by $\underline{x} = A^{-1}\underline{b}$

- Definition: A matrix is in **echelon form** if it has the following properties
  - Every non-zero row begins with a 1 (called a **leading** 1)
  - Every leading one in a lower row is further to the right of the leading one above it.
  - If there are zero rows, they are at the end of the matrix

  A matrix is in **reduced echelon form** if in addition to the above three properties it also has the following property:
  - Every other entry in a column containing a leading one is zero

- Methods for finding Solutions of Equations:
  - Using Row Operations: Recall that when we are solving simultaneous equations, the system of equations remains unchanged if we perform the following operations:

    - Multiply an equation by a non-zero constant
    - Add a multiple of one equation to another equation
    - Interchange two equations.

    We have seen that any system of equations can be written as a matrix system. i.e. the two systems are equivalent.
    So, given a system ***Ax = b*** we can form the **augmented matrix** (***Ab***) by attaching an additional column at the end of the matrix A with entries from matrix b. Since the original system of equations remains unchanged as described above, the system described by the augmented matrix (***Ab***) also remains unchanged under the following row operations:

    - Multiply a row by a non-zero constant
    - Add a multiple of one row to another row
    - Interchange two rows.

    Using row operations, we will change the matrix (***Ab***) to an Echelon form or a reduced Echelon form. Once that is achieved, the solution will be easily found.

    Example: Solve the following system of equations:

    $$x + 2y + z = 1$$
    $$2x + 2y = 2$$
    $$3x + 5y + 4z = 1$$

    Solution: The system can be written in matrix form as:

    $$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 0 \\ 3 & 5 & 4 \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{z} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

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The augmented matrix is

$$(\boldsymbol{Ab}) = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 2 & 0 & 2 \\ 3 & 5 & 4 & 1 \end{bmatrix} \underset{\boldsymbol{R}_3-3\boldsymbol{R}_1}{\overset{\boldsymbol{R}_2-2\boldsymbol{R}_1}{\sim}} \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & -2 & -2 & 0 \\ 0 & -1 & 1 & -2 \end{bmatrix} \overset{\boldsymbol{R}_{23}}{\sim} \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & -1 & 1 & -2 \\ 0 & -2 & -2 & 0 \end{bmatrix}$$

$$\overset{-\boldsymbol{R}_2}{\sim} \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & -2 & -2 & 0 \end{bmatrix} \underset{\boldsymbol{R}_3+2\boldsymbol{R}_2}{\overset{\boldsymbol{R}_1-2\boldsymbol{R}_2}{\sim}} \begin{bmatrix} 1 & 0 & 3 & -3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -4 & 4 \end{bmatrix} \overset{-\boldsymbol{R}_3/4}{\sim} \begin{bmatrix} 1 & 0 & 3 & -3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\underset{\boldsymbol{R}_2+\boldsymbol{R}_3}{\overset{\boldsymbol{R}_1-3\boldsymbol{R}_3}{\sim}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

This gives the solution: $x = 0, y = 1, z = -1$

- Solving Equations using Inverse: If we could find A^{-1}, we could also solve the system by using $\underline{x} = A^{-1}\underline{b}$ One way to find inverse is as follows:

- **Theorem**: Given a *n*x*n* matrix *A*, if the augmented matrix (*AI*), where *I* is the *n*x*n* identity matrix, can be row reduced to a matrix (*IB*), then *B* is the inverse of *A*. If (*AI*) cannot be reduced to (*IB*), then A does not have an inverse.

- Example: Solve the system of equations:

$$2x - 2y + 2z = 1$$
$$2y - z = 1$$
$$2x + 3y = -1$$

Solution: The matrix *A* of the coefficients is

$$\begin{bmatrix} 2 & -2 & 2 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{bmatrix}$$

To find A^{-1} we use the above theorem

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$$(\boldsymbol{AI}) = \begin{bmatrix} 2 & -2 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 2 & 3 & 0 & 0 & 0 & 1 \end{bmatrix} \overset{R_3 - R_1}{\sim} \begin{bmatrix} 2 & -2 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 0 & 5 & -2 & -1 & 0 & 1 \end{bmatrix}$$

$$\overset{\substack{R_1/2 \\ \sim \\ R_{23}}}{\begin{bmatrix} 1 & -1 & 1 & 1/2 & 0 & 0 \\ 0 & 5 & -2 & -1 & 0 & 1 \\ 0 & 2 & -1 & 0 & 1 & 0 \end{bmatrix}} \overset{R_2 - 2R_3}{\sim} \begin{bmatrix} 1 & -1 & 1 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 1 \\ 0 & 2 & -1 & 0 & 1 & 0 \end{bmatrix}$$

$$\overset{\substack{R_1 + R_2 \\ \sim \\ R_3 - 2R_2}}{\begin{bmatrix} 1 & 0 & 1 & -1/2 & -2 & 1 \\ 0 & 1 & 0 & -1 & -2 & 1 \\ 0 & 0 & -1 & 2 & 5 & -2 \end{bmatrix}} \overset{\substack{R_1 + R_3 \\ \sim \\ -R_3}}{\begin{bmatrix} 1 & 0 & 0 & 3/2 & 3 & -1 \\ 0 & 1 & 0 & -1 & -2 & 1 \\ 0 & 0 & 1 & -2 & -5 & 2 \end{bmatrix}}$$

So,

$$\boldsymbol{A}^{-1} = \begin{bmatrix} 3/2 & 3 & -1 \\ -1 & -2 & 1 \\ -2 & -5 & 2 \end{bmatrix}$$

This gives the solution

$$\underline{\boldsymbol{x}} = \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{z} \end{pmatrix} = \boldsymbol{A}^{-1}\underline{\boldsymbol{b}} = \begin{bmatrix} 3/2 & 3 & -1 \\ -1 & -2 & 1 \\ -2 & -5 & 2 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3/2 + 3 + 1 \\ -1 - 2 - 1 \\ -2 - 5 - 2 \end{pmatrix} = \begin{pmatrix} 11/2 \\ -4 \\ -9 \end{pmatrix}$$

- Definition: Let M be the set of all square matrices. Then the **Determinant** is a function from M to the set of real numbers. i.e. the determinant is a process of attaching a real number to every square matrix.
- Notation: If A is a square matrix, then determinant of A is denoted by **detA** or **|A|**
- Example**:** $\det\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$
- Second Order Determinant**:** A 2 x 2 determinant can be found using the following method:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{22}$$

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- Example: $\begin{vmatrix} -1 & 2 \\ -3 & -4 \end{vmatrix} = (-1)(-4) - (-3)(2) = 4 + 6 = 10$

- Definition: The **Minor of an element** in a third-order determinant is a second-order determinant obtained by deleting the row and column that contains the element.

- Example**:** Given $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

  The Minor of $a_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$

  The Minor of $a_{32} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$

- Definition: The **Cofactor** of $a_{ij} = (-1)^{i+j}$ (Minor of a_{ij})

- The cofactor of a_{ij} is denoted by C_{ij}

- Thus, the cofactor of an element is nothing more than a signed minor.

- Example**:** Given $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

  The Cofactor of $a_{23} = C_{23} = (-1)^{2+3} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$

  The Cofactor of $a_{11} = C_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

- Theorem: The value of a determinant of order 3 is the sum of three products obtained by multiplying each element of any one row (or each element of any one column) by its cofactors.

- ***Note***: The above theorem and definitions of minor and cofactor generalize completely for determinants of order higher than 3

- Example**:** Evaluate $\begin{vmatrix} 2 & -2 & 0 \\ -3 & 1 & 2 \\ 1 & -3 & -1 \end{vmatrix}$

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Solution**:** We expand by the first row to get:

$$\begin{vmatrix} 2 & -2 & 0 \\ -3 & 1 & 2 \\ 1 & -3 & -1 \end{vmatrix} =$$

$$2\left[(-1)^{1+1}\begin{vmatrix} 1 & 2 \\ -3 & -1 \end{vmatrix}\right] + (-2)\left[(-1)^{1+2}\begin{vmatrix} -3 & 2 \\ 1 & -1 \end{vmatrix}\right] + (0)\left[(-1)^{1+1}\begin{vmatrix} -3 & 1 \\ 1 & -3 \end{vmatrix}\right] =$$

$$(2)(1)[(1)(-1)-(-3)(2)] + (-2)(-1)[(-3)(-1)-(1)(2)] + 0 =$$

$$(2)(5) + (2)(1) = 12$$

- Finding Inverse Using Determinants
- Definition: Given a matrix *A*, the **transpose** of *A* denoted A^{T} is the matrix obtained by interchanging the rows of *A* with its columns.
- Example: The transpose of $A = \begin{bmatrix} 8 & -1 & 3 \\ 0 & 0 & 2 \\ 10 & 4 & -3 \end{bmatrix}$ is $A^T = \begin{bmatrix} 8 & 0 & 10 \\ -1 & 0 & 4 \\ 3 & 2 & -3 \end{bmatrix}$
- Definition: Given a matrix *A*, calculate all the cofactors of *A*. We then form the matrix (C_{ij}) of the cofactors. The **Adjoint** or **Adjugate** of *A* is the transpose of the matrix of the cofactors. i.e. $adj(A) = (C_{ij})^T = (C_{ji})$
- The inverse of A is then found by the formula: $A^{-1} = \frac{1}{|A|} adj(A)$
- Example: Find the inverse of the following matrix using determinant.

$$\text{A} = \begin{bmatrix} -2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

Solution: We need to first find all the cofactors.

$$C_{11} = (-1)^{1+1}\begin{vmatrix} -1 & 1 \\ 2 & 0 \end{vmatrix} = -2 \qquad C_{12} = (-1)^{1+2}\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1 \qquad C_{13} = (-1)^{1+3}\begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} = 1$$

$$C_{21} = (-1)^{2+1}\begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = 6 \qquad C_{22} = (-1)^{2+2}\begin{vmatrix} -2 & 3 \\ 1 & 0 \end{vmatrix} = -3 \qquad C_{23} = (-1)^{2+3}\begin{vmatrix} -2 & 1 \\ 1 & 2 \end{vmatrix} = 5$$

$$C_{31} = (-1)^{3+1}\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4 \qquad C_{32} = (-1)^{3+2}\begin{vmatrix} -2 & 3 \\ 0 & 1 \end{vmatrix} = 2 \qquad C_{33} = (-1)^{3+3}\begin{vmatrix} -2 & 1 \\ 0 & -1 \end{vmatrix} = 2$$

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Now, $adj(A) = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \begin{bmatrix} -2 & 6 & 4 \\ 1 & -3 & 2 \\ 1 & 5 & 2 \end{bmatrix}$. And

$$\det(A) = \begin{vmatrix} -2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix} = -2\begin{vmatrix} -1 & 1 \\ 2 & 0 \end{vmatrix} - 1\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + 3\begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} = -2(-2) - 1(-1) + 3(1) = 8$$

Therefore, $A^{-1} = \frac{1}{\det(A)} adj(A) = \frac{1}{8}\begin{bmatrix} -2 & 6 & 4 \\ 1 & -3 & 2 \\ 1 & 5 & 2 \end{bmatrix}$

- Cramer's Rule for Solving Systems of Equations: Suppose we are given a system of 2 equations in 2 unknowns as follows:

  $a_{11}x_1 + a_{12}x_2 = b_1$
  $a_{21}x_1 + a_{22}x_2 = b_2$

  Then, the solution is given by: $x_1 = \frac{\Delta_1}{\Delta}$ and $x_2 = \frac{\Delta_2}{\Delta}$, where $\Delta = \det(A)$, $\Delta_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$

  Similarly, for a system involving three unknowns and three equations:

  $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$
  $a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$
  $a_{31}x_1 + a_{32}x_2 + a_{33}x_2 = b_3$

  The solution is given by $x_1 = \frac{\Delta_1}{\Delta}$, $x_2 = \frac{\Delta_2}{\Delta}$ and $x_3 = \frac{\Delta_3}{\Delta}$, where $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$,

  $\Delta_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$ and $\Delta_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$

- Example: Solve using Cramer's Rule:
  $x_1 + 2x_2 + x_3 = 1$
  $2x_1 + 2x_2 = 2$
  $3x_1 + 5x_2 + 4x_3 = 1$

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Solution:

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 0 \\ 3 & 5 & 4 \end{vmatrix} = 1\begin{vmatrix} 2 & 0 \\ 5 & 4 \end{vmatrix} - 2\begin{vmatrix} 2 & 0 \\ 3 & 4 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 3 & 5 \end{vmatrix},$$

$$= 1(8) - 2(8) + 1(4) = 8 - 16 + 4 = -4$$

$$\Delta_1 = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 5 & 4 \end{vmatrix} = 1\begin{vmatrix} 2 & 0 \\ 5 & 4 \end{vmatrix} - 2\begin{vmatrix} 2 & 0 \\ 1 & 4 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 1 & 5 \end{vmatrix},$$

$$= 1(8) - 2(8) + 1(8) = 8 - 16 + 8 = 0$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 0 \\ 3 & 1 & 4 \end{vmatrix} = 1\begin{vmatrix} 2 & 0 \\ 1 & 4 \end{vmatrix} - 1\begin{vmatrix} 2 & 0 \\ 3 & 4 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix}, \text{ and}$$

$$= 1(8) - 1(8) + 1(-4) = 8 - 8 - 4 = -4$$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 2 \\ 3 & 5 & 1 \end{vmatrix} = 1\begin{vmatrix} 2 & 2 \\ 5 & 1 \end{vmatrix} - 2\begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 3 & 5 \end{vmatrix}$$

$$= 1(-8) - 2(-4) + 1(4) = -8 + 8 + 4 = 4$$

So, by Cramer's Rule: $x_1 = \frac{\Delta_1}{\Delta} = \frac{0}{-4} = 0$, $x_2 = \frac{\Delta_2}{\Delta} = \frac{-4}{-4} = 1$, $x_3 = \frac{\Delta_3}{\Delta} = \frac{4}{-4} = -1$.

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Lecture 10-12

Sequences and Series Topic 4

- Definition: Rows of numbers are called **sequences**, and the separate numbers are called **terms** of the sequence.

- Notation: Usually the terms of the sequence are denoted by
  a_1 = First term of the sequence
  a_2 = second term of the sequence
  …
  a_n = nth term of the sequence

- Definition: An **Arithmetic Sequence** (or **Arithmetic Progression**) is a sequence in which each term after the first term is found by adding a constant, called the common difference (d), to the previous term.

- The inductive definition of an arithmetic sequence has the form
  $a_1 = a$ and $a_n = a_{n-1} + d$

- Formula: The formula for finding any term in an arithmetic sequence is $\boldsymbol{a_n = a + (n-1)d.}$

  All you need to know to find any term is the first term in the sequence (a) and the common difference, d.

- Example: Given the sequence, 1500, 3000, 4500, 6000, …, find the 16th term.

  Solution: $a = \boldsymbol{1500}$, $d = \boldsymbol{1500}$, $n - 1 = 16 - 1 = \boldsymbol{15}$.

  So $a_{16} = 1500 + 1500(15) = 24000$

- Definition**:** An **Arithmetic Series** is the sum of the terms in an arithmetic sequence.

- Example**:** Arithmetic sequence: 2, 4, 6, 8, 10,

  Corresponding arithmetic series: 2 + 4 + 6 + 8 + 10

- To find the sum of the first n terms on arithmetic sequence, we can use the formula:

  $\boldsymbol{S_n = {}^{n}/_{2}(a + a_n) = {}^{n}/_{2}(a + a + (n-1)d) = {}^{n}/_{2}(2a + (n-1)d)}$

- Example: Find the sum of the first 50 terms of an arithmetic series with $a = 28$ and $d = -4$

  Solution: We need to know n, a, and a_{50}. $n = 50$, $a = 28$, $d = -4$. The formula is $S_n = {}^{n}/_{2}(2a + (n-1)d)$ So, $S_{50} = {}^{50}/_{2}(2(28) + (50-1)(-4)) = 25(56 + 49\ (-4)) = 25(56 - 196) = 25(-140) = -3500$

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- Definition: A sequence in which each term after the first is found by multiplying the previous term by a constant value called the common ratio, is called a **Geometric Sequence** (or **Geometric Progression**).

- The formula for finding any term of a geometric sequence is $a_n = ar^{n-1}$

- Example: Find the 10th term of the geometric sequence with $a = 2000$ and a common ratio of $^1/_2$.
- Solution: $a_{10} = 2000 \bullet (^1/_2)^9 = 2000 \bullet {}^1/_{512} = {}^{2000}/_{512} = {}^{500}/_{128} = {}^{250}/_{64} = {}^{\mathbf{125}}/_{\mathbf{32}}$

- Definition**:** A **Geometric Series** is the sum of the terms in a arithmetic sequence.

- The formula for fining the sum of the first n terms of a geometric sequence is given by

$$S_n = \frac{a(1-r^n)}{1-r}$$

- Example: Find the sum $\sum_{n=1}^{4} -3(2)^{n-1}$

  Solution: This is a geometric series with first term -3 and common ratio 2.

$$S_4 = \frac{-3(1-2^4)}{1-2} = \frac{-3(1-16)}{-1} = -45$$

- Definition: If a sequence of numbers approaches (or converges) to a finite number, we say that the sequence is **convergent.** If a sequence does not converge to a finite number it is called divergent.

- For the sequence of the geometric series, we know that $S_n = \frac{a(1-r^n)}{1-r}$ and it is the expression r^n that determines if the series converges or diverges: If $|r| > 1$ then $|r^n|$ increases indefinitely and the series is divergent. However, if $|r| < 1$ then $|r^n|$ tends to zero as n tends to infinity. So the sum tends to $S_n = \frac{a(1-0)}{1-r} = \frac{a}{1-r}$. It is conventional to write $S_\infty = \frac{a}{1-r}$

- Example: Evaluate $\sum_{n=1}^{\infty} 35\left(-\frac{1}{4}\right)^{n-1}$

  Solution: This is an infinite geometric series with $a = 35$ and $r = -1/4$. So,

$$S_\infty = \frac{35}{1-(-1/4)} = \frac{35}{(5/4)} = 28$$

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Lecture 13 - 14

Permutations and Combinations

- Multiplication Principle: If two operations A and B are performed in order, with *n* possible outcomes for A and *m* possible outcomes for B, then there are n x m possible combined outcomes of the first operation followed by the second.
- Example: Basket A contains a mango (m) and a banana (b). Basket B contains an apple (a), an orange (o) and a grapefruit (g). You draw one fruit from A and then one fruit from B. How many different pairs of fruits can you have?

  Solution: ma, mo, mg, ba, bo, bg = 6. Or, (2 from A) x (3 from B) = 6 according to the multiplication principle.
- Difference between permutations and combinations: Both are ways to count the possibilities. The difference between them is whether order matters or not. Consider a poker hand:

  A♦, 5♥, 7♣, 10♠, K♠

  Is that the same hand as:

  K♠, 10♠, 7♣, 5♥, A♦

  Does the order the cards are handed out matter?

  - If yes, then we are dealing with permutations
  - If no, then we are dealing with combinations
- Permutation Rules:
  - To arrange *n* distinct objects in a row, the number of different arrangements is *n! = n(n-1)(n-2)…3.2.1*
  - The number of different permutations of *r* objects which can be made from *n* distinct objects is given by
  - The number of different permutations of *n* objects of which *p* are identical to each other, and then *q* of the remainder are identical, and *r* of the remainder are identical is
  - To arrange *n* objects in a line in which *r* of the objects have to be together we have ***r!(n – r + 1)!*** Permutations
  - To arranging *n* objects in a circle in which arrangements are considered to be the same if they can be obtained from each other by rotation, we get ***n!/n = (n – 1)!*** Permutations.

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- Example: Suppose you have 4 pictured cards that have the pictures of the letters A, B, C and D, and you want to arrange them in a row to form "words". How many 4-letter words are there?

  Solution: here we are arranging four distinct objects in a line. The number of permutations is 4! = 24

- Example: Eight runners are hoping to take part in a race, but the track has only six lanes. In how many ways can six of the eight runners be assigned to lanes.

  Solution: This is a permutation of six lane assignments from 8 people:

  $${}^{8}P_{6} = \frac{8!}{(8-6)!} = \frac{8!}{2!} = \frac{8.7.6.5.4.3.2.1}{2.1} = 20160$$

- Example: Find the number of distinct permutations of the letters of the word MISSISSIPPI

  Solution: The total number of letters = 11, Number of S's = 4, Number of I's = 4, Number of P's = 2. The total number of distinct permutations is $\frac{11!}{4!4!2!} = 34650$

- Example: Five people, A, B, C, D and E are arranged randomly in a line. Find the possible permutations when A and B are next to each other

  Solution: Imagine A and B are stuck together in the order AB. Treat them as one unit. Then there are 4 unites to permute (AB, C, D and E) in a line and we know there are 4! = 24 ways to arrange 4 units. But A and B could also be stuck together in the order BA, and there will be another 4! arrangements in that case. Therefore, there will be a total of 2 x 4! = 48 arrangements of the 5 people in the line where A and B are always together.

- Example: How many ways are there to sit 6 people around a circular table, where seatings are considered to be the same if they can be obtained from each other by rotating the table?

  Solution: First, place the first person in the north-most chair. This has only one possibility. Then place the other 5 people. There are 5P5 = 5! = 120 ways to do that By the product rule, we get 1x120 =120.

  Alternatively, there are 6P6 = 720 ways to seat the 6 people around the table. For each seating, there are 6 "rotations" of the seating. Thus, the final answer is 720/6 = 120

- Formula for Combinations: In general, to find the number of **combinations** of *r* objects taken from *n* objects, we divide the number of permutations *nPr* by *r!*. The total number of combinations is given by: $${}^{n}C_{r} = \binom{n}{r} = \frac{{}^{n}P_{r}}{r!} = \frac{n!}{r!(n-r)!}$$

- Example: The manager of a football team has a squad of 16 players. He needs to choose 11 to play in a match. How many possible teams can be chosen?

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Solution: This is a combination problem as the order in which the teams are chosen is not important. The number of combinations is ${}^{16}C_{11} = \frac{16!}{11!5!} = 4368$

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Lecture 15

Binomial Theorem

- In mathematics we are always looking for generalization of ideas. For example, we know the formula for

  $(x + y)^2 = x^2 + 2xy + y^2$

  $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$

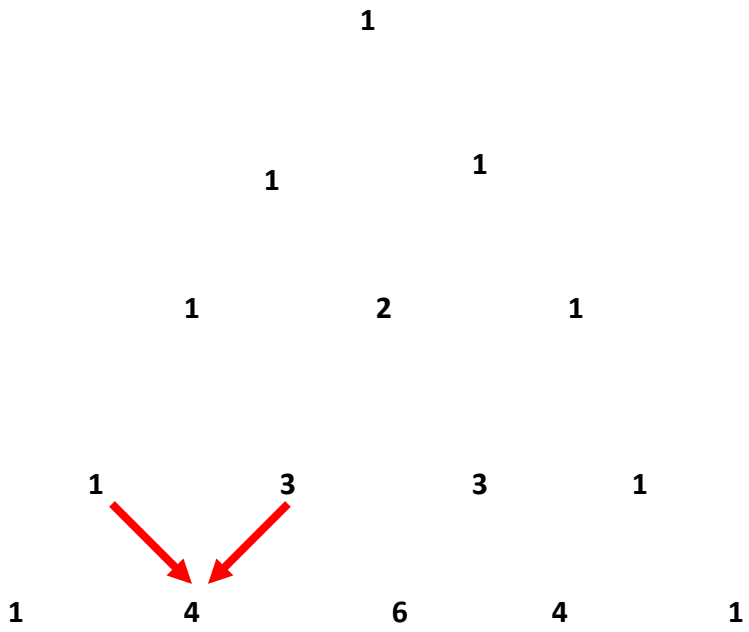
  But it is hard to remember formulas for higher powers. We need a mechanism that would help us expand $(x + y)^n$ for any values of n

- Pascal's Triangle:

Expression	Coefficients
$(x + y)^1 = x + y$	1 1
$(x + y)^2 = x^2 + 2xy + y^2$	1 2 1
$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$	1 3 3 1
$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$	1 4 6 4 1
$(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$	1 5 10 10 5 1

  Each value inside the triangle is obtained by adding the two values above it

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- Example: Expand $(a + x)^6$

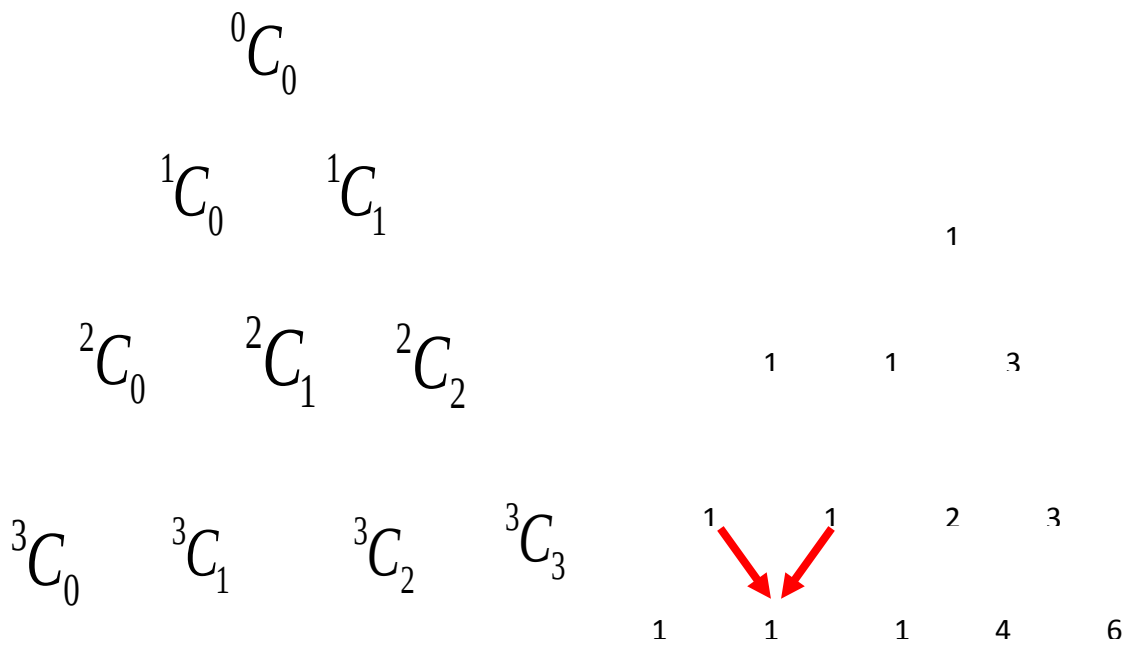
  Solution: Remember the 5th row of the Pascal's triangle: 1 5 10 10 5 1

  Coefficients of the 6th row will be will be: 1, 6, 15, 20, 15, 6 and 1

  So expansion will be $(a + x)^6 = a^6 + 6a^5x + 15a^4x^2 + 20a^3x^3 + 15a^2x^4 + 6ax^5 + x^6$

- We note that the Pascal's triangle is inefficient in finding the coefficient of big expansions such as the coefficient of x^9y^6 in the expansion of $(x + y)^{15}$. What is needed is a formula in terms of n and r for the coefficient of $x^{n-r}y^r$ in the expansion of $(x + y)^n$. We note the following connection between the binomial coefficients and the Pascal's triangle:

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The n-th row is ^nC_k , $k = 0, 1, \ldots, n$

- Binomial Theorem: If n is a natural number, then

$$(x+y)^n = {}^nC_0\,x^n + {}^nC_1\,x^{n-1}y + {}^nC_2\,x^{n-2}y^2 + {}^nC_3\,x^{n-3}y^3 + {}^nC_4\,x^{n-4}y^4 + \ldots + {}^nC_n\,y^n = \sum_{k=0}^{n} {}^nC_k\,x^{n-k}y^k$$

The binomial coefficients are given by $^nC_r = \dfrac{n!}{r!(n-r)!}$

- Example: Calculate the coefficient of $x^{11}y^4$ in the expansion of $(x+y)^{15}$

Solution: By the binomial theorem the coefficient is given by

$$^{15}C_4 = \frac{15!}{4!11!} = \frac{15.14.13.12}{4.3.2.1} = 1365$$

- Example: Use the Binomial Theorem to calculate 1.01^8

Solution: $1.01^8 = (1 + 0.01)^8 = 1 + 8(0.01) + 28(0.01)^2 + 56(0.01)^3 + 70(0.01)^4 + 56(0.01)^5 + 28(0.01)^6 + 8(0.01)^7 + (0.01)^8 = 1 + .08 + .0028 + .000056 + .00000070 + .0000000056 + .000000000028 + .00000000000008 + .0000000000000001 = 1.0828567056280801.$

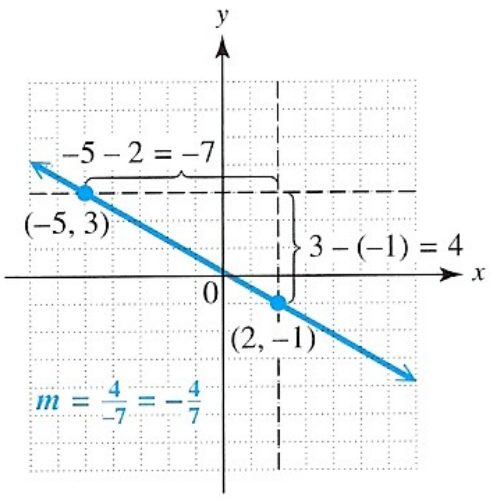
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Lecture 16-17

Coordinate Geometry

- Distance Between Two Points: Given any two points $A(x_1, y_1)$ and $B(x_2, y_2)$, the distance between them is given by $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Example: Given points $A(2, 3)$ and $B(5, 7)$, find the distance between A and B.
  Solution: $AB = \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{(3)^2 + (4)^2} = 5$
- Mid-Point between two points: Given any two points $A(x_1, y_1)$ and $B(x_2, y_2)$, the mid-point M has coordinates $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$
- Example: Given points $A(4, 3)$ and $B(10, 7)$. Let M be the mid-point of AB. Find the coordinates of M
  Solution: $M = (\frac{4+10}{2}, \frac{3+7}{2}) = (7, 5)$
- Definition: Given any two points $A(x_1, y_1)$ and $B(x_2, y_2)$, the **gradient** or **slope** of the line segment joining A and B is the ratio of the change in y with respect to the change in x. It is denoted by m, and is defined as $m = \frac{y_2 - y_1}{x_2 - x_1}$.
- Properties of Gradient:
  - The bigger the gradient's magnitude is, the steeper the line segment.
  - Negative gradient means line is facing downwards.
  - Positive gradient means the line is facing upwards.
  - The slope gives the average rate of change in y per unit change in x, where the value of y depends on x.
  - Two line segments that are parallel will have the same slope.
- Example: Find the slope of the line through the points (2,-1) and (-5,3)
  Solution: $\text{slope} = m = \frac{rise}{run} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-(-1)}{(-5)-2} = \frac{4}{-7} = -\frac{4}{7}$

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- Equation of a Line: The equation of a line or a curve is a rule for determining whether or not the point with coordinates (x, y) lies on the line or curve. The equation of a line through a fixed point (x_1, y_1) with gradient m is given by

$$\frac{y-y_1}{x-x_1}=m$$

$$\Leftrightarrow y-y_1=m(x-x_1)$$

$$\Leftrightarrow y=mx+y_1-mx_1$$

$$\Leftrightarrow y=mx+c$$

where $c = y_1 - mx_1$

- The last form of the line is called the **slope-intercept** form of a straight line where m is the slope and c is the y-intercept.

- Example: Find the slope of the line given $3x - 4y = 12$

  Solution: Writing the equation in the slope-intercept form, we get:

$$3x-4y=12$$

$$-4y=-3x+12$$

$$y=\frac{3}{4}x-3$$

  Therefore, the slope is ¾.

- Two Lines in a Plane: In a plane, two lines either intersect or are parallel. If the gradients are the same, the lines are parallel and do not intersect. If the gradients are not the same, the lines intersect.

- Example: Given the equations of the lines $2x - y = 4$ and $3x + 2y = -1$. Do these lines intersect? If they do, find the point of intersection.

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Solution: Since these two lines have different gradients, they must intersect. To find the point of intersection we need *(x, y)* which lie on both the lines, i.e. which satisfy the two equations simultaneously. We need to solve the equations simultaneously. Solving the equations simultaneously we get $x = 1$ and $y = -2$.

- Perpendicular Lines: If a line has gradient *m*, then the gradient of a line perpendicular to this line is *-1/m.* Two lines with gradients m_1 and m_2 are perpendicular if $m_1.m_2 = -1$, *or* $m_1 = -1/m_2$

- Example: Are the lines $3x + 5y = 6$ and $5x - 3y = 2$ parallel, perpendicular, or neither?

  Solution: For Line 1: $x + 5y = 6 \Leftrightarrow 5y = -3x + 6 \Leftrightarrow y = -\frac{3}{5}x + \frac{6}{5}$

  So, its slope is -3/5.

  For Line 2: $5x - 3y = 2 \Leftrightarrow -3y = -5x + 2 \Leftrightarrow y = \frac{5}{3}x - \frac{2}{3}$

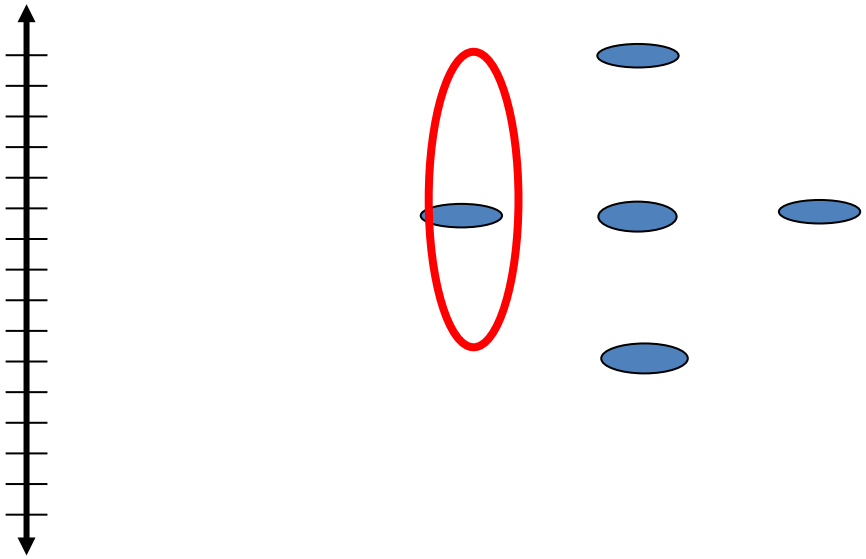
  So, its slope is 5/3, which is the negative reciprocal of the slope of the first line. Therefore, the two lines are perpendicular.

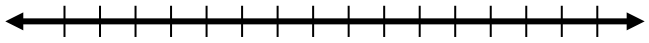
  - Equation of a Circle: The equation of a circle with center *(h, k)* and radius *r* in standard form is: $(x - h)^2 + (y - k)^2 = r^2$

  - Example: Identify the center and radius of the given circle and sketch the graph:
    $(x+4)^2 + (y-3)^2 = 25$

    Solution: Comparing the given equation with standard form, we see that its center is (-4, 3) and radius is 5. The graph is as follows:

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- Example: Find the center and radius of the circle with equation $x^2 + y^2 + 6x - 4y = 23$

  Solution: We transform the equation into the form $(x - h)^2 + (y - k)^2 = r^2$ by completing the square relative to x and relative to y. Then from this standard form we can determine the center and radius.

  We want to write $x^2 + y^2 + 6x - 4y = 23$ as
  $(x^2 + 6x \quad) + (y^2 - 4y \quad) = 23$
  $(x^2 + 6x + \mathbf{9}) + (y^2 - 4y + \mathbf{4}) = 23 + \mathbf{9} + \mathbf{4}$
  $(x+3)^2 + (y-2)^2 = 36$
  $(x - (-3))^2 + (y-2)^2 = 6^2$

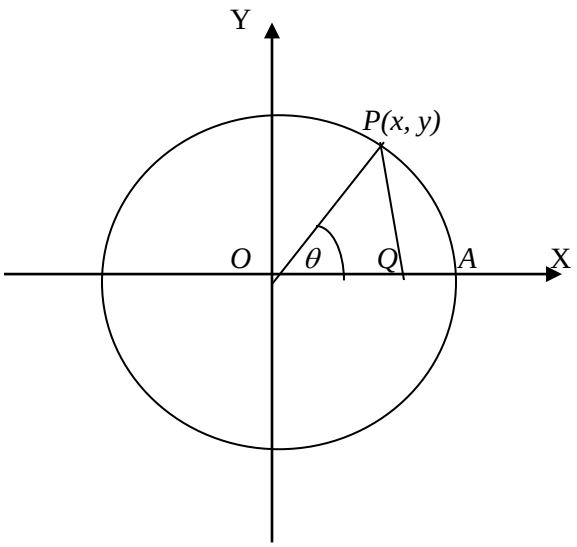
  The center is $(-3, 2)$ and the radius is 6

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Lecture 18-22

Trigonometry

- Basic Functions:
  - Cosine Function: Consider a circle of radius 1.



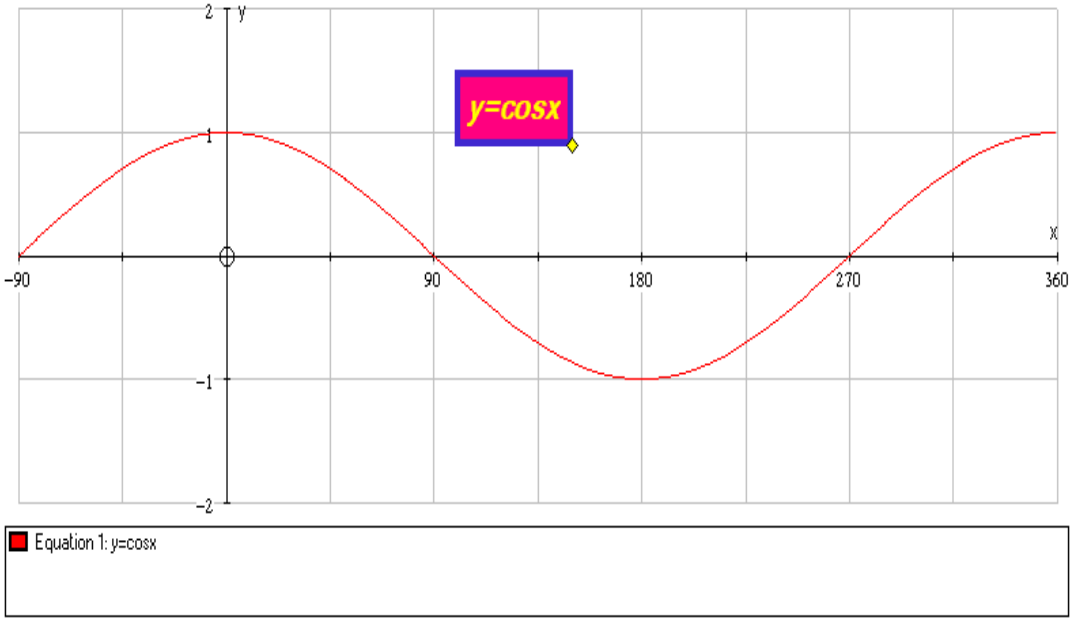
Let $P(x, y)$ be any point making an angle θ with the horizontal x-axis.
Then $Cos\theta = OQ/OP = x/1 = x$
i.e. the value of $Cos\theta$ is the x-coordinate of P as P travels along the circumference of the circle, starting from the point A.

- Properties of the Cosine Function:
  - The cosine function has "period" 360° as it repeats itself after each revolution of 360°
  - $Cos(-\theta) = Cos\theta$ as the x-coordinate of P doesn't change when we reflect across the x-axis.
  - $Cos(180 - \theta) = -Cos\theta$ as the x-coordinate changes signs when reflected across the y-axis.
  - $Cos(\theta - 180) = -Cos\theta$ as the x-coordinate changes signs when reflected across the origin.
  - $Cos\theta$ is positive in the first and the fourth quadrant (as the x coordinate of P is positive), and negative in the second and the third quadrant as the x-axis is negative there.
  - The range of the cosine function is between -1 and 1. The maximum value of 1 is taken when $\theta = 0^o, \pm 360^o, \pm 720^o, \ldots$, and the minimum value of -1 is at $\theta = \pm 180^o, \pm 540^o, \ldots$,

- Definition: The functions with the property that they keep repeating themselves are called **periodic functions**. The smallest interval for which the function repeats itself is called its **period**.

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- Graph of the Cosine Function:

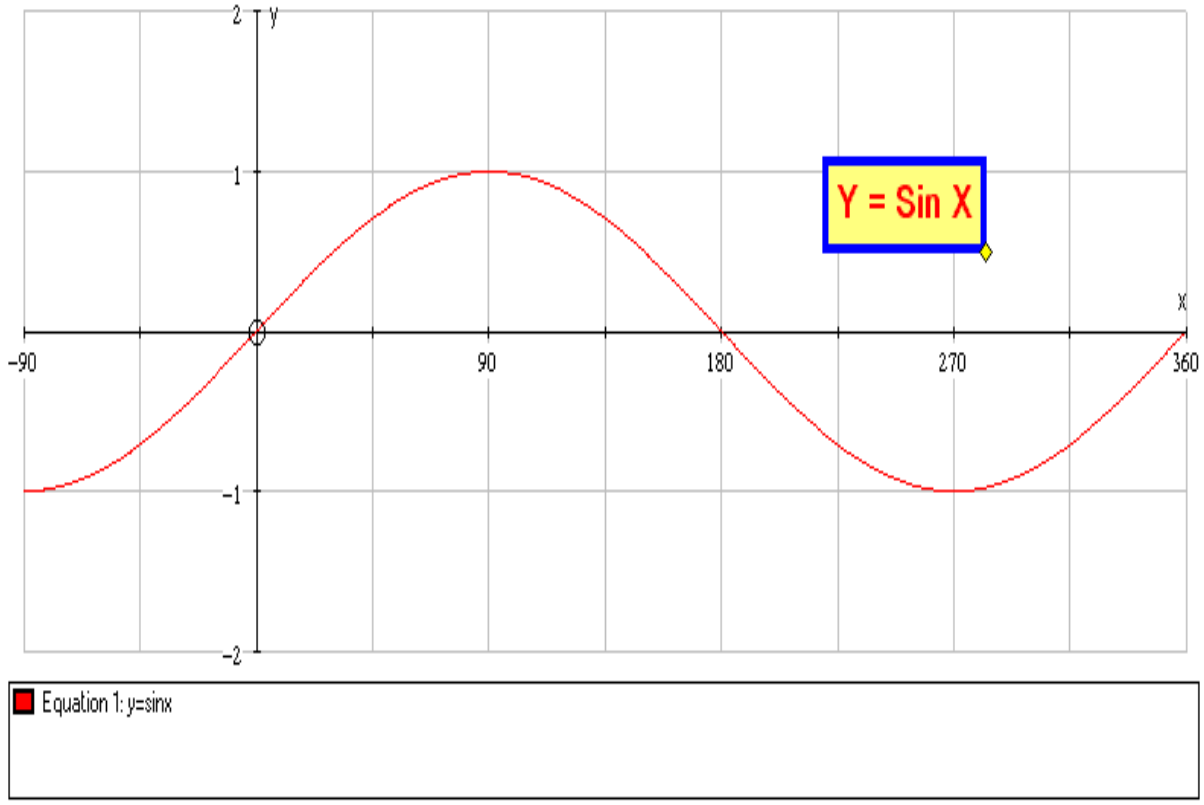


- Sine Function: Using the same diagram as before, we consider a circle of radius 1. Let P(x, y) be any point making an angle θ with the horizontal x-axis. Then Sinθ = PQ/OP = y/1 = y
  i.e. the value of Sinθ is the y-coordinate of P as P travels along the circumference of the circle, starting from the point A.

- Properties of the Sine Function:
  - Sin(-θ) = -Sinθ as the y-coordinate of P changes sign when we reflect across the x-axis.
  - Sinθ is positive in the first and the second quadrant (as the y coordinate of P is positive), and negative in the third and the fourth quadrant as the y coordinate is negative there.
  - Sin(180 - θ) = Sin θ because as we reflect across the y-axis the y-coordinate doesn't change.
  - Sin(θ - 180) = - Sinθ as the y-coordinate changes signs when reflected across the origin.
  - Like the cosine function, the sine function is also periodic, with period 360 degrees, and range between -1 and 1.

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- Graph of the Sine Function:



- Tangent Function: Using the same diagram as before, we consider a circle of radius 1. Let $P(x, y)$ be any point making an angle θ with the horizontal x-axis.

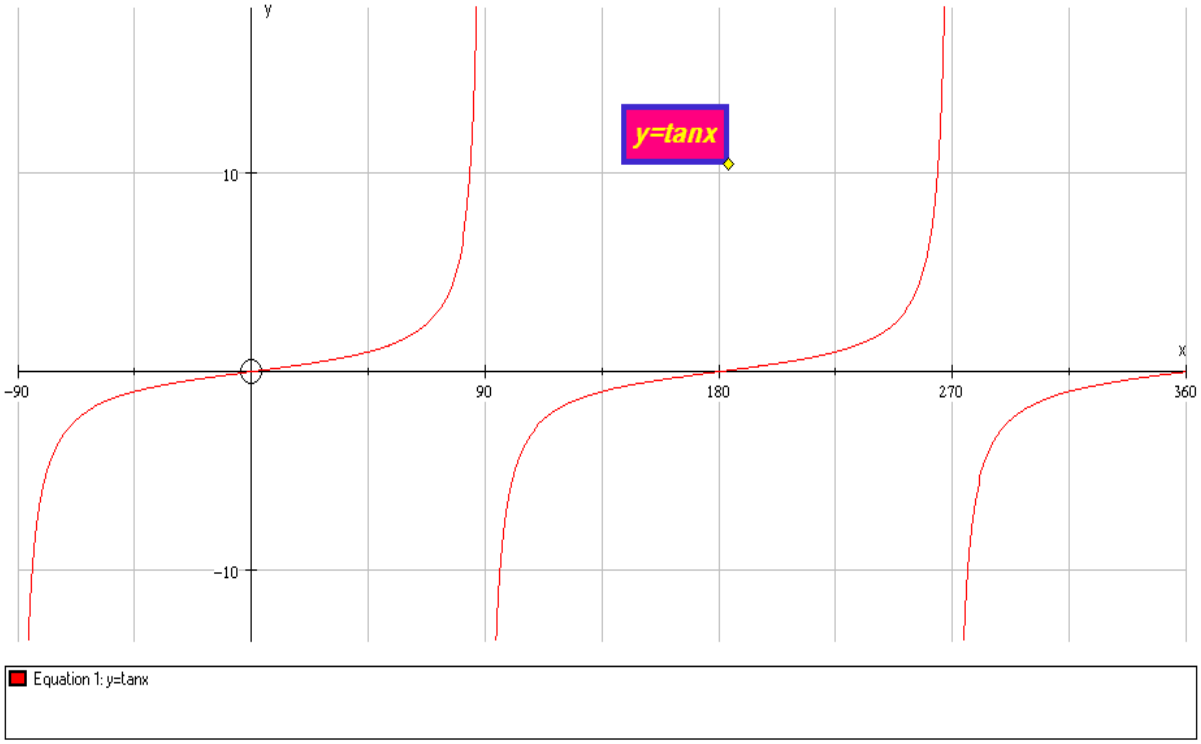
  Then $Tan\theta = PQ/OQ = y/x = \sin\theta / \cos\theta$

- Properties of the Tangent Function:
  - Tan(-θ) = -Tanθ as the y-coordinate of P changes sign when we reflect across the x-axis but the x-coordinate doesn't change sign.
  - Tanθ is positive in the first and the third quadrant (as the x and y coordinates of P have the same signs in these quadrants), and negative in the second and the fourth quadrant as the x and y coordinates have opposite signs in these quadrants.
  - Tan(180 -θ) = -Tanθ as the x-coordinate of P changes sign when we reflect across the y-axis but the y-coordinate doesn't change sign
  - The domain of tanθ does **not** include the angles for which x is 0, namely, for θ = ±90, ±270, …

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- Like the cosine and sine functions, the tangent function is also periodic, but its period is 180. i.e. $\tan(\theta + 180) = \tan\theta$ and $\tan(\theta - 180) = \tan\theta$

o Graph of the Tangent Function:



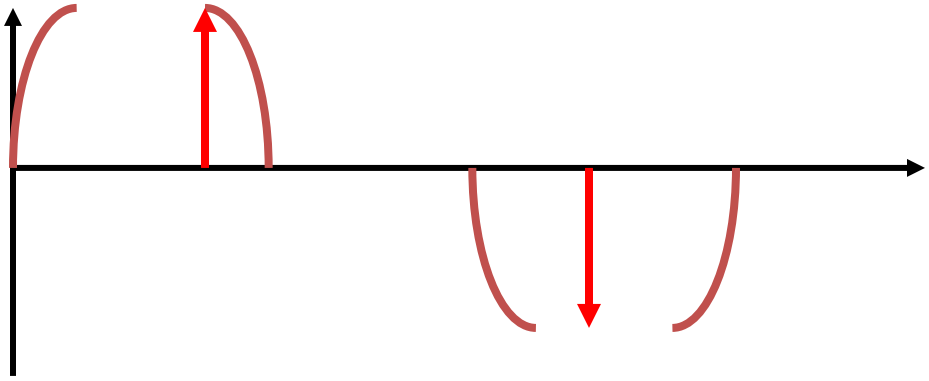
- Basic Properties of Trigonometric Functions:

Function	Periodic Properties	Odd/Even Property	Translation Properties
Cosine	$\cos(\theta \pm 360) = \cos\theta$	$\cos(-\theta) = \cos\theta$	$\cos(\theta - 180) = -\cos\theta$ $\cos(180 - \theta) = -\cos\theta$
Sine	$\sin(\theta \pm 360) = \sin\theta$	$\sin(-\theta) = -\sin\theta$	$\sin(\theta - 180) = -\sin\theta$ $\sin(180 - \theta) = \sin\theta$
Tangent	$\tan(\theta \pm 180) = \tan\theta$	$\tan(-\theta) = -\tan\theta$	$\tan(180 - \theta) = -\tan\theta$

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- Definition: The **Amplitude** of a function is the height from the mean (or the rest) value of the function to its maximum or minimum value.
    - The amplitude of the function $f(x) = A\,Sin\,Bx$ is $|A|$ and the period is $2\pi/|B|$
    - The amplitude of the function $g(x) = A\,Cos\,Bx$ is $|A|$ and the period is $2\pi/|B|$

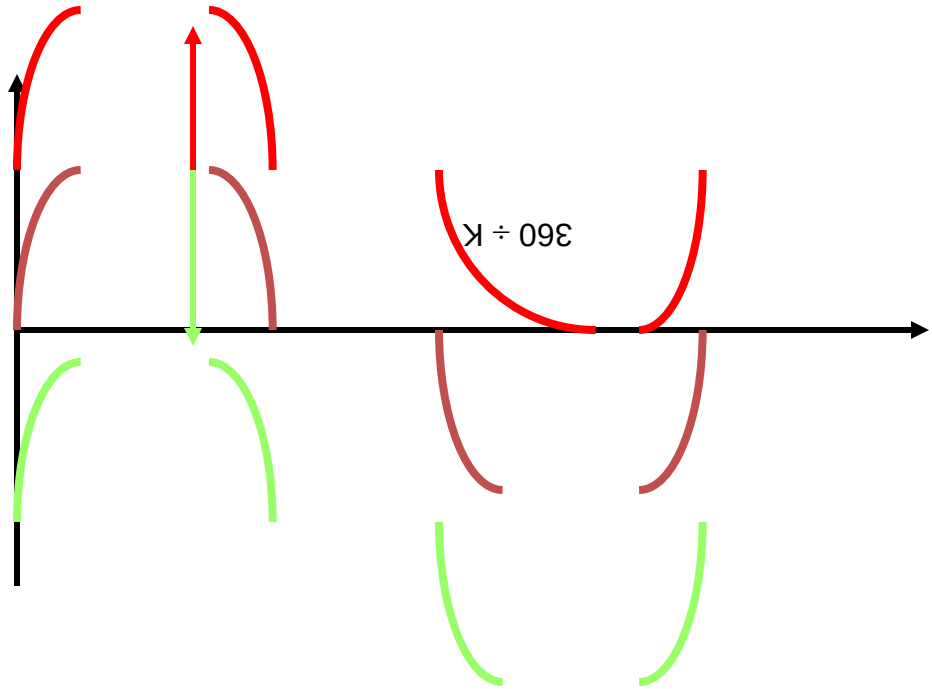
- Changing Trigonometric Graphs:
    - $y = k\,Cos\,x$ & $y = k\,Sin\,x$: The amplitude of the function is "k" .



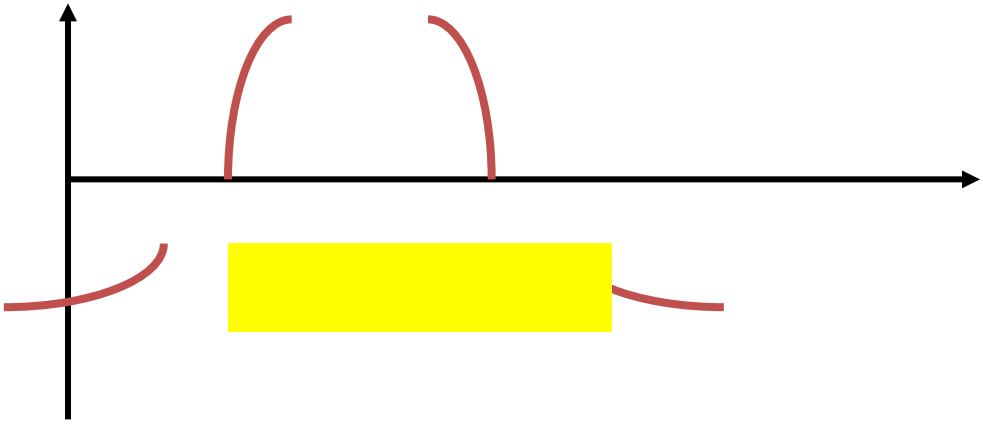
    - $y = Cos\,kx$ & $y = Sin\,kx$: The period of the function is "$360 \div k$".

    - $y = Cos\,x + k$ & $y = Sin\,x + k$: Translates the graph $+ k$ or $- k$ parallel to the y-axis.

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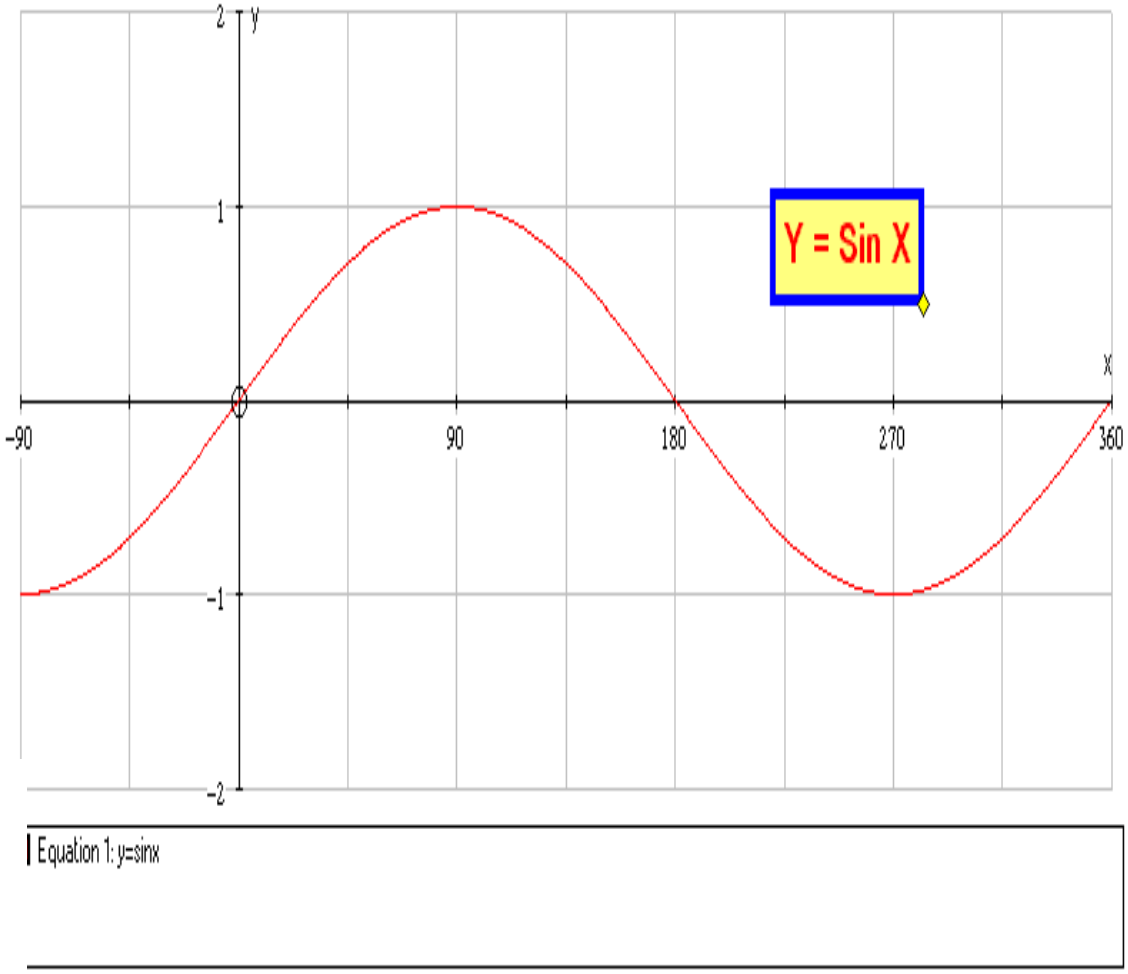
- $y = -\cos x$ & $y = -\sin x$: Reflects the graph in the x-axis.



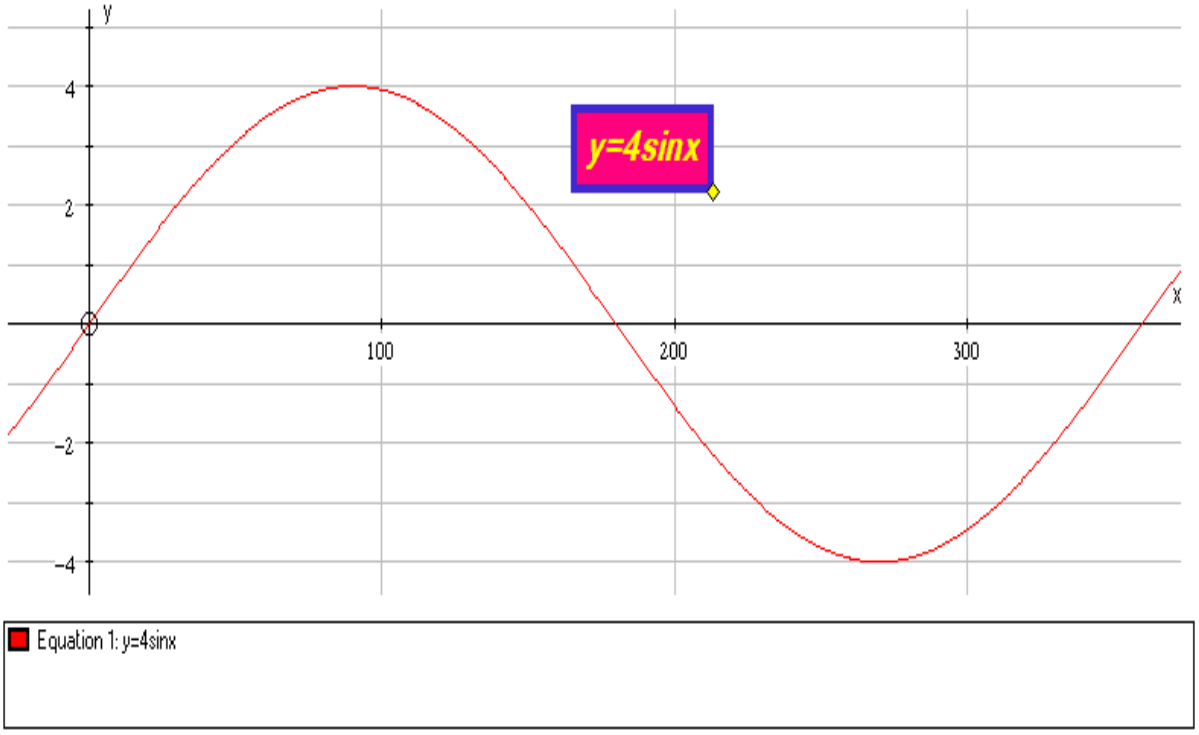
- Combining the Effects: Draw the graph of $y = 4\sin 2x + 3$.

  - Step 1: Draw the graph of $y = \sin x$

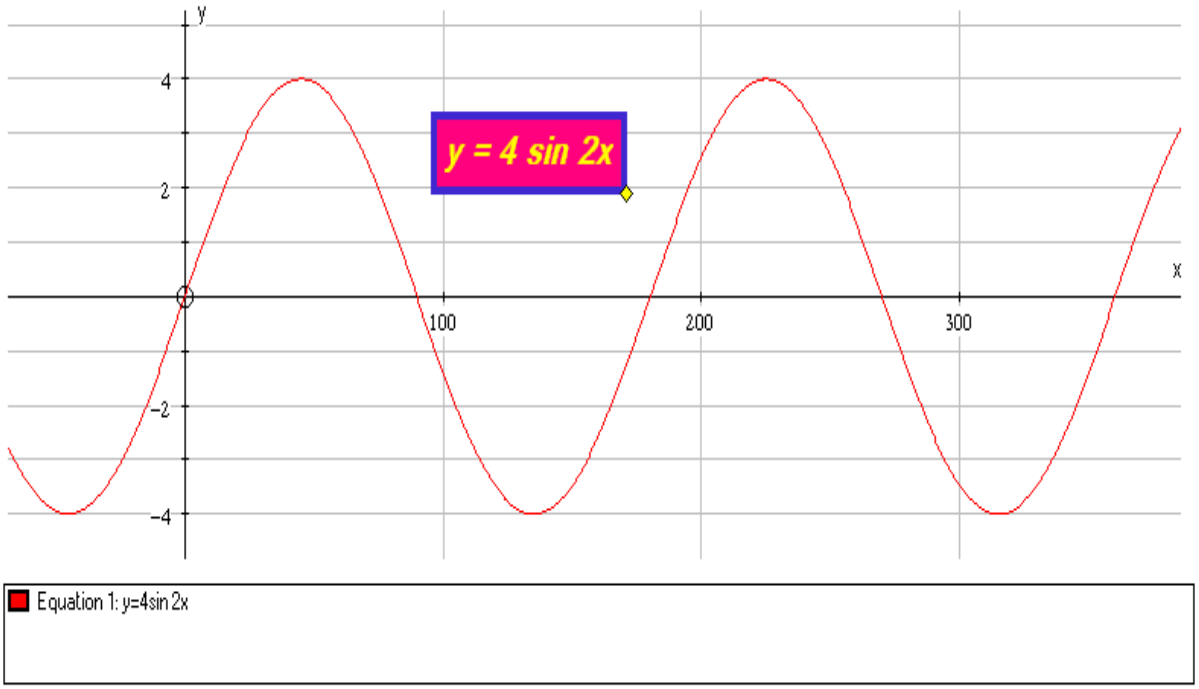
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- Step 2: Draw the graph of $y = 4 \sin x$

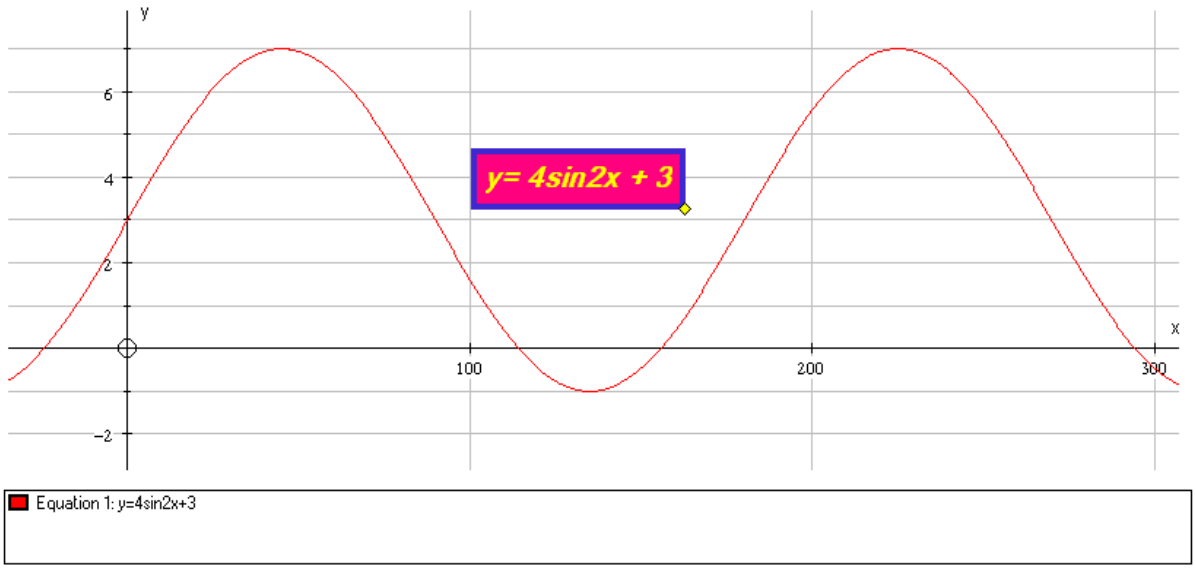


- Step 3: Draw the graph of $y = 4 \sin 2x$

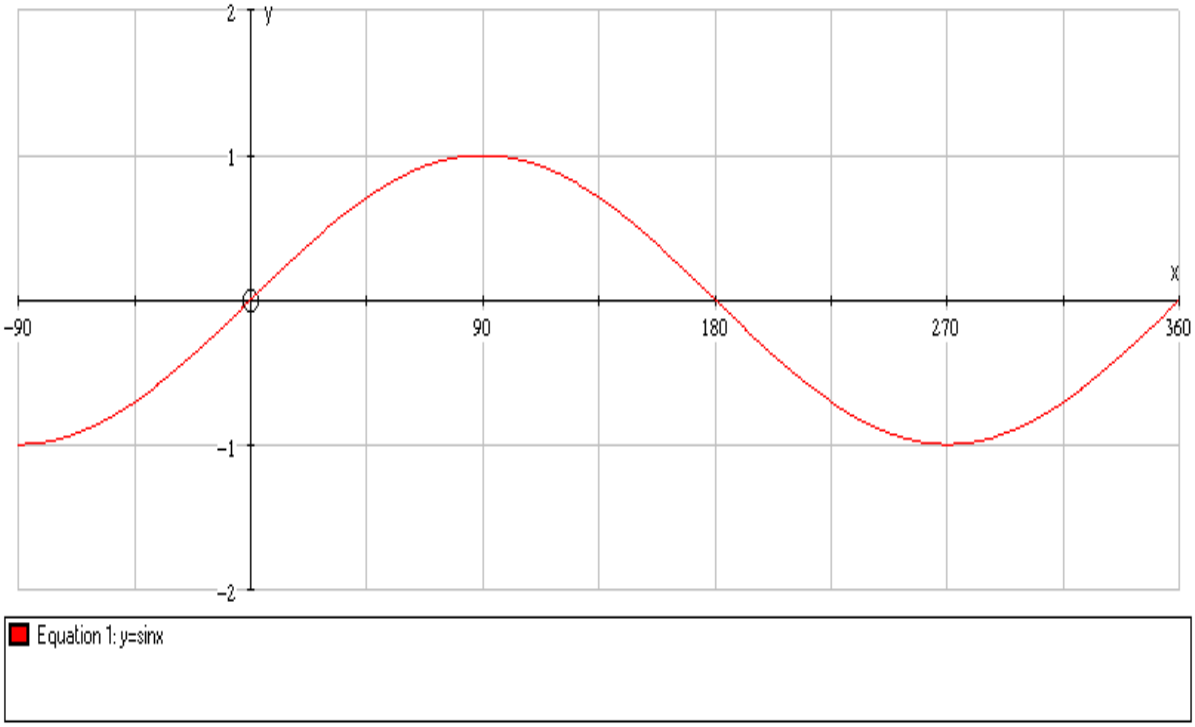


- Step 4: Draw the graph of $y = 4 \sin 2x + 3$

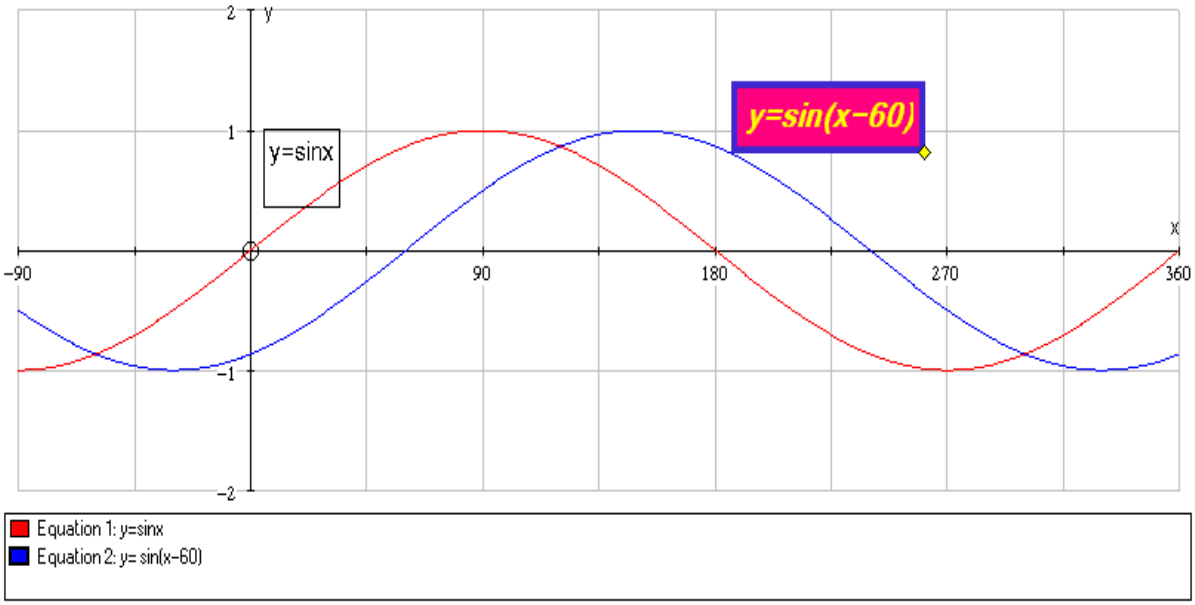
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- Phase Shift:
  - Shown below is the graph of $y = \sin x^o$



  - Now compare it with the graph of $y = \sin(x - 60^o)$



The graph is translated 60° to the right parallel to the x axis.

- From the previous example we can now see that the equation $y = cos\ (x - k)$ and $y = sin\ (x - k)$ translate the graphs k^o to the right parallel to the x-axis and the equations $y = cos\ (x + k)$ and $y = sin\ (x + k)$ translate the graph k^o to the left parallel to the x-axis.

- Note that the functions Sine, Cosine and Tangent are not one-to-one. So, they don't have inverses unless we restrict the domains of the definitions.

- Definition: The **inverse sine function** is defined by $y = \arcsin x$ if and only if $\sin y = x$. The domain of $y = \arcsin x$ is $[-1, 1]$. The range of $y = \arcsin x$ is $[-\pi/ , \pi/2]$.

- Example: $\arcsin \frac{1}{2} = \frac{\pi}{6}$ since $\frac{\pi}{6}$ is the angle whose sine is $\frac{1}{2}$.

- Example: $\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$ since $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$

- Definition: The **inverse cosine function** is defined by $y = \arccos$ if and only if $\cos y = x$. The domain of $y = \arccos x$ is $[-1, 1]$. The range of $y = \arccos x$ is $[0, \pi]$.

- Definition: The **inverse tangent function** is defined by $y = \arctan x$ if and only if $\tan y = x$. The domain of $y = \arctan x$ is $(-\infty, \infty)$. The range of $y = \arctan x$ is $[-\pi/2, \pi/2]$.

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- Solving Trigonometric Equations
  - To solve $\cos \theta = k$
    - Find $\cos^{-1}(k) = \theta$
      - Use symmetry property to get $\cos(-\theta) = \cos(\theta)$ to get $-\theta$ as a solution.
      - Use the periodic property [$\cos(\theta \pm 360) = \cos\theta$] to find all the solutions in the required interval.

    - Example: Given that $\cos(70.52) = 1/3$, solve $\cos\theta = 1/3$, giving all the roots in $0 \leq \theta \leq 360$
      - Find $\cos^{-1}(1/3) = 70.52$
      - Use symmetry property to get $\cos(-70.52) = \cos(70.52)$ to get -70.52 as a solution.
      - Note that -70.52 is not in the required interval. Now use the periodic property $\cos(-70.52) = \cos(-70.52 + 360) = \cos(289.48)$. So another solution is 289.48 degrees
      - All the solutions are 70.52° and 289.48°

    - To solve $\sin\theta = k$
      - Find $\sin^{-1}(k) = \theta$
      - Use symmetry property to get $\sin(180 - \theta) = \sin(\theta)$ to get another solution.
      - Use the periodic property [$\sin(\theta \pm 360) = \cos\theta$] to find all the solutions in the required interval.

    - Example: Given that $\sin(44.42^\circ) = 0.7$, solve $\sin\theta = -0.7$, giving all the roots in $-180 \leq \theta \leq 180$
      - Find $\sin^{-1}(-0.7) = -44.42$
      - Use symmetry property to get $\sin(180 - (-44.42)) = \sin(-44.42)$ to get 224.42 as a solution.
      - Note that 224.42 is not in the required interval. Now use the periodic property $\sin(224.42) = \sin(224.42 - 360)$. So another solution is -135.58 degrees
      - All the solutions are -44.42° and -135.58°

    - To solve $\tan\theta = k$
      - Find $\tan^{-1}(k) = \theta$
      - Use the periodic property $\tan(180 \pm \theta) = \tan(\theta)$ to get other solutions in the required interval.

    - Example: Given that $\tan(63.43^\circ) = 2$, Solve $\tan\theta = -2$, giving all the roots $in\ 0 \leq \theta \leq 360$
      - Find $\tan^{-1}(-2) = -63.43^\circ$,
      - Note that -63.43° is not in our range. Now use the periodic property $\tan(-63.43^\circ) = \tan(-63.43^\circ - 180^\circ)$ to get 116.56. Now use the periodic property $\tan(116.56^\circ) = \tan(116.56^\circ + 180^\circ)$ to get 296.56°.
      - The solutions are is 116.6° and 296.6°

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- Definition: Two functions *f* and *g* are said to be **identically equal** if *f(x) = g(x)* for **every value** of x for which both functions are defined. Such an equation is referred to as an **identity**. An equation that is not an identity is called a **conditional equation.**

- Basic Trigonometric Identities:

  - Reciprocal Identities:

    $$\csc\theta = \frac{1}{\sin\theta} \quad \sec\theta = \frac{1}{\cos\theta} \quad \cot\theta = \frac{1}{\tan\theta}$$

  - Quotient Identities

    $$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

  - Periodic Properties

    $$\sin(\theta + 2\pi) = \sin\theta \qquad \csc(\theta + 2\pi) = \csc\theta$$
    $$\cos(\theta + 2\pi) = \cos\theta \qquad \sec(\theta + 2\pi) = \sec\theta$$
    $$\tan(\theta + \pi) = \tan\theta \qquad \cot(\theta + \pi) = \cot\theta$$

  - Even-Odd Properties

    $$\sin(-\theta) = -\sin\theta \qquad \csc(-\theta) = -\csc\theta$$
    $$\cos(-\theta) = \cos\theta \qquad \sec(-\theta) = \sec\theta$$
    $$\tan(-\theta) = -\tan\theta \qquad \cot(-\theta) = -\cot\theta$$

  - Pythagorean Identities

    $$\sin^2\theta + \cos^2\theta = 1$$
    $$\tan^2\theta + 1 = \sec^2\theta$$
    $$1 + \cot^2\theta = \csc^2\theta$$

- Example: Verify that the following equation is an identity:

  $$\frac{\tan t - \cot t}{\sin t \cos t} = \sec^2 t - \csc^2 t$$

  Solution: LHS = $\frac{\tan t - \cot t}{\sin t \cos t} = \frac{\tan t}{\sin t \cos t} - \frac{\cot t}{\sin t \cos t}$

  $$= \tan t \cdot \frac{1}{\sin t \cos t} - \cot t \cdot \frac{1}{\sin t \cos t}$$

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$$= \frac{\sin t}{\cos t} \cdot \frac{1}{\sin t \cos t} - \frac{\cos t}{\sin t} \cdot \frac{1}{\sin t \cos t}$$
$$= \frac{1}{\cos^2 t} - \frac{1}{\sin^2 t}$$
$$= \sec^2 t - \csc^2 t$$
= RHS

- Sum and Difference Formulas:
  - cos(A-B) = cosAcosB + sinAsinB
    cos(A+B) = cosAcosB - sinAsinB
  - sin(A+B) = sinAcosB + cosAsinB
    sin(A-B) = sinAcosB – cosAsinB
  - $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
    $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

- Double Angle Formulas:
  - $\sin 2A = \sin(A+A) = \sin A\cos A + \sin A\cos A = 2\sin A\cos A$
  - $\cos 2A = \cos(A+A) = \cos A\cos A - \sin A\sin A = \cos^2 A - \sin^2 A$
  - $\tan 2A = \frac{\tan A + \tan A}{1 - \tan A \tan A} = \frac{2\tan A}{1 - \tan^2 A}$

- Example: Find the exact value of Cos 165°.

  Solution: $\cos(165°) = \cos(210° - 45°) = \cos 210° \cos 45° + \sin 210° \sin 45°$
  $$= \frac{-\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + -\frac{1}{2} \cdot \frac{\sqrt{2}}{2} = \frac{-\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{-(\sqrt{6} - \sqrt{2})}{4}$$

Using the trigonometric identities find the value of $\cos(75°)$

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Virtual University of Pakistan
General Mathematics and Biostatistics

WHAT IS STATISTICS?

- **That science which enables us to draw conclusions about various phenomena on the basis of real data collected on sample-basis**
- **A tool for data-based research**
- **Also known as Quantitative Analysis**
- **A lot of application in a wide variety of disciplines … Agriculture, Anthropology, Astronomy, Biology, Economic, Engineering, Environment, Geology, Genetics, Medicine, Physics, Psychology, Sociology, Zoology …. Virtually every single subject from Anthropology to Zoology …. A to Z!**
- **Any scientific enquiry in which you would like to base your conclusions and decisions on real-life data, you need to employ statistical techniques!**
- **Now a day, in the developed countries of the world, there is an active movement for of Statistical Literacy.**

THE NATURE OF THIS DISCIPLINE:

DESCRIPTIVE STATISTICS



PROBABILITY



INFERENTIAL STATISTICS

MEANINGS OF 'STATISTICS':

The word "Statistics" which comes from the Latin words *status*, meaning a political state, originally meant information useful to the state, for example, information about the sizes of population sand armed forces. But this word has now acquired different meanings.

- In the *first place*, the word *statistics* refers to "numerical facts systematically arranged". In this sense, the word statistics is always used in plural. We have, for instance, statistics of

prices, statistics of road accidents, statistics of crimes, statistics of births, statistics of educational institutions, etc. In all these examples, the word statistics denotes a set of numerical data in the respective fields. This is the meaning the man in the street gives to the word *Statistics* and most people usually use the word *data* instead.

- In the *second place*, the word *statistics* is defined as a discipline that includes procedures and techniques used to collect, process and analyze numerical data to make inferences and to research decisions in the face of uncertainty. It should of course be borne in mind that uncertainty does not imply ignorance but it refers to the incompleteness and the instability of data available. In this sense, the word statistics is used in the singular. As it embodies more of less all stages of the general process of learning, sometimes called *scientific method*, statistics is characterized as a science. Thus the word *statistics* used in the plural refers to a set of numerical information and in the singular, denotes the science of basing decision on numerical data. It should be noted that statistics as a subject is mathematical in character.

- *Thirdly*, the word statistics are numerical quantities calculated from sample observations; a single quantity that has been so collected is called a *statistic*. The mean of a sample for instance is a statistic. The word statistics is plural when used in this sense.

CHARACTERISTICS OF THE SCIENCE OF STATISTICS:

Statistics is a discipline in its own right. It would therefore be desirable to know the characteristic features of statistics in order to appreciate and understand its general nature. Some of its important characteristics are given below:

i) Statistics deals with the behaviour of aggregates or large groups of data. It has nothing to do with what is happening to a particular individual or object of the aggregate.

ii) Statistics deals with aggregates of observations of the same kind rather than isolated figures.

iii) Statistics deals with variability that obscures underlying patterns. No two objects in this universe are exactly alike. If they were, there would have been no statistical problem.

iv) Statistics deals with uncertainties as every process of getting observations whether controlled or uncontrolled, involves deficiencies or chance variation. That is why we have to talk in terms of probability.

v) Statistics deals with those characteristics or aspects of things which can be described numerically either by counts or by measurements.

vi) Statistics deals with those aggregates which are subject to a number of random causes, *e.g.* the heights of persons are subject to a number of causes such as race, ancestry, age, diet, habits, climate and so forth.

vii) Statistical laws are valid *on the average* or in the long run. There is n guarantee that a certain law will hold in all cases. Statistical inference is therefore made in the face of uncertainty.

viii) Statistical results might be misleading the incorrect if sufficient care in collecting, processing and interpreting the data is not exercised or if the statistical data are handled by a person who is not well versed in the subject mater of statistics.

THE WAY IN WHICH STATISTICS WORKS:

As it is such an important area of knowledge, it is definitely useful to have a fairly good idea about the way in which it works, and this is exactly the purpose of this introductory course.
The following points indicate some of the main functions of this science:

- Statistics assists in summarizing the larger set of data in a form that is easily understandable.
- Statistics assists in the efficient design of laboratory and field experiments as well as surveys.
- Statistics assists in a sound and effective planning in any field of inquiry.
- Statistics assists in drawing general conclusions and in making predictions of how much of a thing will happen under given conditions.

IMPORTANCE OF STATISTICS
IN VARIOUS FIELDS

As stated earlier, Statistics is a discipline that has finds application in the most diverse fields of activity. It is perhaps a subject that should be used by everybody. Statistical techniques being powerful tools for analyzing numerical data are used in almost every branch of learning. In all areas, statistical techniques are being increasingly used, and are developing very rapidly.

- A modern administrator whether in public or private sector leans on statistical data to provide a factual basis for decision.
- A politician uses statistics advantageously to lend support and credence to his arguments while elucidating the problems he handles.
- A businessman, an industrial and a research worker all employ statistical methods in their work. Banks, Insurance companies and Government all have their statistics departments.
- A social scientist uses statistical methods in various areas of socio-economic life of a nation. It is sometimes said that “a social scientist without an adequate understanding of statistics, is often like the blind man groping in a dark room for a black cat that is not there”.

The Meaning of Data:

The word “data” appears in many contexts and frequently is used in ordinary conversation. Although the word carries something of an aura of scientific mystique, its meaning is quite simple and mundane. It is Latin for “those that are given” (the singular form is “datum”). Data may therefore be thought of as the *results of observation*.

EXAMPLES OF DATA

Data are collected in many aspects of everyday life.

- Statements given to a police officer or physician or psychologist during an interview are data.
- So are the correct and incorrect answers given by a student on a final examination.
- Almost any athletic event produces data.
- The time required by a runner to complete a marathon,
- The number of errors committed by a baseball team in nine innings of play.

And, of course, data are obtained in the course of scientific inquiry:

- the positions of artifacts and fossils in an archaeological site,

- The number of interactions between two members of an animal colony during a period of observation,

- The spectral composition of light emitted by a star.

OBSERVATIONS AND VARIABLES:

Observation:
In statistics, an *observation* often means any sort of numerical recording of information, whether it is a physical measurement such as height or weight; a classification such as heads or tails, or an answer to a question such as yes or no.

Variables:
A characteristic that varies with an individual or an object, is called a *variable*. For example, age is a variable as it varies from person to person. A variable can assume a number of values. The given set of all possible values from which the variable takes on a value is called its Domain. If for a given problem, the domain of a variable contains only one value, then the variable is referred to as a *constant*.

QUANTITATIVE AND
QUALITATIVE VARIABLES:

Variables may be classified into quantitative and qualitative according to the form of the characteristic of interest.

A variable is called a *quantitative variable* when a characteristic can be expressed numerically such as age, weight, income or number of children. On the other hand, if the characteristic is non-numerical such as education, sex, eye-colour, quality, intelligence, poverty, satisfaction, etc. the variable is referred to as a *qualitative variable*. A qualitative characteristic is also called an *attribute*. An individual or an object with such a characteristic can be counted or enumerated after having been assigned to one of the several mutually exclusive classes or categories.

Discrete and Continuous Variables:

A quantitative variable may be classified as discrete or continuous. *A discrete* variable is one that can take only a discrete set of integers or whole numbers, which is the values are taken by jumps or breaks. A discrete variable represents *count* data such as the number of persons in a family, the number of rooms in a house, the number of deaths in an accident, the income of an individual, etc.

A variable is called a *continuous* variable if it can take on any value-fractional or integral—within a given interval, *i.e.* its domain is an interval with all possible values without gaps. A continuous variable represents measurement data such as the age of a person, the height of a plant, the weight of a commodity, the temperature at a place, etc.

A variable whether countable or measurable, is generally denoted by some symbol such as X or Y and X_i or X_j represents the ith or j^{th} value of the variable. The subscript i or j is replaced by a number such as 1,2,3, … when referred to a particular value.

Measurement Scales:

By *measurement*, we usually mean the assigning of number to observations or objects and scaling is a process of measuring. The four scales of measurements are briefly mentioned below:

NOMINAL SCALE:

The classification or grouping of the observations into mutually exclusive qualitative categories or classes is said to constitute a *nominal scale*. For example, students are classified as male and female. Number 1 and 2 may also be used to identify these two categories. Similarly, rainfall may be classified as heavy moderate and light. We may use number 1, 2 and 3 to denote the three classes of rainfall. The numbers when they are used only to identify the categories of the given scale, carry no numerical significance and there is no particular order for the grouping.

ORDINAL OR RANKING SCALE:

It includes the characteristic of a nominal scale and in addition has the property of *ordering* or *ranking* of measurements. For example, the performance of students (or players) is rated as excellent, good fair or poor, etc. Number 1, 2, 3, 4 etc. are also used to indicate ranks. The only relation that holds between any pair of categories is that of “greater than” (or more preferred).

INTERVAL SCALE:

A measurement scale possessing a constant interval size (distance) but not a true zero point, is called an *interval scale*. Temperature measured on either the Celsius or the Fahrenheit scale is an outstanding example of interval scale because the same difference exists between 20° C (68° F) and 30° C (86° F) as between 5° C (41° F) and 15° C (59° F). It cannot be said that a temperature of 40 degrees is twice as hot as a temperature of 20 degree, i.e. the ratio 40/20 has no meaning. The arithmetic operation of addition, subtraction, etc. is meaningful.

RATIO SCALE:

It is a special kind of an interval scale where the sale of measurement has a true *zero* point as its origin. The ratio scale is used to measure weight, volume, distance, money, etc. The, key to differentiating interval and ratio scale is that the zero point is meaningful for ratio scale.

ERRORS OF MEASUREMENT:

Experience has shown that a continuous variable can never be measured with perfect fineness because of certain habits and practices, methods of measurements, instruments used, etc. the measurements are thus always recorded correct to the nearest units and hence are of limited accuracy. The *actual* or *true* values are, however, assumed to exist. For example, if a student's weight is recorded as 60 kg (correct to the nearest kilogram), his true weight in fact lies between 59.5 kg and 60.5 kg, whereas a weight recorded as 60.00 kg means the true weight is known to lie between 59.995 and 60.005 kg. Thus there is a difference, however small it may be between the measured value and the true value. This sort of departure from the true value is technically known as the *error of measurement*. In other words, if the observed value and the true value of a variable are denoted by x and $x + \varepsilon$ respectively, then the difference $(x + \varepsilon) – x$, i.e. ε is the error. This error involves the unit of measurement of x and is therefore called an *absolute error*. An absolute error divided by the true value is called the *relative error*. Thus the relative error $= \frac{\varepsilon}{x + \varepsilon}$, which when multiplied by 100, is *percentage error*. These errors are independent of the units of measurement of x. It ought to be noted that an error has both magnitude and direction and that the word *error* in statistics does not mean mistake which is a chance inaccuracy.

BIASED AND RANDOM ERRORS:

An error is said to be *biased* when the observed value is consistently and constantly higher or lower than the true value. Biased errors arise from the personal limitations of the observer, the imperfection in the instruments used or some other conditions which control the measurements. These errors are not revealed by repeating the measurements. They are cumulative in nature, that is, the greater the number of measurements, the greater would be the magnitude of error. They are thus more troublesome. These errors are also called *cumulative* or *systematic errors*.

An error, on the other hand, is said to be unbiased when the deviations, i.e. the excesses and defects, from the true value tend to occur equally often. Unbiased errors and revealed when measurements are repeated and they tend to cancel out in the long run. These errors are therefore *compensating* and are also known as *random errors* or *accidental errors*.

LECTURE No.24

Steps involved in a Statistical Research-Project

- Collection of Data:
  - Primary Data
  - Secondary Data
- Sampling:
  - Concept of Sampling
  - Non-Random Versus Random Sampling
  - Simple Random Sampling
  - Other Types of Random Sampling

STEPS INVOLVED IN ANY STATISTICAL RESEARCH:

1. Topic and significance of the study
2. Objective of your study
3. Methodology for data-collection
   - Source of your data
   - Sampling methodology
   - Instrument for collecting data

As far as the objectives of your research are concerned, they should be stated in such a way that you are absolutely clear about the goal of your study --- EXACTLY WHAT IT IS THAT YOU ARE TRYING TO FIND OUT?

As far as the *methodology for DATA-COLLECTION* is concerned, you need to consider:

- Source of your data
  (the statistical population)
- Sampling Methodology
- Instrument for collecting data

COLLECTION OF DATA:

The most important part of statistical work is perhaps the collection of data. Statistical data are collected either by a COMPLETE enumeration of the whole field, called CENSUS, which in many cases would be too costly and too time consuming as it requires large number of enumerators and supervisory staff, or by a PARTIAL enumeration associated with a SAMPLE which saves much time and money.

PRIMARY AND SECONDARY DATA:

Data that have been originally collected (raw data) and have not undergone any sort of statistical treatment, are called *PRIMARY data.* Data that have undergone any sort of treatment by statistical methods at least ONCE, i.e. the data that have been collected, classified, tabulated or presented in some form for a certain purpose, are called *SECONDARY data.*

COLLECTION OF PRIMARY DATA:

One or more of the following methods are employed to collect primary data:

i) Direct Personal Investigation.
ii) Indirect Investigation.
iii) Collection through Questionnaires.
iv) Collection through Enumerators.
v) Collection through Local Sources.

DIRECT PERSONAL INVESTIGATION:

In this method, an investigator collects the information personally from the individuals concerned. Since he interviews the informants himself, the information collected is generally considered quite accurate and complete.

This method may prove very costly and time-consuming when the area to be covered is vast.

However, it is useful for laboratory experiments or localized inquiries. Errors are likely to enter the results due to personal bias of the investigator.

INDIRECT INVESTIGATION:

Sometimes the direct sources do not exist or the informants hesitate to respond for some reason or other. In such a case, third parties or witnesses having information are interviewed.

Moreover, due allowance is to be made for the personal bias. This method is useful when the information desired is complex or there is reluctance or indifference on the part of the informants. It can be adopted for extensive inquiries.

COLLECTION THROUGH QUESTIONNAIRES:

A questionnaire is an inquiry form comprising of a number of pertinent questions with space for entering information asked.

The questionnaires are usually sent by mail, and the informants are requested to return the questionnaires to the investigator after doing the needful within a certain period.

This method is cheap, fairly expeditious and good for extensive inquiries.

But the difficulty is that the majority of the respondents (i.e. persons who are required to answer the questions) do not care to fill the questionnaires in, and to return them to the investigators. Sometimes, the questionnaires are returned incomplete and full of errors. Students, in spite of these drawbacks, this method is considered as the STANDARD method for routine business and administrative inquiries.

It is important to note that the questions should be few, brief, very simple, easy for all respondents to answer, clearly worded and not offensive to certain respondents.

COLLECTION THROUGH ENUMERATORS:

Under this method, the information is gathered by employing trained enumerators who assist the informants in making the entries in the schedules or questionnaires correctly.

This method gives the most reliable information if the enumerator is well-trained, experienced and tactful.

Students, it is considered the BEST method when a large-scale governmental inquiry is to be conducted. This method can generally not be adopted by a private individual or institution as its cost would be prohibitive to them.

COLLECTION THROUGH LOCAL SOURCES:

In this method, there is no formal collection of data but the agents or local correspondents are directed to collect and send the required information, using their own judgment as to the best way of obtaining it.

This method is cheap and expeditious, but gives only the estimates.

COLLECTION OF SECONDARY DATA:

The secondary data may be obtained from the following sources:

i) Official, e.g. the publications of the Statistical Division, Ministry of Finance, the Federal and Provincial Bureaus of Statistics, Ministries of Food, Agriculture, Industry, Labour, etc.

ii) Semi-Official, e.g., State Bank of Pakistan, Railway Board, Central Cotton Committee, Boards of Economic Inquiry, District Councils, Municipalities, etc.

iii) Publications of Trade Associations, Chambers of Commerce, etc.

iv) Technical and Trade Journals and Newspapers.

v) Research Organizations such as universities, and other institutions.

Let us now consider the POPULATION from which we will be collecting our data

In this context, the first important question is:

Why do we have to resort to Sampling?

The answer is that:If we have available to us every value of the variable under study, then that would be an ideal and a perfect situation. But, the problem is that this ideal situation is very rarely available --- very rarely do we have access to the entire population.

The census is an exercise in which an attempt is made to cover the entire population.

But, as you might be knowing, even the most developed countries of the world cannot afford to conduct such a huge exercise on an annual basis!

More often than not, we have to conduct our research study on a sample basis.

In fact, the goal of the science of Statistics is to draw conclusions about large populations on the basis of information contained in small samples.

Let us now define some terms in a formal way:

'POPULATION':

A statistical population is the collection of every member of a group possessing the same basic and defined characteristic, but varying in amount or quality from one member to another.

EXAMPLES:

- **Finite population:**
  - IQ's of all children in a school.
- **Infinite population:**
  - Barometric pressure:
    (There are an indefinitely large number of points on the surface of the earth).
  - A flight of migrating ducks in Canada

(Many finite pops are so large that they can be treated as effectively infinite.)

The examples that we have just considered are those of existent populations.

A **hypothetical population** can be defined as the aggregate of all the conceivable ways in which a specified event can happen.

For **Example:**

1)All the possible outcomes from the throw of a die – however long we throw the die and record the results, we could always continue to do so far a still longer period in a theoretical concept – one which has no existence in reality.

2) The No. of ways in which a football team of 11 players can be selected from the 16 possible members named by the Club Manager.

We also need to differentiate between the sampled population and the target population. Sampled population is that from which a sample is chosen whereas the population about which information is sought is called the target population Thus our population will consist of the total no. of students in all the colleges in the Punjab.

Suppose on account of shortage of resources or of time, we are able to conduct such a survey on only 5 colleges scattered throughout the province. In this case, the students of all the colleges will constitute the target pop whereas the students of those 5 colleges from which the sample of students will be selected will constitute the sampled population.

I am sure that from the above discussion regarding the population, you must have realized how important it is to have a very well-defined population.

The next question is: How will we draw a sample from our population?

The answer is that:

In order to draw a random sample from a finite population, the first thing that we need is the complete list of all the elements in our population.

This list is technically called the **FRAME.**

SAMPLING FRAME:

A sampling frame is a complete list of all the elements in the population.

For example:

- The complete list of the BCS students of Virtual University of Pakistan on February 15, 2003.

Speaking of the sampling frame, it must be kept in mind that, as far as possible, our frame should be free from various types of defects:

- does not contain inaccurate elements
- is not incomplete
- is free from duplication, and
- Is not out of date.

Next, let's talk about the sample that we are going to draw from this population.

As you all know, a sample is only a part of a statistical population, and hence it can represent the population to only to some extent. Of course, it is intuitively logical that the larger the sample, the more likely it is to represent the population.

Obviously, the limiting case is that: when the sample size tends to the population size, the sample will tend to be identical to the population. But, of course, in general, the sample is much smaller than the population.

The point is that, in general, statistical sampling seeks to determine how accurate a description of the population the sample and its properties will provide. We may have to compromise on accuracy, but there are certain such advantages of sampling because of which it has an extremely important place in data-based research studies.

ADVANTAGES OF SAMPLING :

1. *Savings in time and money.*

- Although cost per unit in a sample is greater than in a complete investigation, the total cost will be less (because the sample will be so much smaller than the statistical population from which it has been drawn).

- A sample survey can be completed faster than a full investigation so that variations from sample unit to sample unit over time will largely be eliminated.
- Also, the results can be processed and analyzed with increased speed and precision because there are fewer of them.

2. More detailed information may be obtained from each sample unit.
3. Possibility of follow-up:
(After detailed checking, queries and omissions can be followed up --- a procedure which might prove impossible in a complete survey).
4. Sampling is the only feasible possibility where tests to destruction are undertaken or where the population is effectively infinite.
The next two important concepts that need to be considered are those of sampling and non-sampling errors.

Sampling & Non-Sampling Errors:

1. Sampling Error:

The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the parameter) is technically called the sampling error. For example,

Sampling error = $\bar{X} - \mu$

Sampling error arises due to the fact that a sample cannot exactly represent the pop, even if it is drawn in a correct manner

2. Non-Sampling Error:

Besides sampling errors, there are certain errors which are not attributable to sampling but arise in the process of data collection, even if a complete count is carried out.
Main sources of non sampling errors are:
1. The defect in the sampling frame.
2. Faulty reporting of facts due to personal preferences.
3. Negligence or indifference of the investigators
4. Non-response to mail questionnaires.

These (non-sampling) errors can be avoided through
1. Following up the non-response,
2. Proper training of the investigators.
3. Correct manipulation of the collected information,
Etc.
Let us now consider exactly what is meant by 'sampling error':
We can say that there are two types of non-response --- partial non-response and total non-response. *'Partial non-response'* implies that the respondent refuses to answer some of the questions. On the other hand, '*total non-response'* implies that the respondent refuses to answer any of the questions. Of course, the problem of late returns and non-

response of the kind that I have just mentioned occurs in the case of HUMAN populations.

Although refusal of sample units to cooperate is encountered in interview surveys, it is far more of a problem in mail surveys. It is not uncommon to find the response rate to mail questionnaires as low as 15 or 20%.The provision of INFORMATION ABOUT THE PURPOSE OF THE SURVEY helps in stimulating interest, thus increasing the chances of greater response.

Particularly if it can be shown that the work will be to the ADVANTAGE of the respondent IN THE LONG RUN.

Similarly, the respondent will be encouraged to reply if a pre-paid and addressed ENVELOPE is sent out with the questionnaire.

But in spite *of* these ways of reducing non-response, we are bound to have some amount of non-response.

Hence, a decision has to be taken about how many RECALLS should be made.

The term 'recall' implies that we approach the respondent more than once in order to persuade him to respond to our queries.

Another point worth considering is:

How long the process of data collection should be continued?

Obviously, no such process can be carried out for an indefinite period of time!

In fact, the longer the time period over which the survey is conducted, the greater will be the potential VARIATIONS in attitudes and opinions of the respondents.

Hence, a well-defined cut-off date generally needs to be established.

Let us now look at the various ways in which we can select a sample from our population. We begin by looking at the difference between non-random and RANDOM sampling. First of all, what do we mean by non-random sampling?

Nonrandom Sampling:

'Nonrandom sampling' implies that kind of sampling in which the population units are drawn into the sample by using one's personal judgment.

This type of sampling is also known as purposive sampling.

Within this category, one very important type of sampling is known as Quota Sampling.

QUOTA SAMPLING:

In this type of sampling, the selection of the sampling unit from the population is no longer dictated by chance. A sampling frame is not used at all, and the choice of the actual sample units to be interviewed is left to the discretion of the interviewer.

However, the interviewer is restricted by quota controls.

For example, one particular interviewer may be told to interview ten married women between thirty and forty years of age living in town X, whose husbands are professional workers, and five unmarried professional women of the same age living in the same town.

QUOTA SAMPLING:

Quota sampling is often used in commercial surveys such as consumer market-research.

Also, it is often used in public opinion polls.

ADVANTAGES OF QUOTA SAMPLING:

1) There is no need to construct a frame.
2) It is a very quick form of investigation.

3) Cost reduction.

DISADVANTAGES:

1) It is a subjective method. One has to choose between objectivity and convenience.
2) If random sampling is not employed, it is no longer theoretically possible to evaluate the sampling error.

(Since the selection of the elements is not based on probability theory but on the personal judgment of the interviewer, hence the precision and the reliability of the estimates can not be determined objectively i.e. in terms of probability.)

3) Although the purpose of implementing quota controls is to reduce bias, bias creeps in due to the fact that the interviewer is FREE to select particular individuals within the quotas. (Interviewers usually look for persons who either agree with their points of view or are personally known to them or can easily be contacted.)
4) Even if the above is not the case, the interviewer may still be making unsuitable selection of sample units.

(Although he may put some qualifying questions to a potential respondent in order to determine whether he or she is of the type prescribed by the quota controls, some features must necessarily be decided arbitrarily by the interviewer, the most difficult of these being social class.)

If mistakes are being made, it is almost impossible for the organizers to detect these, because follow-ups are not possible unless a detailed record of the respondents' names, addresses etc. has been kept.

Falsification of returns is therefore more of a danger in quota sampling than in random sampling. In spite of the above limitations, it has been shown by F. Edwards that a well-organized quota survey with well-trained interviewers can produce quite adequate results.

Next, let us consider the concept of random sampling.

Random Sampling:

The theory of statistical sampling rests on the assumption that the selection of the sample units has been carried out in a random manner.

By random sampling we mean sampling that has been done by adopting the lottery method.

TYPES OF RANDOM SAMPLING

- Simple Random Sampling
- Stratified Random Sampling
- Systematic Sampling
- Cluster Sampling
- Multi-stage Sampling, etc.

In this course, I will discuss with you the simplest type of random sampling i.e. simple random sampling.

SIMPLE RANDOM SAMPLING:

In this type of sampling, the chance of any one element of the parent pop being included in the sample is the same as for any other element. By extension, it follows that, in simple random sampling, the chance of any one sample appearing is the same as for any other.

There exists quite a lot of misconception regarding the concept of random sampling: Many a time, haphazard selection is considered to be equivalent to simple random sampling.

For example, a market research interviewer may select women shoppers to find their attitude to brand X of a product by stopping one and then another as they pass along a busy shopping area --- and he may think that he has accomplished simple random sampling!

Actually, there is a strong possibility of bias as the interviewer may tend to ask his questions of young attractive women rather than older housewives, or he may stop women who have packets of brand X prominently on show in their shopping bags!.
In this example, there is no suggestion of INTENTIONAL bias! From experience, it is known that the human being is a poor random selector --- one who is very subject to bias. Fundamental psychological traits prevent complete objectivity, and no amount of training or conscious effort can eradicate them. As stated earlier, random sampling is that in which population units are selected by the lottery method.

As you know, the traditional method of writing people's names on small pieces of paper, folding these pieces of paper and shuffling them is very cumbersome!
A much more convenient alternative is the use of *RANDOM NUMBERS TABLES.*
A random number table is a page full of digits from zero to 9. These digits are printed on the page in a TOTALLY random manner i.e. there is no systematic pattern of printing these digits on the page.

ONE THOUSAND RANDOM DIGITS:

2 3 1 5 7 5 4 8 5 9 0 1 8 3 7 2 5 9 9 3 7 6 2 4 9 7 0 8 8 6 9 5 2 3 0 3 6 7 4 4
0 5 5 4 5 5 5 0 4 3 1 0 5 3 7 4 3 5 0 8 9 0 6 1 1 8 3 7 4 4 1 0 9 6 2 2 1 3 4 3
1 4 8 7 1 6 0 3 5 0 3 2 4 0 4 3 6 2 2 3 5 0 0 5 1 0 0 3 2 2 1 1 5 4 3 8 0 8 3 4
3 8 9 7 6 7 4 9 5 1 9 4 0 5 1 7 5 8 5 3 7 8 8 0 5 9 0 1 9 4 3 2 4 2 8 7 1 6 9 5
9 7 3 1 2 6 1 7 1 8 9 9 7 5 5 3 0 8 7 0 9 4 2 5 1 2 5 8 4 1 5 4 8 8 2 1 0 5 1 3
1 1 7 4 2 6 9 3 8 1 4 4 3 3 9 3 0 8 7 2 3 2 7 9 7 3 3 1 1 8 2 2 6 4 7 0 6 8 5 0
4 3 3 6 1 2 8 8 5 9 1 1 0 1 6 4 5 6 2 3 9 3 0 0 9 0 0 4 9 9 4 3 6 4 0 7 4 0 3 6
9 3 8 0 6 2 0 4 7 8 3 8 2 6 8 0 4 4 9 1 5 5 7 5 1 1 8 9 3 2 5 8 4 7 5 5 2 5 7 1
4 9 5 4 0 1 3 1 8 1 0 8 4 2 9 8 4 1 8 7 6 9 5 3 8 2 9 6 6 1 7 7 7 3 8 0 9 5 2 7
3 6 7 6 8 7 2 6 3 3 3 7 9 4 8 2 1 5 6 9 4 1 9 5 9 6 8 6 7 0 4 5 2 7 4 8 3 8 8 0
0 7 0 9 2 5 2 3 9 2 2 4 6 2 7 1 2 6 0 7 0 6 5 5 8 4 5 3 4 4 6 7 3 3 8 4 5 3 2 0
4 3 3 1 0 0 1 0 8 1 4 4 8 6 3 8 0 3 0 7 5 2 5 5 5 1 6 1 4 8 8 9 7 4 2 9 4 6 4 7
6 1 5 7 0 0 6 3 6 0 0 6 1 7 3 6 3 7 7 5 6 3 1 4 8 9 5 1 2 3 3 5 0 1 7 4 6 9 9 3
3 1 3 5 2 8 3 7 9 9 1 0 7 7 9 1 8 9 4 1 3 1 5 7 9 7 6 4 4 8 6 2 5 8 4 8 6 9 1 9
5 7 0 4 8 8 6 5 2 6 2 7 7 9 5 9 3 6 8 2 9 0 5 2 9 5 6 5 4 6 3 5 0 6 5 3 2 2 5 4
0 9 2 4 3 4 4 2 0 0 6 8 7 2 1 0 7 1 3 7 3 0 7 2 9 7 5 7 3 6 0 9 2 9 8 2 7 6 5 0
9 7 9 5 5 3 5 0 1 8 4 0 8 9 4 8 8 3 2 9 5 2 2 3 0 8 2 5 2 1 2 2 5 3 2 6 1 5 8 7
9 3 7 3 2 5 9 5 7 0 4 3 7 8 1 9 8 8 8 5 5 6 6 7 1 6 6 8 2 6 9 5 9 9 6 4 4 5 6 9
7 2 6 2 1 1 1 2 2 5 0 0 9 2 2 6 8 2 6 4 3 5 6 6 6 5 9 4 3 4 7 1 6 8 7 5 1 8 6 7
6 1 0 2 0 7 4 4 1 8 4 5 3 7 1 2 0 7 9 4 9 5 9 1 7 3 7 8 6 6 9 9 5 3 6 1 9 3 7 8
9 7 8 3 9 8 5 4 7 4 3 3 0 5 5 9 1 7 1 8 4 5 4 7 3 5 4 1 4 4 2 2 0 3 4 2 3 0 0 0

| | | |

cumulative frequency of the first class is 6, we allocate numbers 000 to 005 to the six students who belong to this class.

AGE X	No. of Students f	cf	Sampling Numbers
13	6	6	000 – 005
14	61	67	
15	270	337	
16	491	828	
17	153	981	
18	15	996	
19	4	1000	
	1000		

As the cumulative frequency of the second class is 67 while that of the first class was 6, therefore we allocate sampling numbers 006 to 066 to the 61 students who belong to this class.

AGE X	**No. of Students f**	**cf**	**Sampling Numbers**
13	**6**	**6**	**000 – 005**
14	**61**	**67**	**006 – 066**
15	**270**	**337**	
16	**491**	**828**	
17	**153**	**981**	
18	**15**	**996**	
19	**4**	**1000**	
	1000		

As the cumulative frequency of the third class is 337 while that of the second class was 67, therefore we allocate sampling numbers 007 to 337 to the 270 students who belong to this class.

AGE X	**No. of Students f**	**cf**	**Sampling Numbers**
13	**6**	**6**	**000 – 005**
14	**61**	**67**	**006 – 066**
15	**270**	**337**	**067 – 336**

Proceeding in this manner, we obtain the column of sampling numbers.

AGE X	No. of Students f	cf	Sampling Numbers
13	6	6	000 – 005
14	61	67	006 – 066
15	270	337	067 – 336
16	491	828	337 – 827
17	153	981	828 – 980
18	15	996	981 – 995
19	4	1000	996 - 999
	1000		

The column implies that the first student of the first class has been allocated the sampling number 000, the second student has been allocated the sampling 001, and, proceeding in this fashion, the last student i.e. the 1000th student has been allocated the sampling number 999.

The question is :

Why did we not allot the number 0001 to the first student and the number 1000 to the 1000th student?

The answer is that we could do that but that would have meant that every student would have been allocated a four-digit number, whereas by shifting the number backward by 1, we are able to allocate to every student a three-digit number --- which is obviously simpler.

The next step is to SELECT 10 RANDOM NUMBERS from the random number table. This is accomplished by closing one's eyes and letting one's finger land anywhere on the random number table.

In this example, since all our sampling numbers are three-digit numbers, hence we will read three digits that are adjacent to each other at that position where our finger

landed. Suppose that we adopt this procedure and our random numbers come out to be 041, 103, 374, 171, 508, 652, 880, 066, 715, 471.
Selected Random Numbers:
041, 103, 374, 171, 508, 652, 880, 066, 715, 471.
Thus the corresponding ages are:
14, 15, 16, 15, 16, 16, 17,
15, 16, 16

***Explanation*:**

Our first selected random number is 041 which means that we have to pick up the 42nd student.

The cumulative frequency of the first class is 6 whereas the cumulative frequency of the second class is 67. This means that definitely the 42nd student does not belong to the first class but does belong to the second class.

AGE X	No. of Students f	cf
13	6	6
14	61	67
15	270	337
16	491	828
17	153	981
18	15	996
19	4	1000
	1000	

The age of each student in this class is 14 years; hence, obviously, the age of the 42nd student is also 14 years. This is how we are able to ascertain the ages of all the students who have been selected in our sampling. You will recall that in this example, our aim was to draw a sample from the population of college students, and to compare the sample's mean age with the population mean age.

The population mean age comes out to be 15.785 years.

AGE X	No. of Students f	fX
13	6	78
14	61	854
15	270	4050
16	491	7856
17	153	2601
18	15	270
19	4	76
	1000	15785

The population mean age is :

$$\mu = \frac{\sum fx}{\sum f} = \frac{15785}{1000}$$
$$= 15.785 \ years$$

The above formula is a slightly modified form of the basic formula that you have done ever-since school-days i.e. the mean is equal to the sum of all the observations divided by the total number of observations.
Next, we compute the sample mean age.
Adding the 10 values and dividing by 10, we obtain:
Ages of students selected in the sample (in years):
14, 15, 16, 15, 16, 16, 17, 15, 16, 16
Hence the sample mean age is:
Comparing the sample mean age of 15.6 years with the population mean age of 15.785 years, we note that the difference is really quite slight, and hence the sampling error is equal to

Sampling Error:

$$\bar{X} - \mu = 15.6 - 15.785$$
$$= -0.185 \text{ years}$$

And the reason for such a small error is that we have adopted the RANDOM sampling method.

The basic advantage of random sampling is that the probability is very high that the sample will be a good representative of the population from which it has been drawn, and any quantity computed from the sample will be a good estimate of the corresponding quantity computed from the population! Actually, a sample is supposed to be a MINIATURE REPLICA of the population. As stated earlier, there are various other types of random sampling.

Other Types of Random Sampling

- *·Stratified sampling*
  *(if the population is heterogeneous)*Systematic sampling
  (practically, more convenient than simple random sampling)
- *Cluster sampling*
  (sometimes the sampling units exist in natural clusters)
- *Multi-stage sampling,etc.*

- All these designs rest upon random or quasi-random sampling. They are various forms of PROBABILITY sampling --- that in which each sampling unit has a known (but not necessarily equal) probability of being selected.

Because of this knowledge, there exist methods by which the precision and the reliability of the estimates can be calculated OBJECTIVELY.

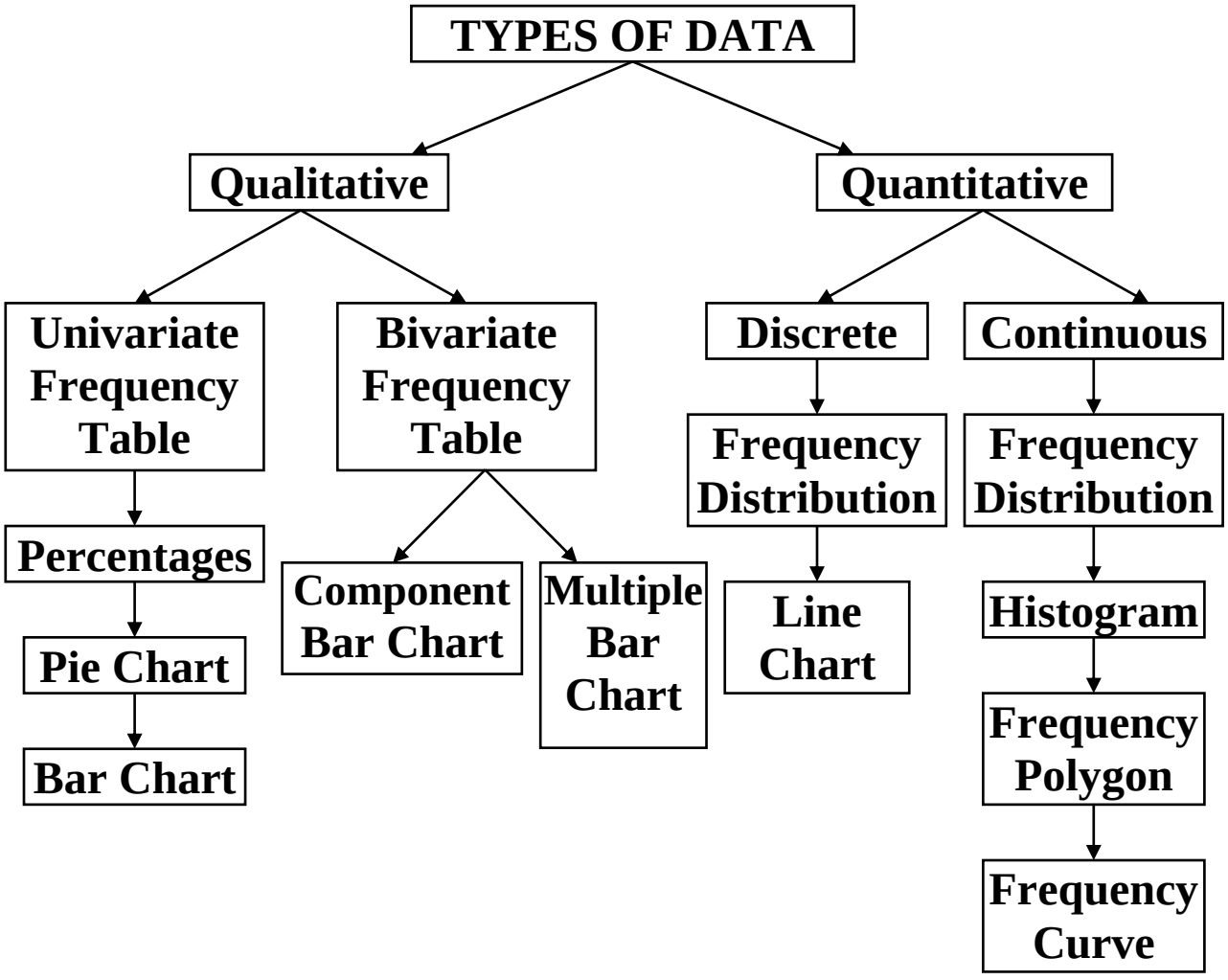
It should be realized that in practice, several sampling techniques are incorporated into each survey design, and only rarely will simple random sample be used, or a multi-stage design be employed, without stratification.

The point to remember is that whatever method be adopted, care should be exercised at every step so as to make the results as reliable as possible.

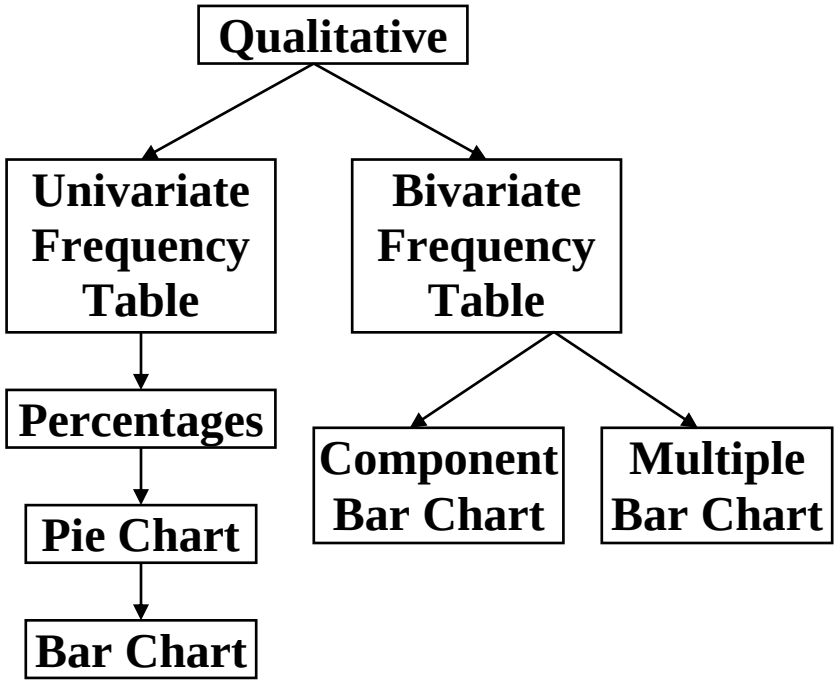
LECTURE No 25

- Tabulation
- Simple bar chart
- Component bar chart
- Multiple bar chart
- Pie chart

As indicated in the last lecture, there are two broad categories of data … qualitative data and quantitative data. A variety of methods exist for summarizing and describing these two types of data. The tree-diagram below presents an outline of the various techniques



In today's lecture, we will be dealing with various techniques for summarizing and describing qualitative data.



We will begin with the univariate situation, and will proceed to the bivariate situation.

EXAMPLE:

Suppose that we are carrying out a survey of the students of first year studying in a co-educational college of Lahore. Suppose that in all there are 1200 students of first year in this large college.

We wish to determine what proportion of these students have come from Urdu medium schools and what proportion has come from English medium schools.

So we will interview the students and we will inquire from each one of them about their schooling.

As a result, we will obtain a set of data as you can now see on the screen.

We will have an array of observations as follows:

U, U, E, U, E, E, E, U, ……

(U : URDU MEDIUM)
(E : ENGLISH MEDIUM)

Now, the question is what should we do with this data?

Obviously, the first thing that comes to mind is to count the number of students who said "Urdu medium" as well as the number of students who said "English medium".

This will result in the following table:

Medium of Institution	No. of Students (f)
Urdu	719
English	481
	1200

The technical term for the numbers given in the second column of this table is "frequency".

It means "how frequently something happens?"

Out of the 1200 students, 719 stated that they had come from Urdu medium schools.

So in this example, the frequency of the first category of responses is 719 whereas the frequency of the second category of responses is 481.

It is evident that this information is not as useful as if we compute the proportion or percentage of students falling in each category.

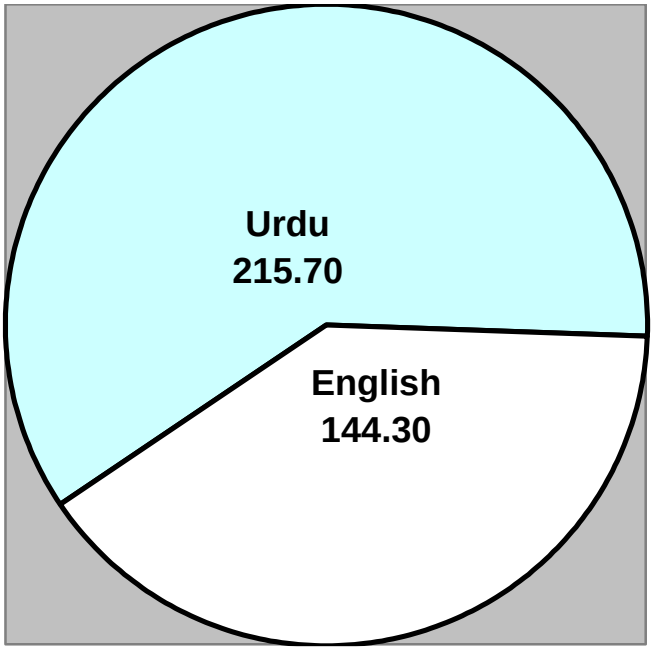
Dividing the cell frequencies by the total frequency and multiplying by 100 we obtain the following:

Medium of Institution	**f**	**%**
Urdu	719	59.9 = 60%
English	481	40.1 = 40%
	1200	

What we have just accomplished is an example of a univariate frequency table pertaining to qualitative data.
Let us now see how we can represent this information in the form of a diagram.
One good way of representing the above information is in the form of a pie chart.
A pie chart consists of a circle which is divided into two or more parts in accordance with the number of distinct categories that we have in our data.
For the example that we have just considered, the circle is divided into two sectors, the larger sector pertaining to students coming from Urdu medium schools and the smaller sector pertaining to students coming from English medium schools.
How do we decide where to cut the circle?
The answer is very simple! All we have to do is to divide the cell frequency by the total frequency and multiply by 360.
This process will give us the exact value of the angle at which we should cut the circle.

PIE CHART

Medium of Institution	**f**	**Angle**
Urdu	719	215.7^0
English	481	144.3^0
	1200	



SIMPLE BAR CHART:

The next diagram to be considered is the simple bar chart.

A simple bar chart consists of horizontal or vertical bars of equal width and lengths proportional to values they represent.

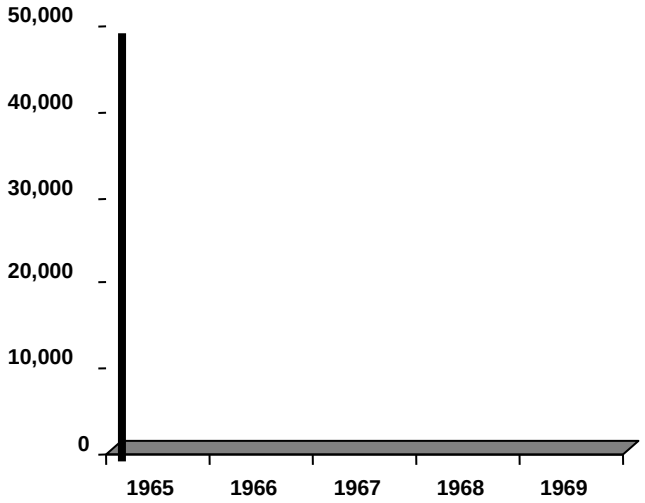
As the basis of comparison is one-dimensional, the widths of these bars have no mathematical significance but are taken in order to make the chart look attractive.
Let us consider an example.

Suppose we have available to us information regarding the turnover of a company for 5 years as given in the table below:

Years	1965	1966	1967	1968	1969
Turnover (Rupees)	35,000	42,000	43,500	48,000	48,500

In order to represent the above information in the form of a bar chart, all we have to do is to take the year along the x-axis and construct a scale for turnover along the y-axis.

Next, against each year, we will draw vertical bars of equal width and different heights

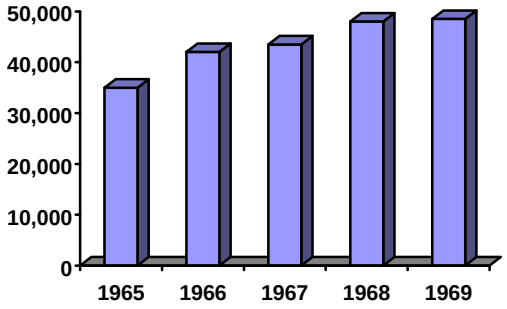


in accordance with the
turn-over figures that we have in our table.

As a result we obtain a simple and attractive diagram as shown below.
When our values do not relate to time, they should be arranged in ascending or descending order before-charting.

BIVARIATE FREQUENCY TABLE:

What we have just considered was the univariate situation.
In each of the two examples, we were dealing with one single variable.
In the example of the first year students of a college, our lone variable of interest was 'medium of schooling'.



And in the second example, our one single variable of interest was turnover.
Now let us expand the discussion a little, and consider the bivariate situation.
Going back to the example of the first year students, suppose that alongwith the enquiry about the Medium of Institution, you are also recording the sex of the student.
Suppose that our survey results in the following information:

Student No.	**Medium**	**Gender**
1	U	F
2	U	M
3	E	M
4	U	F
5	E	M
6	E	F
7	U	M
8	E	M
:	:	:
:	:	:

Now this is a bivariate situation; we have two variables, medium of schooling and sex of the student.
In order to summarize the above information, we will construct a table containing a box head and a stub as shown below:

Sex **Med.**	**Male**	**Female**	**Total**
Urdu			

English			
Total			

The top row of this kind of a table is known as the boxhead and the first column of the table is known as stub.

Next, we will count the number of students falling in each of the following four categories:

1. Male student coming from an Urdu medium school.
2. Female student coming from an Urdu medium school.
3. Male student coming from an English medium school.
4. Female student coming from an English medium school.

As a result, suppose we obtain the following figures:

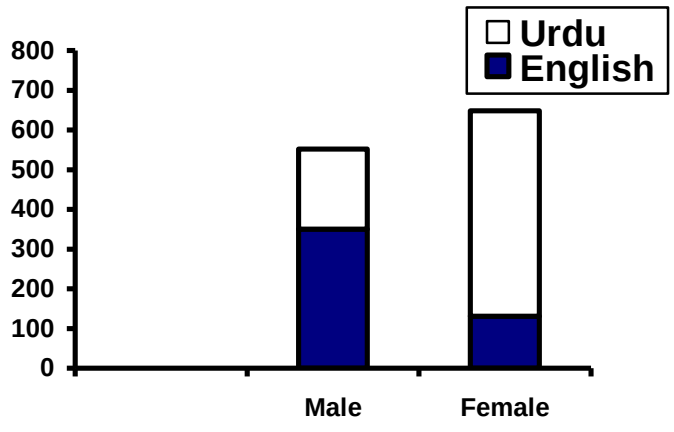
Sex / Med.	**Male**	**Female**	**Total**
Urdu	202	517	719
English	350	131	481
Total	552	648	1200

What we have just accomplished is an example of a bivariate frequency table pertaining to two qualitative variables.

COMPONENT BAR CHAR:

Let us now consider how we will depict the above information diagrammatically.

This can be accomplished by constructing the component bar chart (also known as the subdivided bar chart) as shown below:



In the above figure, each bar has been divided into two parts. The first bar represents the total number of male students whereas the second bar represents the total number of female students.

As far as the medium of schooling is concerned, the lower part of each bar represents the students coming from English medium schools. Whereas the upper part of each bar represents the students coming from the Urdu medium schools.The advantage of this kind of a diagram is that we are able to ascertain the situation of both the variables at a glance.

We can compare the number of male students in the college with the number of female students, and at the same time we can compare the number of English medium students among the males with the number of English medium students among the females.

MULTIPLE BAR CHART

The next diagram to be considered is the multiple bar chart.

Let us consider an example.

Suppose we have information regarding the imports and exports of Pakistan for the years 1970-71 to 1974-75 as shown in the table below:

Years	**Imports (Crores of Rs.)**	**Exports (Crores of Rs.)**
1970-71	370	200
1971-72	350	337
1972-73	840	855
1973-74	1438	1016
1974-75	2092	1029

Source: State Bank of Pakistan

A multiple bar chart is a very useful and effective way of presenting this kind of information.

This kind of a chart consists of a set of grouped bars, the lengths of which are proportionate to the values of our variables, and each of which is shaded or coloured differently in order to aid identification.

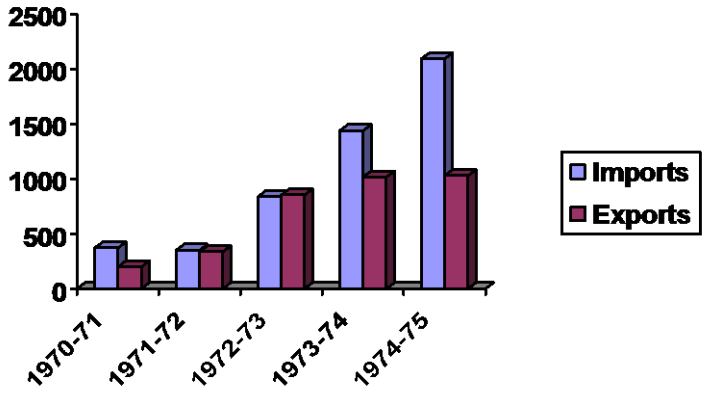
With reference to the above example, we obtain the multiple bar chart shown below:

Multiple Bar Chart Showing

Imports & Exports

of Pakistan 1970-71 to 1974-75

This is a very good device for the comparison of two different kinds of information.



If, in addition to information regarding imports and exports, we also had information regarding production, we could

have compared them from year to year by grouping the three bars together.
The question is, what is the basic difference between a component bar chart and a multiple bar chart?
The component bar chart should be used when we have available to us information regarding totals and their components.

For example, the total number of male students out of which some are Urdu medium and some are English medium. The number of Urdu medium male students and the number of English medium male students add up to give us the total number of male students.
On the contrary, in the example of exports and imports, the imports and exports do not add up to give us the totality of some one thing!

LECTURE No. 26

Today's lecture is in continuation with the last lecture, and today we will begin with various types of frequency curves that are encountered in practice. Also, we will discuss the cumulative frequency distribution and cumulative frequency polygon for a continuous variable.

In the last lecture, it was mentioned that:

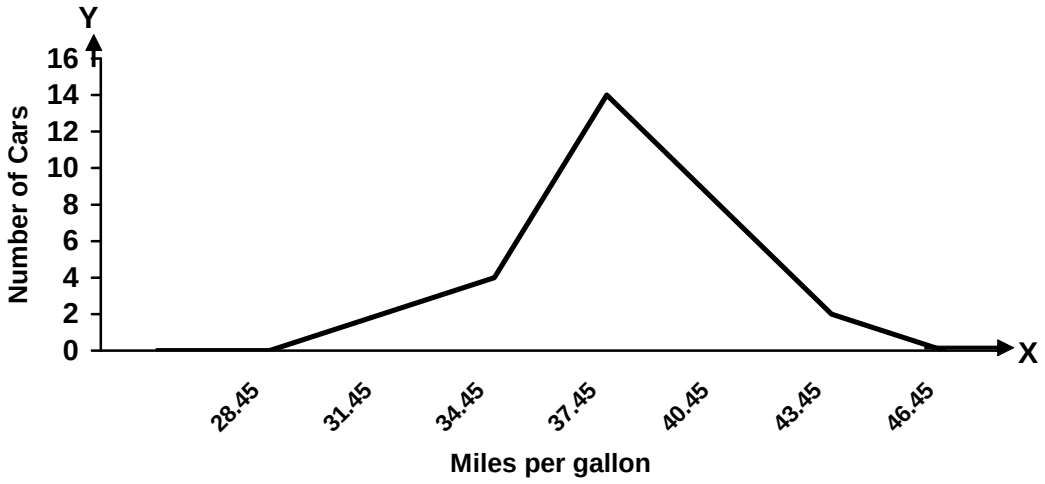
FREQUENCY POLYGON:

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments.

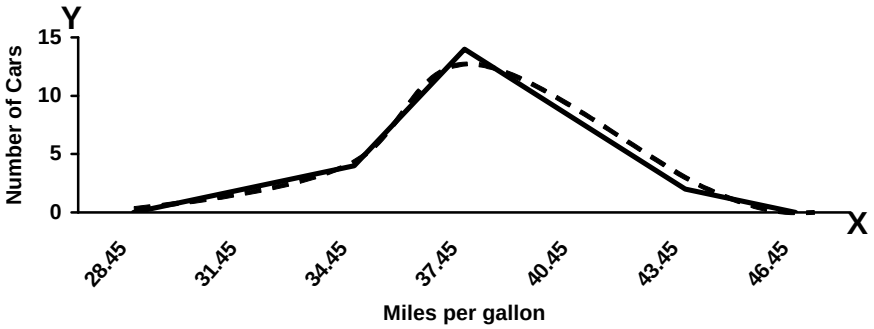
In our example of the EPA mileage ratings, the classes were:

Class Boundaries	Mid-Point (X)	Frequency (f)
26.95 – 29.95	28.45	
29.95 – 32.95	31.45	2
32.95 – 35.95	34.45	4
35.95 – 38.95	37.45	14
38.95 – 41.95	40.45	8
41.95 – 44.95	43.45	2
44.95 – 47.95	46.45	

And our frequency polygon came out to be:



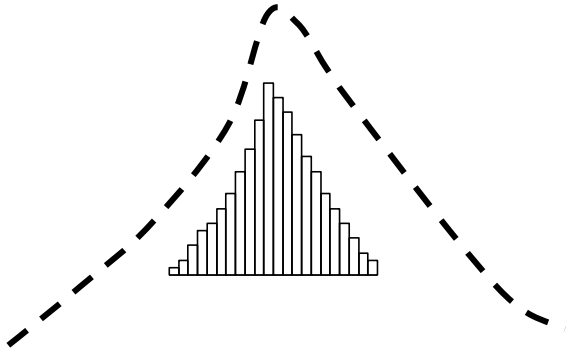
Also, it was mentioned that, when the frequency polygon is smoothed, we obtain what may be called the **FREQUENCY CURVE.**
In our example:



In the above figure, the dotted line represents the frequency curve. It should be noted that it is not necessary that our frequency curve must touch all the points. The purpose of the frequency curve is simply to display the overall pattern of the distribution. Hence we draw the curve by the free-hand method, and hence it does not have to touch all the plotted points. It should be realized that the frequency curve is actually a theoretical concept.
If the class interval of a histogram is made very small, and the number of classes is very large, the rectangles of the histogram will be narrow as shown below:



The smaller the class interval and the larger the number of classes, the narrower the rectangles will be. In this way, the histogram approaches a smooth curve as shown below:



In spite of the fact that the frequency curve is a theoretical concept, it is useful in analyzing real-world problems. The reason is that very close approximations to theoretical curves are often generated in the real world so close that it is quite valid to utilize the properties of various types of mathematical curves in order to aid analysis of the real-world problem at hand.

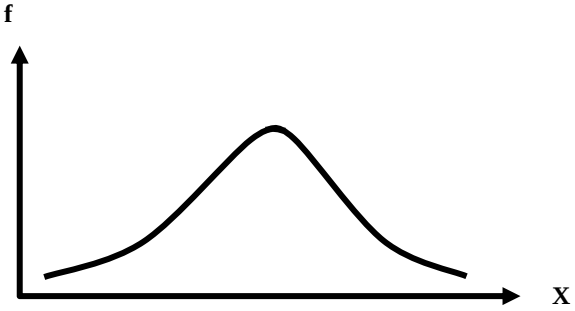
VARIOUS TYPES OF FREQUENCY CURVES:

- the symmetrical frequency curve
- the moderately skewed frequency curve
- the extremely skewed frequency curve
- the U-shaped frequency curve

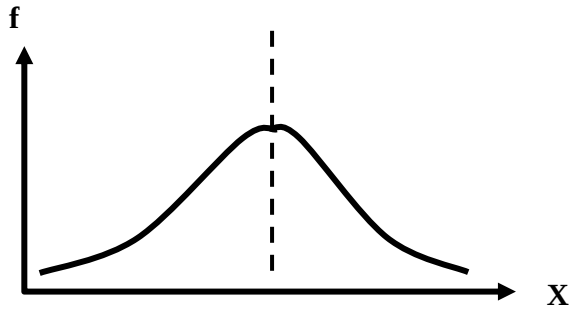
Let us discuss them one by one.

First of all, the symmetrical frequency curve is of the following shape:

THE SYMMETRIC CURVE

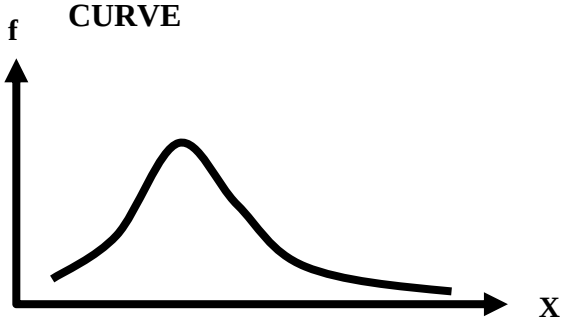


If we place a vertical mirror in the centre of this graph, the left hand side will be the mirror image of the right hand side.



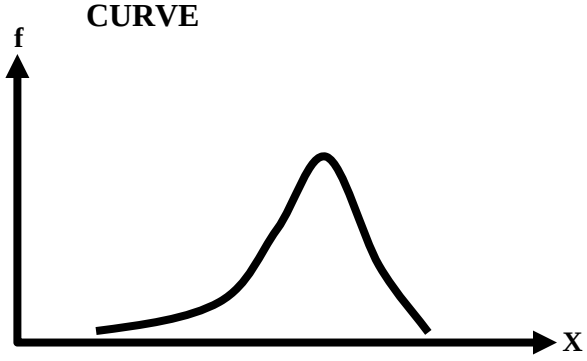
Next, we consider the moderately skewed frequency curve. We have the positively skewed curve and the negatively skewed curve. The positively skewed curve is that one whose right tail is longer than its left tail, as shown below

THE POSITIVELY SKEWED CURVE



On the other hand, the negatively skewed frequency curve is the one for which the left tail is longer than the right tail.

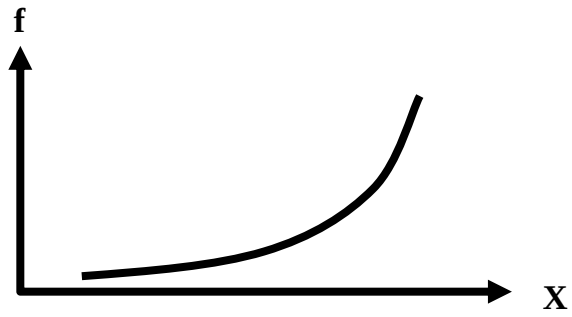
THE NEGATIVELY SKEWED CURVE



Both of these that we have just consider are moderately positively and negatively skewed.

Sometimes, we have the extreme case when we obtain the EXTREMELY skewed frequency curve. An extremely negatively skewed curve is of the type shown below:

THE EXTREMELY NEGATIVELY SKEWED (J-SHAPED) CURVE



This is the case when the maximum frequency occurs at the end of the frequency table.

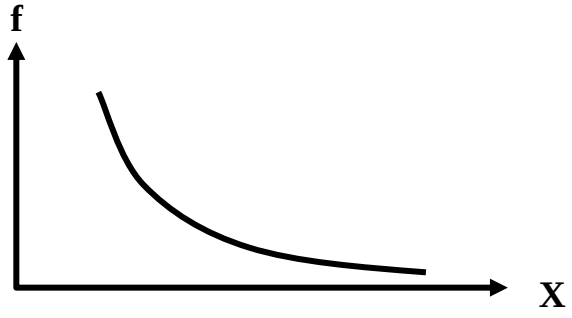
For example, if we think of the death rates of adult males of various age groups starting from age 20 and going up to age 79 years, we might obtain something like this:

DEATH RATES BY AGE GROUP

Age Group	No. of deaths per thousand
20 – 29	2.1
30 – 39	4.3
40 – 49	5.7
50 – 59	8.9
60 – 69	12.4
70 – 79	16.7

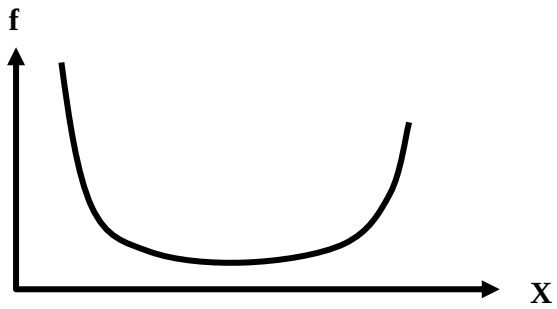
This will result in a J-shaped distribution similar to the one shown above.
Similarly, the extremely positively skewed distribution is known as the REVERSE J-shaped distribution.

THE EXTREMELY POSITIVELY SKEWED (REVERSE J-SHAPED) CURVE



A relatively LESS frequently encountered frequency distribution is the U-shaped distribution.

THE U-SHAPED CURVE



If we consider the example of the death rates not for only the adult population but for the population of ALL the age groups, we will obtain the **U-shaped distribution.**

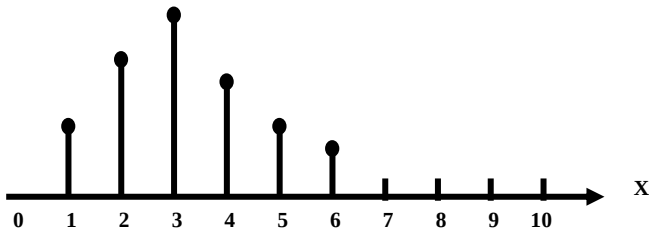
Out of all these curves, the MOST frequently encountered frequency distribution is the moderately skewed frequency distribution. There are thousands of natural and social phenomena which yield the moderately skewed frequency distribution. Suppose that we walk into a school and collect data of the weights, heights, marks, shoulder-lengths, finger-lengths or any other such variable pertaining to the children of any one class.

If we construct a frequency distribution of this data, and draw its histogram and its frequency curve, we will find that our data will generate a moderately skewed distribution. Until now, we have discussed the various possible shapes of the frequency distribution of a continuous variable.

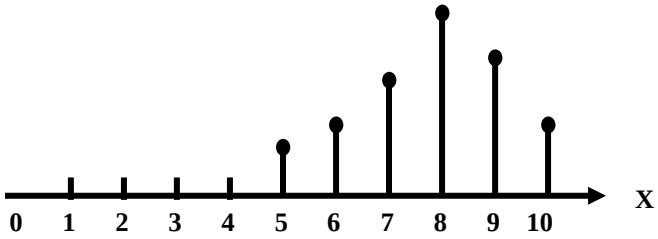
Similar shapes are possible for the frequency distribution of a discrete variable.

VARIOUS TYPES OF DISCRETE FREQUENCY DISTRIBUTION

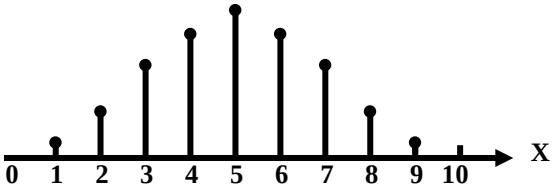
I. Positively Skewed Distribution



II. Negatively Skewed Distribution



III. Symmetric Distribution



Let us now consider another aspect of the frequency distribution i.e.

CUMULATIVE FREQUENCY DISTRIBUTION.

As in the case of the frequency distribution of a discrete variable, if we start adding the frequencies of our frequency table column-wise, we obtain the column of cumulative frequencies.

In our example, we obtain the cumulative frequencies shown below:

CUMULATIVE FREQUENCY DISTRIBUTION

Class Boundaries	Frequency	Cumulative Frequency
29.95 – 32.95	2	2
32.95 – 35.95	4	2+4 = 6
35.95 – 38.95	14	6+14 = 20
38.95 – 41.95	8	20+8 = 28
41.95 – 44.95	2	28+2 = 30
	30	

In the above table, 2+4 gives 6, 6+14 gives 20, and so on.

The question arises: “What is the purpose of making this column?”

You will recall that, when we were discussing the frequency distribution of a discrete variable, any particular cumulative frequency meant that we were counting the number of observations starting from the very first value of X and going up to THAT particular value of X against which that particular cumulative frequency was falling.

In case of a the distribution of a continuous variable, each of these cumulative frequencies represents the total frequency of a frequency distribution from the lower class boundary of the lowest class to the UPPER class boundary of THAT class whose cumulative frequency we are considering.

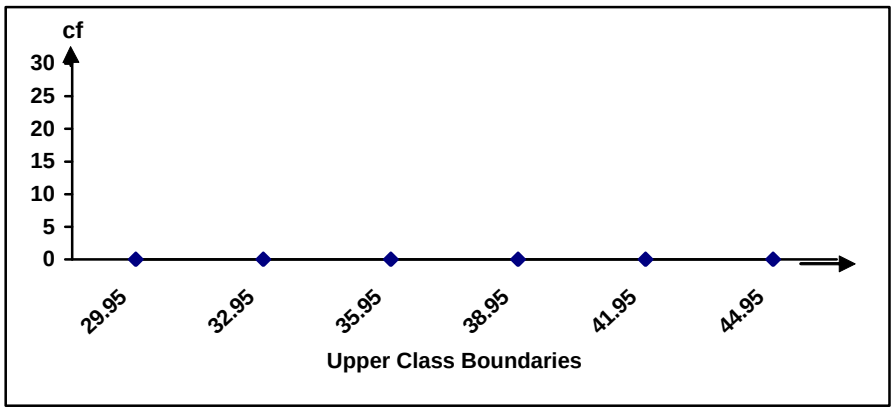
In the above table, the total number of cars showing mileage less than 35.95 miles per gallon is 6, the total number of car showing mileage less than 41.95 miles per gallon is 28, etc.

CUMULATIVE FREQUENCY DISTRIBUTION

Class Boundaries	Frequency	Cumulative Frequency
29.95 – 32.95	2	2
32.95 – 35.95	4	2+4 = 6
35.95 – 38.95	14	6+14 = 20
38.95 – 41.95	8	20+8 = 28
41.95 – 44.95	2	28+2 = 30
	30	

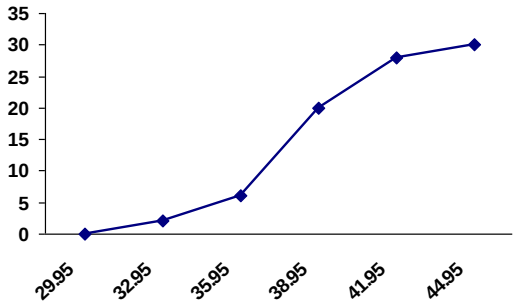
Such a cumulative frequency distribution is called a "**less than**" type of a cumulative frequency distribution. The graph of a cumulative frequency distribution is called a **CUMULATIVE FREQUENCY POLYGON or OGIVE.**

A "less than" type ogive is obtained by marking off the upper class boundaries of the various classes along the X-axis and the cumulative frequencies along the y-axis, as shown below:



The cumulative frequencies are plotted on the graph paper against the upper class boundaries, and the points so obtained are joined by means of straight line segments. Hence we obtain the cumulative frequency polygon shown below:

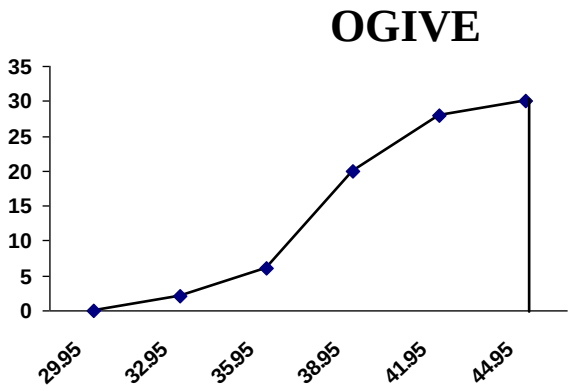
Cumulative Frequency Polygon or OGIVE



It should be noted that this graph is touching the X-Axis on the left-hand side. This is achieved by ADDING a class having zero frequency in the beginning of our frequency distribution, as shown below:

Class Boundaries	Frequency	Cumulative Frequency
26.95 – 29.95	0	0
29.95 – 32.95	2	0+2 = 2
32.95 – 35.95	4	2+4 = 6
35.95 – 38.95	14	6+14 = 20
38.95 – 41.95	8	20+8 = 28
41.95 – 44.95	2	28+2 = 30
	30	

Since the frequency of the first class is zero, hence the cumulative frequency of the first class will also be zero, and hence, automatically, the cumulative frequency polygon will touch the X-Axis from the left hand side. If we want our cumulative frequency polygon to be closed from the right-hand side also , we can achieve this by connecting the last point on our graph paper with the X-axis by means of a vertical line, as shown below:



In the example of EPA mileage ratings, all the data-values were correct to one decimal place.

Let us now consider another example:

EXAMPLE:

For a sample of 40 pizza products, the following data represent cost of a slice in dollars (S Cost).

PRODUCT	S cost
Pizza Hut Hand Tossed	**1.51**
Domino's Deep Dish	**1.53**
Pizza Hut Pan Pizza	**1.51**
Domino's Hand Tossed	**1.90**
Little Caesars Pan! Pizza!	**1.23**

PRODUCT	S Cost
Boboli crust with Boboli sauce	1.00
Jack's Super Cheese	0.69
Pappalo's Three Cheese	0.75
Tombstone Original Extra Cheese	0.81
Master Choice Gourmet Four Cheese	0.90
Celeste Pizza For One	0.92
Totino's Party	0.64
The New Weight Watchers Extra Cheese	1.54
Jeno's Crisp'N Tasty	0.72
Stouffer's French Bread 2-Cheese	1.15

PRODUCT	S Cost
Ellio's 9-slice	0.52
Kroger	0.72
Healthy Choice French Bread	1.50
Lean Cuisine French Bread	1.49
DiGiorno Rising Crust	0.87
Tombstone Special Order	0.81
Pappalo's	0.73
Jack's New More Cheese!	0.64
Tombstone Original	0.77
Red Baron Premium	0.80

PRODUCT	Scost
Tony's Italian Style Pastry Cruse	0.83
Red Baron Deep Dish Singles	1.13
Totino's Party	0.62
The New Weight Watchers	1.52
Jeno's Crisp'N Tasty	0.71
Stouffer's French Bread	1.14
Celeste Pizza For One	1.11
Tombstone For One French Bread	1.11
Healthy Choice French Bread	1.46
Lean Cuisine French Bread	1.71

PRODUCT	Scost
Little Caesars Pizza! Pizza!	**1.28**
Pizza Hut Stuffed Crust	**1.23**
DiGiorno Rising Crust Four Cheese	**0.90**
Tombstone Speical Order Four Cheese	**0.85**
Red Baron Premium 4-Cheese	**0.80**

Source: "Pizza," Copyright 1997 by Consumers Union of United States, Inc., Yonkers, N.Y. 10703.

Example taken from "Business Statistics – A First Course" by Mark L. Berenson & David M. Levine (International Edition), Prentice-Hall International, Inc., Copyright © 1998.

In order to construct the frequency distribution of the above data, the first thing to note is that, in this example, all our data values are correct to two decimal places. As such, we should construct the class limits correct to TWO decimal places, and the class boundaries correct to three decimal places.

As in the last example, first of all, let us find the maximum and the minimum values in our data, and compute the RANGE.

Minimum value X0 = 0.52
Maximum value Xm = 1.90
Hence:

Range = 1.90 - 0.52
= 1.38

Desired number of classes = 8
Hence:

Class interval h = **RANGE/No. of classes**

= 1.38 / 8 = 0.1725 ~ 0.20

Lower limit of the first class = 0.51
Hence, our successive class limits come out to be:

Class Limits
0.51 – 0.70
0.71 – 0.90
0.91 – 1.10
1.11 – 1.30
1.31 – 1.50
1.51 – 1.70
1.71 – 1.90

Stretching the class limits to the left and to the right, we obtain class boundaries as shown below:

Class Limits	Class Boundaries
0.51 – 0.70	0.505 – 0.705
0.71 – 0.90	0.705 – 0.905
0.91 – 1.10	0.905 – 1.105
1.11 – 1.30	1.105 – 1.305
1.31 – 1.50	1.305 – 1.505
1.51 – 1.70	1.505 – 1.705
1.71 – 1.90	1.705 – 1.905

By tallying the data-values in the appropriate classes, we will obtain a frequency distribution similar to the one that we obtained in the examples of the EPA mileage ratings.

By constructing the histogram of this data-set, we will be able to decide whether our distribution is symmetric, positively skewed or negatively skewed. This may please be attempted as an exercise.

LECTURE No. 27

In **today's lecture**, we will begin with a diagram called **STEM AND LEAF PLOT.**

This plot was introduced by the famous statistician *John Tukey in 1977*.

A frequency table has the disadvantage that the identity of individual observations is lost in grouping process. To overcome this drawback, John Tukey (1977) introduced this particular technique (**known as the Stem-and-Leaf Display).**

This technique offers a quick and novel way for simultaneously sorting and displaying data sets where each number in the data set is divided into two parts, a Stem and a Leaf.

A stem is the leading digit(s) of each number and is used in sorting, while a leaf is the rest of the number or the trailing digit(s) and shown in display. A vertical line separates the leaf (or leaves) from the stem.

For example, the number 243 could be split in two ways:

Leading Digit	Trailing Digits
2	43
Stem	Leaf

OR

Leading Digit	Trailing Digit
24	3
Stem	Leaf

How do we construct a stem and leaf display when we have a whole set of values?

This is explained by way of the following example:

Example:

The ages of 30 patients admitted to a certain hospital during a particular week were as follows:

48, 31, 54, 37, 18, 64, 61, 43,
40, 71, 51, 12, 52, 65, 53, 42,
39, 62, 74, 48, 29, 67, 30, 49,
68, 35, 57, 26, 27, 58.

Construct a stem-and-leaf display from the data and list the data in an array.

A scan of the data indicates that the observations range (in age) from 12 to 74. We use the first (or leading) digit as the stem and the second (or trailing) digit as the leaf. The first observation is 48, which has a stem of 4 and a leaf of 8, the second a stem of 3 and a leaf of 1, etc. Placing the leaves in the order in which they APPEAR in the data, we get the stem-and-leaf display as shown below:

Stem (Leading Digit)	Leaf (Trailing Digit)
1	8 2
2	9 6 7
3	1 7 9 0 5
4	8 3 0 2 8 9
5	4 1 2 3 7 8
6	4 1 5 2 7 8
7	1 4

But it is a common practice to ARRANGE the trailing digits in each row from smallest to highest. In this example, in order to obtain an array, we associate the leaves in order of size with the stems as shown below:

DATA IN THE FORM OF AN ARRAY :
(in ascending order):
12, 18, 26, 27, 29, 30, 31, 35,
37, 39, 40, 42, 43, 48, 48, 49,
51, 52, 53, 54, 57, 58, 61, 62,
64, 65, 67, 68, 71, 74.

Hence we obtain the stem and leaf plot shown below:

STEM AND LEAF
DISPLAY

Stem (Leading Digit)	Leaf (Trailing Digit)
1	2 8
2	6 7 9
3	0 1 5 7 9
4	0 2 3 8 8 9
5	1 2 3 4 7 8
6	1 2 4 5 7 8
7	1 4

The stem-and-leaf table provides a useful description of the data set and, if we so desire, can easily be converted to a frequency table.

In this example:

The frequency of the class 10-19 is 2, the frequency of the class 20-29 is 3, and the frequency of the class 30-39 is 5, and so on.

STEM AND LEAF
DISPLAY

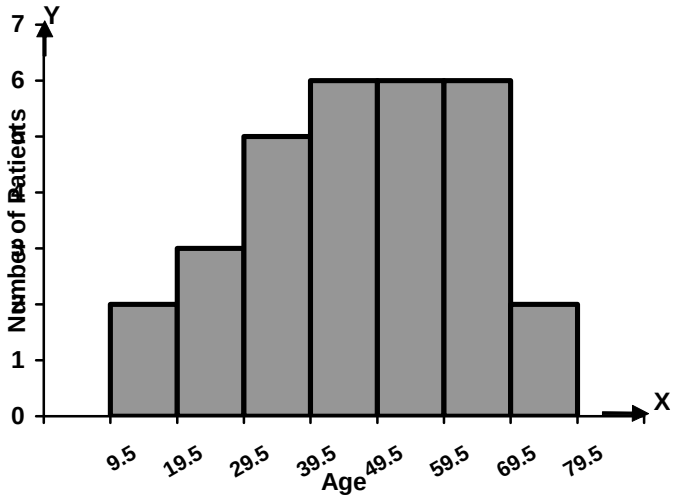
Stem (Leading Digit)	Leaf (Trailing Digit)
1	2 8
2	6 7 9
3	0 1 5 7 9
4	0 2 3 8 8 9
5	1 2 3 4 7 8
6	1 2 4 5 7 8
7	1 4

Hence, this stem and leaf plot conveniently converts into the frequency distribution shown below:

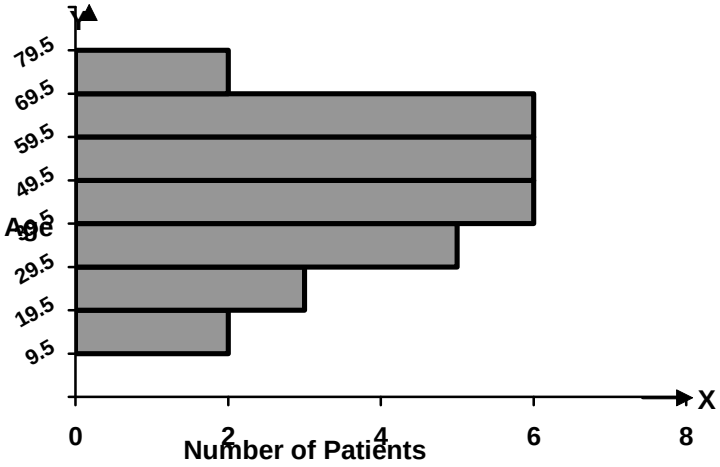
FREQUENCY DISTRIBUTION

Class Limits	Class Boundaries	Tally Marks	Frequency
10 – 19	9.5 – 19.5	//	2
20 – 29	19.5 – 29.5	///	3
30 – 39	29.5 – 39.5	////	5
40 – 49	39.5 – 49.5	//// /	6
50 – 59	49.5 – 59.5	//// /	6
60 – 69	59.5 – 69.5	//// /	6
70 - 79	69.5 – 79.5	//	2

Converting this frequency **distribution into a histogram,** we obtain:



If we rotate this histogram by 90 degrees, we will obtain:



Let us re-consider the stem and leaf plot that we obtained a short while ago.

STEM AND LEAF DISPLAY

Stem (Leading Digit)	Leaf (Trailing Digit)
7	1 4
6	1 2 4 5 7 8
5	1 2 3 4 7 8
4	0 2 3 8 8 9
3	0 1 5 7 9
2	6 7 9
1	2 8

It is noteworthy that the shape of the stem and leaf display is exactly like the shape of our histogram. Let us now consider another example.

Example:

Construct a stem-and-leaf display for the data of mean annual death rates per thousand at ages 20-65 given below:

7.5, 8.2, 7.2, 8.9, 7.8, 5.4, 9.4, 9.9, 10.9, 10.8, 7.4, 9.7, 11.6, 12.6, 5.0, 10.2, 9.2, 12.0, 9.9, 7.3, 7.3, 8.4, 10.3, 10.1, 10.0, 11.1, 6.5, 12.5, 7.8, 6.5, 8.7, 9.3, 12.4, 10.6, 9.1, 9.7, 9.3, 6.2, 10.3, 6.6, 7.4, 8.6, 7.7, 9.4, 7.7, 12.8, 8.7, 5.5, 8.6, 9.6, 11.9, 10.4, 7.8, 7.6,

12.1, 4.6, 14.0, 8.1, 11.4, 10.6, 11.6, 10.4, 8.1, 4.6, 6.6, 12.8, 6.8, 7.1, 6.6, 8.8, 8.8, 10.7, 10.8, 6.0, 7.9, 7.3, 9.3, 9.3, 8.9, 10.1, 3.9, 6.0, 6.9, 9.0, 8.8, 9.4, 11.4, 10.9

Using the decimal part in each number as the leaf and the rest of the digits as the stem, we get the ordered stem-and-leaf display shown below:

STEM AND LEAF DISPLAY

Stem	Leaf
3	9
4	6 6
5	0 4 5
6	0 0 2 2 5 5 6 6 6 8 9
7	1 3 3 3 4 4 5 6 7 7 8 8 8 9
8	1 1 2 4 6 6 7 7 8 8 8 9 9
9	0 1 2 3 3 3 3 4 4 4 6 7 7 9 9
10	0 1 1 2 3 3 4 4 6 6 7 8 8 9 9
11	1 4 4 6 6 9
12	0 1 4 5 6 8 8
14	0

EXERCISE:

1) The above data may be converted into a stem and leaf plot (so as to verify that the one shown above is correct).

2) Various variations of the stem and leaf display may be studied on your own.

The next concept that we are going to consider is the concept of the central tendency of a data-set. In this context, the first thing to note is that in any data-based study, our data is always going to be variable, and hence, first of all, we will need to describe the data that is available to us.

DESCRIPTION OF VARIABLE DATA:

Regarding any statistical enquiry, primarily we need some means of describing the situation with which we are confronted. A concise numerical description is often preferable to a lengthy tabulation, and if this form of description also enables us to form a mental image of the data and interpret its significance, so much the better.

MEASURES OF CENTRAL TENDENCY AND MEASURES OF DISPERSION

- **Averages enable us to measure the central tendency of variable data**

- **Measures of dispersion enable us to measure its variability.**

AVERAGES (I.E. MEASURES OF CENTRAL TENDENCY)

An average is a single value which is intended to represent a set of data or a distribution as a whole. It is more or less CENTRAL value ROUND which the observations in the set of data or distribution usually tend to cluster.

As a measure of central tendency (i.e. an average) indicates the location or general position of the distribution on the X-axis, it is also known as a measure of location or position.

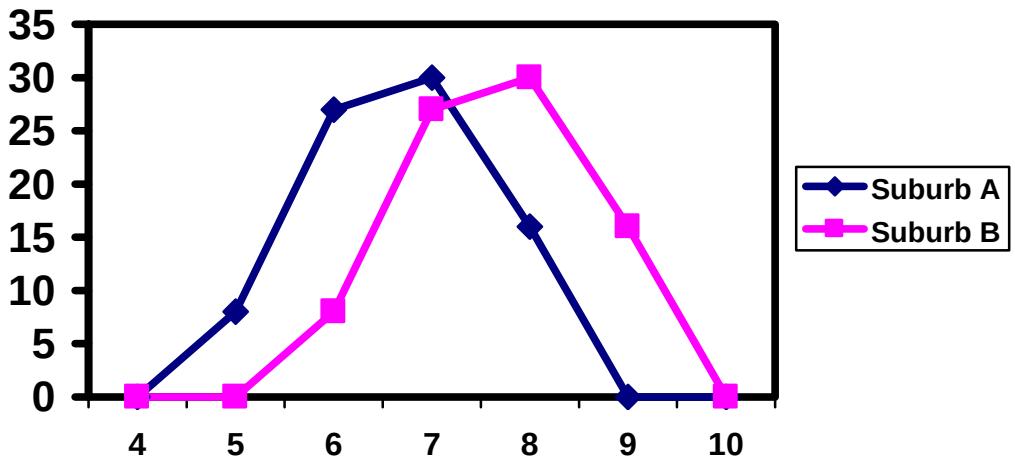
Let us consider an example:

Suppose that we have the following two frequency distributions:

EXAMPLE:

Looking at these two frequency distributions, we should ask ourselves what exactly is the distinguishing feature?

If we draw the frequency polygon of the two frequency distributions, we obtain



Inspection of these frequency polygons shows that they have exactly the same shape. It is their position relative to the horizontal axis (X-axis) which distinguishes them.

If we compute the mean number of rooms per house for each of the two suburbs, we will find that the average number of rooms per house in A is 6.67 while in B it is 7.67.

This difference of 1 is equivalent to the difference in position of the two frequency polygons.

Our interpretation of the above situation would be that there are LARGER houses in suburb B than in suburb A, to the extent that there are on the *average.* ONE More Room in each house.

Various TYPES of Averages:

There are several types of averages each of which has a use in specifically defined circumstances.

VARIOUS TYPES OF AVERAGES:

The most common types of averages are:

1) **The arithmetic mean,**
2) **The geometric mean,**
3) **The harmonic mean**
4) **The median, and**
5) **The mode**

The ***Arithmetic, Geometric and Harmonic*** means are averages that are mathematical in character, and give an indication of the magnitude of the observed values.

The ***Median*** indicates the middle position while the mode provides information about the most frequent value in the distribution or the set of data. THE MODE:

The ***Mode*** is defined as that value which occurs most frequently in a set of data i.e. it indicates the most common result.

EXAMPLE:

Suppose that the marks of eight students in a particular test are as follows:

2, 7, 9, 5, 8, 9, 10, 9

Obviously, the most common mark is 9.

In other words,

Mode = 9.

MODE IN CASE OF RAW DATA
PERTAINING TO A CONTINUOUS VARIABLE

In case of a set of values (pertaining to a continuous variable) that have not been grouped into a frequency distribution (i.e. in case of raw data pertaining to a continuous variable), the mode is obtained by counting the number of times each value occurs.

Example:

Suppose that the government of a country collected data regarding the percentages of revenues spent on Research and Development by 49 different companies, and obtained the following figures:

Percentage of Revenues Spent on Research and Development

Company	Percentage	Company	Percentage
1	13.5	14	9.5
2	8.4	15	8.1
3	10.5	16	13.5
4	9.0	17	9.9
5	9.2	18	6.9
6	9.7	19	7.5
7	6.6	20	11.1
8	10.6	21	8.2
9	10.1	22	8.0
10	7.1	23	7.7
11	8.0	24	7.4
12	7.9	25	6.5
13	6.8	26	9.5

Company	Percentage	Company	Percentage
27	8.2	39	6.5
28	6.9	40	7.5
29	7.2	41	7.1
30	8.2	42	13.2
31	9.6	43	7.7
32	7.2	44	5.9
33	8.8	45	5.2
34	11.3	46	5.6
35	8.5	47	11.7
36	9.4	48	6.0
37	10.5	49	7.8
38	6.9		

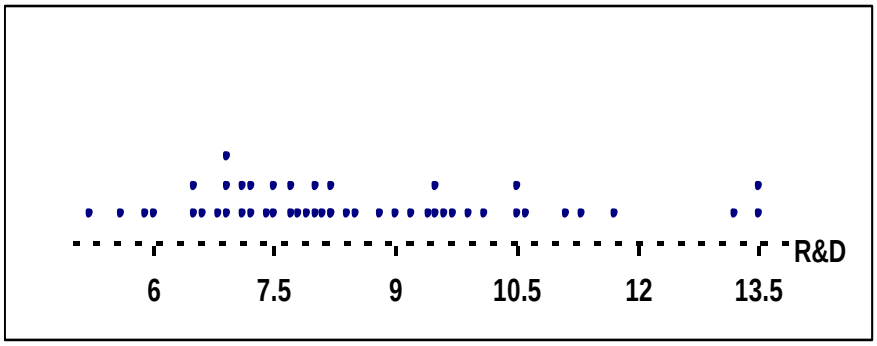
We can represent this data by means of a plot that is called dot plot.

DOT PLOT:

The horizontal axis of a dot plot contains a scale for the quantitative variable that we want to represent. The numerical value of each measurement in the data set is

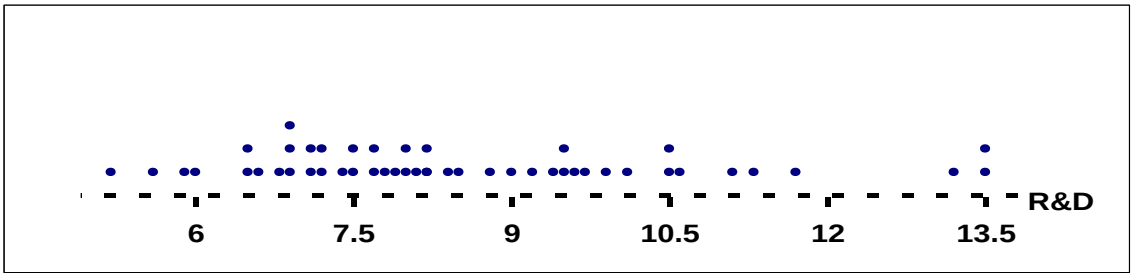
located on the horizontal scale by a dot. When data values repeat, the dots are placed above one another, forming a pile at that particular numerical location.
In this example

Dot Plot



The above material has been taken from
*"Statistics for Business and Economics" by James T. McClave, P. George Benson and Terry Sincich (Seventh Edition), © 1998 Prentice-Hal International***, Inc.**

As is obvious from the above diagram, the value 6.9 occurs 3 times whereas all the other values are occurring either once or twice.
Hence the modal value is 6.9.



Dot plot:

Dot Plot

$\hat{X}$ **= 6.9**

Also, this dot plot shows that

- almost all of the R&D percentages are falling between 6% and 12%,
- most of the percentages are falling between 7% and 9%.

THE MODE IN CASE OF A DISCRETE FREQUENCY DISTRIBUTION:

In case of a discrete frequency distribution, identification of the mode is immediate; one simply finds that value which has the highest frequency.

Example:

An airline found the following numbers of passengers in fifty flights of a forty-seated plane.

No. of Passengers X	No. of Flights f
28	1
33	1
34	2
35	3
36	5
37	7
38	10
39	13
40	8
Total	**50**

Highest Frequency fm = 13
Occurs against the X value 13.
Hence:

Mode = x = 13

The mode is obviously 39 passengers and the company should be quite satisfied that a 40 seater is the correct-size aircraft for this particular route.

THE MODE IN CASE OF THE *FREQUENCY DISTRIBUTION* OF A CONTINUOUS VARIABLE:

In case of grouped data, the modal group is easily recognizable (the one that has the highest frequency).

At what point within the modal group does the mode lie?

The answer is contained in the following formula:

Mode:

$$\hat{X} = 1 + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \text{ x } h$$

Where

l = lower class boundary of the modal class,

f_m = frequency of the modal class,

f_1 = frequency of the class preceding the modal class,

f_2 = frequency of the class following modal class, and

h = length of class interval of the modal class

Going back to the example of EPA mileage ratings, we have:

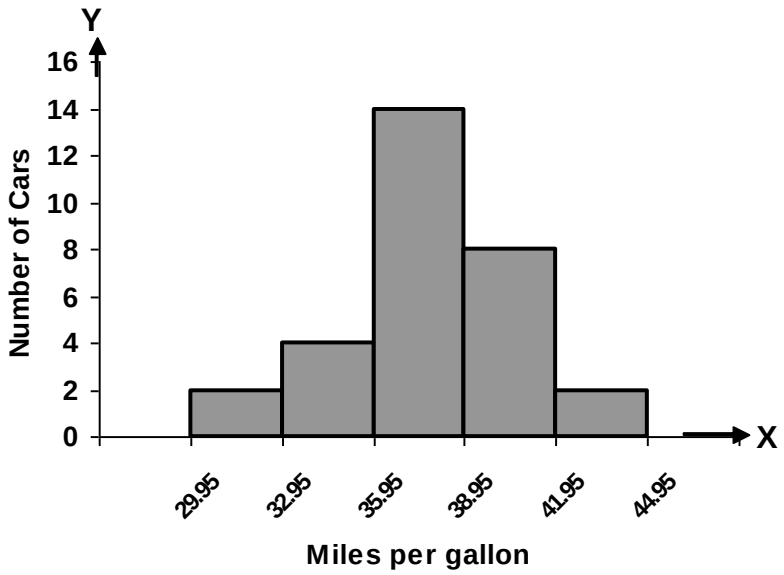
EPA MILEAGE RATINGS

Mileage Rating	Class Boundaries	No. of Cars
30.0 – 32.9	29.95 – 32.95	2
33.0 – 35.9	32.95 – 35.95	4 = f_1
36.0 – 38.9	35.95 – 38.95	14 = f_m
39.0 – 41.9	38.95 – 41.95	8 = f_2
42.0 – 44.9	41.95 – 44.95	2

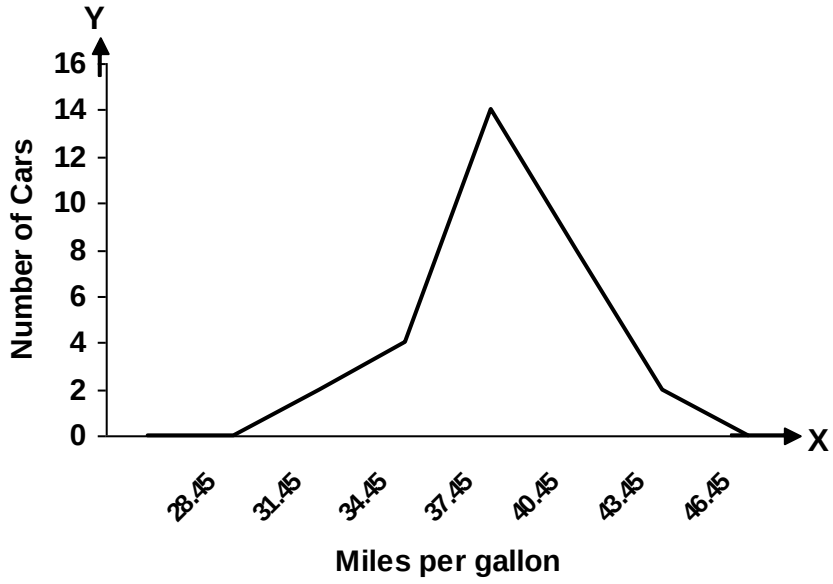
It is evident that the third class is the modal class. The mode lies somewhere between 35.95 and 38.95.
In order to apply the formula for the mode, we note that fm = 14, f1 = 4 and f2 = 8.
Hence we obtain:

$$\begin{aligned}\hat{X} &= 35.95 + \frac{14-4}{(14-4)+(14-8)} \times 3 \\ &= 35.95 + \frac{10}{10+6} \times 3 \\ &= 35.95 + 1.875 \\ &= 37.825\end{aligned}$$

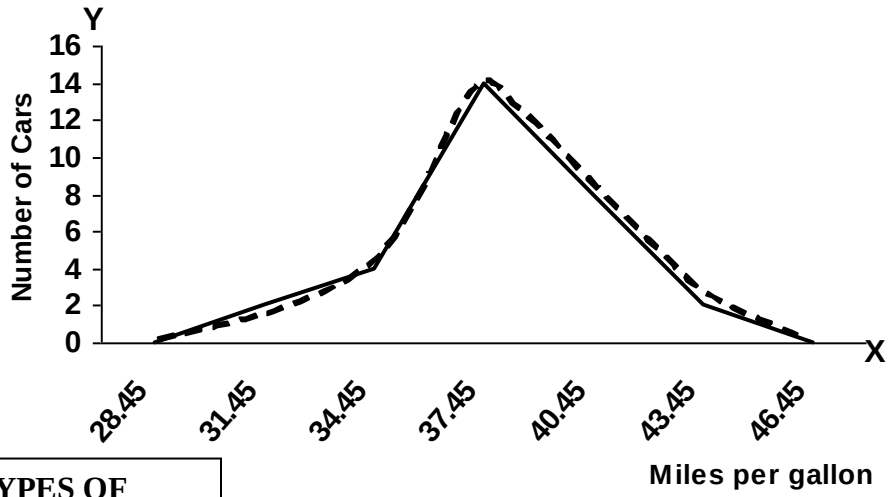
Let us now perceive the mode by considering the graphical representation of our frequency distribution. You will recall that, for the example of EPA Mileage Ratings, the histogram was as shown below:



The frequency polygon of the same distribution was:



And the frequency curve was as indicated by the dotted line in the following figure:

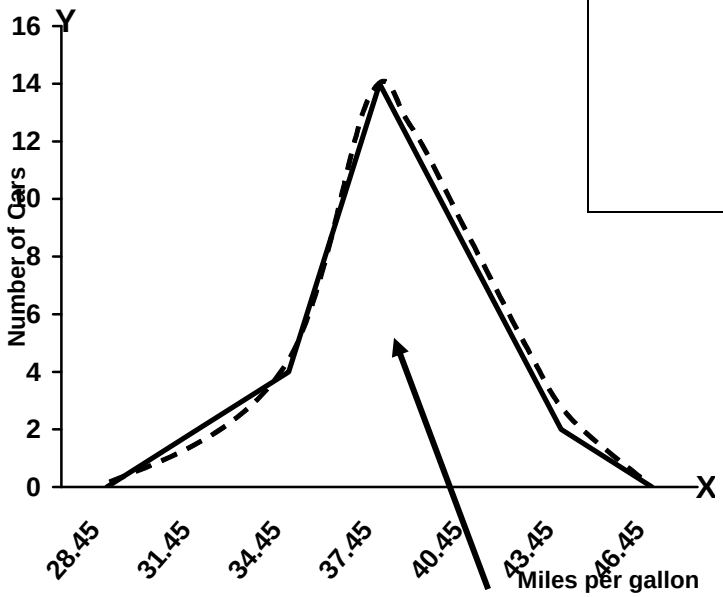


VARIOUS TYPES OF AVERAGES.

In this example, the mode is 37.825, and if we locate this value on the X-axis, we obtain the following picture:

The most common types of averages are:

1) **the arithmetic mean,**
2) **the geometric mean,**
3) **the harmonic mean**
4) **the median, and**
5) **the mode**



$\hat{X}$ **= 37.825**

Since, in most of the situations the mode exists somewhere in the middle of our data-values, hence it is thought of as a measure of central tendency.

LECTURE NO 28

- Geometric mean
- Harmonic mean
- Relation between the arithmetic, geometric and harmonic means
- Some other measures of central tendency

GEOMETRIC MEAN:

The geometric mean, G, of a set of n positive values X1, X2,…,Xn is defined as the positive nth root of their product.

$$G = \sqrt[n]{X_1 X_2 ... X_n}$$

(Where $X_i > 0$)

When n is large, the computation of the geometric mean becomes laborious as we have to extract the nth root of the product of all the values.

The arithmetic is simplified by the use of logarithms.

Taking logarithms to the base 10, we get

$$\log G = \frac{1}{n}\left[\log X_1 + \log X_2 + ... + \log X_n\right]$$

Hence

$$= \frac{\sum \log X}{n}$$

$$G = \text{anti}\log\left[\frac{\sum \log X}{n}\right]$$

Example:

Find the geometric mean of numbers:

45, 32, 37, 46, 39,
36, 41, 48, 36.

Solution:

We need to compute the numerical value of

$$= \sqrt[9]{45\times 32\times 37\times 46\times 39\times 36\times 41\times 48\times 36}$$

But, obviously, it is a bit cumbersome to find the ninth root of a quantity. So we make use of logarithms, as shown below:

X	log X
45	1.6532
32	1.5052
37	1.5682
46	1.6628
39	1.5911
36	1.5563
41	1.6128
48	1.6812
36	1.5563
	14.3870

$$\log G = \frac{\sum \log X}{n}$$

$$= \frac{14.3870}{9} = 1.5986$$

$$\text{Hence } G = \text{anti log } 1.5986$$

$$= 39.68$$

The above example pertained to the computation of the geometric mean in case of raw data.
Next, we consider the computation of the geometric mean in the case of grouped data.

GEOMETRICMEAN
FOR GROUPED DATA:

In case of a frequency distribution having k classes with midpoints X1, X2, …,Xk and the corresponding frequencies f1, f2, …, fk (such that ∑fi = n), the geometric mean is given by

$$G = \sqrt[n]{X_1^{f_1} X_2^{f_2} \ldots . X_k^{f_k}}$$

Each value of X thus has to be multiplied by itself f times, and the whole procedure becomes quite a formidable task!
In terms of logarithms, the formula becomes

$$\log G = \frac{1}{n}\left[f_1 \log X_1 + f_2 \log X_2 + \ldots + f_k \log X_k\right]$$

$$= \frac{\sum f \log X}{n}$$

Hence

$$G = anti\log\left[\frac{\sum f \log X}{n}\right]$$

Obviously, the above formula is much easier to handle.
Let us now apply it to an example.
Going back to the example of the EPA mileage ratings, we have:

Mileage Rating	No. of Cars	Class-mark (midpoint) X	log X	f log X
30.0 - 32.9	2	31.45	1.4976	2.9952
33.0 - 35.9	4	34.45	1.5372	6.1488
36.0 - 38.9	14	37.45	1.5735	22.0290
39.0 - 41.9	8	40.45	1.6069	12.8552
42.0 - 44.9	2	43.45	1.6380	3.2760
	30			47.3042

$$G = \text{antilog } \frac{47.3042}{30}$$

= antilog 1.5768 = 37.74

This means that, if we use the geometric mean to measures the central tendency of this data set, then the central value of the mileage of those 30 cars comes out to be 37.74 miles per gallon.

The question is, "When should we use the geometric mean?"

The answer to this question is that when relative changes in some variable quantity are averaged, we prefer the geometric mean.

EXAMPLE:

Suppose it is discovered that a firm's turnover has increased during 4 years by the following amounts:

Year	Turnover	Percentage Compared With Year Earlier
1958	£ 2,000	–
1959	£ 2,500	125
1960	£ 5,000	200
1961	£ 7,500	150
1962	£ 10,500	140

The yearly increase is shown in a percentage form in the right-hand column i.e. the turnover of 1959 is 125 percent of the turnover of 1958, the turnover of 1960 is 200 percent of the turnover of 1959, and so on. The firm's owner may be interested in knowing his average rate of turnover growth.

If the arithmetic mean is adopted he finds his answer to be:

Arithmetic Mean:

$$\frac{125+200+150+140}{4}$$

$$=153.75$$

i.e. we are concluding that the turnover for any year is 153.75% of the turnover for the previous year. In other words, the turnover in each of the years considered appears to be 53.75 per cent higher than in the previous year.
If *this* percentage is used to calculate the turnover from 1958 to 1962 inclusive, we obtain:

153.75% of £ 2,000 = £ 3,075
153.75% of £ 3,075 = £ 4,728
153.75% of £ 4,728 = £ 7,269
153.75% of £ 7,269 = £ 11,176

Whereas the actual turnover figures were

Year	Turnover
1958	£ 2,000
1959	£ 2,500
1960	£ 5,000
1961	£ 7,500
1962	£ 10,500

It seems that both the individual figures and, more important, the total at the end of the period, are incorrect. Using the arithmetic mean has exaggerated the 'average' annual rate of increase in the turnover of this firm. Obviously, we would like to rectify this false impression. The geometric mean enables us to do so:
Geometric mean of the turnover figures:

$$\sqrt[4]{(125\times 200\times 150\times 140)}$$
$$= \sqrt[4]{525000000}$$
$$= 151.37\%$$

Now, if we utilize this particular value to obtain the individual turnover figures, we find that:
151.37% of £2,000 = £3,027
151.37% of £3,027 = £4,583
151.37% of £4,583 = £6,937
151.37% of £6,937 = £10,500
So that the turnover figure of 1962 is exactly the same as what we had in the original data.

Interpretation:

If the turnover of this company were to increase annually at a constant rate, then the annual increase would have been 51.37 percent.(On the average, each year's turnover is 51.37% higher than that in the previous year.) The above example clearly indicates the significance of the geometric mean in a situation when relative changes in a variable quantity are to be averaged.

But we should bear in mind that such situations are not encountered too often, and that the occasion to calculate the geometric mean arises less frequently than the arithmetic mean.(The most frequently used measure of central tendency is the arithmetic mean.)

The next measure of central tendency that we will discuss is the harmonic mean.

HARMONIC MEAN;

The harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the values. **HARMONIC MEAN**

In case of raw data:

$$H.M. = \frac{n}{\sum\left(\frac{1}{X}\right)}$$

In case of grouped data (data grouped into a frequency distribution):

$$H.M. = \frac{n}{\sum f\left(\frac{1}{X}\right)}$$

(where X represents the midpoints of the various classes).

EXAMPLE:

Suppose a car travels 100 miles with 10 stops, each stop after an interval of 10 miles. Suppose that the speeds at which the car travels these 10 intervals are 30, 35, 40, 40, 45, 40, 50, 55, 55 and 30 miles per hours respectively.

What is the average speed with which the car traveled the total distance of 100 miles?

If we find the arithmetic mean of the 10 speeds, we obtain:

Arithmetic mean of the 10 speeds:

$$\frac{30+35+....+30}{10}$$

$$= \frac{420}{10} = 42 \text{ miles per hour.}$$

But, if we study the problem carefully, we find that the above answer is incorrect.

By definition, the average speed is the speed with which the car would have traveled the 100 mile distance if it had maintained a constant speed throughout the 10 intervals of 10 miles each.

$$\textbf{Average speed} = \frac{\textbf{Total distance travelled}}{\textbf{Total time taken}}$$

Now, *total distance traveled* = 100 miles.

Total time taken will be computed as shown below:

Interval	Distance	Speed = $\frac{\text{Distance}}{\text{Time}}$	Time = $\frac{\text{Distance}}{\text{Speed}}$
1	**10 miles**	**30 mph**	**10/30 = 0.3333 hrs**
2	**10 miles**	**35 mph**	**10/35 = 0.2857 hrs**
3	**10 miles**	**40 mph**	**10/40 = 0.2500 hrs**
4	**10 miles**	**40 mph**	**10/40 = 0.2500 hrs**
5	**10 miles**	**45 mph**	**10/45 = 0.2222 hrs**
6	**10 miles**	**40 mph**	**10/40 = 0.2500 hrs**
7	**10 miles**	**50 mph**	**10/50 = 0.2000 hrs**
8	**10 miles**	**55 mph**	**10/55 = 0.1818 hrs**
9	**10 miles**	**55 mph**	**10/55 = 0.1818 hrs**
10	**10 miles**	**30 mph**	**10/30 = 0.333 hrs**
Total = 100 miles		**Total Time = 2.4881 hrs**	

Hence
Average speed = $\frac{100}{2.4881} = 40.2\ mph$

which is not the same as 42 miles per hour.
Let us now try the harmonic mean to find the average speed of the car.

$$H.M. = \frac{n}{\sum\left(\frac{1}{X}\right)}$$

where n is the no. of terms)
We have:

X	1/X
30	**1/30 = 0.0333**
35	**1/35 = 0.0286**
40	**1/40 = 0.0250**
40	**1/40 = 0.0250**
45	**1/45 = 0.0222**
40	**1/40 = 0.0250**
50	**1/50 = 0.0200**
55	**1/55 = 0.0182**
55	**1/55 = 0.0182**
30	**1/30 = 0.0333**
	$\sum\frac{1}{X} = 0.2488$

$$\text{H.M.} = \frac{n}{\sum\frac{1}{X}}$$

$$= \frac{10}{0.2488}$$

= 40.2 mph

Hence it is clear that the harmonic mean gives the totally correct result.

The key question is, "When should we compute the harmonic mean of a data set?"
The answer to this question will be easy to understand if we consider the following rules:

RULES

1. When values are given as x per y where x is constant and y is variable, the Harmonic Mean is the appropriate average to use.
2. When values are given as x per y where y is constant and x is variable, the Arithmetic Mean is the appropriate average to use.
3. When relative changes in some variable quantity are to be averaged, the geometric mean is the appropriate average to use.

We have already discussed the geometric and the harmonic means.
Let us now try to understand Rule No. 1 with the help of an example:

EXAMPLE:

If 10 students have obtained the following marks (in a test) out of 20:
13, 11, 9, 9, 6,
5, 19, 17, 12, 9
Then the average marks (by the formula of the arithmetic mean) are:

$$\frac{13+11+9+9+6+5+19+17+12+9}{10}$$

$$=\frac{110}{10}=11$$

This is equivalent to

$$\frac{\frac{13}{20}+\frac{11}{20}+\frac{9}{20}+\frac{9}{20}+\frac{6}{20}+\frac{5}{20}+\frac{19}{20}+\frac{17}{20}+\frac{12}{20}+\frac{9}{20}}{10}$$

$$=\frac{\frac{110}{20}}{10}=\frac{110}{10\times 20}=\frac{11}{20}$$

(i.e. the average marks of this group of students are 11 out of 20).
In the above example, the point to be noted was that all the marks were expressible as x per y where the denominator y was constant i.e. equal to 20, and hence, it was appropriate to compute the arithmetic mean.

Let us now consider a mathematical relationship exists between these three measures of central tendency.

RELATION BETWEEN ARITHMETIC, GEOMETRIC AND HARMONIC MEANS:

Arithmetic Mean ≥ Geometric Mean ≥Harmonic Mean

We have considered the five most well-known measures of central tendency i.e. arithmetic mean, median, mode, geometric mean and harmonic mean.
It is interesting to note that there are some other measures of central tendency as well.

Two of these are the mid range, and the mid quartile range.

Let us consider these one by one:

MID-RANGE:

If there are n observations with x0 and xm as their smallest and largest observations respectively, then their mid-range is defined as

$$\text{mid} - \text{range} = \frac{x_0 + x_m}{2}$$

It is obvious that if we add the smallest value with the largest, and divide by 2, we will get a value which is more or less in the middle of the data-set.

MID-QUARTILE RANGE:

If x1, x2… xn are n observations with Q1 and Q3 as their first and third quartiles respectively, then their mid-quartile range is defined as

$$\text{mid} - \text{quartile range} = \frac{Q_1 + Q_3}{2}$$

Similar to the case of the mid-range, if we take the arithmetic mean of the upper and lower quartiles, we will obtain a value that is somewhere in the middle of the data-set.

The mid-quartile range is also known as the mid-hinge.

Let us now revise briefly the core concept of central tendency:

Masses of data are usually expressed in the form of frequency tables so that it becomes easy to comprehend the data.

Usually, a statistician would like to go a step ahead and to compute a number that will represent the data in some definite way.

Any such single number that represents a whole set of data is called **'Average'**.

Technically speaking, there are many kinds of averages (i.e. there are several ways to compute them). These quantities that represent the data-set are called "measures of central tendency".

LECTURE NO 29

- Concept of dispersion
- Absolute and relative measures of dispersion
- Range
- Coefficient of dispersion
- Quartile deviation
- Coefficient of quartile deviation

Let us begin the concept of DISPERSION.

Just as variable series differ with respect to their location on the horizontal axis (having different 'average' values); similarly, they differ in terms of the amount of variability which they exhibit.

Let us understand this point with the help of an example:

EXAMPLE:
In a technical college, it may well be the case that the ages of a group of first-year students are quite consistent, e.g. 17, 18, 18, 19, 18, 19, 19, 18, 17, 18 and 18 years.
A class of evening students undertaking a course of study in their spare time may show just the opposite situation, e.g. 35, 23, 19, 48, 32, 24, 29, 37, 58, 18, 21 and 30.
It is very clear from this example that the variation that exists between the various values of a data-set is of substantial importance. We obviously need to be aware of the amount of variability present in a data-set if we are to come to useful conclusions about the situation under review. This is perhaps best seen from studying the two frequency distributions given below:

EXAMPLE:
The sizes of the classes in two comprehensive schools in different areas are as follows:

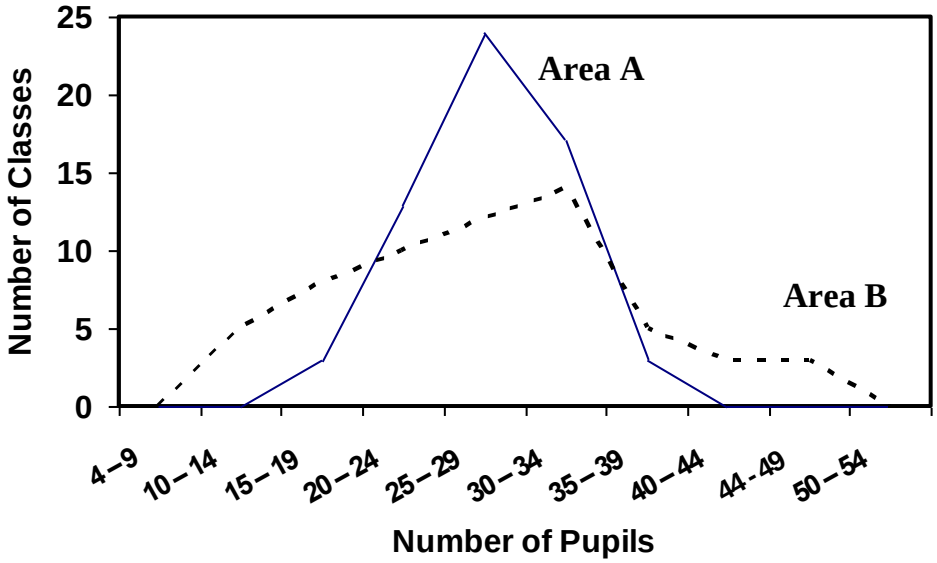
Number of Pupils	Number of Classes	
	Area A	Area B
10 – 14	0	5
15 – 19	3	8
20 – 24	13	10
25 – 29	24	12
30 – 34	17	14
35 – 39	3	5
40 – 44	0	3
45 - 49	0	3
	60	60

If the arithmetic mean size of class is calculated, we discover that the answer is identical: 27.33 pupils in both areas.

Average class-size of each school

$$\overline{X} = 27.33$$

Even though these two distributions share a common average, it can readily be seen that they are entirely DIFFERENT.
And the graphs of the two distributions (given below) clearly indicate this fact.



The question which must be posed and answered is 'In what way can these two situations be distinguished?'

We need a measure of variability or DISPERSION to accompany the relevant measure of position or 'average' used.

The word 'relevant' is important here for we shall find one measure of dispersion which expresses the scatter of values round the arithmetic mean, another the scatter of values round the median, and so forth. Each measure of dispersion is associated with a particular 'average'.

Absolute versus Relative Measures of Dispersion:

There are two types of measurements of dispersion: absolute and relative.

An ***absolute*** measure of dispersion is one that measures the dispersion in terms of the same units or in the square of units, as the units of the data.

For example, if the units of the data are rupees, meters, kilograms, etc., the units of the measures of dispersion will also be rupees, meters, kilograms, etc.

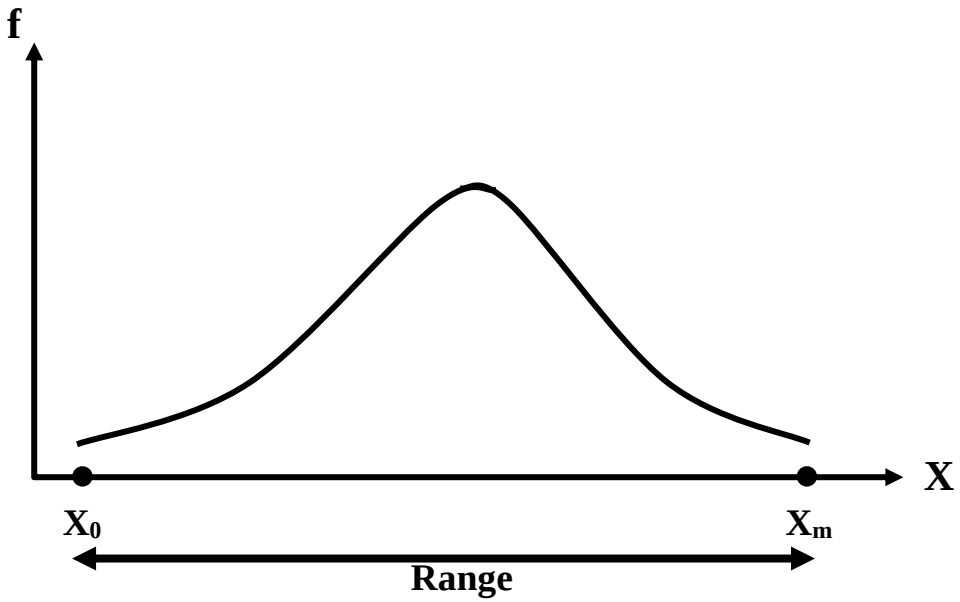
On the other hand, ***relative*** measure of dispersion is one that is expressed in the form of a ratio, co-efficient of percentage and is independent of the units of measurement.

A *relative* measure of dispersion is useful for comparison of data of different nature. A measure of central tendency together with a measure of dispersion gives an adequate description of data. We will be discussing FOUR measures of dispersion i.e. the range, the quartile deviation, the mean deviation, and the standard deviation.

RANGE:

The range is defined as the difference between the two extreme values of a data-set, i.e. R = Xm – X0 where Xm represents the highest value and X0 the lowest.
Evidently, the calculation of the range is a simple question of MENTAL arithmetic.

The simplicity of the concept does not necessarily invalidate it, but in general it gives no idea of the DISTRIBUTION of the observations between the two ends of the series. For this reason it is used principally as a supplementary aid in the description of variable data, in conjunction with other measures of dispersion. When the data are grouped into a frequency distribution, the range is estimated by finding the difference between the upper boundary of the highest class and the lower boundary of the lowest class.
We now consider the graphical representation of the range:



Obviously, the greater the difference between the largest and the smallest values, the greater will be the range. As stated earlier, the range is a simple concept and is easy to compute. However, because of the fact that it is computed from only the two extreme values in a data-set, it has two serious disadvantages.
1. It *ignores* all the INFORMATION available from the intermediate observations.
2. It might give a MISLEADING picture of the spread in the data.

From THIS point of view, it is an unsatisfactory measure of dispersion. However, it is APPROPRIATELY used in statistical quality control charts of manufactured products, daily temperatures, stock prices, etc.

It is interesting to note that the range can also be viewed in the following way:
It is twice of the arithmetic mean of the deviations of the smallest and largest values round the mid-range i.e.

$$\frac{(\text{Midrange} - X_0) + (X_m - \text{Midrange})}{2}$$

$$= \frac{\text{Midrange} - X_0 + X_m - \text{Midrange}}{2}$$

$$= \frac{X_m - X_0}{2}$$

Because of what has been just explained, the range can be regarded as that measure of dispersion which is associated with the mid-range. As such, the range may be employed to indicate dispersion when the mid-range has been adopted as the most appropriate average.

The range is an *absolute* measure of dispersion. Its *relative* measure is known as the CO-EFFICIENT OF DISPERSION, and is defined by the relation given below:

Coefficient of Dispersion:

$$= \frac{\frac{1}{2}(\text{Range})}{\text{Mid} - \text{Range}}$$

$$= \frac{\frac{X_m - X_0}{2}}{\frac{X_m + X_0}{2}} = \frac{X_m - X_0}{X_m + X_0}$$

This is a *pure* (i.e. *dimensionless) number* and is used for the purposes of *COMPARISON*. (This is so because a pure number *can* be compared with another pure number.)

For example, if the coefficient of dispersion for one data-set comes out to be 0.6 whereas the coefficient of dispersion for another data-set comes out to be 0.4, then it is obvious that there is greater amount of dispersion in the first data-set as compared with the second.

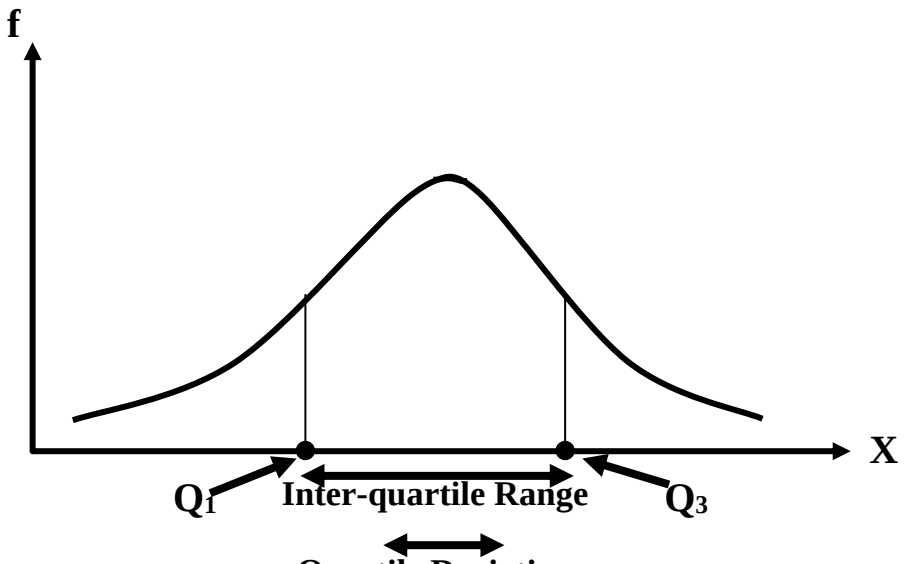
QUARTILE DEVIATION:

The quartile deviation is defined as half of the difference between the third and first quartiles i.e.

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

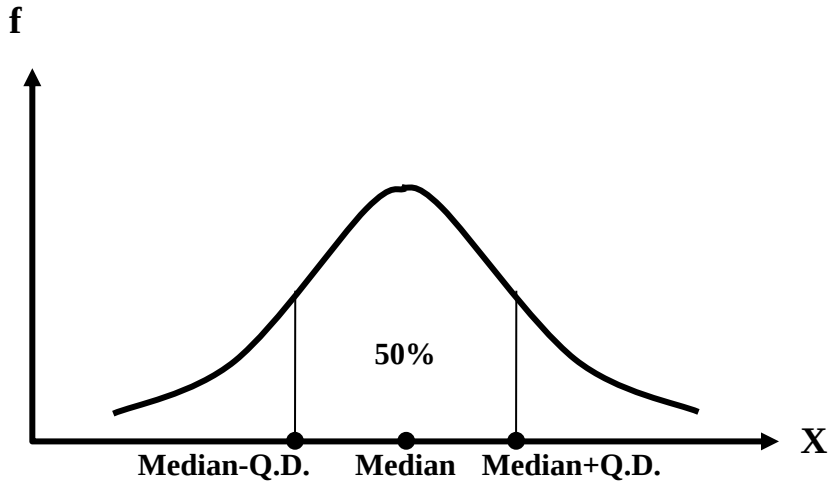
It is also known as **semi-interquartile range.**

Let us now consider the graphical representation of the quartile deviation:



Although simple to compute, it is NOT an extremely satisfactory measure of dispersion because it takes into account the spread of only two values of the variable round the median, and this gives no idea of the rest of the dispersion within the distribution.

The quartile deviation has an attractive feature that the range "Median + Q.D." contains approximately 50% of the data.
This is illustrated in the figure given below:



Let us now apply the concept of quartile deviation to the following example:
EXAMPLE:

The shareholding structure of two companies is given below:

	Company X	**Company Y**
1st quartile	60 shares	165 shares
Median	185 shares	185 shares
3rd quartile	270 shares	210 shares

The quartile deviation for company X is

$$\frac{270-60}{2}=105 \qquad \text{Shares}$$

For company Y, it is

$$\frac{210-165}{2}=22 \qquad \text{Shares}$$

A comparison of the above two results indicate that there is a considerable concentration of shareholders about the MEDIAN number of shares in company Y, whereas in company X, there does not exist this kind of a concentration around the median. (In company X, there is approximately the SAME numbers of small, medium and large shareholders.)

From the above example, it is obvious that the larger the quartile deviation, the greater is the scatter of values within the series. The quartile deviation is superior to range as it is not affected by extremely large or small observations. It is simple to understand and easy to calculate.

The mean deviation can also be viewed in another way:

It is the arithmetic mean of the deviations of the first and third quartiles round the median i.e.

$$\frac{(M-Q_1)+(Q_3-M)}{2}$$

$$=\frac{M-Q_1+Q_3-M}{2}$$

$$=\frac{Q_3-Q_1}{2}$$

Because of what has been just explained, the quartile deviation is regarded as that measure of dispersion which is associated with the median. As such, the quartile deviation should always be employed to indicate dispersion when the median has been adopted as the most appropriate average.

The quartile deviation is also an *absolute* measure of dispersion. Its *relative* measure called the CO-EFFICIENT OF QUARTILE DEVIATION or of Semi-Inter-quartile Range, is defined by the relation:

Coefficient of Quartile Deviation:

$$= \frac{Quartile\ Deviation}{Mid - Quartile\ Range}$$

$$= \frac{\frac{Q_3 - Q_1}{2}}{\frac{Q_3 + Q_1}{2}} = \frac{Q_3 - Q_1}{Q_3 + Q_1},$$

The Coefficient of Quartile Deviation is a pure number and is used for *COMPARING* the variation in two or more sets of data.

The next two measures of dispersion to be discussed are the Mean Deviation and the Standard Deviation. In this regard, the first thing to note is that, whereas the range as well as the quartile deviation are two such measures of dispersion which are NOT based on all the values, the mean deviation and the standard deviation are two such measures of dispersion that involve each and every data-value in their computation.

The range measures the dispersion of the data-set around the *mid-range*, whereas the quartile deviation measures the dispersion of the data-set around the *median*.

How are we to decide upon the amount of dispersion round the *arithmetic mean*?

It would seem reasonable to compute the *DISTANCE* of each observed value in the series from the arithmetic mean of the series.

But the problem is that the sum of the deviations of the values from the mean is ZERO! (No matter what the amount of dispersion in a data-set is, this quantity will *always* be zero, and hence it cannot be used to measure the dispersion in the data-set.)

Then, the question arises, 'HOW will we be able to measure the dispersion present in our data-set?'

In an attempt to answer this question, we might look at the numerical differences between the mean and the data values WITHOUT considering whether these are positive or negative. By ignoring the sign of the deviations we will achieve a NON-ZERO sum, and averaging these absolute differences, again, we obtain a non-zero quantity which can be used as a measure of dispersion. (The larger this quantity, the greater is the dispersion in the data-set).

This quantity is known as the **MEAN DEVIATION.**

Let us denote these absolute differences
by 'modulus of d'
or 'mod d'. Then, the mean deviation is given by

MEAN DEVIATION:

$$\text{M.D.} = \frac{\sum |\,\text{d}\,|}{\text{n}}$$

As the absolute deviations of the observations from their mean are being averaged, therefore the complete name of this measure is Mean Absolute Deviation --- but generally, it is simply called “Mean Deviation”. In the next lecture, this concept will be discussed in detail. (The case of raw data as well as the case of grouped data will be considered.)Next, we will discuss the most important and the most widely used measure of dispersion i.e. the Standard Deviation.

LECTURE NO. 30

- Mean Deviation
- Standard Deviation and Variance
- Coefficient of variation

First, we will discuss it for the case of raw data, and then we will go on to the case of a frequency distribution. The first thing to note is that, whereas the range as well as the quartile deviation are two such measures of dispersion which are NOT based on all the values, the mean deviation and the standard deviation are two such measures of dispersion that involve each and every data-value in their computation.

You must have noted that the range was measuring the dispersion of the data-set around the mid-range, whereas the quartile deviation was measuring the dispersion of the data-set around the median.

How are we to decide upon the amount of dispersion round the arithmetic mean? It would seem reasonable to compute the DISTANCE of each observed value in the series from the arithmetic mean of the series.

Let us do this for a simple data-set shown below:

THE NUMBER OF FATALITIES IN MOTORWAY ACCIDENTS IN ONE WEEK

Day	Number of fatalities X
Sunday	4
Monday	6
Tuesday	2
Wednesday	0
Thursday	3
Friday	5
Saturday	8
Total	28

Let us do this for a simple data-set shown below:

THE NUMBER OF FATALITIES IN MOTORWAY ACCIDENTS IN ONE WEEK

Day	Number of fatalities X
Sunday	4
Monday	6
Tuesday	2
Wednesday	0
Thursday	3
Friday	5
Saturday	8
Total	28

The arithmetic mean number of fatalities per day is

$$\overline{X} = \frac{\sum X}{n} = \frac{28}{7} = 4$$

In order to determine the distances of the data-values from the mean, we subtract our value of the arithmetic mean from each daily figure, and this gives us the deviations that occur in the third column of the table below

Day	Number of fatalities X	$X - \overline{X}$
Sunday	4	0
Monday	6	+ 2
Tuesday	2	– 2
Wednesday	0	– 4
Thursday	3	– 1
Friday	5	+ 1
Saturday	8	+ 4
TOTAL	28	0

The deviations are negative when the daily figure is less than the mean (4 accidents) and positive when the figure is higher than the mean. It does seem, however, that our efforts for computing the dispersion of this data set have been in vain, for we find that the total amount of dispersion obtained by summing the $(x - \bar{x})$ column comes out to be zero! In fact, this should be no surprise, for it is a basic property of the arithmetic means that: The sum of the deviations of the values from the mean is zero. The question arises:

How will we measure the dispersion that is actually present in our data-set?

Our problem might at first sight seem irresolvable, for by this criterion it appears that no series has any dispersion. Yet we know that this is absolutely incorrect, and we must think of some other way of handling this situation. Surely, we might look at the numerical difference between the mean and the daily fatality figures without considering whether these are positive or negative. Let us denote these absolute differences by 'modulus of d' or 'mod d'.

This is evident from the third column of the table below

X	$X - \overline{X} = d$	| d |
4	0	0
6	2	2
2	–2	2
0	–4	4
3	–1	1
5	1	1
8	4	4
	Total	14

By ignoring the sign of the deviations we have achieved a non-zero sum in our second column. Averaging these absolute differences, we obtain a measure of dispersion known as the mean deviation.
In other words, the mean deviation is given by the formula:

MEAN DEVIATION

$$M.D. = \frac{\sum | d_i |}{n}$$

As we are averaging the absolute deviations of the observations from their mean, therefore the complete name of this measure is mean absolute deviation --- but generally we just say “mean deviation”. Applying this formula in our example, we find that, the mean deviation of the number of fatalities is

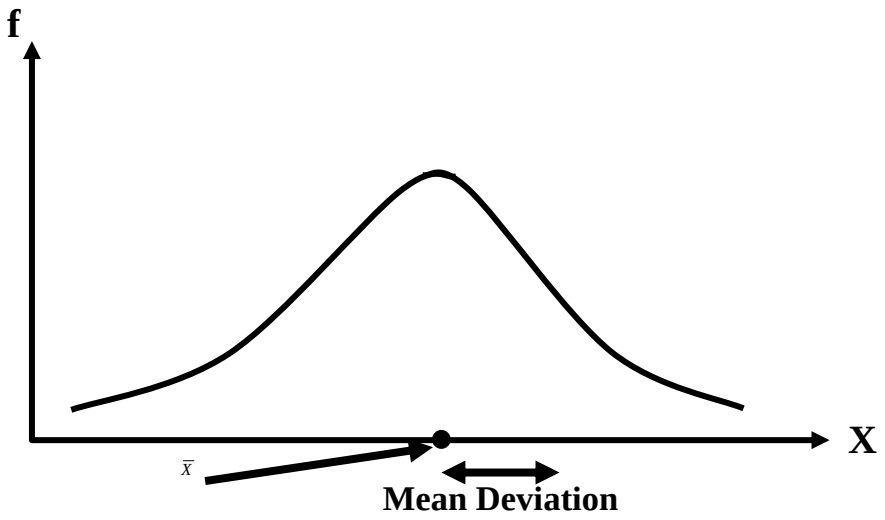
$$M.D. = \frac{14}{7} = 2.$$

The formula that we have just considered is valid in the case of raw data. In case of grouped data i.e. a frequency distribution, the formula becomes

MEAN DEVIATION FOR GROUPED DATA

$$M.D. = \frac{\sum f_i \left| x_i - \bar{x} \right|}{n} = \frac{\sum f_i \left| d_i \right|}{n}$$

As far as the graphical representation of the mean deviation is concerned, it can be depicted by a horizontal line segment drawn below the X-axis on the graph of the frequency distribution, as shown below



The approach which we have adopted in the concept of the mean deviation is both quick and simple. But the problem is that we introduce a kind of artificiality in its calculation by ignoring the algebraic signs of the deviations.

In problems involving descriptions and comparisons alone, the mean deviation can validly be applied; but because the negative signs have been discarded, further theoretical development or application of the concept is impossible.

Mean deviation is an absolute measure of dispersion. Its relative measure, known as the co-efficient of mean deviation, is obtained by dividing the mean deviation by the average used in the calculation of deviations i.e. the arithmetic mean. Thus

CO-EFFICIENT OF M.D

Sometimes, the mean deviation is computed by averaging the absolute deviations of the data-values from the median i.e.

$$\text{Mean deviation} = \frac{\sum|x - \tilde{x}|}{n}$$

And when will we have a situation when we will be using the median instead of the mean? As discussed earlier, the median will be more appropriate than the mean in those cases where our data-set contains a few very high or very low values. In such a situation, the coefficient of mean deviation is given by:

Co-efficient of M.D:

$$= \frac{\text{M.D.}}{\text{Median}} \qquad = \frac{\text{M.D.}}{\text{Mean}}$$

Let us now consider the ***standard deviation*** --- that statistic which is the most important and the most widely used measure of dispersion.

The point that made earlier that from the mathematical point of view, it is not very preferable to take the absolute values of the deviations, *This problem is overcome by computing the standard deviation.*

In order to compute the standard deviation, rather than taking the absolute values of the deviations, we square the deviations.

Averaging these squared deviations, we obtain a statistic that is known as the variance.

VARIANCE

$$= \frac{\sum (x - \bar{x})^2}{n}$$

Let us compute this quantity for the data of the above example.
Our X-values were:

X
4
6
2
0
3
5
8

Taking the deviations of the X-values from their mean, and then squaring these deviations, we obtain:

X	$(x - \bar{x})$	$(x - \bar{x})^2$
4	0	0
6	+ 2	4
2	– 2	4
0	– 4	16
3	– 1	1
5	+ 1	1
8	+ 4	16
		42

Obviously, both (– 2)2 and (2)2 equal 4, both (– 4)2 and (4)2 equal 16, and both (– 1)2 and (1)2 = 1

Hence $\sum(x - \bar{x})^2 = 42$ is now positive, and this positive value has been achieved without 'bending' the rules of mathematics. Averaging these squared deviations, the variance is given by:

Variance:

$$= \frac{\sum(x-\bar{x})^2}{n}$$

$$= \frac{42}{7} = 6$$

The variance is frequently employed in statistical work, but it should be noted that the figure achieved is in 'squared' units of measurement.
In the example that we have just considered, the variance has come out to be "6 squared fatalities", which does not seem to make much sense! In order to obtain an answer which is in the original unit of measurement, we take the positive square root of the variance. The result is known as the standard deviation.

STANDARD DEVIATION

$$S = \sqrt{\frac{\sum(x-\bar{x})^2}{n}}$$

Hence, in this example, our standard deviation has come out to be 2.45 fatalities.

In computing the standard deviation (or variance) it can be tedious to first ascertain the arithmetic mean of a series, then subtract it from each value of the variable in the series, and finally to square each deviation and then sum.
It is very much more straight-forward to use the short cut formula given below:

SHORT CUT FORMULA FOR THE STANDARD DEVIATION

$$S = \sqrt{\left\{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2\right\}}$$

In order to apply the short cut formula, we require only the aggregate of the series ($\sum x$) and the aggregate of the squares of the individual values in the series ($\sum x2$).
In other words, only two columns of figures are called for. The number of individual calculations is also considerably reduced, as seen below:

	X	X^2
	4	16
	6	36
	2	4
	0	0
	3	9
	5	25
	8	64
Total	28	154

Therefore

$$S = \sqrt{\left\{\frac{154}{7} - \left(\frac{28}{7}\right)^2\right\}} = \sqrt{(22 - 16)}$$

$$= \sqrt{6} = 2.45 \text{ fatalities}$$

The formulae that we have just discussed are valid in case of raw data. In case of grouped data i.e. a frequency distribution, each squared deviation round the mean must be multiplied by the appropriate frequency figure i.e.

STANDARD DEVIATION IN CASE OF GROUPED DATA

$$S = \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$$

And the short cut formula in case of a frequency distribution is:

SHORT CUT FORMULA OF THE STANDARD DEVIATION IN CASE OF GROUPED DATA

$$S = \sqrt{\left\{\frac{\sum fx^2}{n} - \left(\frac{\sum fx}{n}\right)^2\right\}}$$

Which is again preferred from the computational standpoint
For example, the standard deviation life of a batch of electric light bulbs would be calculated as follows:

EXAMPLE

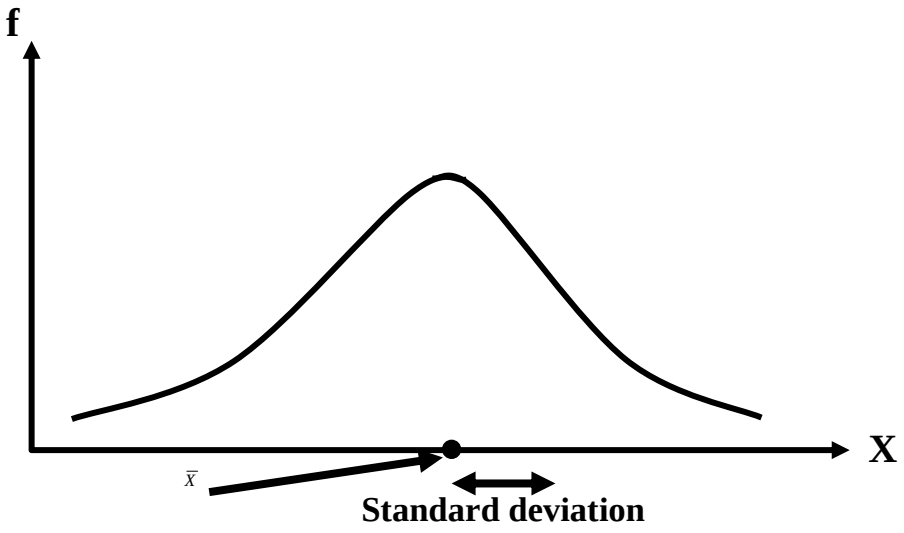
Life (in Hundreds of Hours)	No. of Bulbs f	Mid-point x	fx	fx^2
0 – 5	4	2.5	10.0	25.0
5 – 10	9	7.5	67.5	506.25
10 – 20	38	15.0	570.0	8550.0
20 – 40	33	30.0	990.0	29700.0
40 and over	16	50.0	800.0	40000.0
	100		2437.5	78781.25

Therefore, standard deviation:

$$S = \sqrt{\left\{\frac{78781.25}{100} - \left(\frac{2437.5}{100}\right)^2\right\}}$$

=13.9hundredhours
= 1390 hours

As far as the graphical representation of the standard deviation is concerned, a horizontal line segment is drawn below the X-axis on the graph of the frequency distribution --- just as in the case of the mean deviation.



The standard deviation is an absolute measure of dispersion. Its relative measure called coefficient of standard deviation is defined as:

COEFFICIENT OF S.D

$$= \frac{\text{Standard Deviation}}{\text{Mean}}$$

And, multiplying this quantity by 100, we obtain a very important and well-known measure called the coefficient of variation.

COEFFICIENT OF VARIATION

$$C.V. = \frac{S}{\overline{X}} \times 100$$

As mentioned earlier, the standard deviation is expressed in absolute terms and is given in the same unit of measurement as the variable itself.
There are occasions, however, when this absolute measure of dispersion is inadequate and a relative form becomes preferable. For example, if a comparison between the variability of distributions with different variables is required, or when we need to compare the dispersion of distributions with the same variable but with very different arithmetic means. To illustrate the usefulness of the coefficient of variation, let us consider the following two examples.

EXAMPLE-1

Suppose that, in a particular year, the mean weekly earnings of skilled factory workers in one particular country were $ 19.50 with a standard deviation of $ 4, while for its neighboring country the figures were Rs. 75 and Rs. 28 respectively.

From these figures, it is not immediately apparent which country has the GREATER VARIABILITY in earnings. The coefficient of variation quickly provides the answer:

For country No. 1:

$$\frac{4}{19.5} \times 100 = 20.5 \text{ per cent,}$$

And for country No. 2:

$$\frac{28}{75} \times 100 = 37.3 \text{ per cent.}$$

From these calculations, it is immediately obvious that the spread of earnings in country No. 2 is greater than that in country No. 1, and the reasons for this could then be sought.

EXAMPLE-2:

The crop yield from 20 acre plots of wheat-land cultivated by ordinary methods averages 35 bushels with a standard deviation of 10 bushels. The yield from similar land treated with a new fertilizer averages 58 bushels, also with a standard deviation of 10 bushels. At first glance, the yield variability may seem to be the same, but in fact it has improved (i.e. decreased) in view of the higher average to which it relates.

Again, the coefficient of variation shows this very clearly:

Untreated land:

$$\frac{10}{35} \times 100 = 28.57 \text{ per cent}$$

Treated land:

$$\frac{10}{58} \times 100 = 17.24 \text{ per cent}$$

The coefficient of variation for the untreated land has come out to be 28.57 percent, whereas the coefficient of variation for the treated land is only 17.24 percent.

LECTURE NO. 31

- Box and Whisker Plot
- Pearson's Coefficient of Skewness

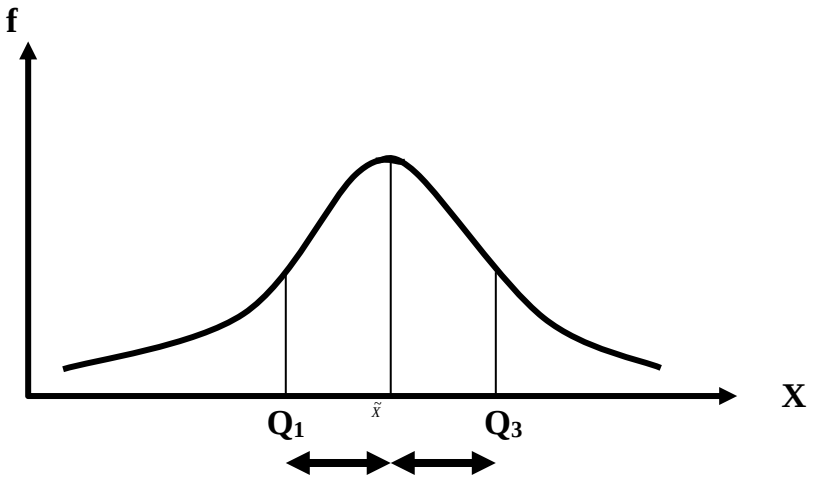
Prior to discussing the THE BOX-AND-WHISKER PLOT, let us review the concept of THE FIVE-NUMBER SUMMARY. As indicated in the last lecture, once we have studied the three major properties of numerical data (i.e. central tendency, variation, and shape), it is important that we identify and describe the major features of the data in a SUMMARIZED format. One way of doing this is to develop a five-number summary.

FIVE-NUMBER SUMMARY

A five-number summary consists of X_0, Q_1, Median, Q_3; Xm.It provides us a better idea as to the SHAPE of the distribution, as explained below:
If the data were perfectly symmetrical, the following would be true:

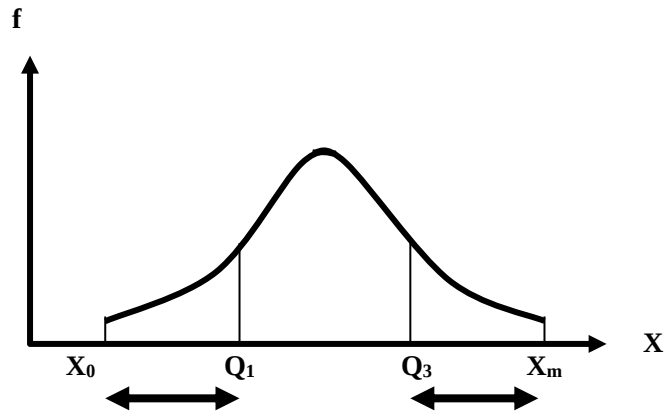
1. The distance from Q1 to the median would be equal to the distance from the median to Q3, as shown below:

THE SYMMETRIC CURVE



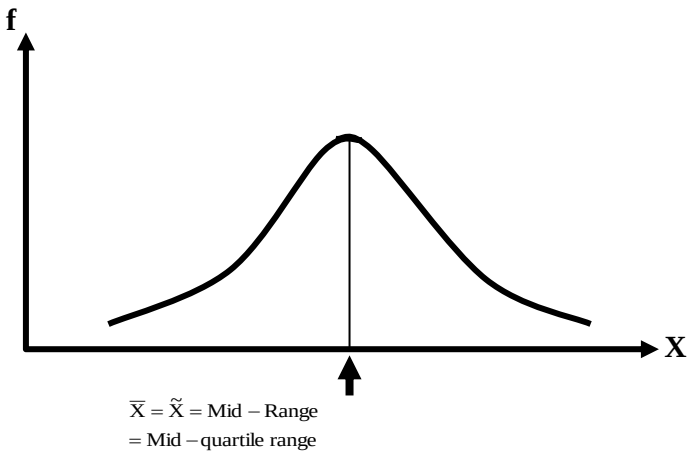
2. The distance from X0 to Q1 would be equal to the distance from Q3 to Xm, as shown below:

THE SYMMETRIC CURVE



3. The median, the mid-quartile range, and the midrange would ALL be equal.
These measures would also be equal to the arithmetic mean of the data, as shown below:

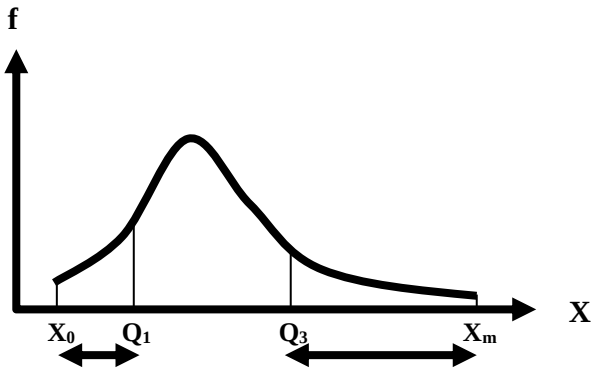
THE SYMMETRIC CURVE



On the other hand, for non-symmetrical distributions, the following would be true:

1. In right-skewed (positively-skewed) distributions the distance from Q3 to Xm greatly EXCEEDS the distance from X0 to Q1, as shown below:

THE POSITIVELY SKEWED CURVE

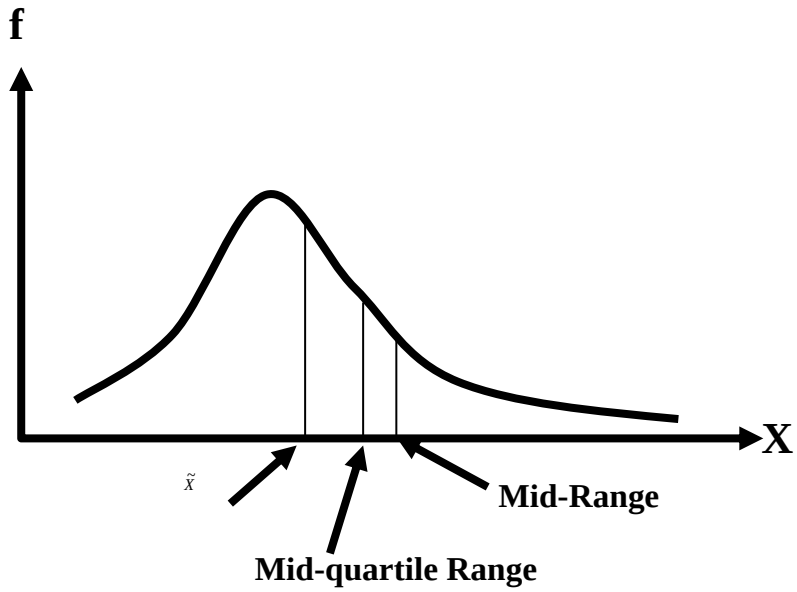


2. In right-skewed distributions,

median < mid-quartile range < midrange

This is indicated in the following figure:

THE POSITIVELY SKEWED CURVE



Similarly, in left-skewed distributions, the distance from X0 to Q1 greatly exceeds the distance from Q3 to Xm.Also, in left-skewed distributions, midrange < mid-quartile range < median.

Let us try to understand this concept with the help of an example:

EXAMPLE

Suppose that a study is being conducted regarding the annual costs incurred by students attending public versus private colleges and universities in the United States of America. In particular, suppose, for exploratory purposes, our sample consists of 10 Universities whose athletic programs are members of the 'Big Ten' Conference? The annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big

Ten Conference are given in the following table; state the five-number summary for these data.

Annual Costs Incurred on Tuition Fees, etc.

Name of University	Annual Costs (in $000)
Indiana University	15.6
Michigan State University	17.0
Ohio State University	15.2
Pennsylvania State University	16.4
Purdue University	15.2
University of Illinois	15.4
University of Iowa	13.0
University of Michigan	23.1
University of Minnesota	14.3
University of Wisconsin	14.9

SOLUTION:
For our sample, the ordered array is

$X_0 = 13.0$	14.3	14.9	15.2	15.2	15.4	15.6	16.4	17.0	$X_m = 23.1$

The median for this data comes out to be 15.30 thousand dollars. The first quartile comes out to be 14.90 thousand dollars, and the third quartile comes out to be 16.40 thousand dollars.
Therefore, the five-number summary is:

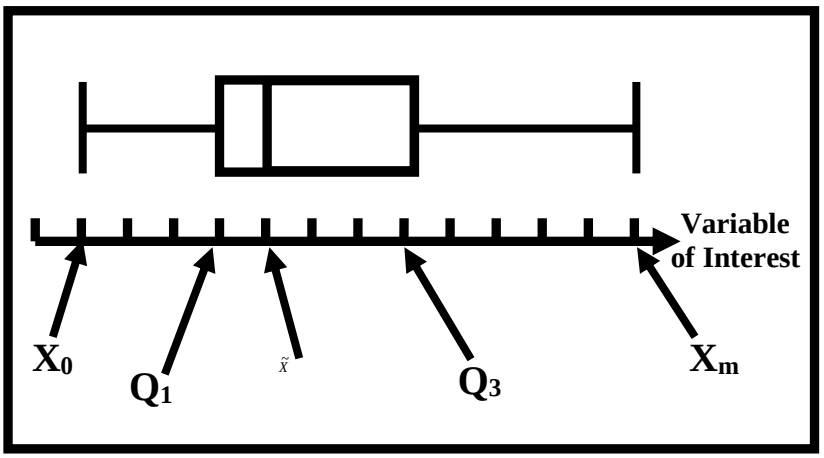
X_0	Q_1	$\tilde{X}$	Q_3	X_m
13.0	14.9	15.3	16.4	23.1

We may now use the five-number summary to study the *shape* of this distribution:
We notice that
1. The distance from Q3 to Xm (i.e., 6.7) greatly exceeds the distance from X0 to Q1 (i.e., 1.9).
2. If we compare the median (which is 15.3), the mid-quartile range (which is 15.65), and the midrange (which is 18.05), we observe that the median < the mid-quartile range < the midrange.
Hence, from the preceding rules, it is clear that the annual cost data for our sample are *right-skewed*. The gist of the above discussion is that the five-number summary is a SIMPLE yet effective way of determining the shape of our frequency distribution --- WITHOUT actually drawing the graph of the frequency distribution. The concept of the five number summary is directly linked with the concept of the box and whisker plot:

BOX AND WHISKER PLOT

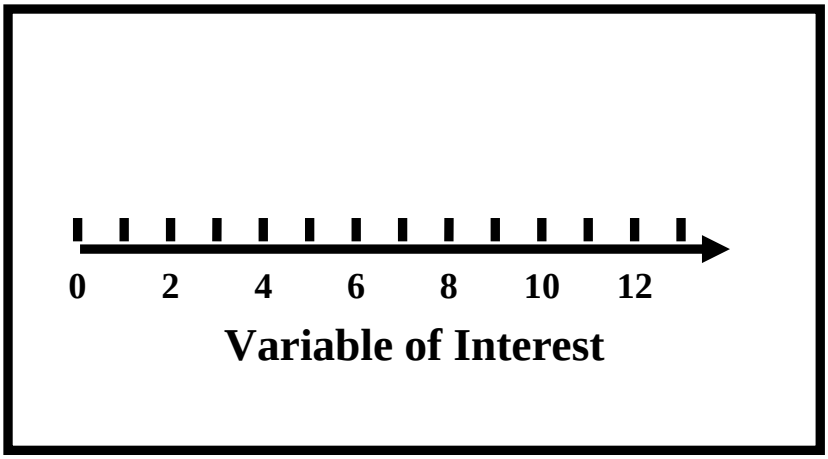
In its simplest form, a box-and-whisker plot provides a graphical representation of the data through its five-number summary.

Box and Whisker Plot

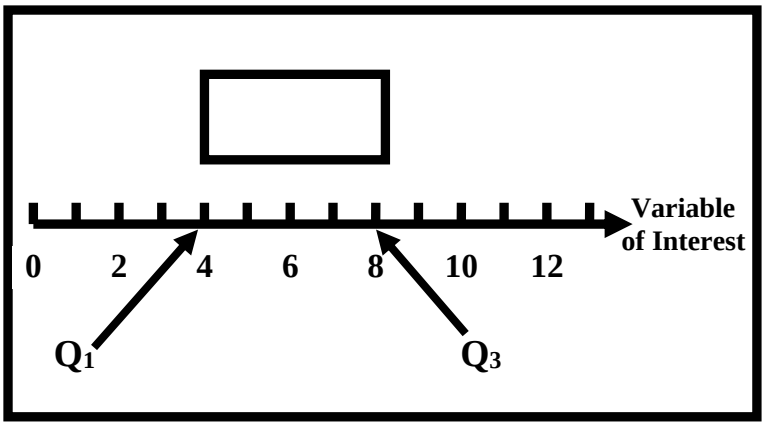


To construct a box-and-whisker plot, we proceed as follows:
Steps involved in the construction of the Box and Whisker Plot:

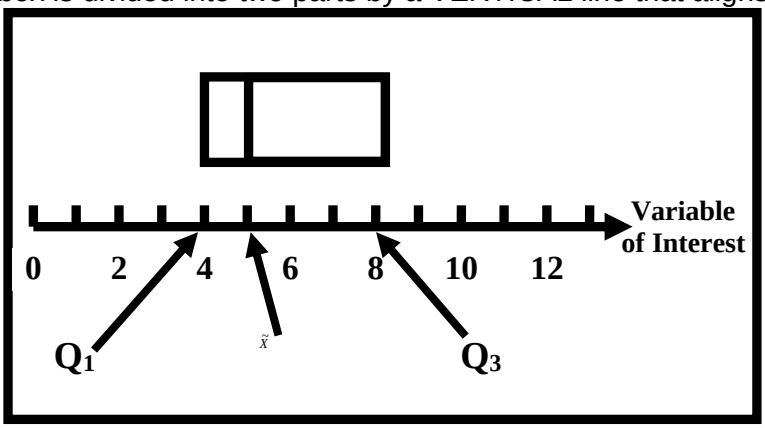
1. The variable of interest in represented on the horizontal axis.



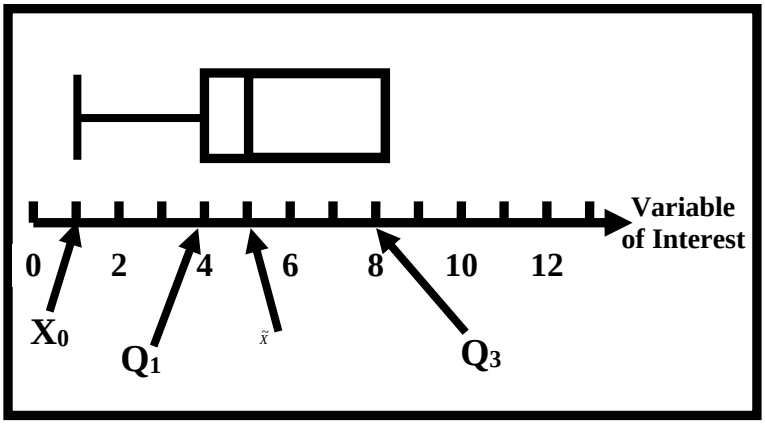
2. A BOX is drawn in the space above the horizontal axis in such a way that the left end of the box aligns with the first quartile Q1 and the right end of the box is aligned with the third quartile Q3.



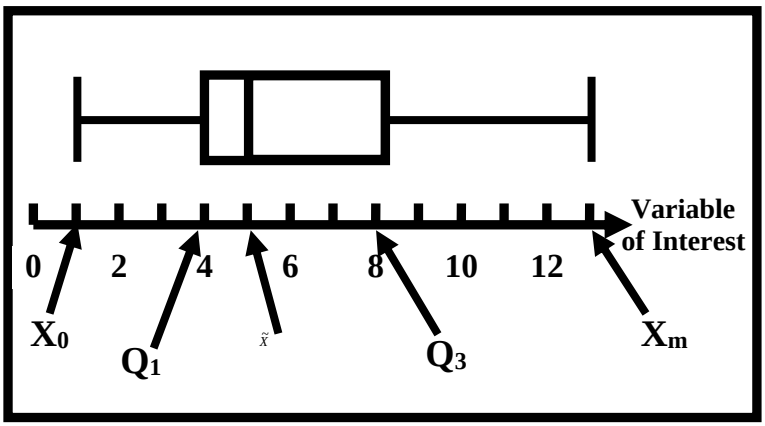
3. The box is divided into two parts by a VERTICAL line that aligns with the MEDIAN.



4. A line, called a whisker, is extended from the LEFT end of the box to a point that aligns with X0, the smallest measurement in the data set.



5. Another line, or whisker, is extended from the RIGHT end of the box to a point that aligns with the LARGEST measurement in the data set.



Let us understand the construction of the box-and-whisker plot with reference to an example:

EXAMPLE

The following table shows the downtime, in hours, recorded for 30 machines owned by a large manufacturing company. The period of time covered was the same for all machines.

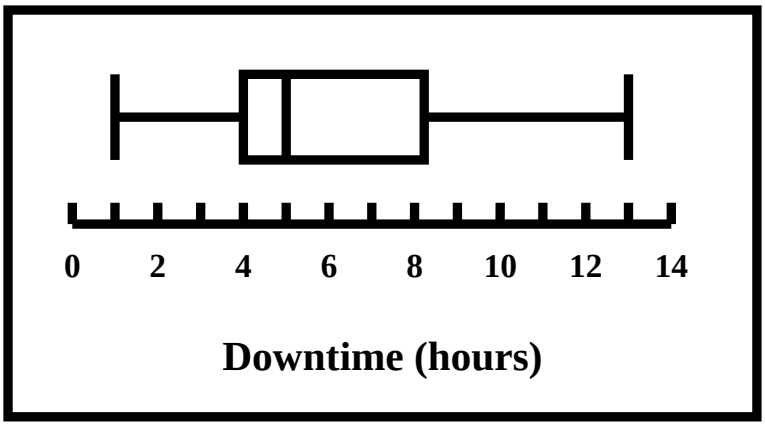
DOWNTIME IN HOURS OF 30 MACHINES

4	4	1	4	1	4
6	10	5	5	8	2
1	6	10	1	13	5
8	4	3	9	4	9
1	4	4	11	8	9

In order to construct a box-and-whisker plot for these data, we proceed as follows:
First of all, we determine the two extreme values in our data-set:
The smallest and largest values are X0 = 1 and Xm = 13, respectively.
As far as the computation of the quartiles is concerned, we note that, in this example, we are dealing with raw data.

The first quartile is the (30 + 1)/4 = 7.75th ordered measurement and is equal to 4.
The median is the (30 + 1)/2 = 15.5th measurement, or 5,and
The third quartile is the 3(30 + 1)/4 = 23.25th ordered measurement, which is 8.25.
As a result, we obtain the following box and whisker plot:

Box and Whisker Plot



INTERPRETATION OF THE BOX AND WHISKER PLOT

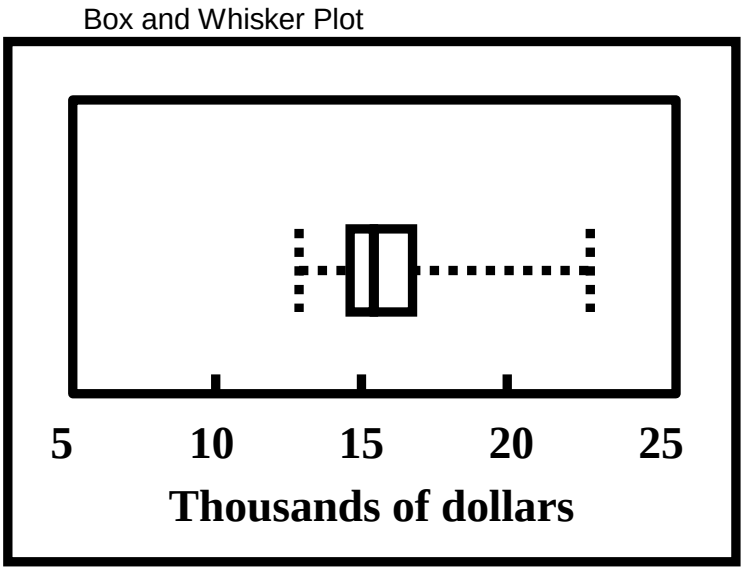
With regard to the *interpretation* of the Box and Whisker Plot, it should be noted that, by looking at a box-and-whisker plot, one can quickly form an impression regarding the amount of SPREAD, location of CONCENTRATION, and SYMMETRY of our data set.
A glance at the box and whisker plot of the example that we just considered reveals that:

- 50% of the measurements are between 4 and 8.25.
- The median is 5, and the range is 12. and, most importantly:
- Since the median line is closer to the *left* end of the box, hence the data are SKEWED to the RIGHT.(The fundamental point is that in a perfectly symmetrical data set, the median line will be EXACTLY HALFWAY between the two ends of the box, and in a data set that is skewed to the LEFT, the median line will be CLOSER TO THE RIGHT END of the box.)

Let us consolidate all the above ideas by going back to the example of the Big Ten Universities in which the annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference were given as follows:

Name of University	Annual Costs (in $000)
Indiana University	15.6
Michigan State University	17.0
Ohio State University	15.2
Pennsylvania State University	16.4
Purdue University	15.2
University of Illinois	15.4
University of Iowa	13.0
University of Michigan	23.1
University of Minnesota	14.3
University of Wisconsin	14.9

As stated earlier, the Five-Number Summary of this data-set is :
For this data, the Box and Whisker Plot is of the form given below:



As indicated earlier, the vertical line drawn within the box represents the location of the median value in the data; the vertical line at the LEFT side of the box represents the location of Q1, and the vertical line at the RIGHT side of the box represents the location of Q3. Therefore, the BOX contains the middle 50% of the observations in the distribution. The lower 25% of the data are represented by the whisker that connects the *left* side of the box to the location of the *smallest* value, X0, and the upper 25% of the data are represented by the whisker connecting the *right* side of the box to Xm.

INTERPRETATION OF THE BOX AND WHISKER PLOT

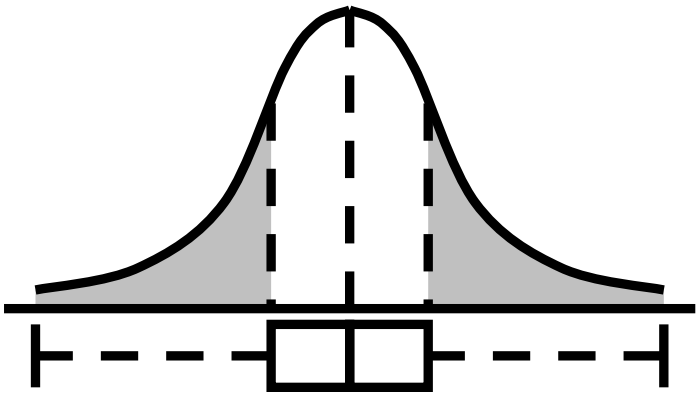
We note that (1) the vertical median line is CLOSER to the left side of the box, and (2) the left side whisker length is clearly SMALLER than the right side whisker length .Because of these observations, we The gist of the above discussion is that if the median line is at a greater distance from the left side of the box as compared with its distance from the right side of the box, our distribution will be skewed to the left.

In this situation, the whisker appearing on the left side of the box and whisker plot will be longer than the whisker of the right side. Conclude that the data-set of the annual costs is RIGHT-skewed.

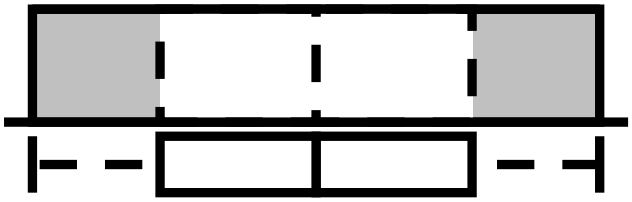
The gist of the above discussion is that if the median line is at a greater distance from the left side of the box as compared with its distance from the right side of the box, our distribution will be skewed to the left. In this situation, the whisker appearing on the left side of the box and whisker plot will be longer than the whisker of the right side. The Box and Whisker Plot comes under the realm of “exploratory data analysis” (EDA) which is a relatively new area of statistics. The following figures provide a comparison between the Box and Whisker Plot and the traditional procedures such as the frequency polygon and the frequency curve with reference to the SKEWNESS present in the data-set.

Four different types of hypothetical distributions are depicted through their box-and-whisker plots and corresponding frequency curves.

1) When a data set is perfectly symmetrical, as is the case in the following two figures, the mean, median, midrange, and mid-quartile range will be the SAME:



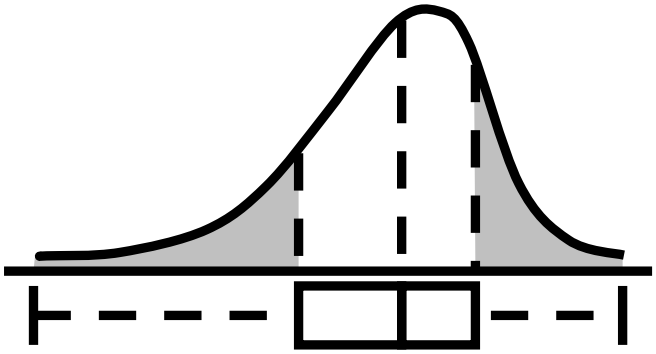
(a) Bell-shaped distribution



(b) Rectangular distribution

In ADDITION, the length of the left whisker will be equal to the length of the right whisker, and the median line will divide the box in HALF. (In *practice*, it is unlikely that we will observe a data set that is perfectly symmetrical. However, we should be able to state that our data set is *approximately* symmetrical *if* the lengths of the two whiskers are *almost* equal and the *median* line *almost* divides the box in HALF.)

2) When our data set is LEFT-skewed as in the following figure, the few small observations pull the midrange and mean toward the LEFT tail:

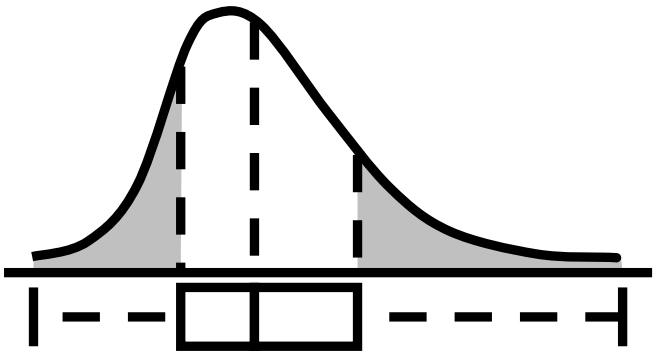


Left-skewed distribution

For this LEFT-skewed distribution, we observe that the skewed nature of the data set indicates that there is a HEAVY CLUSTERING of observations at the HIGH END of the scale (i.e., the RIGHT side).

75% of all data values are found between the left edge of the box (Q1) and the end of the right whisker (Xm). Therefore, the LONG left whisker contains the distribution of only the smallest 25% of the observations, demonstrating the distortion from symmetry in this data set.

3) If the data set is RIGHT-skewed as shown in the following figure, the few large observations PULL the midrange and mean toward the right tail.



Right-skewed distribution

For the right-skewed data set, the concentration of data points is on the LOW end of the scale (i.e., the left side of the box-and-whisker plot). Here, 75% of all data values are found between the beginning of the left whisker (X0) and the RIGHT edge of the box (Q3), and the remaining 25% of the observations are DISPERSED ALONG the LONG right whisker at the upper end of the scale. This brings us to the end of the discussion of the five number summary and the box and whisker plot.

Next, we discuss *another* way of determining the skewness of the data-set and that is the **PEARSON'S**

COEFFICIENT OF SKEWNESS

In this connection, the first thing to note is that, by providing information about the location of a series and the dispersion within that series it might appear that we have achieved a PERFECTLY adequate overall description of the data. But, the fact of the matter is that, it is quite possible that two series are decidedly dissimilar and yet have exactly the same arithmetic mean AND standard deviation: Let us understand this point with the help of an example:

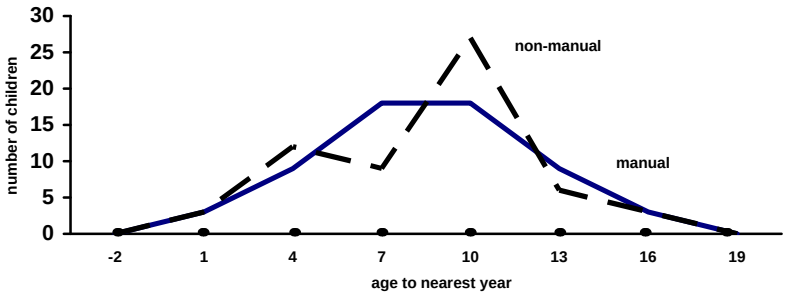
EXAMPLE:

Age of Onset of Nervous Asthma in Children (to Nearest Year)	Children of Manual Workers	Children of Non-Manual Workers
0 – 2	3	3
3 – 5	9	12
6 – 8	18	9
9 – 11	18	27
12 – 14	9	6
15 – 17	3	3
	60	60

In order to compute the mean and standard deviation for each distribution, we carry out the following calculations:

Age of Onset of Nervous Asthma in Children (to Nearest Year)		Children of Manual Workers			Children of Non-Manual Workers		
Age Group	X	f_1	f_1X	f_1X^2	f_2	f_2X	f_2X^2
0 – 2	1	3	3	3	3	3	3
3 – 5	4	9	36	144	12	48	192
6 – 8	7	18	126	882	9	63	441
9 – 11	10	18	180	1800	27	270	2700
12 – 14	13	9	117	1521	6	78	1014
15 – 17	16	3	48	768	3	48	768
	51	60	510	5118	60	510	5118

We find that, for each of the two distributions, the mean is 8.5 years and the standard deviation is 3.61 years.
The frequency polygons of the two distributions are as follows:



By inspecting these, it can be seen that one distribution is symmetrical while the other is quite different. The distinguishing feature here is the degree of asymmetry or *SKEWNESS* in the two polygons. In order to *measure* the skewness in our distribution, we compute the **PEARSON's COEFFICIENT OF SKEWNESS** which is defined as:
Pearson's Coefficient of Skewness:

$$\frac{mean - mode}{standard\ deviation}$$

Applying the *empirical relation* between the mean, median and the mode, the Pearson's Coefficient of Skewness is given by:

Pearson's Coefficient of Skewness

$$= \frac{3\,(\text{mean} - \text{median})}{\text{standard deviation}}$$

For a *symmetrical* distribution the coefficient will always be ZERO, for a distribution skewed to the RIGHT the answer will always be positive, and for one skewed to the LEFT the answer will always be negative.
Let us now calculate this coefficient for the example of the children of the manual and non-manual workers.

Sample statistics pertaining to the ages of these children are as follows:

	Children of Manual Workers	Children of Non-Manual Workers
Mean	8.50 years	8.50 years
Standard deviation	3.61 years	3.61 years
Median	8.50 years	9.16 years
Q_1	6.00 years	5.50 years
Q_3	11.00 years	10.83 years
Quartile deviation	2.50 years	2.66 years

The Pearson's Coefficient of Skewness is calculated for each of the two categories of children, as shown below: Pearson's Coefficient of Skewness (Modified):

$$\text{Ages of Children of Manual Workers} = \frac{3(8.50 - 8.50)}{3.61} = 0$$

$$\text{Ages of Children of Non-Manual Workers} = \frac{3(8.50 - 9.16)}{3.61} = -0.55$$

For the data pertaining to children of manual workers, the coefficient is zero, whereas, for the children of non-manual workers, the coefficient has turned out to be a negative number. This indicates that the distribution of the ages of the children of the manual workers is symmetric whereas the distribution of the ages of the children of the non-manual workers is negatively skewed.

The students are encouraged to draw the frequency polygon and the frequency curve for each of the two distributions, and to compare the results that have just been obtained with the *shapes* of the two distributions.

LECTURE NO. 32

On numerous occasions, our interest lies not in just one single variable but in two, three, four or more variables. For example, if we talk about the yield of a crop, we realize that the yield of any crop depends on a variety of factors --- the fertility of the soil, the type of fertilizer used, the amount of rainfall, and so on.

- Simple Linear Regression
- Standard Error of Estimate
- Correlation

Let me begin the discussion of the bivariate situation by picking up an example.

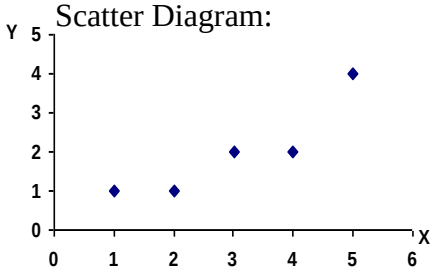
EXAMPLE:

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus.
Suppose the company administers this drug on 5 subjects and obtains the following information:

Subject	Percentage of drug	Reaction Time (milli-seconds)
	X	Y
A	1	1
B	2	1
C	3	2
D	4	2
E	5	4

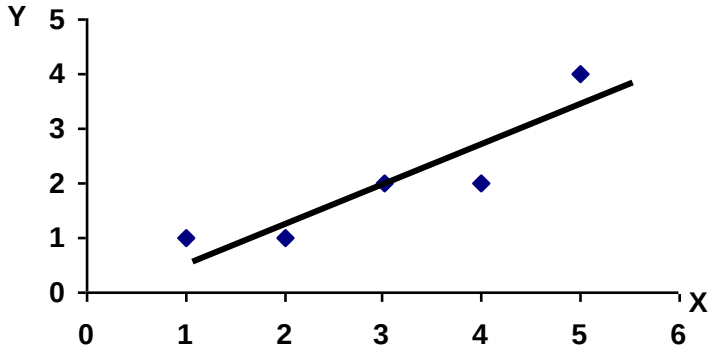
In this example, the reaction time to the stimulus will DEPEND on the amount of drug in the blood-stream. As you must know, the dependent variable is denoted by Y, and the independent variable is denoted by X.In this example, the reaction time will be denoted by Y, and the percentage of drug in the blood stream by X. Going back to the example that we were just considering, it is obvious that we are interested in determining the nature of the relationship between the amount of drug in the blood stream and the time it takes to react to a stimulus.
In order to ascertain the nature of the relationship between these two variables, the first step is to draw a **SCATTER DIAGRAM** --- which is a simple graph of the X-values against the Y-values depicted on the graph paper in the form of points.
In this example, the scatter diagram is as follows:

Scatter Diagram:



As you can see, there is an upward trend in the scatter diagram i.e. it is clear that as X increases, Y also increases. Of course, the points are not all falling on a straight line, but if we look carefully, we find an overall linear pattern as shown below:

Scatter Diagram:



It will be very RARE in the field of behavioral or social sciences to find two sets of data which are related perfectly by a straight line: it is more likely that only a general linear pattern or tendency will be apparent.
WHY is it that we will not get an exact linear relationship?
Let me explain this to you with the help of an example:
Suppose one is studying the relationship between the research and development expenditure and the profit margin on products of a number of firms. While it may be generally true to state that the two will increase together, it is INEVITABLE that some firms' profit margin will be higher than others with the SAME *R and D* expenditure, and vice versa. The reasons for this may be that the conditions under which the various firms are operating may be very different. The goods being produced, the firm's share of the market, the efficiency of the firm etc. will ALL play a part in determining the individual results.

A linear relationship between two variables is a SURPRISINGLY common occurrence, and even where a refined non-linear curve might prove slightly superior, the SIMPLER form will often be quite adequate in the context of the problem under consideration. Having plotted the n pairs of values in the form of a scatter diagram, IF an overall linear pattern emerges, then the object of regression is to superimpose on this pattern the general relationship between y and x in the linear form which will REMOVE

the effect of outside factors. I am sure that you are aware of the equation of a straight line.
Do you not remember the equation **Y = mX + c,** where Y represents the slope of the line, and c represent the Y-intercept?
This equation can also be stated as Y = c + mX, and if we rename c and m as 'a' and 'b', the equation becomes **Y = a + bX.**

EQUATION OF A STRAIGHT LINE

Y = a + bX

Where

- Y represents the dependent variable
- X represents the independent variable
- a represents the Y-intercept

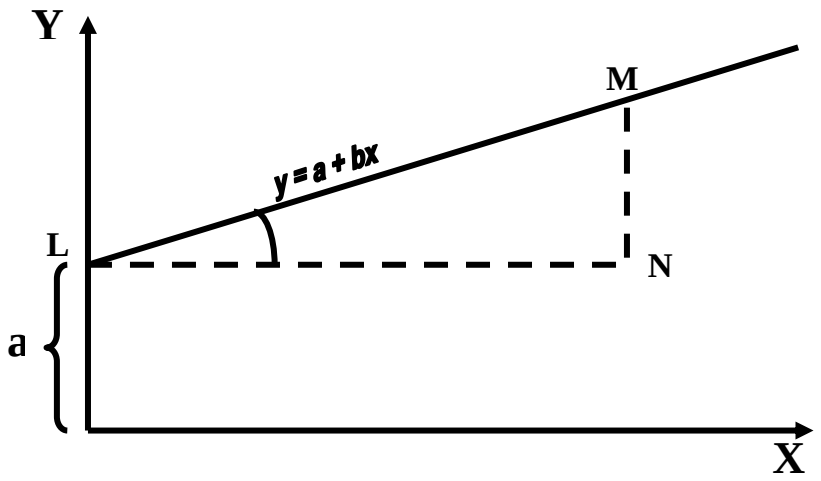
(i.e. the value of Y when X is equal to zero)

- b represents the slope of the line

(i.e. the value of the tan θ, where θ represents the angle between the line and the horizontal axis)

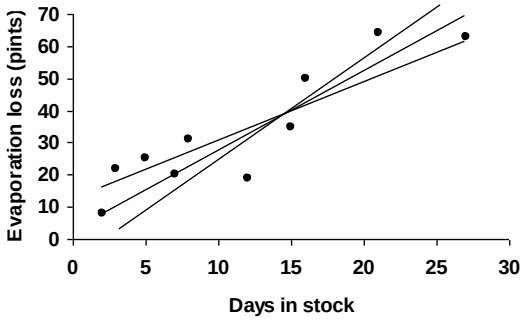
Interpretation of 'a' and 'b':

$$b = \tan\theta = \frac{MN}{NL}$$



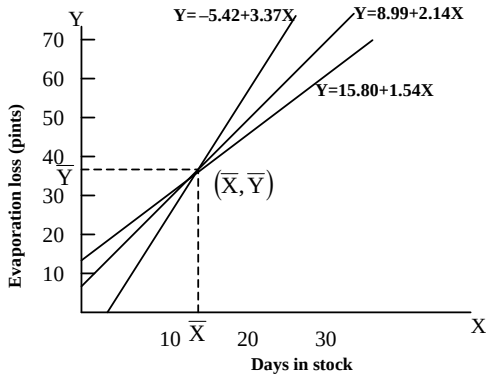
A very important point to note is that MANY lines can be drawn through the same scatter diagram.

THE LINEAR PATTERN:



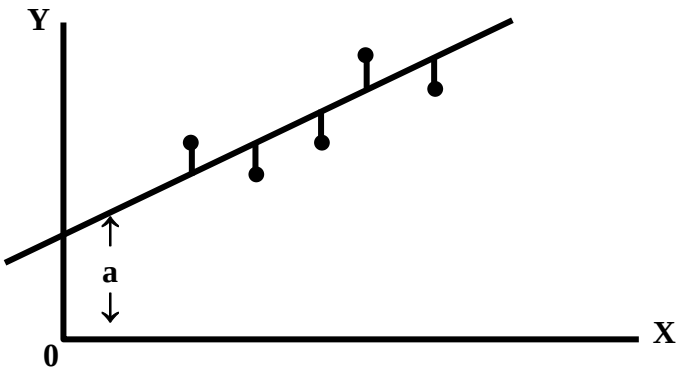
Even with the greatest care and skill, a line drawn between the points with a ruler will be highly SUBJECTIVE, and different individuals will arrive at different lines.
The *real* objective is to find the line of BEST fit. For this, we use a method known as THE METHOD OF LEAST SQUARES. The line of best fit obtained by the method of least squares is called the REGRESSION LINE of Y on X.
And, this whole process is known as simple linear regression. A very important point to note here is that, from the MATHEMATICAL standpoint, simple linear regression requires that X is a NON-RANDOM variable, whereas Y is a RANDOM variable. For example, consider the case of agricultural experiments. If we conduct an experiment to determine the optimal amount of a particular fertilizer to obtain the maximum yield of a certain crop, then the amount of fertilizer is a non-random variable whereas the yield is a random variable. This is so because the amount of fertilizer is in our OWN control. But, the yield is a random variable because it is NOT in our control. In connection with determining the line of BEST fit, the first point is that.
If we use the 'FREE-hand' method of curve-fitting in order to represent the relationship between X and Y as portrayed by the scatter diagram, one tends, consciously or subconsciously, to draw the straight line such that there is EQUAL numbers of points located on either side of the line.
What is more, the eye will automatically try to judge and EQUATE the total distances between the points above and below the line.
This is a recognition of the fact that the line of best fit must be an 'AVERAGE' line in the true sense.
You will recall that the sum of the deviations round the ARITHMETIC MEAN of a data-set is always equal to zero i.e. the positive and negative deviations CANCEL each other.
Similarly, POSITIVE and NEGATIVE deviations round a line of BEST fit must CANCEL out.
This is the first of the conditions or requirements for an optimal line.
But, the point to understand is that there are an INFINITELY large number of straight lines which will satisfy this condition. $(\overline{X},\overline{Y})$ $(\overline{X},\overline{Y})$ Any line that passes through the point will satisfy this condition, and as shown in the following figure, numerous lines can pass through the point

Three equations where $\sum(Y-\hat{Y})=0$:



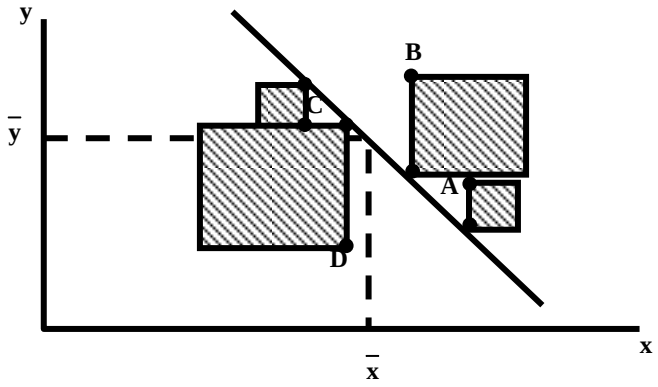
.

For each of the three lines that you see, the SUM of the VERTICAL deviations between the data-points and the line is ZERO. These deviations are depicted by the following diagram:

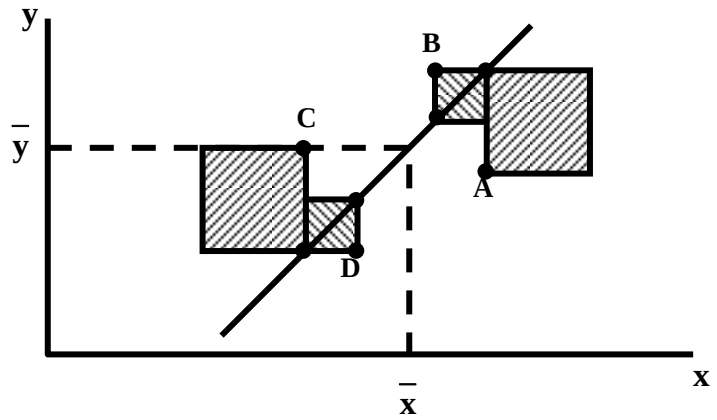


How will we calculate these vertical deviations?
The values of Y obtained from the line are denoted by .
And the deviations of the actual Y-values from the corresponding Y-values obtained from the line are obtained by subtracting from Y.
In all the cases --- as long as our line passes through the point $(\bar{x},\bar{y})$ --- , we find that the sum of the deviations of the *actual* Y-values from the corresponding Y-values obtained from the *line* is zero.
Hence, it appears that we need some SECOND criterion for establishing a *unique* position for the BEST-fitting line.
Our interest is NOT simply in achieving a non-zero sum: it is the MAGNITUDE of the sum which is our main concern. In the two figures that follow, the DIFFERENCES in the sums of squared deviations for two different lines passing through the SAME scatter diagram are clearly portrayed. In position 1, the shaded areas are relatively large, but AS the line is rotated around the point $(\bar{X},\bar{Y})$ in a clockwise direction to position 2, the areas become smaller.

Position 1



Position 2



There would seem to be some UNIQUE position at which the sum of the square deviations is at a MINIMUM. *This* is the position of *least* squares.
If we can ascertain the location of THIS particular straight line in terms of the constants a and b of the linear equation Y = a + bX, then we have found the line of BEST fit.
The rationale is that the SMALLER the sum of the squared deviations round the mean, the LESS dispersed are the data points around the fitted line.

THE PRINCIPAL OF LEAST SQUARES

According to the principal of least squares, the best-fitting line to a set of points is the one for which the sum of the squares of the vertical distances between the points and the line is minimum.
The line Y = a + bX is the one that best fits the given set of points according to the principal of least squares.
And, this best fitting line is obtained by solving simultaneously two equations which are known as the normal equations.

NORMAL EQUATIONS

$$\left.\begin{aligned} \sum Y &= na + b\sum X \\ \sum XY &= a\sum X + b\sum X^2 \end{aligned}\right\}$$

In connection with these two equations, two points should be noted:
1) I will not be discussing the mathematical derivation of these equations.
2) The word "normal" here has nothing to do with the well-known normal distribution.
For any bivariate data-set, obviously we will have available to us two columns, a column of X and a column of Y.
Hence, obviously, we will be in a position to compute sums like ∑X, ∑Y, ∑XY, and so on.
Hence, the only unknown quantities in the two normal equations are a and b, as shown below:

NORMAL EQUATIONS

$$\left.\begin{aligned} \sum Y &= na + b\sum X \\ \sum XY &= a\sum X + b\sum X^2 \end{aligned}\right\}$$

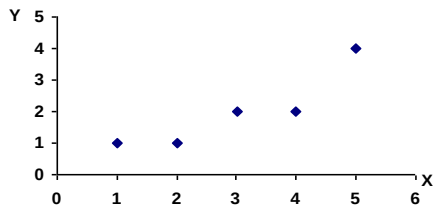
Hence, when we solve the two normal equations simultaneously, we will obtain the values of a and b, and these are EXACTLY the two quantities that we need in order to obtain the BEST-fitting line.
Let me explain this whole concept to you with the help of the same example that I picked up in the beginning of today's lecture:

EXAMPLE

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus. Suppose the company administers this drug on 5 subjects and obtains the following information:

Subject	Percentage of drug	Reaction Time (milli-seconds)
	X	Y
A	1	1
B	2	1
C	3	2
D	4	2
E	5	4

Scatter Diagram:



In order to find a and b, we need to solve the two normal equations, and for this purpose, we will carry out computations as shown below:

X	Y	X^2	XY
1	1	1	1
2	1	4	2
3	2	9	6
4	2	16	8
5	4	25	20
15	10	55	37

$10 = 5a + 15\,b$ — 1

$37 = 15a + 55b$ — 2

$-7 = -10b$

$b = \frac{-7}{-10} = 0.7$

$10 = 5a + 15b$

$\therefore \quad 10 - 15b = 5a$

$\therefore \quad 10 - 15(0.7) = 5a$

$\therefore \quad 10 - 10.5 = 5a$

$\therefore \quad -0.5 = 5a \qquad \therefore \quad a = \frac{-0.5}{5} = -0.1$

Hence our straight line is given by

$Y = -0.1 + 0.7\,X$

A hat is placed on top of the Y so as to differentiate the Y values obtained from the line from the ones that pertain to the actual data-points.

The question is, “What is the advantage of fitting this line?”

The answer to this question is that this line can be used to ESTIMATE the value of the dependent variable corresponding to some particular value of the independent variable. In this example, suppose that we are interested in finding out what will be the reaction time of a person who has 4.33% of the drug in his blood stream? The answer will be obtained by putting X=4.33 in the equation that we just obtained.

Our regression line is

$= -0.1 + 0.7\,X$

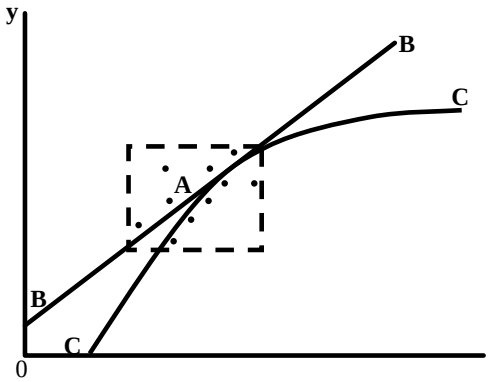
Putting X = 4.33, we obtain

$= -0.1 + 0.7\,(4.33)$

$= -0.1 + 3.031 = 2.931$

Hence we conclude that it can be expected that a person having 4.33% of the drug in his blood stream will take 2.9 milli-seconds to react to the stimulus. A point to be noted here is that this procedure of estimating the value of the dependent variable should not be used for extrapolation. Extrapolation means the making of estimates or predictions outside the range of the experimental data, and in some situations, this can be very unwise.
Let me explain this point with the help of the following diagram:

The extrapolation trap



A = region of interpolation
B = regions of extrapolation
C = true relationship in regions of extrapolation

While a set of observations may show a good linear relationship between the variables, there is NEVER any *guarantee* that the SAME linear form is present over THOSE ranges of the variable NOT under consideration. I would now like to convey to you another point: All the discussion that I have done until now assumes that Y is the dependent variable and X is the independent variable, and therefore, we are regressing Y on X. But, in some situations, we may be interested in just the OPPOSITE --- i.e. we may wish to regress X on Y. In this situation, all we have to do is to interchange the roles of X and Y. I would like to encourage you to work on this on your own, and to establish the normal equations that will be required in this situation. You may be thinking, "Why should we go through this hassle? Can't we use the equation that we have just fitted i.e. Y = a + bX to estimate X from Y?"
It is important to note that this is not the case. If we are confronted with a situation where we require to predict Y from X AND X from Y, then *two* DISTINCT equations need to be found. The regressions of Y on X and X on Y for the same bivariate data are NOT identical.
The next concept that I am going to discuss with you is the STANDARD ERROR OF ESTIMATE.

STANDARD ERROR OF ESTIMATE

The observed values of (X, Y) do not all fall on the regression line but they scatter away from it. The degree of scatter of the observed values about the regression line is measured by what is called the standard deviation of regression or the standard error of estimate of Y on X.
For sample data, the standard error of estimate is obtained from the formula

$$s_{y.x} = \sqrt{\frac{\sum\left(Y - \hat{Y}\right)^2}{n-2}}$$

where Y denotes an observed values, denotes the corresponding values obtained from the least-squares line and n denotes the sample size.
The formula that I just conveyed to you is a bit cumbersome to apply because, in order to apply it, we first need to compute corresponding to all our X-values.
Alternative formula for Sy.x
The standard error of estimate can be more conveniently computed from the alternative formula

$$s_{y.x} = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$$

INTERPRETATION OF Sy.x

The range within which sy.x lies is given by 0 < sy.x < sy (where sy denotes the standard deviation of the y values). sy.x will be zero when all the observed points fall on the regression line (denoting perfect relationship between the two variables). sy.x will be equal to sy when there is no relationship between the two variables. Hence the closer sy.x is to zero (the further away it is from sy), the closer the points are to the line, and the more RELIABLE is our line for purposes of prediction.
Let us apply this concept to the example of the amount of drug in the blood stream and the time taken to react to a stimulus:

X	Y	X^2	Y^2	XY
1	1	1	1	1
2	1	4	1	2
3	2	9	4	6
4	2	16	4	8
5	4	25	16	20
15	10	55	26	37

a = -0.1
b = 0.7

$$Sy.x = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$$

$$= \sqrt{\frac{26 - (-0.1)(10) - (0.7)(37)}{5-2}}$$

$$= \sqrt{\frac{26 + 1 - 25.9}{3}}$$

$$= \sqrt{\frac{1.1}{3}} = \sqrt{0.3667} = 0.61$$

$$\text{Also, } s_y = \sqrt{\frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n}\right)^2}$$

$$= \sqrt{\frac{26}{5} - \left(\frac{10}{5}\right)^2} = \sqrt{5.2 - 4}$$

$$= \sqrt{1.2} \qquad = 1.10$$

INTERPRETATION

Sy.x is NOT very small compared with sy, hence our least-squares line.

$$\hat{Y} = -0.1 + 0.7X$$

is probably NOT very reliable for purposes of prediction. As it explained a short while ago, the smaller our Standard Error of Estimate, the closer the data-points will be to the line --- i.e. the more REPRESENTATIVE our line will be of the data-points --- and, hence, the more RELIABLE our line will be for estimation purposes.
The next concept that I am going to discuss with you is the concept of CORRELATION.
It is a concept that is very closely linked with the concept of linear regression.

CORRELATION

is a measure of the strength or the degree of relationship between two RANDOM variables.

A numerical measure of the strength of the linear relationship between two random variables X and Y is known as Pearson's Product-Moment Coefficient of Correlation.

PEARSON'S COEFFICIENT OF CORRELATION

where, covariance of X and Y is defined as

$$r = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}}$$

$$Cov(X,Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n}$$

This formula is a bit cumbersome to apply. Therefore, we may use the following short cut formula:

SHORT CUT FORMULA FOR THE PEARSON'S COEFFICIENT OF CORRELATION

$$r = \frac{\sum XY - \left(\sum X\right)\left(\sum Y\right)/n}{\sqrt{\left[\sum X^2 - \left(\sum X\right)^2/n\right]\left[\sum Y^2 - \left(\sum Y\right)^2/n\right]}}$$

It should be noted that
r is a pure number that lies between –1 and 1 i.e.

$-1 < r < 1$

Actually, the mathematical expressions that you have just seen is a combination of three different mathematical expressions:

Case 1:

Positive correlation: $0 < r < 1$

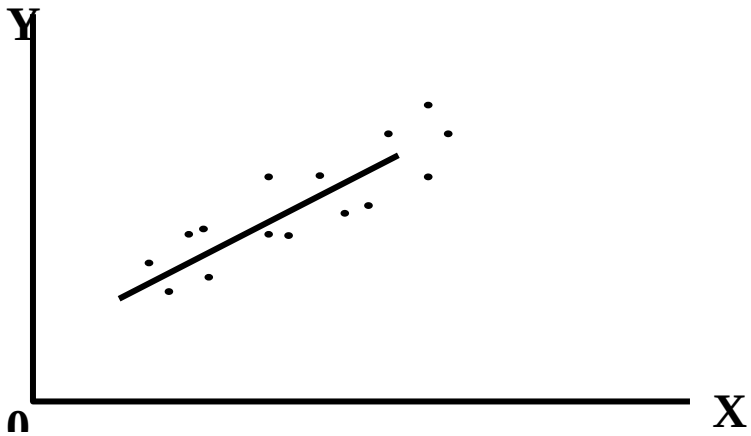
Case 2:

No correlation: $r = 0$

Case 3:

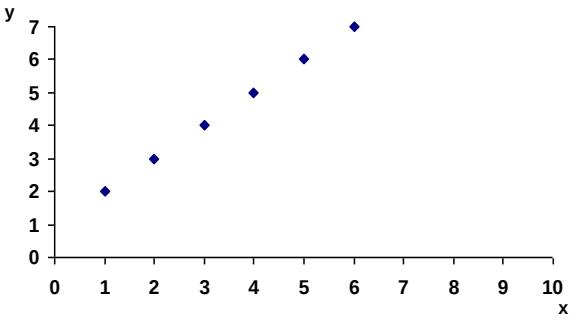
Negative correlation: $-1 < r < 0$

Case 1:
In case of a positive linear relationship, r lies between 0 and 1.



In this case, the closer the points are to the UPWARD-going line, the STRONGER is the positive linear relationship, and the closer r is to 1.

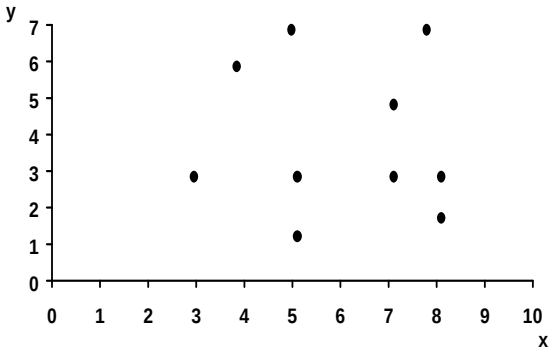
***Perfect* Positive Linear Correlation (r = 1):**



In this case, the closer the points are to the DOWNWARD-going line, the stronger is the linear relationship, and the closer r is to –1.

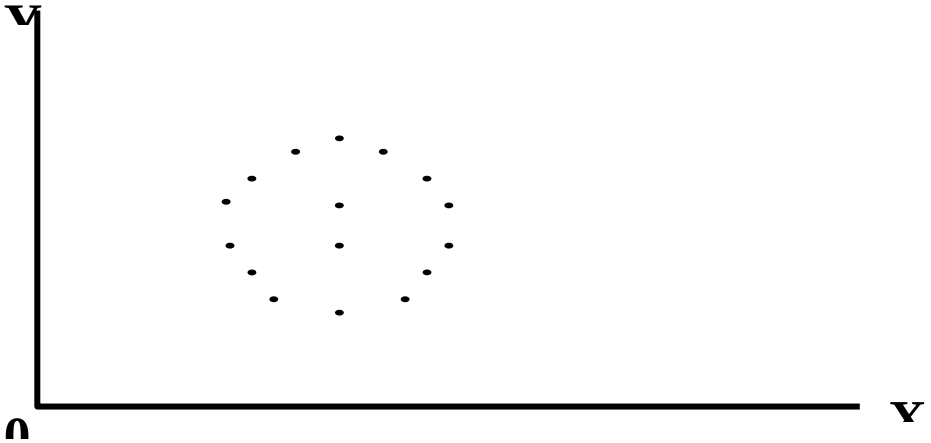
Case 3:

In a situation where neither an upward linear trend nor a downward linear trend can be visualized, r ~ 0



Here, the bivariate data seem to be *completely* random.

The *extreme* of dissociation (*zero* correlation (r = 0)):



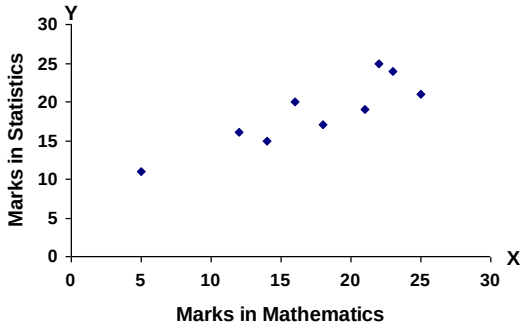
In such a situation, X and Y are said to be *uncorrelated*.

EXAMPLE

Suppose that the principal of a college wants to know if there exists any correlation between grades in Mathematics and grades in Statistics. Suppose that he selects a random sample of 9 students out of all those who take this combination of subjects. The following information is obtained:

Student	Marks in Mathematics (Total Marks: 25)	Marks in Statistics (Total Marks: 25)
	X	Y
A	5	11
B	12	16
C	14	15
D	16	20
E	18	17
F	21	19
G	22	25
H	23	24
I	25	21

SCATTER DIAGRAM



In order to compute the correlation coefficient, we carry out the following computations,

X	Y	X^2	Y^2	XY
5	11	25	121	55
12	16	144	256	192
14	15	196	225	210
16	20	256	400	320
18	17	324	289	306
21	19	441	361	399
22	25	484	625	550
23	24	529	576	552
25	21	625	441	525
156	168	3024	3294	3109

$$r = \frac{\sum XY - (\sum X)(\sum Y)/n}{\sqrt{\left[\sum X^2 - (\sum X)^2/n\right]\left[\sum Y^2 - (\sum Y)^2/n\right]}}$$

$$= \frac{3109 - (156)(168)/9}{\sqrt{\left[3024 - (156)^2/9\right]\left[3294 - (168)^2/9\right]}}$$

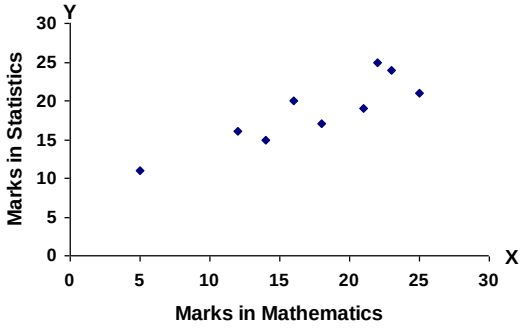
$$= \frac{3109 - 2912}{\sqrt{[3024 - 2704][3294 - 3136]}}$$

$$= \frac{197}{\sqrt{320 \times 158}} = \frac{197}{224.86} = 0.88$$

INTERPRETATION

There exists a strong *positive* linear correlation between marks in Mathematics and marks in Statistics for these 9 students who have been taken into consideration.
The conclusion that we have just drawn i.e. strong positive linear correlation --- this conclusion is supported by the scatter diagram.

SCATTER DIAGRAM



As you can see in the scatter diagram, the data-points appear to follow a linear pattern quite strongly.
In today's lecture, I have discussed with you the concept of regression and correlation. Although I have conveyed to you a number of interesting concepts, believe me, this is only the BEGINNING of a very vast and important area of Statistics. You can study this concept further, and, if possible, to study a little bit about MULTIPLE regression and correlation as well --- the situation when we try to study the relationship between three or more variables.
This brings us to the end of the FIRST part of this course i.e. Descriptive Statistics.
This brings us to the end of the FIRST part of this course i.e. Descriptive Statistics.

LECTURE NO. 33

- Set Theory
- Counting Rules:
- The Rule of Multiplication

"SET"

A set is any well-defined collection or list of distinct objects, e.g. a group of students, the books in a library, the integers between 1 and 100, all human beings on the earth, etc. The term well-defined here means that any object must be classified as either belonging or not belonging to the set under consideration, and the term distinct implies that each object must appear only once. The objects that are in a set, are called members or elements of that set. Sets are usually denoted by capital letters such as A, B, C, Y, etc., while their elements are represented by small letters such as, a, b, c, y, etc. Elements are enclosed by parentheses to represent a set.
For example:

EXAMPLES OF SETS:

A = {a, b, c, d} or
B = {1, 2, 3, 7}

The Number of a set A, written as n(A), is defined as the number of elements in A.
If x is an element of a set A, we write $x \in A$ which is read as "x belongs to A" or x is in A.
If x does not belong to A, i.e. x is not an element of A, we write $x \notin A$.
A set that has no elements is called an empty or a null set and is denoted by the symbolϕ. (It must be noted that {0} is not an empty set as it contains an element 0.)
If a set contains only one element, it is called a unit set or a singleton set.
It is also important to note the difference between an element "x" and a unit set {x}.
A set may be specified in two ways:
1. We may give a list of all the elements of a set (the "Roster" method),
e.g.

A = {1, 3, 5, 7, 9, 11} ;
B = {a book, a city, a clock, a teacher};

2. We may state a rule that enables us to determine whether or not a given object is a member of the set(the "Rule" method or the "Set Builder" method),
e.g.
A = {x | x is an odd number and x < 12} meaning that A is a set of all elements x such that x is an odd number and x is less than 12. (The vertical line is read as "such that".).
An important point to note is that: The repetition or the order in which the elements of a set occur, does not change the nature of the set. The size of a set is given by the number of elements present in it. This number may be finite or infinite. Thus a set is finite when it contains a finite number of elements; otherwise it is an infinite set.
The Empty set is regarded as a Finite set.

EXAMPLES OF FINITE SETS

i) A = {1, 2, 3……., 99, 100};

ii) B = {x | x is a month of the year};

iii) C = {x | x is a printing mistake in a book};

iv) D = {x | x is a living citizen of Pakistan};

Examples of infinite sets:

i) A = {x | x is an even integer};

ii) B = {x | x is a real number between 0 and 1 inclusive}, i.e. B = (x | x 0 < x < 1}

iii) C = {x | x is a point on a line};

iv) D = {x | x is a sentence in a English language}; etc

SUBSETS

A set that consists of some elements of another set, is called a subset of that set. For example, if B is a subset of A, then every member of set B is also a member of set A. If B is a subset of A, we write:

$B \subset A$ or equivalently:

$A \supset B$

'B is a subset of A' is also read as 'B is contained in A', or 'A contains B'.

EXAMPLE

If A = {1, 2, 3, 4, 5, 10}

and B {1, 3, 5}

then $B \subset A$,

i.e. B is contained in A.

It should be noted that any set is always regarded a subset of itself.

and an empty set ϕ is considered to be a subset of every set.

Two sets A and B are Equal or Identical, if and only if they contain exactly the same elements.

In other words, A = B if and only if $A \subset B$ and $B \subset A$.

PROPER SUBSET

If a set B contains some but not all of the elements of another set A, while A contains each element of B, i.e. if

$B \subset A$ and $B \neq A$

then the set B is defined to be a proper subset of A.

Universal Set:

The original set of which all the sets we talk about, are subsets, is called the universal set (or the space) and is generally denoted by S or Ω.

The universal set thus contains all possible elements under consideration.

A set S with n elements will produce 2n subsets, including S and ϕ.

EXAMPLE;

Consider the set A = {1, 2, 3}.

All possible subsets of this set are:

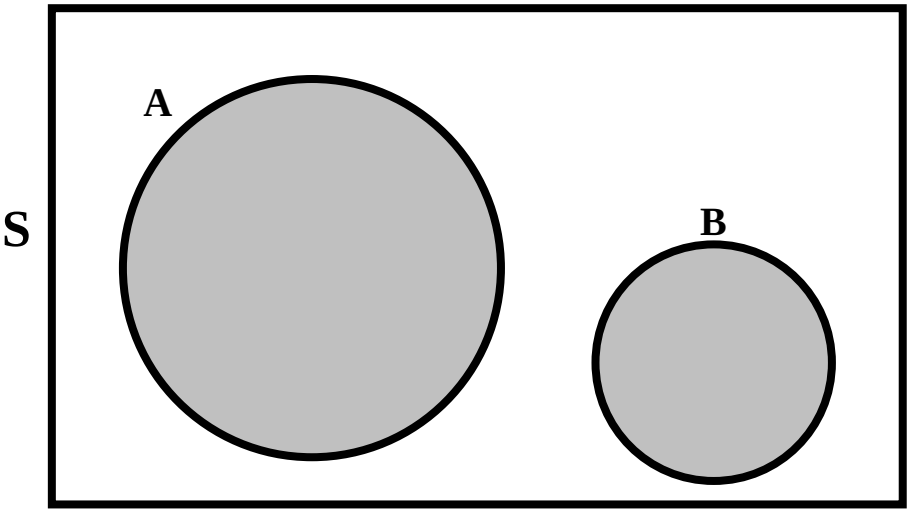
ϕ, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3} and {1, 2, 3}

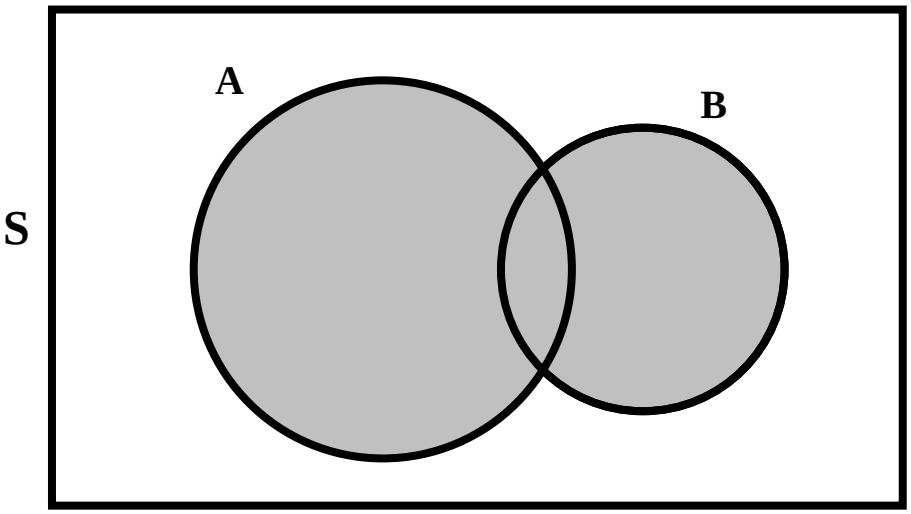
Hence, there are 23 = 8 subsets of the set A.

VENN DIAGRAM

A diagram that is understood to represent sets by circular regions, parts of circular regions or their complements with respect to a rectangle representing the space S is called a Venn diagram, named after the English logician John Venn (1834-1923).
The Venn diagrams are used to represent sets and subsets in a pictorial way and to verify the relationship among sets and subsets.
A Simple Venn diagram:

Disjoint Sets



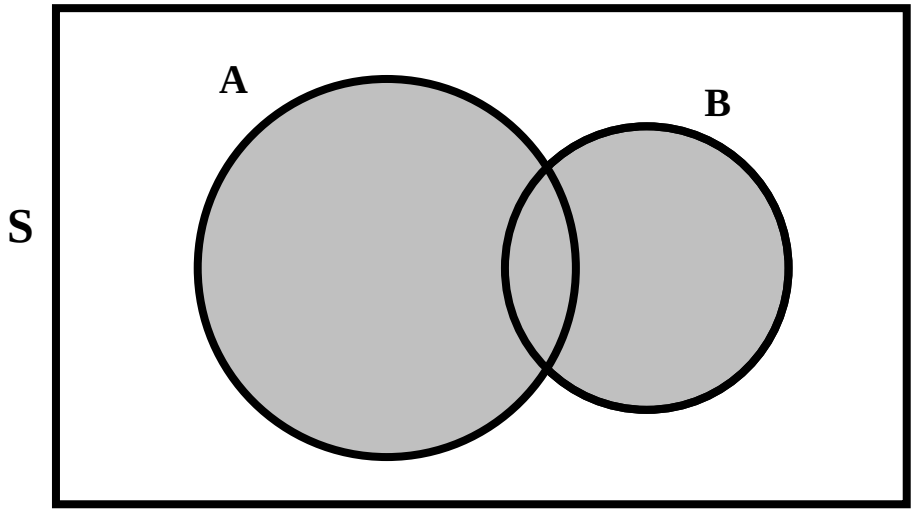


OPERATIONS ON SETS

Let the sets A and B be the subsets of some universal set S. Then these sets may be combined and operated on in various ways to form new sets which are also subsets of S.
The basic operations are union, intersection, difference and complementation.

UNION OF SETS

The union or sum of two sets A and B, denoted by $A \cup B$, and read as "A union B", means the set of all elements that belong to at least one of the sets A and B, that is $A \cup B = \{ x \mid x \in A \text{ or } x \in B\}$
By means of a Venn Diagram, $A \cup B$ is shown by the shaded area as below:



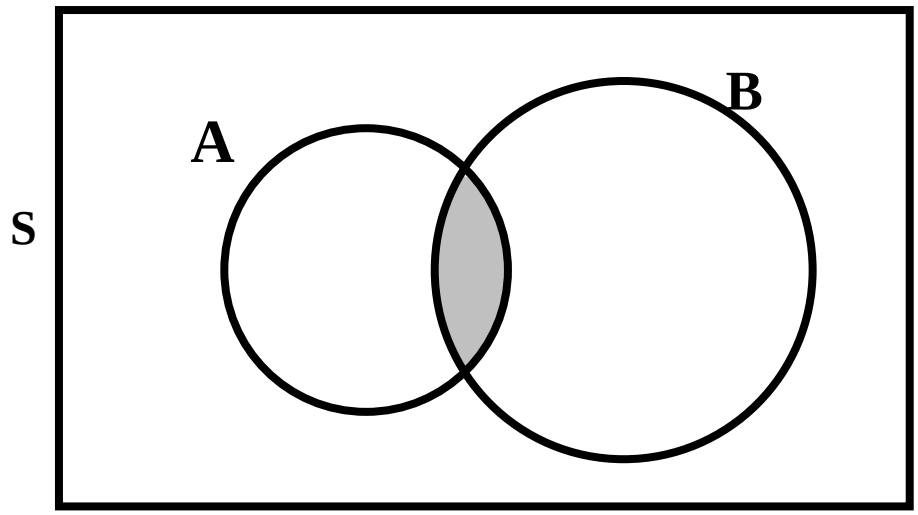
$A \cup B$ is shaded

EXAMPLE

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$
Then $A \cup B = \{1, 2, 3, 4, 5, 6\}$

INTERSECTION OF SETS

The intersection of two sets A and B, denoted by $A \cap B$, and read as "A intersection B", means that the set of all elements that belong to both A and B; that is $A \cap B = \{x \mid x \in \text{ and } x \in B\}$.
Diagrammatically, $A \cap B$ is shown by the shaded area as below:



$A \cap B$ is shaded

EXAMPLE

Let A = {1, 2, 3, 4} and B = {3, 4, 5, 6}
Then $A \cap B = \{3, 4\}$
The operations of union and intersection that have been defined for two sets may conveniently be extended to any finite number of sets.

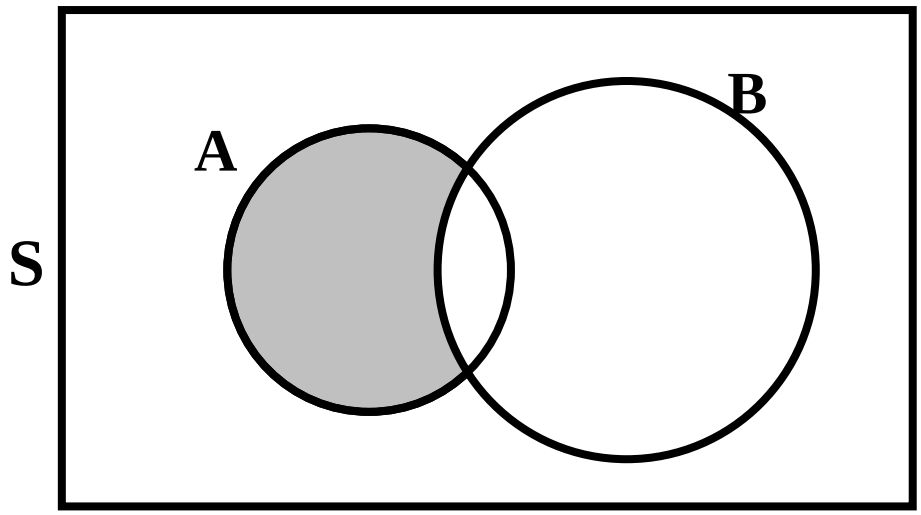
DISJOINT SETS

Two sets A and B are defined to be disjoint or mutually exclusive or non-overlapping when they have no elements in common, i.e. when their intersection is an empty set
i.e. $A \cap B = \phi$.
On the other hand, two sets A and B are said to be conjoint when the have at least one element in common.

SET DIFFERENCE

The difference of two sets A and B, denoted by A – B or by $A - (A \cap B)$, is the set of all elements of A which do not belong to B.
Symbolically,
$A - B = \{x \mid x \in A \text{ and } x \notin B\}$

It is to be pointed out that in general $A - B \neq B - A$.
The shaded area of the following Venn diagram shows the difference A – B:



Difference A – B is shaded

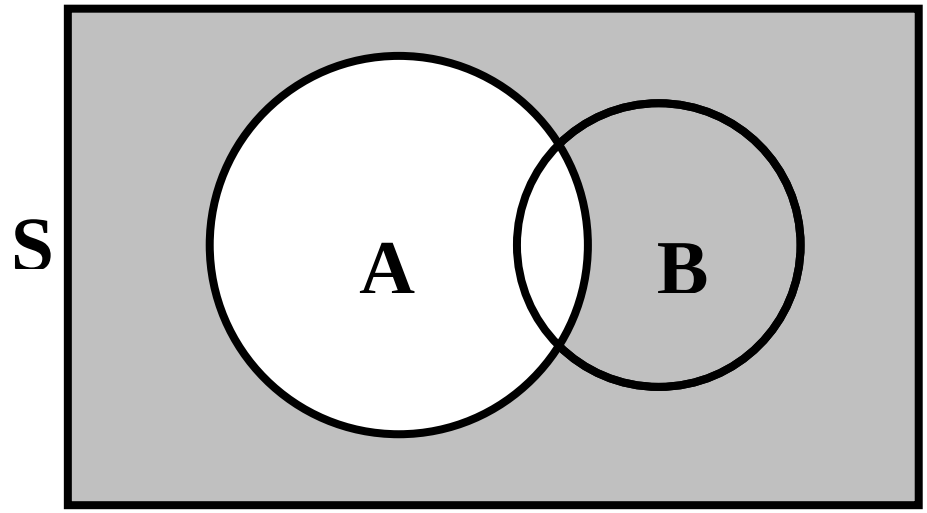
It is to be noted that A – B and B are disjoint sets. If A and B are disjoint, then the difference A – B coincides with the set A.

COMPLEMENTATION

The particular difference S – A, that is, the set of all those elements of S which do not belong to A, is called the complement of A and is denoted by $\bar{A}$ or by Ac. In symbols:

$$\bar{A} = \{x \mid x \in S \text{ and } s \notin A\}$$

The complement of S is the empty set ϕ.
The complement of A is shown by the shaded portion in the following Venn diagram.



$\bar{A}$ is shaded

It should be noted that A – B and A $\cap$ $\bar{B}$, where $\bar{B}$ is the complement of set B, are the same set. Next, we consider the Algebra of Sets. The algebra of sets provides us with laws which can be used to solve many problems in probability calculations.
Let A, B and C be any subsets of the universal set S. Then, we have:

1. Commutative laws:

$A \cup B = B \cup A$

$A \cap B = B \cap A$

2. Associative laws:

$(A \cup B) \cup C = A \cup (B \cup C)$

$(A \cap B) \cap C = A \cap (B \cap C)$

3. Distributive laws

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

4. Idempotent laws

$A \cup A = A$
$A \cap A = A$

5. Identity laws

$A \cup S = S$,
$A \cap S = A$,
$A \cup \phi = A$, and
$A \cap \phi = \phi$.

6. Complementation laws

$A \cup \bar{A} = S$,
$A \cap \bar{A} = \phi$,

$(\bar{A}) = A$,

$\bar{S} = \phi$, and
$\bar{\phi} = S$

7. De Morgan's laws:

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B},$$

and $$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

PARTITION OF SETS

A partition of a set S is a sub-division of the set into non-empty subsets that are disjoint and exhaustive, i.e. their union is the set S itself.
This implies that:

- i)$A_i \cap A_j = \phi$, where $i \neq j$;
- ii)$A_1 \cap A_2 \cup \ldots \cup A_n = S$.

The subsets in a partition are called cells.

EXAMPLE

Let us consider a set S = {a, b, c, d, e}.
Then {a, b}, and {c, d, e} is a partition of S as each element of S belongs to exactly one cell.

CLASS OF SETS

A set of sets is called a class. For example, in a set of lines, each line is a set of points.

POWER SET

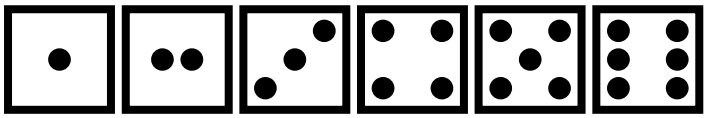
The class of ALL subsets of a set A is called the Power Set of A and is denoted by P(A). For example, if A = {H, T}, then P(A) = {ϕ, {H}, {T}, {H, T}}.

CARTESIAN PRODUCT OF SETS

The Cartesian product of sets A and B, denoted by A $\times$ B, (read as "A cross B"), is a set that contains all ordered pairs (x, y), where x belongs to A and y belongs to B.
Symbolically, we write
A $\times$ B = {(x, y) | x $\in$ A and y $\in$ B}
This set is also called the Cartesian set of A and B set of A and B, named after the French mathematician Rene' Descartes (1596-1605).
The product of a set A by itself is denoted by A2.
This concept of product may be extended to any finite number of sets.

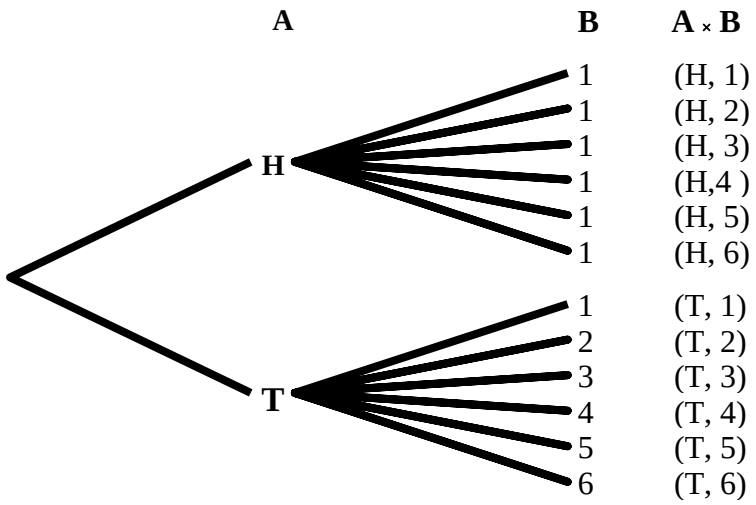
EXAMPLE

Let A = {H, T} and B = {1, 2, 3, 4, 5, 6}. Then the Cartesian product set is the collection of the following twelve (2 $\times$ 6) ordered pairs:
A$\times$B = {(H, 1); (H, 2);(H, 3); (H, 4);(H, 6); (H, 6);(T, 1); (T, 2); (T, 3); (T, 4); (T, 5); (T, 6) }
Clearly, these twelve elements together make up the universal set S when a COIN and a DIE are tossed together.
A die is a cube of wood or ivory whose six faces are marked with dots are shown below:



The plural of the word 'die' is 'dice'.
The product A $\times$ B may conveniently be found by means of the so-called tree diagram shown below:

Tree Diagram



TREE DIAGRAM

The "tree" is constructed from the left to the right. A "tree diagram" is a useful device for enumerating all the possible outcomes of two or more sequential events.
The possible outcomes are represented by the individual paths or branches of the tree.
It is relevant to note that, in general

$$A \times B \neq B \times A.$$

Having reviewed the basics of set theory, let us now review the COUNTING RULES that facilitate the computation of probabilities in a number of problems.

RULE OF MULTIPLICATION

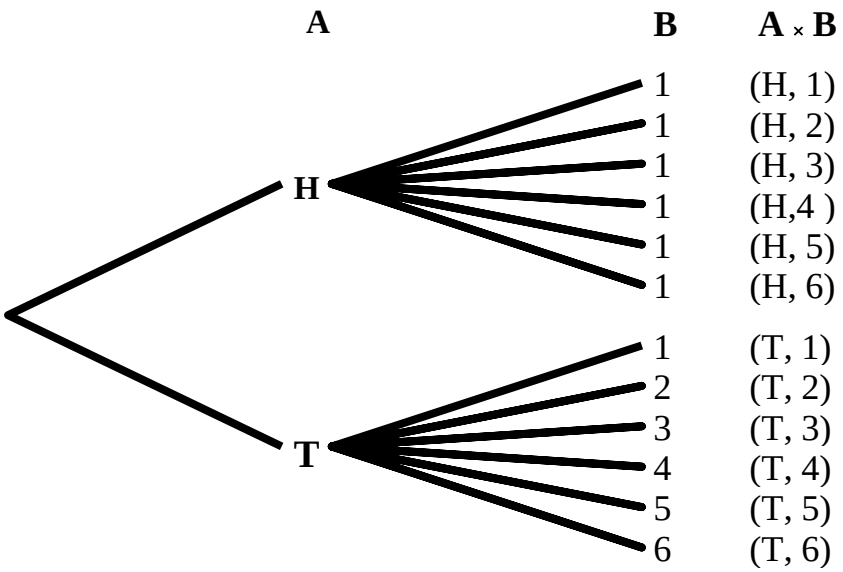
If a compound experiment consists of two experiments which that the first experiment has exactly m distinct outcomes and, if corresponding to each outcome of the first experiment there can be n distinct outcomes of the second experiment, then the compound experiment has exactly mn outcomes.

EXAMPLE

The compound experiment of tossing a coin and throwing a die together consists of two experiments. The coin-tossing experiment consists of two distinct outcomes (H, T), and the die-throwing experiment consists of six distinct outcomes (1, 2, 3, 4, 5, 6).
The total number of possible distinct outcomes of the compound experiment is therefore $2 \times 6 = 12$ as each of the two outcomes of the coin-tossing experiment can occur with each of the six outcomes of die-throwing experiment. As stated earlier, if A = {H, T} and B = {1, 2, 3, 4, 5, 6}, then the Cartesian product set is the collection of the following twelve (2×6) ordered pairs:

$A \times B$ = { (H, 1); (H, 2);(H, 3); (H, 4);
(H, 6); (H, 6);(T, 1); (T, 2);
(T, 3); (T, 4); (T, 5); (T, 6) }

Tree Diagram



The rule of multiplication can be readily extended to compound experiments consisting of any number of experiments performed in a given sequence.

This rule can also be called the Multiple Choice Rule, as illustrated by the following example:

EXAMPLE

Suppose that a restaurant offers three types of soups, four types of sandwiches, and two types of desserts. Then, a customer can order any one out of $3 \times 4 \times 2 = 24$ different meals.

EXAMPLE

Suppose that we have a combination lock on which there are eight rings.
In how many ways can the lock be adjusted?

Solution:

The logical way to look at this problem is to see that there are eight rings on the lock, each of which can have any of the 10 figures 0 to 9:

$$\overline{A}\ \overline{B}\ \overline{C}\ \overline{D}\ \overline{E}\ \overline{F}\ \overline{G}\ \overline{H}$$

ring A can have any of the digits 0 to 9 and ring B can have any of the digits 0 to 9 and ring C can have any of the digits 0 to 9 and

.
.
.

ring H can have any of the digits 0 to 9
Hence the total No. of ways in which these 8 rings can be filled is 8
$10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$
$= 10$
i.e. 100,000,000 — one hundred million.

LECTURE NO. 34

DEFINITIONS OF PROBABILITY

- Subjective Approach to Probability
- Objective Approach:
- Classical Definition of Probability

RELATIVE FREQUENCY DEFINITION OF PROBABILITY

Before we begin the various definitions of probability, let us revise the concepts of:

- Mutually Exclusive Events
- Exhaustive Events
- Equally Likely Events

MUTUALLY EXCLUSIVE EVENTS

Two events A and B of a single experiment are said to be mutually exclusive or disjoint if and only if they cannot both occur at the same time i.e. they have no points in common.

EXAMPLE-1

When we toss a coin, we get *either* a head *or* a tail, but *not* both at the same time. The two events head and tail are therefore mutually exclusive.

EXAMPLE-2

When a die is rolled, the events 'even number' and 'odd number' are mutually exclusive as we can get either an even number or an odd number in one throw, not both at the same time. Similarly, a student *either* qualifies *or* fails, a person is either a teenager or not a teenager, etc., etc.

Three or more events originating from the same experiment are mutually exclusive if pair wise they are mutually exclusive. If the two events *can* occur at the same time, they are not mutually exclusive, e.g., if we draw a card from an ordinary deck of 52 playing cars, it *can* be both a king and a diamond.

Therefore, kings and diamonds are not mutually exclusive. Speaking of playing cards, it is to be remembered that an ordinary deck of playing cards contains 52 cards arranged in 4 suits of 13 each. The four suits are called diamonds, hearts, clubs and spades; the first two are red and the last two are black. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, …, 10, jack, queen and king. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, …, 10, jack, queen and king.

We have discussed the concepts of mutually exclusive events.

Another important concept is that of exhaustive events.

EXHAUSTIVE EVENTS

Events are said to be collectively exhaustive, when the union of mutually exclusive events is equal to the entire sample space S.

EXAMPLES

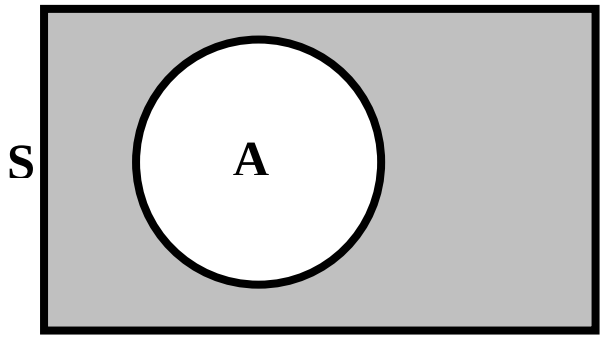
- In the coin-tossing experiment, 'head' and 'tail' are collectively exhaustive events.
- In the die-tossing experiment, 'even number' and 'odd number' are collectively exhaustive events.

In conformity with what was discussed in the last lecture:

PARTITION OF THE SAMPLE SPACE

A group of mutually exclusive and exhaustive events belonging to a sample space is called a partition of the sample space. With reference to any sample space S, events A and $\bar{A}$ form a partition as they are mutually exclusive and their union is the entire sample space. The Venn Diagram below clearly indicates this point.

Venn Diagram



$\bar{A}$ **is shaded**

EQUALLY LIKELY EVENTS

Two events A and B are said to be equally likely, when one event is as likely to occur as the other. In other words, each event should occur in equal number in repeated trials.

EXAMPLE

When a fair coin is tossed, the head is as likely to appear as the tail, and the proportion of times each side is expected to appear is 1/2.

EXAMPLE

If a card is drawn out of a deck of well-shuffled cards, each card is equally likely to be drawn, and the proportion of times each card can be expected to be drawn in a *very* large number of draws is 1/52.Having discussed basic concepts related to probability theory, we now begin the discussion of THE CONCEPT AND DEFINITIONS OF PROBABILITY. Probability can be discussed from two points of view: the subjective approach, and the objective approach.

SUBJECTIVE OR PERSONALISTIC PROBABILITY

As its name suggests, the subjective or personality probability is a measure of the strength of a person's belief regarding the occurrence of an event A. Probability in this sense is purely subjective, and is based on whatever evidence is available to the individual. It has a disadvantage that two or more persons faced with the same evidence may arrive at different probabilities.
For example, suppose that a panel of three judges is hearing a trial. It is possible that, based on the evidence that is presented, two of them arrive at the conclusion that the

accused is guilty while one of them decides that the evidence is NOT strong enough to draw this conclusion. On the other hand, objective probability relates to those situations where everyone will arrive at the same conclusion.
It can be classified into two broad categories, each of which is briefly described as follows:

1. THE CLASSICAL OR 'A PRIORI' DEFINITION OF PROBABILITY

f a random experiment can produce n mutually exclusive and equally likely outcomes, and if m out to these outcomes are considered favorable to the occurrence of a certain event A, then the probability of the event A, denoted by P(A), is defined as the ratio m/n.
Symbolically, we write

$$P(A) = \frac{m}{n}$$
$$= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

This definition was formulated by the French mathematician P.S. Laplace (1949-1827) and can be very conveniently used in experiments where the total number of possible outcomes and the number of outcomes favorable to an event can be DETERMINED.
Let us now consider a few examples to illustrate the classical definition of probability:

EXAMPLE-1

If a card is drawn from an ordinary deck of 52 playing cards, find the probability that i) the card is a red card, ii) the card is a 10.

SOLUTION:
The total number of possible outcomes is 13+13+13+13 = 52, and we assume that all possible outcomes are equally likely.(It is well-known that an ordinary deck of cards contains 13 cards of diamonds, 13 cards of hearts, 13 cards of clubs, and 13 cards of spades.)
(i) Let A represent the event that the card drawn is a red card.
Then the number of outcomes favorable to the event A is 26 (since the 13 cards of diamonds and the 13 cards of hearts are *red*).

Hence

$$P(A) = \frac{m}{n}$$
$$= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$
$$= \frac{26}{52} = \frac{1}{2}$$

Thus $P(B) = \frac{4}{52} = \frac{1}{13}$.

EXAMPLE-2

A fair coin is tossed three times. What is the probability that at least one head appears?

SOLUTION

The sample space for this experiment is

S = {HHH, HHT, HTH, THH,
HTT, THT, TTH, TTT}

and thus the total number of sample points is 8 i.e. n(S) = 8.Let A denote the event that at least one head appears. Then

A = {HHH, HHT, HTH, THH, HTT, THT, TTH}

Therefore n(A) = 7.

Hence

$$P(A)=\frac{n(A)}{n(S)}=\frac{7}{8}.$$

EXAMPLE-3

Four items are taken at random from a box of 12 items and inspected. The box is *rejected* if more than 1 item is found to be faulty. If there are 3 faulty items in the box, find the probability that the box is *accepted*.

SOLUTION

The sample space S contains $\binom{12}{4}=495$ Sample points

(Because there are $\binom{12}{4}$ ways of selecting four items out of twelve)

The box contains 3 faulty and 9 good items. The box is accepted if there is (i) no faulty items, or (ii) one faulty item in the sample of 4 items *selected*.
Let A denote the event the number of faulty items chosen is 0 or 1.
Then

$$n(A) = \binom{3}{0}\binom{9}{4}+\binom{3}{1}\binom{9}{3}$$

$$=126+252=378 \ \ sample \ \ point s.$$

$$\therefore \quad P(A)=\frac{m}{n}=\frac{378}{495}=0.76$$

Hence the probability that the box is accepted is 76% , (in spite of the fact that the box *contains* 3 faulty items).

THE CLASSICAL DEFINITION HAS THE FOLLOWING SHORTCOMINGS

- This definition is said to involve circular reasoning as the term equally likely really means equally probable.
- Thus probability is defined by introducing concepts that presume a *prior* knowledge of the *meaning* of probability.
- This definition becomes vague when the possible outcomes are INFINITE in number, or uncountable.
- This definition is NOT applicable when the assumption of equally likely does *not* hold. And the fact of the matter is that there are NUMEROUS situations where the assumption of equally likely cannot hold.

And these are the situations where we have to look for another definition of probability!

THE **RELATIVE FREQUENCY DEFINITION OF PROBABILITY**

The essence of this definition is that if an experiment is repeated a large number of times under (more or less) identical conditions, and if the event of our interest occurs a certain number of times, then the *proportion* in which this event occurs is regarded as the probability of that event.

For example, we know that a large number of students sit for the matric examination every year. Also, we know that a certain proportion of these students will obtain the first division, a certain proportion will obtain the second division, --- and a certain proportion of the students will fail.

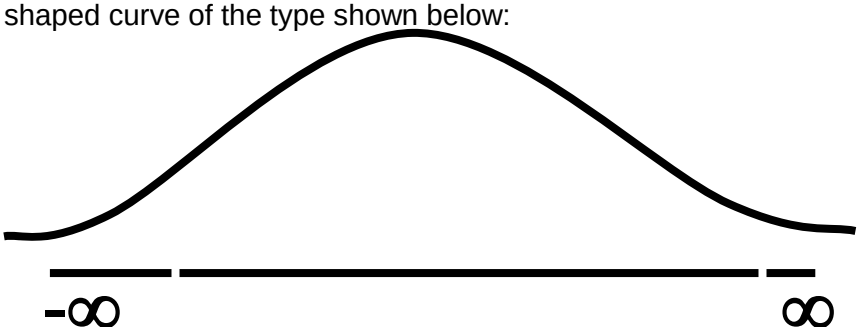
Since the total number of students appearing for the matric exam is very large, hence:

- The proportion of students who obtain the first division --- this proportion can be regarded as the *probability* of obtaining the first division,
- The proportion of students who obtain the second division --- this proportion can be regarded as the *probability* of obtaining the second division, and so on.

LECTURE NO. 35

- Normal Distribution.
  - Mathematical Definition
  - Important Properties
- The Standard Normal Distribution
  - Direct Use of the Area Table
  - Inverse Use of the Area Table
- Normal Approximation to the Binomial Distribution

The normal distribution was discovered in 1733. The normal distribution has a bell-shaped curve of the type shown below:



Let us begin its detailed discussion by considering its formal MATHEMATICAL DEFINITION, and its main
PROPERTIES.

NORMAL DISTRIBUTION

A continuous random variable is said to be normally distributed with mean μ and standard deviation σ if its probability density function is given by

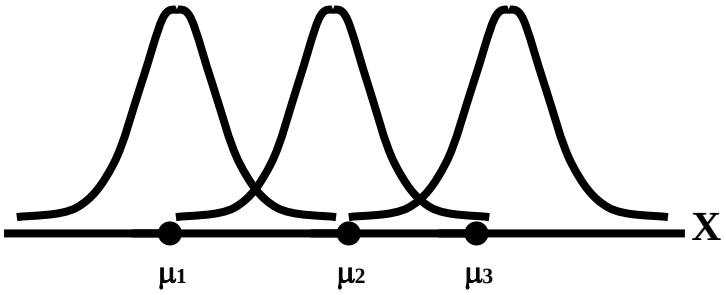
$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}, \quad -\infty<x<\infty \qquad \begin{pmatrix}\text{where}\\ \pi=3.1416\simeq 22/7,\\ e\simeq 2.71828\end{pmatrix}$$

For any particular value of μ and any particular value of σ, giving different values to x and we obtain a set of ordered pairs (x, f(x)) that yield the bell-shaped curve given above. The formula of the normal distribution defines a *FAMILY* of distributions depending on the values of the two *parameters* μ and σ (as these are the two values that determine the shape of the distribution).

PROPERTIES OF THE NORMAL DISTRIBUTION

Property No. 1

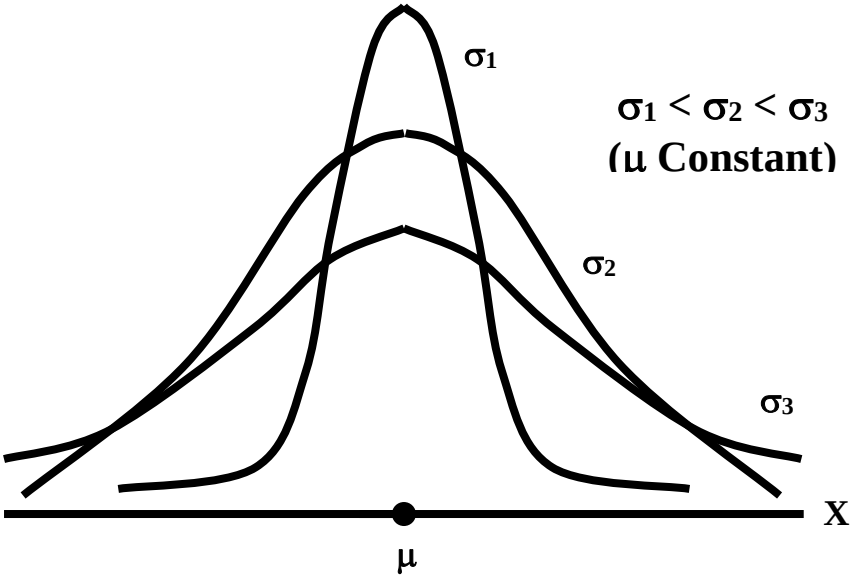
It can be mathematically proved that, for the normal distribution N(μ,σ2), μ represents the *mean*, and σ represents the *standard deviation* of the normal distribution. A change in the mean μ *shifts* the distribution to the left or to the right along the x-axis:



$\mu_1 < \mu_2 < \mu_3$

(σ Constant)

The different values of the standard deviationσ, (which is a measure of *dispersion*), determine the *flatness* or *peakedness* of the normal curve. In other words, a change in the standard deviation on σ *flattens* it or *compresses* it while leaving its centre in the same position:



$\sigma_1 < \sigma_2 < \sigma_3$

(μ Constant)

Property No. 2

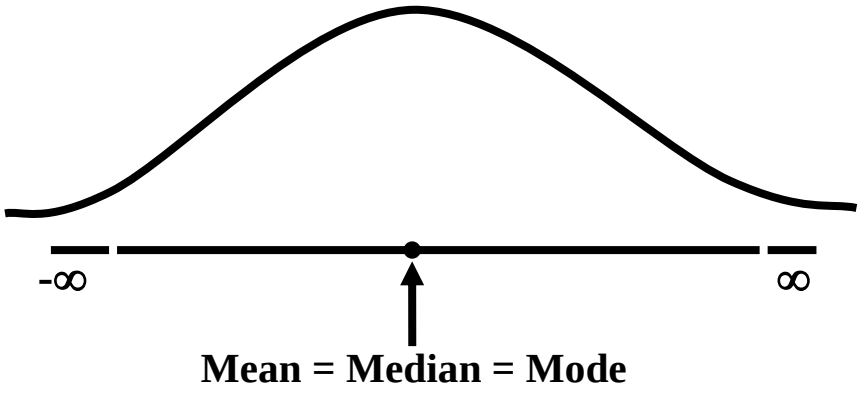
The normal curve is asymptotic to the x-axis as $x \rightarrow \pm \infty$.

Property No. 3

Because of the *symmetry* of the normal curve, 50% of the area is to the right of a vertical line erected at the mean, and 50% is to the left.(Since the total area under the normal curve from $-\infty$ to $+\infty$ is unity, therefore the area to the left of μ is 0.5 and the area to the right of μ is also 0.5.)

Property No. 4

The density function attains its maximum value at $x = \mu$ and falls off symmetrically on each side of μ. This is why the mean, median and mode of the normal distribution are all equal to μ.



Property No. 5

Since the normal distribution is absolutely symmetrical, hence $\mu 3$, the third moment about the mean is zero.

Property No. 6

For the normal distribution, it can be mathematically proved that $\mu 4 = 3\ \sigma 4$

Property No. 7
The moment ratios of the normal distribution come out to be 0 and 3 respectively:

Moment Ratios:

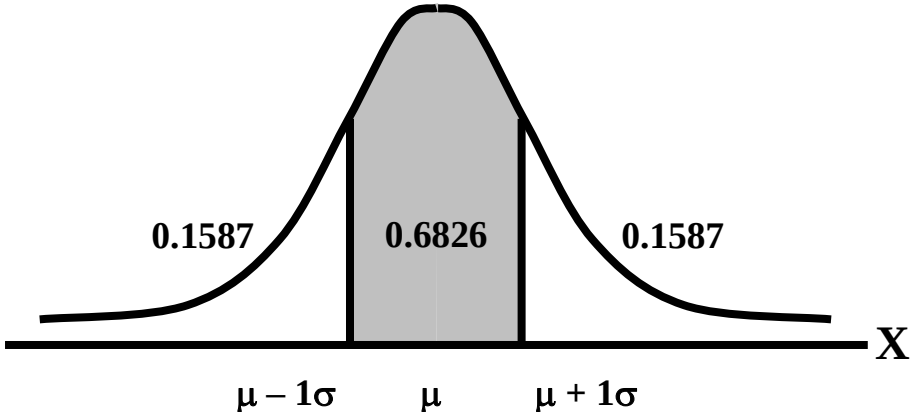
$$\beta_1 = \frac{\mu^2{}_3}{\mu_2{}^3} = \frac{0^2}{\left(\sigma^2\right)^3} = 0,$$

$$\beta_2 = \frac{\mu^4}{\mu_2{}^2} = \frac{3\sigma^4}{\left(\sigma^2\right)^2} = 3$$

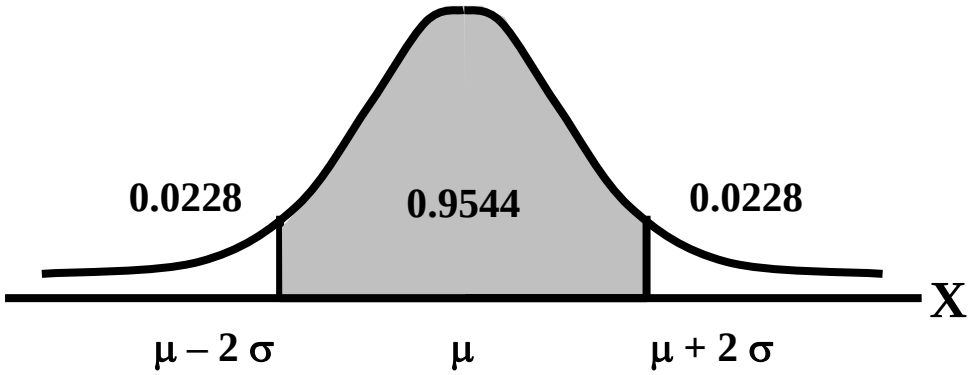
NOTE
Because of the fact that, for the normal distribution, $\beta 2$ comes out to be 3, this is why this value has been taken as a criterion for measuring the kurtosis of any distribution: The amount of peakedness of the *normal* curve has been taken as a *standard*, and we say that this particular distribution is masochistic. Any distribution for which $\beta 2$ is greater than 3 is more peaked than the normal curve, and is called leptokurtic; Any distribution for which $\beta 2$ is less than 3 is less peaked than the normal curve, and is called platykurtic.

Property No. 8
No matter what the values of μ and σ are, areas under the normal curve remain in certain *fixed* proportions within a *specified* number of standard deviations on either side of μ.
For the normal distribution:

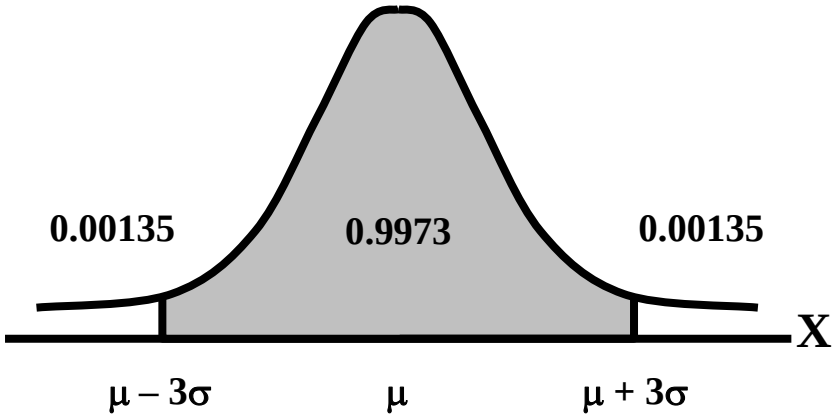
- The interval $\mu \pm \sigma$ will always contain 68.26% of the total area.



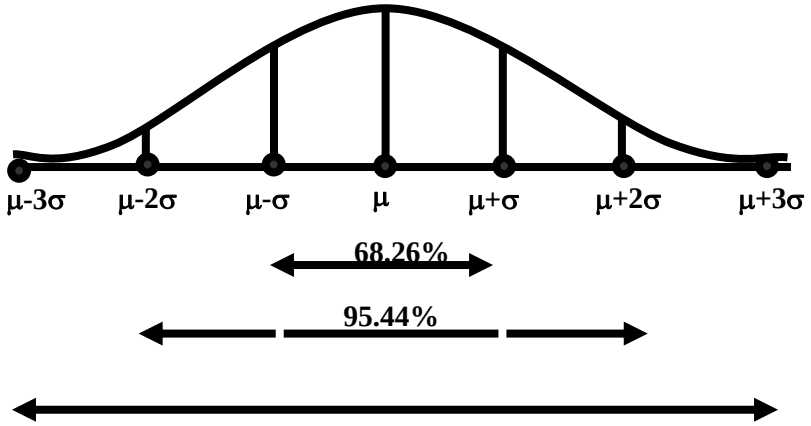
- The interval $\mu + 2\sigma$ will always contain 95.44% of the total area.



- The interval $\mu \pm 3\sigma$ will always contain 99.73% of the total area.



Combining the above three results, we have:



At this point, the student are reminded of the Empirical Rule that was discussed during the first part of this course --- that on descriptive statistics. You will recall that, in the case of any approximately symmetric hump-shaped frequency distribution, approximately 68% of the data-values lie between $\bar{X}$ + S, approximately 95% between the $\bar{X}$ + 2S, and approximately 100% between $\bar{X}$ + 3S.You can now recognize the similarity between the empirical rule and the property given above. (In case a distribution is absolutely normal, the areas in the above-mentioned ranges are 68.26%, 95.44% and 99.73%; in case a distribution approximately normal, the areas in these ranges will be *approximately* equal to these percentages.)

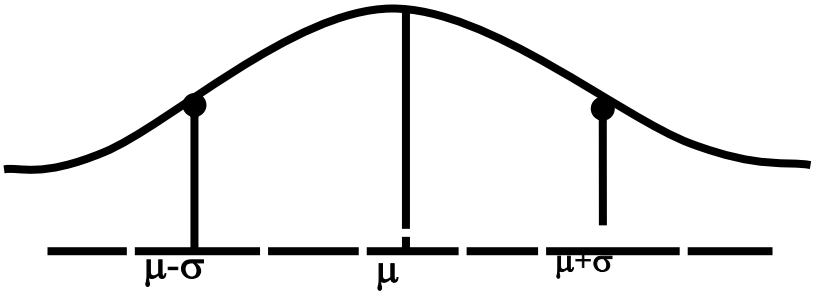
Property No. 9

The normal curve contains points of inflection (where the direction of concavity changes) which are equidistant from the mean. Their coordinates on the XY-plane are

$$\left(\mu-\sigma,\frac{1}{\sigma\sqrt{2\pi e}}\right) and \left(\mu+\sigma,\frac{1}{\sigma\sqrt{2\pi e}}\right)$$

respectively.

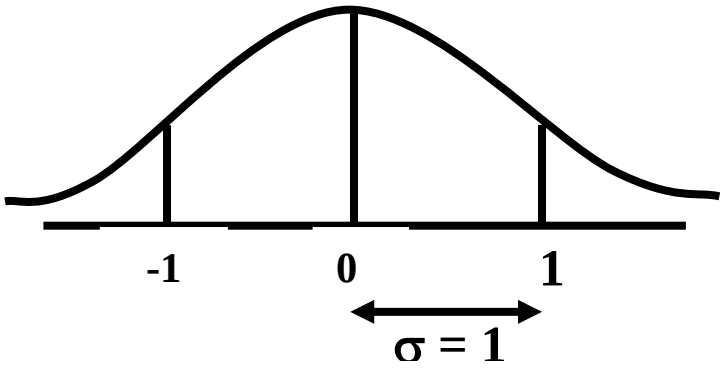
Points of Inflection



Next, we consider the concept of the Standard Normal Distribution:

THE STANDARD NORMAL DISTRIBUTION

A normal distribution whose mean is zero and whose standard deviation is 1 is known as the standard normal distribution.



This distribution has a very important role in *computing areas* under the normal curve. The *reason* is that the mathematical equation of the normal distribution is so complicated that it is not possible to find areas under the normal curve by ordinary integration. Areas under the normal curve have to be found by the more advanced method of *numerical integration*. The point to be noted is that areas under the normal curve have been computed for *that* particular normal distribution whose mean is zero and whose standard deviation is equal to 1, i.e. the standard normal distribution.

Areas under the Standard Normal Curve

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0159	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2083	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2380	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3880
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3990	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4430	0.4441
1.6	0.4452	0.4463	0.4474	0.4485	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4690	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4758	0.4762	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4865	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4980	0.4980	0.4981
2.9	0.4981	0.4982	0.4983	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.49865	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.49903	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993

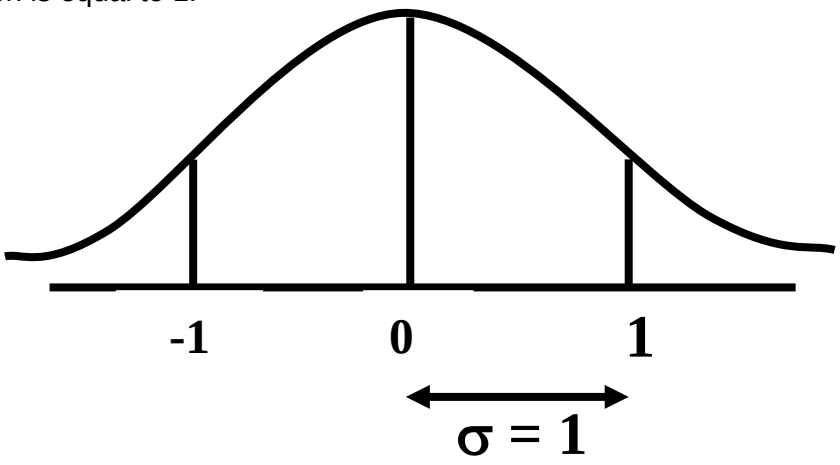
In any problem involving the normal distribution, the generally established procedure is that the normal distribution under consideration is *converted to* the standard normal distribution. This process is called *standardization*. The formula for converting N (μ, σ) to N (0, 1) is:

THE PROCESS OF STANDARDIZATION

The standardization formula is:

$$Z = \frac{X - \mu}{\sigma}$$

If X is N (μ, σ), then Z is N (0, 1). In other words, the standardization formula given above converts our normal distribution to the one whose mean is 0 and whose standard deviation is equal to 1.



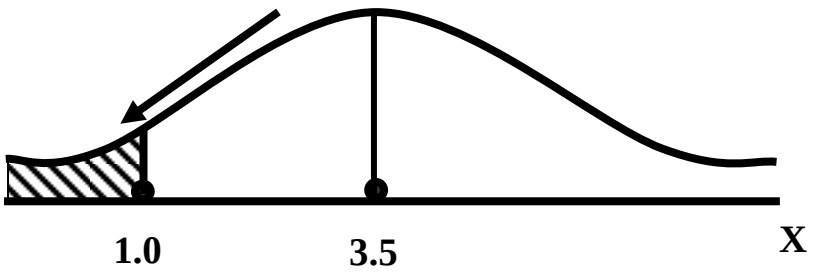
We illustrate this concept with the help of an interesting example:

EXAMPLE

The length of life for an automatic dishwasher is approximately normally distributed with a mean life of 3.5 years and a standard deviation of 1.0 years. If this type of dishwasher is guaranteed for 12 months, what fraction of the sales will require replacement?

SOLUTION

Since 12 months equal one year, hence we need to compute the fraction or *proportion* of dishwashers that will cease to function before a time-span of one year. In other words, we need to find the *probability* that a dishwasher fails before one year.



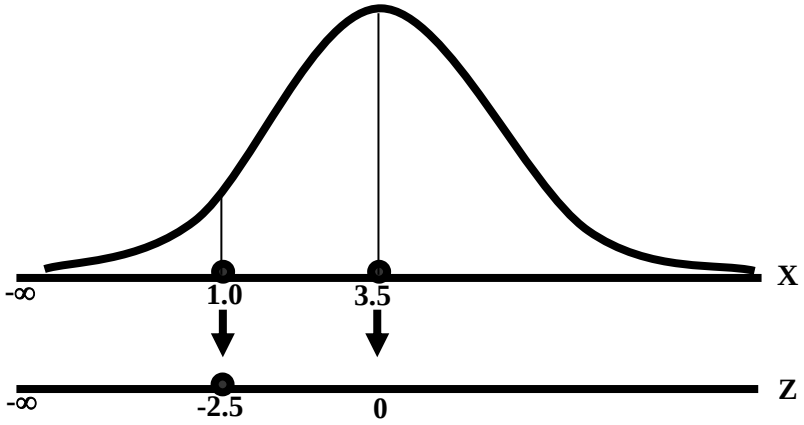
In order to find this area we nee to standardize normal distribution i.e. to convert N(3.5, 1) to N(0, 1):

The method is

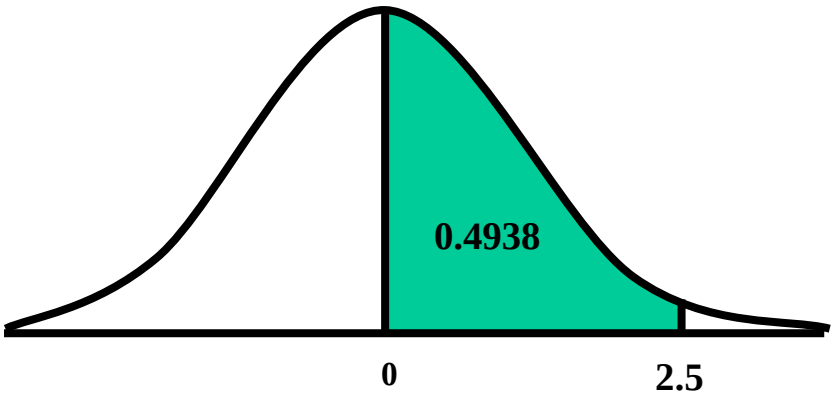
$$Z = \frac{X - \mu}{\sigma} = \frac{X - 3.5}{1.0}$$

The X-value representing the warranty
period is 1.0 so

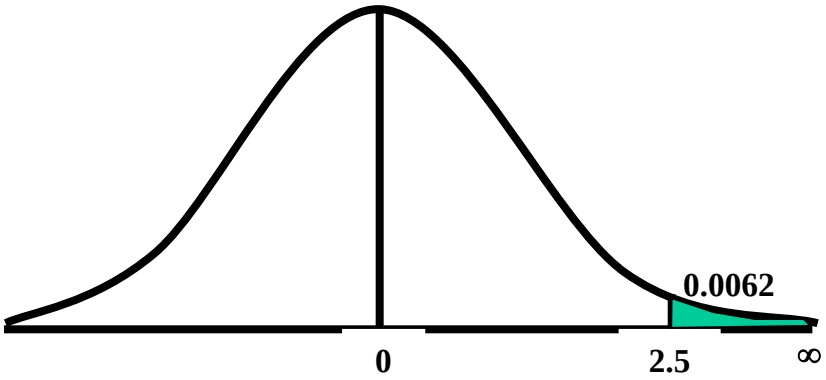
$$Z = \frac{1.0 - 3.5}{1.0} = \frac{-2.5}{1} = -2.5$$



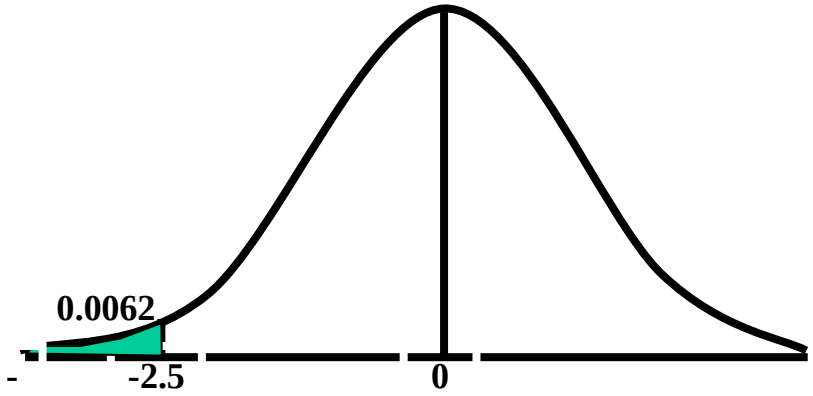
Now we need to find the area under the normal curve from z= -∞ to Z = -2.5. Looking at the area table of the standard normal distribution, we find that Area from 0 to 2.5 = 0.4938



Hence: The area from X = 2.5 to ∞ is 0.0062



But, this means that the area from -∞ to -2.5 is *also* 0.0062, as shown in the following figure:



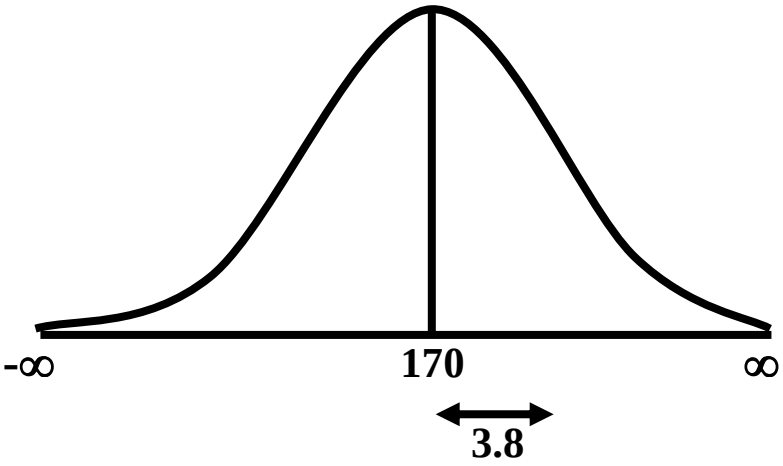
This means that the probability of a dishwasher lasting less than a year is 0.0062 i.e. 0.62% --- even less than 1%.Hence, the owner of the factory should be quite happy with the decision of placing a twelve-month guarantee on the dishwasher! Next, we discuss the Inverse use of the Table of Areas under the Normal Curve. In the above example, we were required to find a certain area against a given x-value. In some situations, we are confronted with just the opposite --- we are given certain areas, and we are required to find the corresponding x-values. We illustrate this point with the help of the following example:

EXAMPLE

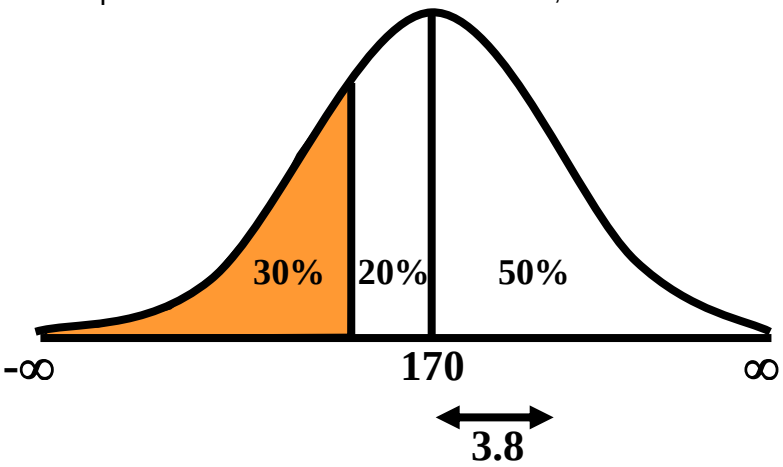
The heights of applicants to the police force in a certain country are normally distributed with mean 170 cm and standard deviation 3.8 cm. If 1000 persons apply for being inducted into the police force, and it has been decided that not more than 70% of these applicants will be accepted, (and the shortest 30% of the applicant are to be rejected), what is the minimum acceptable height for the police force?

SOLUTION:

We have:



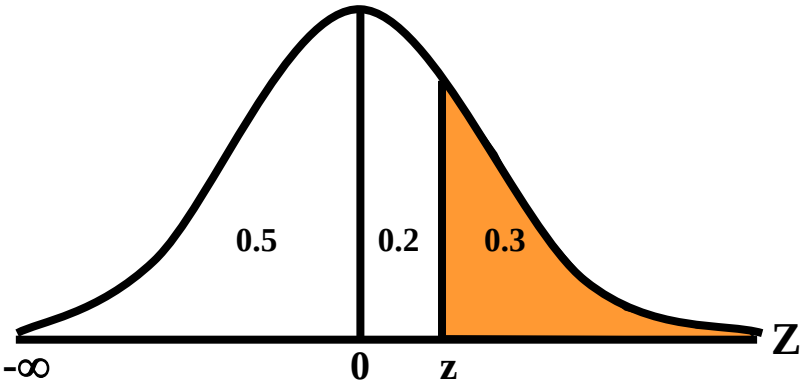
We need to compute the x-value to the left of which, there exists 30% area



The standardization formula can be re-written as

$$Z = \frac{X - \mu}{\sigma}$$

The Z value to the left of which there exists 30% area is obtained as follows.

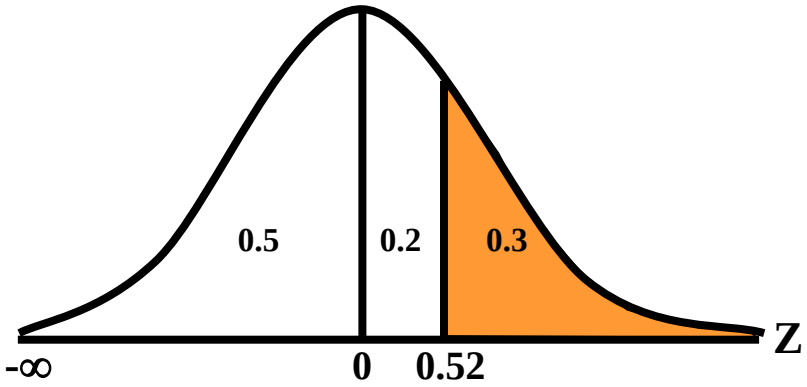


By studying the figures inside the body of the area table of the standard normal distribution, we find that:

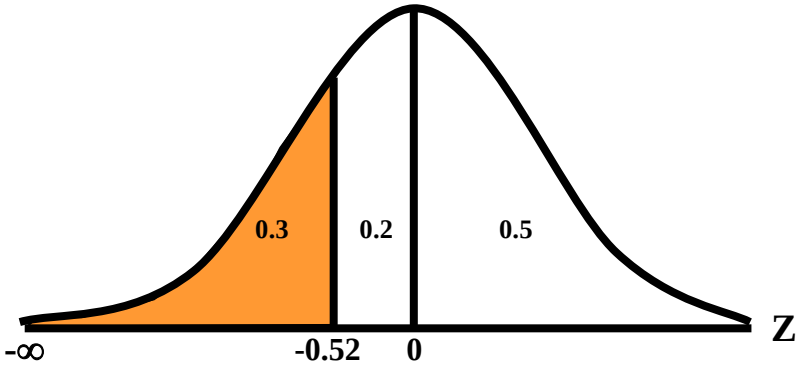
- The area between z = 0
  and z = 0.52 is 0.1985, and

- The area between z = 0
  and z = 2.53 is 0.2019

Since 0.1985 is closer to 0.2000 than 0.2019, hence 0.52 is taken as the appropriate z-value.



But, we are interested not in the upper 30% but the lower 30% of the applicants.
Hence, we have:



Since the normal distribution is absolutely symmetrical, hence the z-value to the left of which there exists 30% area (on the left-hand-side of the mean) will be at exactly the same distance from the mean as the z-value to the right of which there exists 30% area (on the right-hand-side of the mean).
Substituting z = -0.52 in the standardization formula, we obtain:

X = 170 + 3.8 Z
= 170 + 3.8 (-0.52)
= 170 - 1.976
= 168.024 168 cm

Hence, the minimum acceptable height for the police force is 168 cm. Just as binomial, Poisson and other discrete distributions can be *fitted* to *real-life* data; similarly, the normal distribution can also be *FITTED* to real data.
This can be done by equating μ to $\bar{X}$, the mean computed from the observed frequency distribution (based on sample data), and σ to S, the standard deviation of the observed frequency distribution. Of course, this should be done only if we are reasonably sure that the shape of the observed frequency distribution is quite similar to that of the normal distribution. (As indicated in the case of the fitting of the binomial distribution to real data), in order to *decide* whether or not our fitted normal distribution is a *reasonably good fit*, the *proper* statistical procedure is the Chi-square Test of Goodness of Fit.

NORMAL APPROXIMATION TO THE BINOMIAL DISTRIBUTION

The probability for a binomial random variable X to take the value x is

$$f(x)=\binom{n}{x}p^{x}q^{n-x},$$

$$\text{for } 0 \le x \le n \text{ and } q+p=1.$$

The above formula becomes cumbersome to apply if n is LARGE. In such a situation, *as long as neither p nor q is close to zero*, we can compute the required probabilities by applying the normal approximation to the binomial distribution. The binomial distribution can be quite closely approximated by the normal distribution *when n is sufficiently large and neither p nor q is close to zero*. As a rule of *thumb*, the normal distribution provides a reasonable approximation to the binomial distribution *if both np and nq are equal to or greater than 5*, i.e.

np > 5 and nq > 5

EXAMPLE:

Suppose that a past record indicate that, in a particular province of an under-developed country, the death rate from Malaria is 20%. Find the probability that in a particular village of that particular province, the number of deaths is between 70 and 80 (inclusive) out of a total of 500 patients of Malaria.

SOLUTION:

Regarding 'death from Malaria' as success, we have

n = 500

and p = 0.20.

It is obvious that it is very cumbersome to apply the binomial formula in order to compute P(70 < X < 80).

In this problem,

np = 500(0.2) = 100 > > > 5, and nq = 500(0.8) = 400 > > > 5,

therefore we can *happily* apply the normal approximation to the binomial distribution. In order to apply the normal approximation to the binomial, we need to keep in mind the following two points:

1) The first point is: The mean and variance of the binomial distribution valid in our problem will be regarded as the mean and variance of the normal distribution that will be used to approximate the binomial distribution.

In this problem, we have:

and $\mu = np = 500 \times 0.20 = 100$

$\sigma^2 = npq = 500 \times 0.20 \times 0.80 = 80$

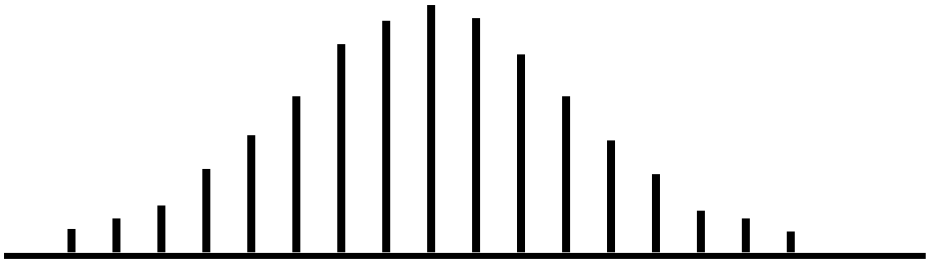
Hence $\sigma = \sqrt{npq} = \sqrt{80} = 8.94$

2) The second important point is:

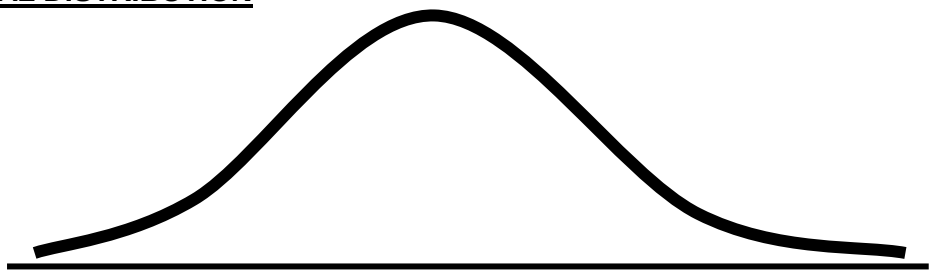
We need to apply a correction that is known as the Continuity Correction. The rationale for this correction is as follows:

The binomial distribution is essentially a discrete distribution whereas the normal distribution is a continuous distribution i.e.:

BINOMIAL DISTRIBUTION

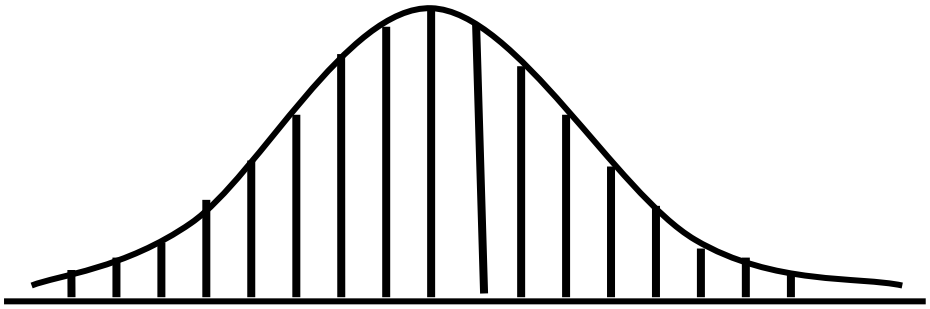


NORMAL DISTRIBUTION



In applying the normal approximation to the binomial, we have the following situation:

THE NORMAL DISTRIBUTION *SUPERIMPOSED* ON THE BINOMIAL DISTRIBUTION



But, the question arises: “How can a set of distinct vertical lines be replaced by a continuous curve?”
In order to overcome this problem, what we do is to replace every integral value x of our binomial random variable by an interval x - 0.5 to x + 0.5. By doing so, we will have the following situation. The x-value 70 is replaced by the interval 69.5 - 70.5, The x-value 71 is replaced by the interval 70.5 - 71.The x-value 72is replaced by the interval 71.5 - 72.5 ……..The x-value 80 is replaced by the interval 79.5 - 80.5

Hence:

Applying the continuity correction,

$P(70 < X < 80)$

is replaced by

$P(69.5 < X < 80.5)$.

Accordingly, the area that we need to compute is the area under the normal curve between the values 69.5 and 80.5.

It is left to the *students* to compute this area, and thus determine the required probability. (This computation involves a few steps.)

By doing so, the students will find that, in that particular village of that province, the probability that the number of deaths from Malaria in a sample of 500 lies between 70 and 80 (inclusive) is 0.0145 i.e. 1½%.

This brings us to the end of the second part of this course i.e. *Probability Theory*.

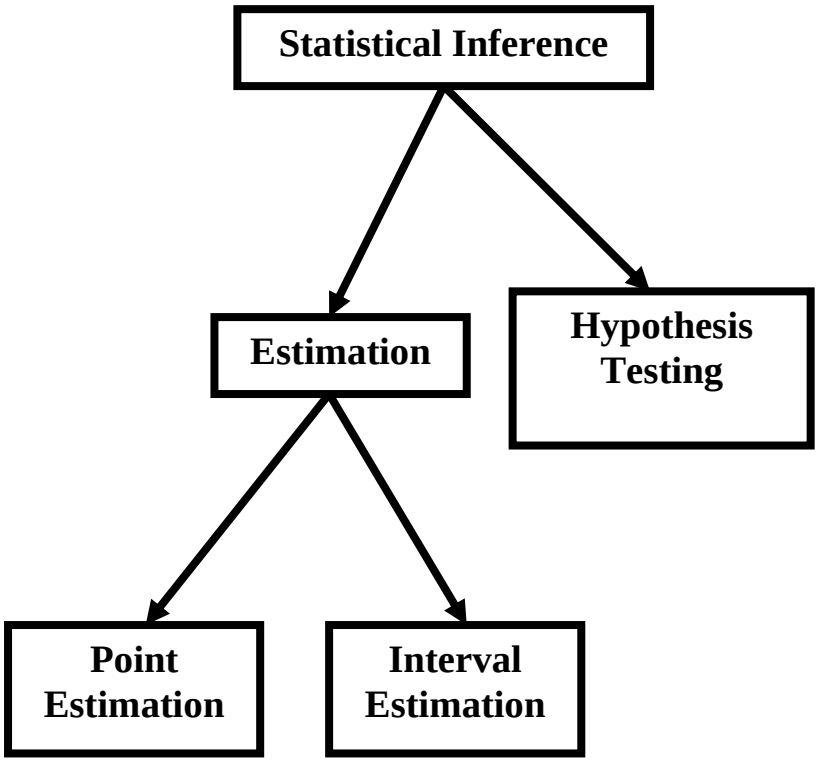
In the next lecture, we will begin the third and last portion of this course i.e. *Inferential Statistics* --- that area of Statistics which enables us to draw conclusions about various phenomena on the basis of data collected on sample basis.

LECTURE NO. 36

- Sampling Distribution of $\overline{X}$
- Mean and Standard Deviation of the Sampling Distribution of $\overline{X}$
- Central Limit Theorem

INFERENTIAL STATISTICS

That branch of Statistics which enables us to draw conclusions or inferences about various phenomena on the basis of real data collected on sample basis. In this regard, the first point to be noted is that statistical inference can be divided into two main branches --- estimation, and hypothesis-testing. Estimation itself can be further divided into two branches --- point estimation, and interval estimation



The second important point is that the concept of sampling distributions forms the basis for both estimation and hypothesis-testing,

SAMPLING DISTRIBUTION

The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in a sample, etc.) is known as its sampling distribution. In this regard, the first point to be noted is that there are two ways of sampling --- sampling with replacement, and sampling without replacement. In case of a finite population containing N elements, the total number of possible samples of size n that can be drawn from this population with replacement is Nn.

In case of a finite population containing N elements, the total number of possible samples of size n that can be drawn from this population without replacement.

We illustrate the concept of the sampling distribution of $\overline{X}$ with the help of the following example:

EXAMPLE

Let us examine the case of an annual Ministry of Transport test to which all cars, irrespective of age, have to be submitted. The test looks for faulty breaks, steering, lights and suspension, and it is discovered after the first year that approximately the same numbers of cars have 0, 1, 2, 3, or 4 faults.
The above situation is equivalent to the following:
Let X denotes the number of faults in a car. Then X can take the values 0, 1, 2, 3, and 4, the probability of each of these X values is 1/5. Hence, we have the following probability distribution:

No. of Faulty Items (X)	Probability f(x)
0	1/5
1	1/5
2	1/5
3	1/5
4	1/5
Total	1

In order to compute the mean and standard deviation of this probability distribution, we carry out the following computations,

MEAN AND VARIANCE OF THE POPULATION DISTRIBUTION

$$\mu = E(X) = \sum xf(x) = 2$$

$$\sigma^2 = Var(X) = E(X)^2 - [E(X)]^2$$
$$= \sum x^2 f(x) - [\sum x f(x)]^2$$
$$= 6 - 2^2 = 6 - 4 = 2$$

Practically speaking, only a sample of the cars will be tested at any one occasion, and, as such, we are interested in considering the results that would be obtained if a sample of vehicles is tested. Let us consider the situation when only two cars are tested after being selected at the roadside by a mobile testing station. The following table gives all the possible situations:

NO. OF FAULTY ITEMS

Second Car / First Car	0	1	2	3	4
0	(0,0)	(0,1)	(0,2)	(0,3)	(0,4)
1	(1,0)	(1,1)	(1,2)	(1,3)	(1,4)
2	(2,0)	(2,1)	(2,2)	(2,3)	(2,4)
3	(3,0)	(3,1)	(3,2)	(3,3)	(3,4)
4	(4,0)	(4,1)	(4,2)	(4,3)	(4,4)

The above situation is equivalent to drawing all possible samples of size 2 from this probability distribution (i.e. the population) WITH REPLACEMENT. From the above list of 25 samples, we can work out all the possible sample means. These are indicated in the following table:

SAMPLE MEANS

Second Car / First Car	0	1	2	3	4
0	0.0	0.5	1.0	1.5	2.0
1	0.5	1.0	1.5	2.0	2.5
2.	1.0	1.5	2.0	2.5	3.0
3	1.5	2.0	2.5	3.0	3.5
4	2.0	2.5	3.0	3.5	4.0

It is immediately evident that some of these possible samples mean occur several times. In view of this, it would seem reasonable and sensible to construct a frequency distribution from the sample means. This is given in the following table:

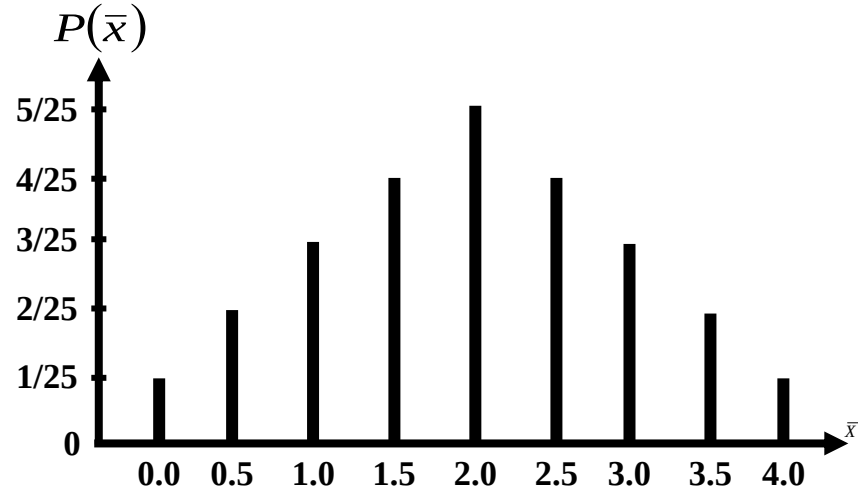
Sample Mean	No. of Samples
$\bar{x}$	f
0.0	1
0.5	2
1.0	3
1.5	4
2.0	5
2.5	4
3.0	3
3.5	2
4.0	1
Total	25

If we divide each of the above frequencies by the total frequency 25, we obtain the probabilities of the various values of $\bar{X}$.(This is so because every one of the 25 possible situations is equally likely to occur, and hence the probabilities of the various possible values of $\bar{X}$ can be computed using the classical definition of probability i.e. m/n --- number of favorable outcomes divided by total number of possible outcomes). Hence, we obtain the following probability distribution:

Sample Mean	No. of Samples	Probability
$\bar{x}$	f	$P(\bar{X} = \bar{x})$
0.0	1	1/25
0.5	2	2/25
1.0	3	3/25
1.5	4	4/25
2.0	5	5/25
2.5	4	4/25
3.0	3	3/25
3.5	2	2/25
4.0	1	1/25
Total	25	25/25=1

The above is referred to as the SAMPLING DISTRIBUTION of the mean. The visual picture of the sampling distribution is as follows:

Sampling Distribution of $\bar{X}$ for n = 2



Next, we wish to compute the mean and standard deviation of this distribution.

As we are already aware, for the probability distribution of a random variable X, the mean is given by

$\mu = E(X) = \sum x\, f(x)$ and the variance is given by $\sigma^2 = Var(X) = E(X2) - [E(X)]2$

The point to be noted is that, in case of the sampling distribution of $\bar{X}$, our random variable is not X but $\bar{X}$.

Hence, the mean and variance of our sampling distribution are given by

MEAN AND VARIANCE OF THE SAMPLING DISTRIBUTION OF $\bar{X}$

$$\mu_{\bar{x}} = E(\bar{X}) = \sum \bar{x} f(\bar{x})$$

$$\sigma^2_{\bar{x}} = Var(\bar{X}) = E(\bar{X})^2 - [E(\bar{X})]^2$$
$$= \sum \bar{x}^2 f(\bar{x}) - [\sum \bar{x}\, f(\bar{x})]^2$$

The square root of the variance is the standard deviation, and the standard deviation of a sampling distribution is termed as its standard error. In order to find the mean and standard error of the sampling distribution of $\bar{X}$ in this example, we carry out the following computations:

In order to find the mean and standard error of the sampling distribution of $\bar{X}$ in this example, we carry out the following computations:

Sample Mean	**Probability**		
$\bar{x}$	$f(\bar{x}) = P(\bar{X} = \bar{x})$	$\bar{x}\, f(\bar{x})$	$(\bar{x})^2 f(\bar{x})$
0.0	1/25	0	0
0.5	2/25	1/25	1/50
1.0	3/25	3/25	6/50
1.5	4/25	6/25	18/50
2.0	5/25	10/25	40/50
2.5	4/25	10/25	50/50
3.0	3/25	9/25	54/50
3.5	2/25	7/25	49/50

4.0	1/25	4/25	32/50
Total	25/25=1	50/25=2	250/50=5

Hence, in this example, we have:

$$\mu_{\bar{x}} = E(\bar{X}) = \sum \bar{x} f(\bar{x})$$
$$= 50/25 = 2$$

And

$$\sigma^2{}_{\bar{x}} = Var(\bar{X}) = E(\bar{X})^2 - [E(\bar{X})]^2$$
$$= \sum \bar{x}^2 f(\bar{x}) - [\sum \bar{x} f(\bar{x})]^2$$
$$= 5 - 2^2 = 5 - 4 = 1$$

$$\sigma_{\bar{x}} = \sqrt{\sigma^2{}_{\bar{x}}} = \sqrt{1} = 1$$

These computations lead to the following two very important properties of the sampling distribution of $\bar{X}$

Property No.1

In the case of sampling with replacement as well as in the case of sampling without replacement, we have:

$$\mu_{\bar{x}} = \mu$$

In this example:

$$\mu = 2$$

and

$$\mu_{\bar{x}} = 2$$

Hence

$$\mu_{\bar{x}} = \mu$$

Property No.2

In case of sampling with replacement:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

In this example:

$$\sigma = \sqrt{2}$$
$$\therefore \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{2}}{\sqrt{2}} = 1$$
$$and\ \sigma_{\bar{x}} = 1$$
$$Hence\ \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

NOTE:

In case of sampling without replacement from a finite population:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

The factor $\sqrt{\frac{N-n}{N-1}}$

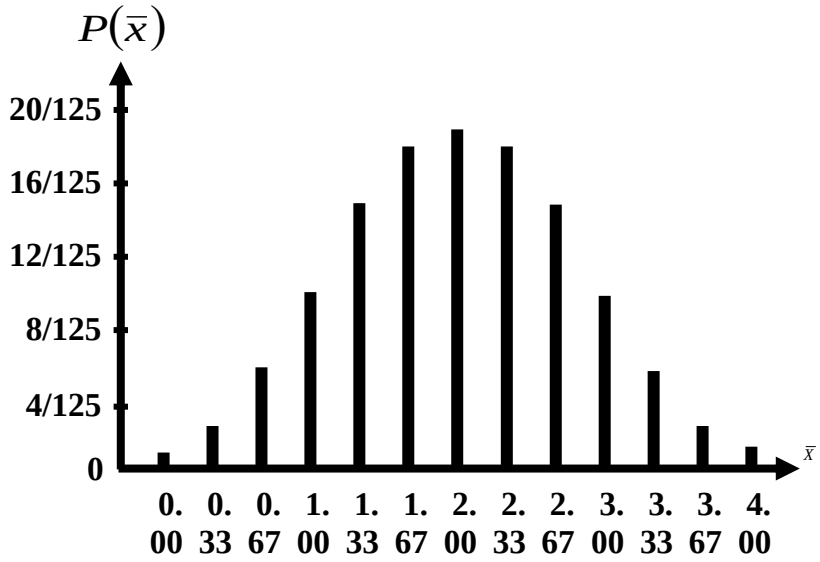
is known as the finite population correction (fpc). The point to be noted is that, if the sample size n is much smaller than the population size N, then is approximately equal to 1, and, as such, the fpc is not required. Hence, in sampling from a finite population, we apply the fpc only if the sample size is greater than 5% of the population size. Next, we consider the shape of the sampling distribution of $\bar{X}$. As indicated by the line chart, the above sampling distribution is absolutely symmetric and triangular. But let us consider what will happen to the shape of the sampling distribution with if the sample size is increased. If in the car tests instead of taking samples of 2 we had taken all possible samples of size 3, our sampling distribution would contain 53 = 125 sample means, and it would be in the following form:

SAMPLING DISTRIBUTION
FOR SAMPLES OF SIZE **3**

$\bar{x}$	No. of Samples	f($\bar{x}$)
0.00	1	1/125
0.33	3	3/125
0.67	6	6/125
1.00	10	10/125
1.33	15	15/125
1.67	18	18/125
2.00	19	19/125
2.33	18	18/125
2.67	15	15/125
3.00	10	10/125
3.33	6	6/125
3.67	3	3/125
4.00	1	1/125
	125	1

The graph of this distribution is as follows:

Sampling Distribution of $\bar{X}$ for n = 3

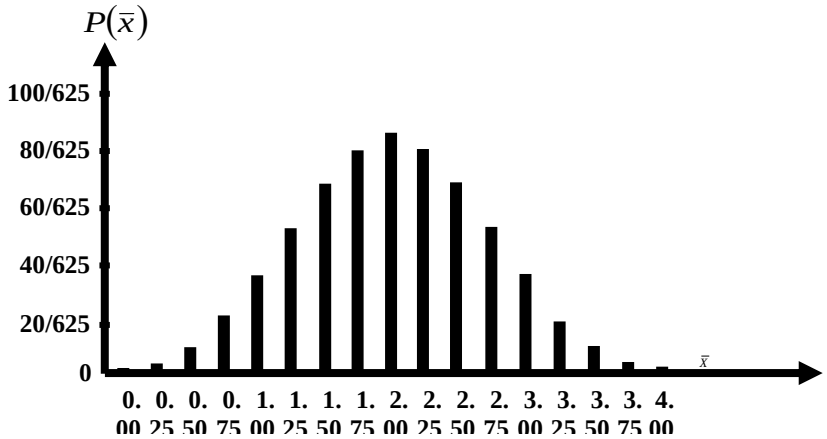


If in the car tests instead of taking samples of 2 we had taken all possible samples of size 4, our sampling distributions would contain 54 = 625 sample means, and it would be in the following form:

SAMPLING DISTRIBUTION
FOR SAMPLES OF SIZE 4

$\bar{x}$	No. of Samples	$f(\bar{x})$
0.00	1	1/625
0.25	4	4/625
0.50	10	10/625
0.75	20	20/625
1.00	35	35/625
1.25	52	52/625
1.50	68	68/625
1.75	80	80/625
2.00	85	85/625
2.25	80	80/625
2.50	68	68/625
2.75	52	52/625
3.00	35	35/625
3.25	20	20/625
3.50	10	10/625
3.75	4	4/625
4.00	1	1/625
	625	1

The graph of this distribution is as follows, Sampling Distribution of $\bar{X}$ for n = 4



As in the case of the sampling distribution of $\bar{X}$ based on samples of size 2, each of these two distributions has a mean of 2 defective items. It is clear from the above figures that as larger samples are taken, the shape of the sampling distribution undergoes discernible changes.

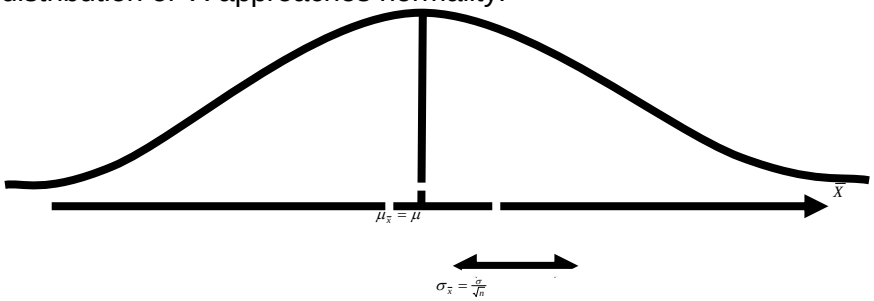
In all three cases the line charts are symmetrical, but as the sample size increases, the overall configuration changes from a triangular distribution to a bell-shaped distribution. When relatively large samples are taken, this bell-shaped distribution assumes the form of a 'normal' distribution (also called the 'Gaussian' distribution), and this happens irrespective of the form of the parent population. (For example, in the problem currently under consideration, the population of defective items in a car is rectangular.)

This leads us to the following fundamentally important theorem:

CENTRAL LIMIT THEOREM

The theorem states that:

"If a variable X from a population has mean μ and finite variance $\sigma 2$, then the sampling distribution of the sample mean $\bar{X}$ approaches a normal distribution with mean μ and variance $\sigma 2/n$ as the sample size n approaches infinity.**"** As $n \to \infty$, the sampling distribution of $\bar{X}$ approaches normality.



Due to the Central Limit Theorem, the normal distribution has found a central place in the theory of statistical inference.(Since, in many situations, the sample is large enough for our sampling distribution to be approximately normal, therefore we can utilize the mathematical properties of the normal distribution to draw inferences about the variable of interest). The rule of thumb in this regard is that if the sample size, n, is greater than or equal to 30, then we can assume that the sampling distribution of $\bar{X}$ is approximately normally distributed. On the other hand, If the POPULATION sampled is normally distributed, then the sampling distribution of $\bar{X}$ will also be normal regardless of sample size. In other words, $\bar{X}$ will be normally distributed with mean μ and variance $\sigma 2/n$.

LECTURE NO. 37

- Sampling Distribution of $\hat{p}$
- Sampling Distribution of $\bar{X}_1 - \bar{X}_2$

We discussed the mean and the standard deviation of the sampling distribution, and, towards the end of the lecture, we consider the very important theorem known as the Central Limit Theorem. Let us now consider the *real-life* application of this concept with the help of an example:

EXAMPLE

A construction company has 310 employees who have an average annual salary of Rs.24,000.The standard deviation of annual salaries is Rs.5,000.
Suppose that the employees of this company launch a demand that the government should institute a law by which their average salary should be at least Rs. 24500, and, suppose that the government decides to check the validity of this demand by drawing a random sample of 100 employees of this company, and acquiring information regarding their present salaries. What is the probability that, in a random sample of 100 employees, the average salary will exceed Rs.24,500 (so that the government decides that the demand of the employees of this company is unfounded, and hence does not pay attention to the demand(although, in reality, it was justified))?

SOLUTION

The sample size (n = 100) is large enough to assume that the sampling distribution of $\bar{X}$ is approximately *normally* distributed with the following mean and standard deviation: and standard deviation

$$\mu_{\bar{x}} = \mu = Rs.24{,}000.$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\cdot\sqrt{\frac{N-n}{N-1}} = \frac{5000}{\sqrt{100}}\sqrt{\frac{310-100}{310-1}}$$
$$= \text{Rs.}412.20$$

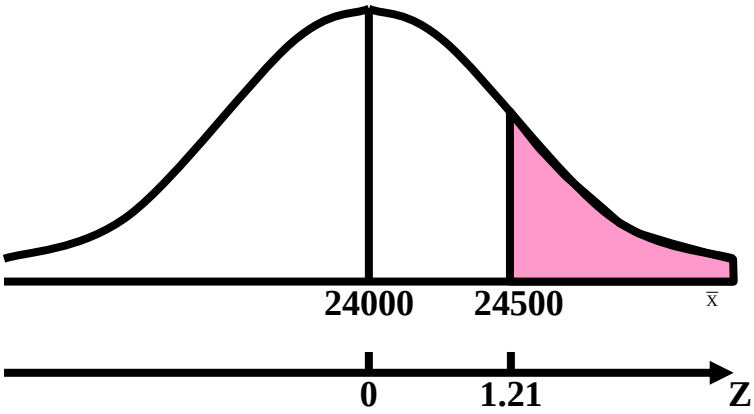
NOTE:

Here we have used finite population correction factor (fpc), because the sample size n = 100 is *greater than 5 percent* of the population size N = 310. Since $\bar{X}$ is approximately N (24000, 412.20), therefore

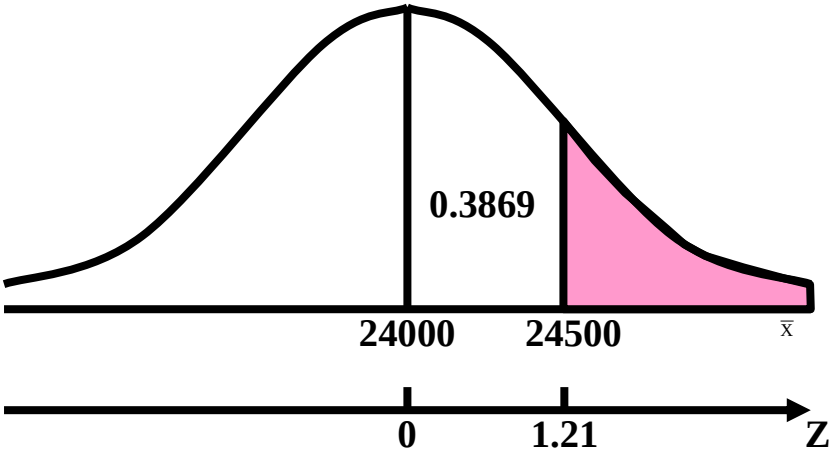
$$Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{X} - 24000}{412.20}$$

is approximately N(0, 1).We are required to evaluate P($\bar{X}$ > 24,500).
At $\bar{x}$ = 24,500, we find that

$$z = \frac{24500 - 24000}{412.20} = 1.21$$



Using the table of areas under the standard normal curve, we find that the area between z = 0 and z = 1.21 is 0.3869.



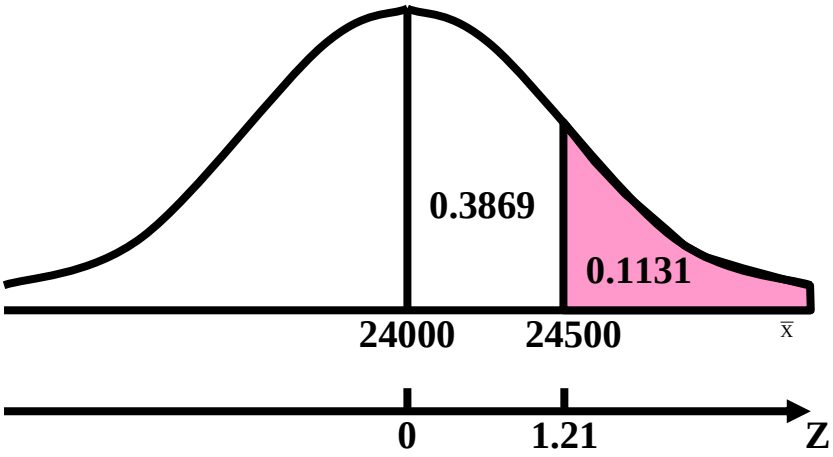
Hence,

P($\bar{X}$ > 24,500)

= P(Z > 1.21)

= 0.5 – P(0 < Z < 1.21)

= 0.5 – 0.3869 = 0.1131.



Hence, the chances are only 11% that in a random sample of 100 employees from this particular construction company, the average salary will exceed Rs.24,500.In other words, the chances are 89% that, in such a sample, the average salary will *not* exceed Rs.24,500.

Hence, the chances are considerably *high* that the government *might* pay attention to the employees' demand.

SAMPLING DISTRIBUTION OF THE SAMPLE PROPORTION

In this regard, the first point to be noted is that, whenever the elements of a population can be classified into *two* categories, technically called "success" and "failure", we may be interested in *the proportion of "successes"* in the population. If X denotes the number of successes in the population, then the proportion of successes in the population is given by $p = \frac{X}{N}$.

Similarly, if we draw a sample of size n from the population, the proportion of successes in the sample is given by

$$\hat{p} = \frac{X}{n},$$

where X represents the number of successes in the sample.
It is interesting to note that X is a *binomial* random variable and the binomial parameter p is being called a proportion of successes here. The sample proportion has different values in different samples. It is obviously a random variable and has a probability distribution.
This probability distribution of the proportions of successes in all possible random samples of size n, is called the sampling distribution of $\hat{p}$.
We illustrate this sampling distribution with the help of the following examples:

EXAMPLE-1

A population consists of six values 1, 3, 6, 8, 9 and 12.Draw all possible samples of size n = 3 *without* replacement from the population and find the proportion of even numbers in each sample. Construct the sampling distribution of sample proportions and verify that

i) $\mu_{\hat{p}} = p$

ii) $\mathrm{Var}(\hat{p}) = \frac{pq}{n} \cdot \frac{N-n}{N-1}.$

SOLUTION

The number of possible samples of size n = 3 that could be selected without replacement from a population of size N is

$$\binom{6}{3} = 20.$$

Let $\hat{p}$ represent the proportion of even numbers in the sample. Then the 20 possible samples and the proportion of even numbers are given as follows:

Sample No.	Sample Data	Sample Proportion $(\hat{p})$
1	1, 3, 6	1/3
2	1, 3, 8	1/3
3	1, 3, 9	0
4	1, 3, 12	1/3
5	1, 6, 8	2/3
6	1, 6, 9	1/3
7	1, 6, 12	2/3
8	1, 8, 9	1/3
9	1, 8, 12	2/3
10	1, 9, 12	1/3
11	3, 6, 8	2/3
12	3, 6, 9	1/3
13	3, 6, 12	2/3
14	3, 8, 9	1/3
15	3, 8, 12	2/3
16	3, 9, 12	1/3
17	6, 8, 9	2/3
18	6, 8, 12	1
19	6, 9, 12	2/3
20	8, 9, 12	2/3

The sampling distribution of sample proportion is given below;

SAMPLING DISTRIBUTION OF $\hat{p}$**:**

$(\hat{p})$	No. of Samples	Probability $f(\hat{p})$	$\hat{p}\, f(\hat{p})$	$\hat{p}^2 f(\hat{p})$
0	1	1/20	0	0
1/3	9	9/20	3/20	1/20
2/3	9	9/20	6/20	4/20
1	1	1/20	1/20	1/20
$\sum$	20	1	10/20	6/20

Now

$$\mu_{\hat{p}} = \sum \hat{p}\, f(\hat{p}) = \frac{10}{20} = 0.5, \text{ and}$$

$$\sigma^2_{\hat{p}} = \sum \hat{p}^2 f(\hat{p}) - [\sum \hat{p}\, f(\hat{p})]^2$$

$$= \frac{2}{60} - \left(\frac{10}{20}\right)^2 = \frac{1}{20} = 0.05.$$

To verify the given relations, we first calculate the population proportion p.Thus:

$p = \frac{X}{N}$;Where X represents the number of even numbers in the population. In other words,

$p = \frac{3}{6} = 0.5$

Hence, we find that

$\mu_{\hat{p}} = 0.5 = p$,

$$\frac{pq}{n} \cdot \frac{N-n}{N-1} = \frac{0.25}{3} \cdot \frac{6-3}{6-1}$$

$$= \frac{0.25}{5} = 0.05 = \mathrm{Var}(\hat{p})$$

Hence, two properties of the sampling distribution of $\hat{p}$ are verified.

$$\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}\frac{N-n}{N-1}},$$

The sampling distribution of $\hat{p}$ has the following important properties.

PROPERTIES OF THE SAMPLING DISTRIBUTION OF $\hat{p}$

Property No. 1

The mean of the sampling distribution of proportions, denoted by $\mu_{\hat{p}}$, is equal to the population proportion p, that is $\mu_{\hat{p}} = p$

Property No. 2

The standard deviation of the sampling distribution of proportions, called the *standard error* of $\hat{p}$ and denoted by $\sigma_{\hat{p}}$

is given as:

- a) $\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}}$,

- when the sampling is performed *with* replacement
- b) when sampling is done *without* replacement from a *finite* population

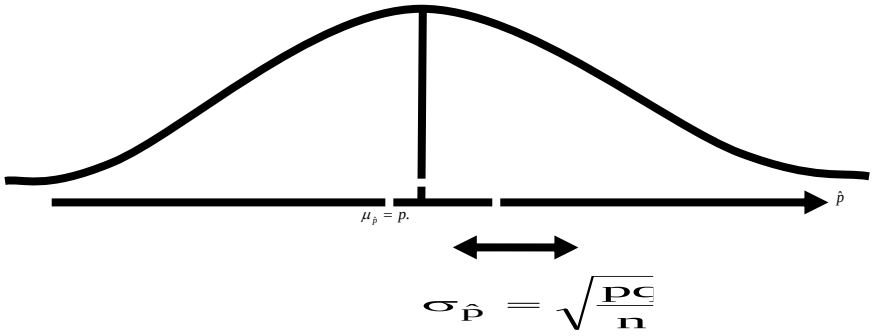
(As in the case of the sampling distribution of $\bar{X}$,is known as the finite population correction factor (fpc)

$$\sqrt{\frac{N-n}{N-1}},$$

Property No. 3

SHAPE OF THE DISTRIBUTION

The sampling distribution of $\hat{p}$ is the *binomial* distribution. However, for sufficiently *large* sample sizes, the sampling distribution of is approximately normal. As $n \to \infty$, the sampling distribution of approaches normality.



As a rule of thumb, the sampling distribution of $\hat{p}$ will be approximately *normal* whenever both np and nq are equal to or greater than 5.Let us apply this concept to a real-world situation:

EXAMPLE-2

Ten percent of the 1-kilogram boxes of sugar in a large warehouse are underweight. Suppose a retailer buys a random sample of 144 of these boxes. What is the probability that at least 5 percent of the sample boxes will be underweight?

SOLUTION

Here the statistic is the sample proportion, the sample size (n = 144) is large enough to assume that the sample proportion is approximately normally distributed with mean

Mean of the sampling distribution of $\hat{p}$

$$\mu_{\hat{p}} = p = 0.10,$$

Standard Error of $\hat{p}$

$$\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.10)(0.90)}{144}}$$

$$= \frac{0.3}{12} = 0.025.$$

Therefore, the sampling distribution of is approximately N(0.10, 0.025); and hence

$$Z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}} = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

$$= \frac{\hat{p} - 0.10}{0.025}$$

is approximately N(0, 1).

We are required to find the probability that the proportion of underweight boxes in the sample is equal to or greater than 5% i.e., we require

$$P(\hat{p} \geq 0.05).$$

In this regard, a very important point to be noted is that, just as we use a *continuity correction* of + ½ whenever we consider the normal approximation to the binomially distributed random variable X, in *this* situation, since

$$\hat{p} = \frac{X}{n}, \qquad \pm\frac{1}{2n}$$

therefore, we need to use the following continuity correction; We need to use a *continuity correction* of $\hat{p}$.
in the case of the sampling distribution of

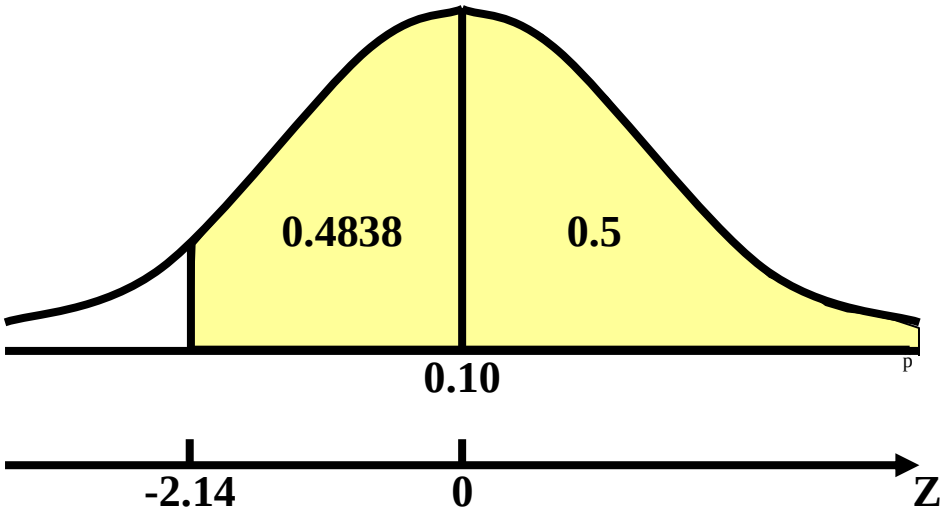
Applying the continuity correction in this problem, we have:

$$P(\hat{p} \geq 0.05) \Rightarrow P\left(\hat{p} \geq 0.05 - \frac{1}{(2)(144)}\right)$$

$$= P\left(\hat{p} \geq 0.05 - \frac{1}{288}\right)$$

$$= P\left(\frac{\hat{p} - 0.10}{0.025} \geq \frac{(0.05 - 1/288) - 0.10}{0.025}\right)$$

$$= P(Z \geq -2.14)$$
$$= P(-2.14 \leq Z \leq 0) + P(0 \leq Z \leq \infty)$$
$$= 0.4838 + 0.5 = 0.9838$$



Hence, the probability that at *least* 5% of the sample boxes are under-weight *is as high as* 98% .
The sampling distributions of $\bar{X}$ and $\hat{p}$ pertain to the situation when we are drawing all possible samples of a

particular size from one particular population. Next, we will discuss the case when we are dealing with all possible samples drawn from *two* populations, such that the samples from the two populations are *independent*.
In this regard, we will consider the sampling distributions of $\bar{X}_1 - \bar{X}_2$ and $\hat{p}_1 - \hat{p}_2$:

We begin with the sampling distribution of $\bar{X}_1 - \bar{X}_2$:

SAMPLING DISTRIBUTION OF DIFFERENCES BETWEEN MEANS

Suppose we have two distinct populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively. Let *independent* random samples of sizes n_1 and n_2 be selected from the respective populations, and the differences $\bar{x}_1 - \bar{x}_2$ between the means of all possible pairs of samples be computed.
Then, a probability distribution of the differences $\bar{X}_1 - \bar{X}_2$ can be obtained. Such a distribution is called the sampling distribution of the differences of sample means $\bar{X}_1 - \bar{X}_2$.
We illustrate the sampling distribution of $\bar{X}_1 - \bar{X}_2$ with the help of the following example.

EXAMPLE

Draw all possible random samples of size n1 = 2 *with replacement* from a finite population consisting of 4, 6, similarly, draw all possible random samples of size n = 2 *with replacement* from another finite population consisting of 1, 2, 3.
a) Find the possible differences between the sample means of the two populations
b) Construct the sampling distribution of $\bar{X}_1 - \bar{X}_2$ and compute its mean and variance
c) Verify that

$$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2 \text{ and } \sigma^2_{\bar{x}_1 - \bar{x}_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_1}.$$

SOLUTION

Whenever we are sampling with replacement from a finite population, the total number of possible samples is Nn (where N is the population size, and n is the sample size).Hence, in this example, there are (3)2 = 9 possible samples which can be drawn with replacement from each population. These two sets of samples and their means are given below:

From Population 1			From Population 2		
Sample No.	Sample Value	$\bar{x}_1$	Sample No.	Sample Value	$\bar{x}_2$
1	4, 4	4	1	1, 1	1.0
2	4, 6	5	2	1, 2	1.5
3	4, 8	6	3	1, 3	2.0
4	6, 4	5	4	2, 1	1.5
5	6, 6	6	5	2, 2	2.0
6	6, 8	7	6	2, 3	2.5
7	8, 4	6	7	3, 1	2.0
8	8, 6	7	8	3, 2	2.5
9	8, 8	8	9	3, 3	3.0

a) Since there are 9 samples from the first population as well as 9 from the second, hence, there are 81 possible combinations of $\bar{x}1$ and $\bar{x}2$
The 81 possible differences $\bar{x}1 - \bar{x}2$ are presented in the following table:

$\bar{x}_2$	$\bar{x}_2$								
	4	5	6	5	6	7	6	7	8
1.0	3.0	4.0	5.0	4.0	5.0	6.0	5.0	6.0	7.0
1.5	2.5	3.5	4.5	3.5	4.5	5.5	4.5	5.5	6.5
2.0	2.0	3.0	4.0	3.0	4.0	5.0	4.0	5.0	6.0
1.5	2.5	3.5	4.5	3.5	4.5	5.5	4.5	5.5	6.5
2.0	2.0	3.0	4.0	3.0	4.0	5.0	4.0	5.0	6.0
2.5	1.5	2.5	3.5	2.5	3.5	4.5	3.5	4.5	5.5
2.0	2.0	3.0	4.0	3.0	4.0	5.0	4.0	5.0	6.0
2.5	1.0	2.5	3.5	2.5	3.5	4.5	3.5	4.5	5.5
3.0	1.0	2.0	3.0	2.0	3.0	4.0	3.0	4.0	5.0

b)The sampling distribution of $\bar{X}_1 - \bar{X}_2$ is as follows:

$\bar{x}_1 - \bar{x}_2 = d$	Tally	f	Probability $f(\bar{x}_1 - \bar{x}_2) = f(d)$	df (d)	d^2 f(d)
1.0	|	1	1/81	1/81	1.0/81
1.5	||	2	2/81	3/81	4.5/81
2.0	𝍸	5	5/81	10/81	20.0/81
2.5	𝍸 |	6	6/81	15/81	37.5/81
3.0	𝍸 𝍸	10	10/81	30/81	90.0/81
3.5	𝍸 𝍸	10	10/81	35/81	122.5/81
4.0	𝍸 𝍸|||	13	13/81	52/81	208.0/81
4.5	𝍸 𝍸	10	10/81	45/81	202.5/81
5.0	𝍸 𝍸	10	10/81	50/81	250.0/81
5.5	𝍸 |	6	6/81	33/81	181.5/81
6.0	𝍸	5	5/81	30/81	180.0/81
6.5	||	2	2/81	13/81	84.5/81
7.0	|	1	1/81	7/81	49.0/81
Total	---	81	1	324/81	1431/81

Thus the mean and the variance are

$$\mu_{\bar{x}_1-\bar{x}_2} = \sum (\bar{x}_1 - \bar{x}_2)\, f(\bar{x}_1 - \bar{x}_2)$$
$$= \sum df(d) = \frac{324}{81} = 4, \quad and$$

$$\sigma^2_{\bar{x}_1-\bar{x}_2} = \sum d^2 f(d) - [\sum df(d)]^2$$
$$= \frac{1431}{81} - \left(\frac{324}{81}\right)^2 = \frac{53}{3} - 16 = \frac{5}{3} = 1.67$$

c) In order to verify the properties of the sampling distribution of $\bar{x}_1 - \bar{x}_2$ we first need to compute the mean and variance of the first population:
The mean and standard deviation of the first population are:

$$\mu_1 = \frac{4+6+8}{3} = 6, \text{ and}$$
$$\sigma_1^2 = \frac{(4-6)^2 + (6-6)^2 + (8-6)^2}{3} = \frac{8}{3}.$$

$$\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}=\frac{8}{3}.\frac{1}{2}+\frac{2}{3}.\frac{1}{2}$$
$$=\frac{4}{3}+\frac{1}{3}=\frac{5}{3}$$
$$=1.67$$
$$=\sigma^2_{\bar{x}_1-\bar{x}_2}$$

The mean and variance of the second population are:

$$\mu_2=\frac{1+2+3}{3}=2\text{, and}$$
$$\sigma_2^2=\frac{(1-2)^2+(2-2)^2+(3-2)^2}{3}=\frac{2}{3}.$$

Now $\mu_{\bar{x}_1-\bar{x}_2}=4=6-2=\mu_1-\mu_2$, *and*

$$\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}=\frac{8}{3}.\frac{1}{2}+\frac{2}{3}.\frac{1}{2}$$
$$=\frac{4}{3}+\frac{1}{3}=\frac{5}{3}$$
$$=1.67$$
$$=\sigma^2_{\bar{x}_1-\bar{x}_2}$$

Hence, two properties of the sampling distribution of $\bar{X}_1-\bar{X}_2$ are satisfied. The sampling distribution of the differences $\bar{X}_1-\bar{X}_2$ has the following properties:

PROPERTIES OF THE SAMPLING DISTRIBUTION OF $\bar{X}_1-\bar{X}_2$

Property No. 1:

The mean of the sampling distribution of $\bar{X}_1-\bar{X}_2$, denoted by $\mu_{\bar{X}_1-\bar{X}_2}$, is equal to the difference between population means, that is

$$\mu_{\bar{X}_1-\bar{X}_2}=\mu_1-\mu_2$$

Property No. 2:

In case of sampling with or without replacement from two *infinite* populations, the standard deviation of the sampling distribution of $\overline{X}_1 - \overline{X}_2$ (i.e. *standard error* of $\overline{X}_1 - \overline{X}_2$), denoted by $\sigma_{\overline{X}_1 - \overline{X}_2}$, is given by

$$\sigma_{\overline{X}_1 - \overline{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

The above expression for the Standard Error of $\overline{X}_1 - \overline{X}_2$ also holds for finite population when sampling is performed *with* replacement. In case of sampling without replacement from a finite population, the formula for the standard error of will be suitably modified.

Property No. 3:

Shape of the distribution:

a) If the POPULATIONS are normally distributed, the sampling distribution of $\overline{X}_1 - \overline{X}_2$, regardless of sample sizes, will be *normal* with mean $\mu_1 - \mu_2$ and variance $\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$.

In other words, the variable

$$Z = \frac{\left(\overline{X}_1 - \overline{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is normally distributed with zero mean and unit variance.

b) If the POPULATIONS are *non-normal* and if *both* sample sizes are *large*, (i.e., greater than or equal to 30), then the sampling distribution of the differences between means is approximately a *normal* distribution by the Central Limit Theorem.

In this case too, the variable

$$Z = \frac{\left(\overline{X}_1 - \overline{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

will be approximately *normally* distributed with mean zero and variance one.

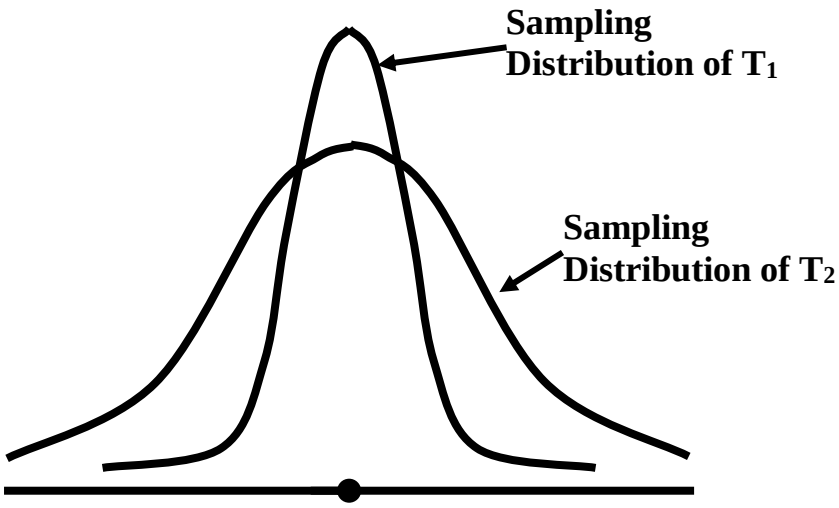
LECTURE NO. 38

- Desirable Qualities of a Good Point Estimator:
  - Efficiency
- Methods of Point Estimation:
  - The Method of Moments
  - The Method of Least Squares
  - The Method of Maximum Likelihood
- Interval Estimation:
  - Confidence Interval for μ

As a sample is only a *part* of the population, it is obvious that the *larger* the sample size, the *more* representative we expect it to be of the population from which it has been drawn. In agreement with the above argument, we will expect our estimator to be close to the corresponding parameter if the sample size is *large*. Hence, we will naturally be happy if the probability of our estimator being close to the parameter *increases* with an increase in the sample size. As *such*, consistency is a desirable property.
Another important desirable quality of a good point estimator is *EFFICIENCY*:

EFFICIENCY

An unbiased estimator is defined to be *efficient* if the variance of its sampling distribution is *smaller than* that of the sampling distribution of *any* other unbiased estimator of the same parameter. In other words, suppose that there are two unbiased estimators T_1 and T_2 of the same parameterθ.Then, the estimator T_1 will be said to be *more efficient* than T_2 if Var (T_1) < Var (T_2).
In the following diagram, since Var (T1) < Var (T2),
hence T1 is *more efficient* than T2 :



The *relative efficiency* of T_1 compared to T_2 (where both T_1 and T_2 are unbiased estimators) is given by the ratio $E_f = \frac{\text{Var}(T_2)}{\text{Var}(T_1)}$.

And, if we multiply the above expression by 100, we obtain the relative efficiency in *percentage* form. It thus provides a criterion for *comparing* different unbiased estimators of a parameter. Both the sample mean and the sample median for a population that has a *normal* distribution, are unbiased and consistent estimators of μ but the variance of the sampling distribution of sample means is *smaller than* the variance of the sampling

distribution of sample medians. Hence, the sample mean is more *efficient* than the sample median as an estimator of μ.The sample mean may therefore be *preferred* as an estimator.
Next, we consider various *methods of point estimation*. A point estimator of a parameter can be obtained by several methods. We shall be presenting a brief account of the following three methods:

METHODS OF POINTESTIMATION

- The Method of Moments
- The Method of Least Squares
- The Method of Maximum Likelihood

These methods give estimates which may *differ* as the methods are based on different *theories* of estimation.

THE METHOD OF MOMENTS

The method of moments which is due to Karl Pearson (1857-1936), consists of calculating a few *moments* of the sample values and *equating* them to the corresponding moments of a population, thus getting *as many* equations as are needed to solve for the unknown parameters. The procedure is described below:
Let X1, X2, …, Xn be a random sample of size n from a population. Then the rth sample moment about zero is

$$m'_r = \frac{\sum X_i^r}{n},\ r = 1,2,...$$

and the corresponding rth population moment is μ'_r .We then *match* these moments and get *as many* equations as we need to solve for the unknown parameters. The following examples illustrate the method:

EXAMPLE-1

Let X be uniformly distributed on the interval (0, θ). Find an estimator of θ by the method of moments.

SOLUTION

The probability density function of the given uniform distribution is

$$f(x) = \frac{1}{\theta},\ \ 0 \le x \le \theta$$

Since the uniform distribution has only one parameter, (i.e. θ), therefore, in order to find the maximum likelihood estimator of θ by the method of moments, we need to consider only *one* equation.
The first sample moment about zero is

$$m'_1 = \frac{\sum X_i}{n}.$$

And, the first population moment about zero is

$$\mu'_1 = \int_0^\theta x.f(x)dx = \int_0^\theta x.\frac{1}{\theta}\,dx = \frac{1}{\theta}\left[\frac{x^2}{2}\right]_0^\theta = \frac{\theta}{2}$$

Matching these moments, we obtain:

$$\frac{\sum X_i}{n} = \frac{\theta}{2} \text{ or } \theta = 2\overline{X}.$$

Hence, the moment estimator of θ is equal to $2\overline{X}$
i.e. $\hat{\theta} = 2\overline{X}.$

In other words, the moment estimator of θ is just twice the sample mean. It should be noted that, for the above uniform distribution, the mean is given by

$$\mu = \frac{\theta}{2}.$$

(This is so due to the *absolute* symmetry of the uniform distribution around the value $\frac{\theta}{2}$.)

Now, $\mu = \frac{\theta}{2}$ implies that $\theta = 2\mu$.

In other words, if we wish to have the *exact* value of θ, all we need to do is to multiply the population mean μ by 2.
Generally, it is not possible to determine μ, and all we can do is to draw a sample from the probability distribution, and compute the *sample* mean $\overline{X}$. Hence, naturally, the equation will be replaced by the equation
(As $2\bar{x}$ provides an *estimate* of θ, hence a 'hat' is placed on top of θ.)
It is interesting to note hat $2\bar{x}$ is *exactly* the same quantity as what we obtained as an estimate of θ by the method of moments!(The result obtained by the method of moments *coincides* with what we obtain through simple logic

EXAMPLE-2

Let X1, X2… Xn be a random sample of size n from a normal population with parameters μ and $\sigma 2$. Find these parameters by the method of moments.

SOLUTION

Here we need two equations as there are *two* unknown parameters, μ and $\sigma 2$. The first two sample moments about zero are

$$m'_1 = \frac{1}{n}\sum X_i = \bar{X} \text{ and } m'_2 = \frac{1}{n}\sum X_i^2.$$

The *corresponding* two moments of a normal distribution are

$'\mu 1 = \mu$ and $'\mu 2 = \sigma 2 + \mu 2$.

($\sigma 2 = '\mu 2 - '\mu 12 = '\mu 2 - \mu 2$)

To get the desired estimators by the method of moments, we *match* them.
Thus, we have :

$$\mu = \frac{1}{n}\sum X_i \text{ and } \sigma^2 + \mu^2 = \frac{1}{n}\sum X_i^2$$

Solving the above equations *simultaneously*, we obtain:

$$\hat{\mu} = \frac{1}{n}\sum X_i = \overline{X}, \text{ and}$$

$$\hat{\sigma}^2 = \frac{\sum X_i^{\,2}}{n} - \overline{X}^2 = \frac{1}{n}\sum\left(X_i - \overline{X}\right)^2 = S^2.$$

as the moment estimators for μ and σ^2. A shortcoming of this method is that the moment estimators are, in general, *inefficient*.

THE METHOD OF LEAST SQUARES

The method of Least Squares, which is due to *Gauss* (1777-1855) and *Markov* (1856-1922), is based on the theory of *linear* estimation. It is regarded as one of the *important* methods of point estimation. An estimator found by *minimizing* the sum of squared deviations of the sample values from some *function* that has been hypothesized as a *fit* for the data, is called the least squares estimator. The method of least-squares has already been discussed in connection with regression analysis that was presented in Lecture No. 15.

You will recall that, when fitting a straight line y = a+bx to real data, 'a' and 'b' were determined by *minimizing* the sum of squared deviations between the fitted line and the data-points.

The y-intercept and the slope of the fitted line i.e. 'a' and 'b' are *least-square estimates* (respectively) of the y-intercept and the slope of the *TRUE* line that would have been obtained by considering the entire population of data-points, and not just a sample.

METHOD OF MAXIMUM LIKELIHOOD

The method of maximum likelihood is regarded as the *MOST important* method of estimation, and is the *most* widely used method. This method was introduced in 1922 by Sir Ronald A. Fisher (1890-1962).The mathematical technique of finding Maximum Likelihood Estimators is a bit *advanced*, and involves the concept of the Likelihood Function.

RATIONALE OF THE METHOD OF MAXIMUM LIKELIHOOD (ML)

"To consider *every* possible value that the parameter might have, and for *each* value, compute the *probability* that the given sample would have occurred if that *were* the true value of the parameter. That value of the parameter for which the probability of a given sample is *greatest*, is chosen as an estimate." An estimate obtained by this method is called the maximum likelihood estimate (MLE). It should be noted that the method of maximum likelihood is applicable to *both* discrete and continuous random variables.

EXAMPLES OF MLE's IN CASE OF DISCRETE DISTRIBUTIONS

Example-1:

For the Poisson distribution given by

$$P(X = x) = \frac{e^{-\mu}\mu^x}{x!},\ x = 0,1,2,.....,$$

the MLE of μ is $\overline{X}$ (the sample mean).

EXAMPLE-2

For the geometric distribution given by the MLE of p is Hence, the MLE of p is equal to the reciprocal of the mean.

EXAMPLE-3

For the Bernoulli distribution given by

$$P(X = x) = p^x q^{1-x},\ x = 0,1\ ,$$

the MLE of p is (the sample mean).

EXAMPLES OF MLE's IN CASE OF *CONTINUOUS* DISTRIBUTIONS

Example-1

For the exponential distribution given by

$$f(x) = \theta e^{-\theta x},\ x > 0,\ \theta > 0\ ,$$

the MLE of θ is (the reciprocal of the sample mean $\frac{1}{\bar{X}}$)

EXAMPLE-2

For the *normal* distribution with parameters μ and $\sigma 2$, the *joint* ML estimators of μ and $\sigma 2$ is the sample mean
and the sample variance S2*(which is not an unbiased estimator of $\sigma 2$).*As indicated many times earlier, the *normal distribution* is encountered frequently in practice, and, in this regard, it is both interesting and important to note that, in the case of this *frequently* encountered distribution, the *simplest* formulae (i.e. the sample *mean* and the sample *variance*) fulfill the criteria of the relatively *advanced* method of maximum likelihood estimation !The last example among the five presented above (the one on the *normal* distribution) points to *another* important fact --- and that is :
The Maximum Likelihood Estimators are consistent and efficient but *not necessarily unbiased*. (As we know, S2 is *not* an unbiased estimator of $\sigma 2$.)

EXAMPLE

It is well-known that human weight is an approximately normally distributed variable. Suppose that we are interested in estimating the mean and the variance of the weights of adult males in one particular province of a country. A random sample of 15 adult males from this particular population yields the following weights (in pounds):

131.5	136.9	133.8	130.1	133.9
135.2	129.6	134.4	130.5	134.2
131.6	136.7	135.8	134.5	132.7

Find the maximum likelihood estimates for $\theta 1 = \mu$ and $\theta 2 = \sigma 2$.

SOLUTION

The above data is that of a random sample of size 15 from N(μ, $\sigma 2$). It has been mathematically proved that the joint maximum likelihood estimators of μ and $\sigma 2$ are $\bar{X}$ and S2. We compute these quantities for this particular sample, and obtain $\bar{X}$ = 133.43, and S2 = 5.10 .These are the Maximum Likelihood Estimates of the mean and variance

of the population of weights in this particular example. Having discussed the concept of point estimation in some detail, we now begin the discussion of the concept of *interval estimation*:

As stated earlier, whenever a *single* quantity computed from the sample acts as an estimate of a population parameter, we call that quantity a *point* estimate e.g. the sample mean is a point estimate of the population mean μ.

The *limitation* of point estimation is that we have no way of ascertaining how close our point estimate is to the true value (the parameter).

For example, we know that is an unbiased estimator of μ i.e. if we had taken all possible samples of a particular size from the population and calculated the mean of each sample, then the mean of the sample means would have been equal to the population mean (μ), but in an actual survey we will be selecting only one sample from the population and will calculate its mean .

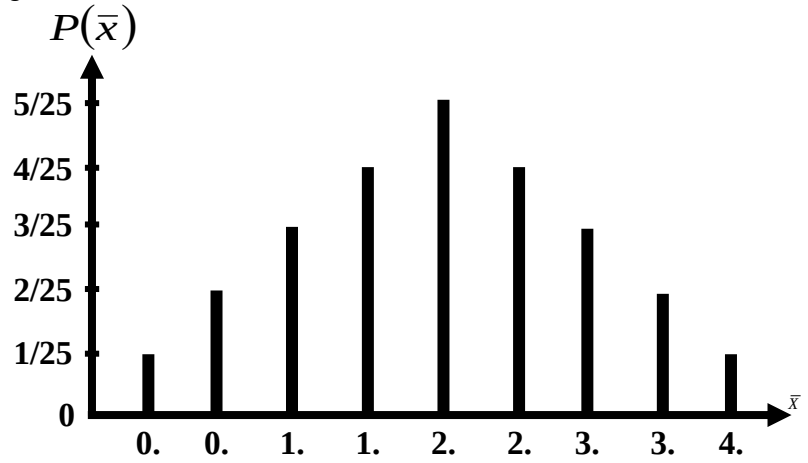
We will have no way of ascertaining how close this particular is toμ. Whereas a point estimate is a single value that acts as an estimate of the population parameter, *interval estimation* is a procedure of estimating the unknown parameter which specifies a *range* of values within which the parameter is expected to lie. A *confidence interval* is an interval computed from the sample observations x1, x2….xn, with a statement of how *confident* we are that the interval *does* contain the population parameter.

We develop the concept of interval estimation with the help of the example of the Ministry of Transport test to which all cars, irrespective of age, have to be submitted.

EXAMPLE

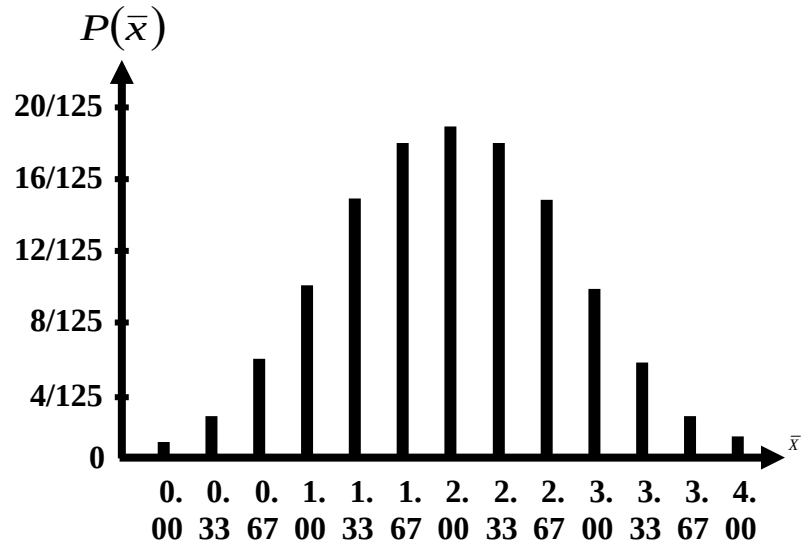
Let us examine the case of an annual Ministry of Transport test to which all cars, irrespective of age, have to be submitted. The test looks for faulty breaks, steering, lights and suspension, and it is discovered after the first year that approximately the *same number* of cars has 0, 1, 2, 3, or 4 faults. You will recall that when we drew all possible samples of size 2 from this uniformly distributed population, the sampling distribution of $\bar{X}$ was *triangular*:

Sampling Distribution of $\bar{X}$ for n = 2

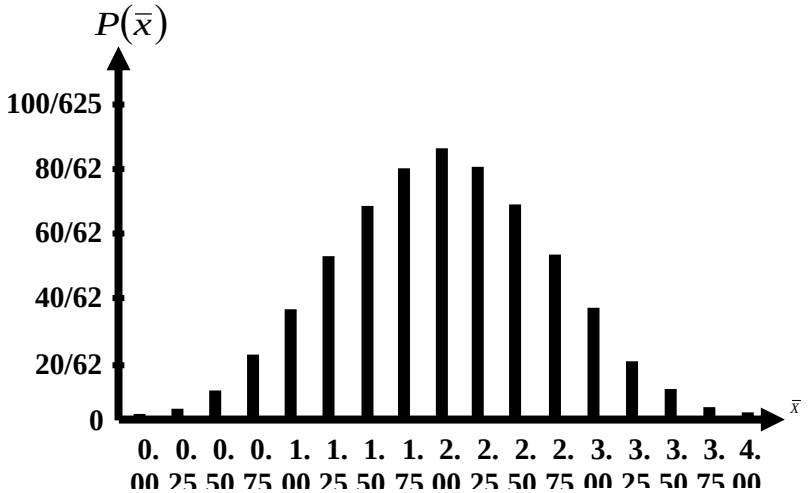


But when we considered what happened to the shape of the sampling distribution with if the sample size is *increased*, we found that it was somewhat like a normal distribution:

Sampling Distribution of $\bar{X}$ for n = 3



And, when we increased the sample size to 4, the sampling distribution resembled a normal distribution even *more* closely, Sampling Distribution of $\bar{X}$ for n = 4



It is clear from the above discussion that as *larger* samples are taken, the shape of the sampling distribution of $\bar{X}$ undergoes *discernible changes*.

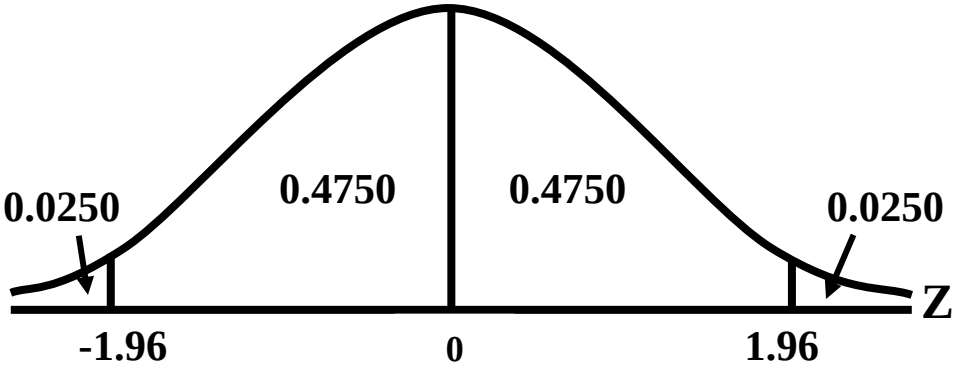
In all three cases the line charts are symmetrical, but as the sample size increases, the overall configuration changed from a triangular distribution to a *bell-shaped* distribution. In other words, for large samples, we are dealing with a *normal* sampling distribution of $\bar{x}$.In other words: When sampling from an infinite population such that the sample size n is large, $\bar{X}$ is *normally distributed* with mean μ and variance $\frac{\sigma^2}{n}$

i.e. $\bar{X}$ is $N\left(\mu, \frac{\sigma^2}{n}\right)$.

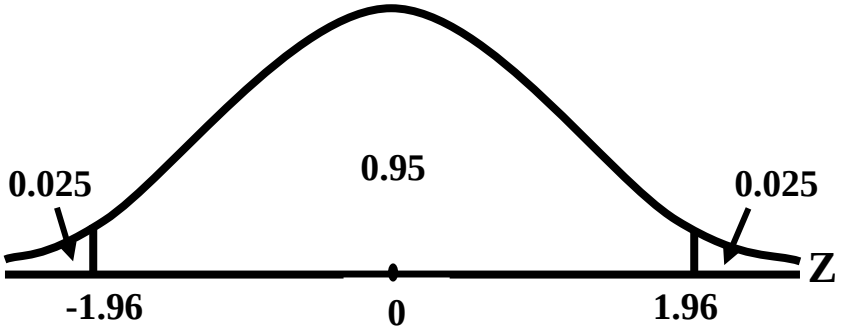
Hence, the standardized version of $\bar{X}$ i.e.

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

is normally distributed with mean 0 and variance 1 i.e. Z is N(0, 1). Now, for the standard normal distribution, we have: For the standard normal distribution, we have:



The above is equivalent to P(-1.96 < Z < 1.96)
= 0.4750 + 0.4750 = 0.95



In other words:

$$P\left(-1.96 \le \frac{\overline{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \le 1.96\right) = 0.95$$

The above can be re-written as:

$$P\left(-1.96\frac{\sigma}{\sqrt{n}} \le \overline{X}-\mu \le 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Or

$$P\left(-\overline{X}-1.96\frac{\sigma}{\sqrt{n}} \le -\mu \le -\overline{X}+1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

or

$$P\left(\overline{X}+1.96\frac{\sigma}{\sqrt{n}} \ge \mu \ge \overline{X}-1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

or

$$P\left(\overline{X}-1.96\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X}+1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

The above equation yields the 95% confidence interval for μ :

The 95% confidence interval for μ is

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} , \bar{X} + 1.96\frac{\sigma}{\sqrt{n}} \right).$$

In other words, the 95% C.I. for μ is given by

$$\overline{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

In a *real-life* situation, the population standard deviation is usually *not known* and hence it has to be *estimated*.
It can be mathematically proved that the quantity

$$s^2 = \frac{\sum (X - \overline{X})^2}{n-1}$$

is an unbiased estimator of $\sigma 2$ (the population variance). (just as the sample mean is an unbiased estimator of μ).
In this situation, the 95% Confidence Interval for μ is given by:

$$P\left(\overline{X} - 1.96\frac{s}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{s}{\sqrt{n}} \right) = 95\%$$

The points

$$\overline{X} - 1.96\frac{s}{\sqrt{n}} \text{ and } \overline{X} + 1.96\frac{s}{\sqrt{n}}$$

are called the lower and upper *limits* of the 95% confidence interval.

LECTURE NO. 39

- Hypothesis Testing Regarding p1-p2 (based on Z-statistic)
- The Student's t-distribution
- Confidence Interval for μ based on the t-distribution

In the last lecture, we discussed hypothesis-testing regarding p, the proportion of successes in a binomial population. Next, we consider the case when we are interested in testing the equality of two population proportions. We illustrate this situation with the help of the following example:

EXAMPLE

A leading perfume company in a western country recently developed a new perfume which they plan to market under the name 'Fragrance'. A number of comparison tests indicate that 'Fragrance' has very good market potential. The Sales Departments of the company want to plan their strategy so as to reach and impress the largest possible segments of the buying public. One of the questions is whether the perfume is preferred by younger or older women. These are two independent populations, a population consisting of the younger women and a population consisting of the older women. A standard scent test will be used where each sampled woman is asked to sniff several perfumes, one of which is 'Fragrance', and indicate the one that she likes best.
A total of 100 young women were selected at random, and each was given the standard scent test. Twenty of the 100 young women chose 'Fragrance' as the perfume they liked best. Two hundred older women were selected at random, and each was given the same standard scent test. Of the 200 older women, 100 preferred 'Fragrance
Test the hypothesis that there is no difference between the proportions of younger and older women who prefer 'Fragrance'.

SOLUTION

We designate p1 as the proportion of younger women who prefer 'Fragrance' and p2 as the proportion of older women who prefer 'Fragrance'.
Hypothesis-Testing Procedure:

Step-1:

H0 : p1 = p2 (i.e. p1-p2 = 0)
(There is no difference between the proportions of young women and older women who prefer 'Fragrance'.)
H1 : p1 $\neq$ p2 (i.e. p1-p2 $\neq$ 0)
(The two proportions are not equal.)

Step-2:

Level of Significance

α = 0.05.

Step-3:

Test Statistic

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c \hat{q}_c \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where the combined or pooled proportion, $\hat{p}_c$, is given by:

$$\hat{p}_c = \frac{\text{Total number of successes in the two samples combined}}{\text{Total number of observations in the two samples combined}}$$

$$= \frac{X_1 + X_2}{n_1 + n_2}$$

This can also be written as

$$\hat{p}_c = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

which means that $\hat{p}_c$ is the weighted mean of $\hat{p}_1$ and $\hat{p}_2$, n1 and n2 acting as the weights.

Important Note:

In this example, as the hypothesized value of p1 - p2 is equal to zero; therefore both $\hat{p}_1$ *and* $\hat{p}_2$ are estimating the common population proportion p. Hence, we use the pooled proportion of the two samples to estimate p.

(The rationale is that the pooled estimator $\hat{p}_c$ is a better estimator of the common Population proportion p (as compared with $\hat{p}_1$ or $\hat{p}_2$), as it is based on n1 + n2 observations (i.e. based on a greater amount of information).

Step-4:

Calculations:

X1is the number of Preferring 'Fragrance' = 20. n1 is the number is the sample = 100.

$$\hat{p}_1 = \frac{X_1}{n_1} = \frac{20}{100} = 0.20$$

X2 is the number of preferring 'Fragrance' = 100. n2 is the number is the sample = 200.

$$\hat{p}_2 = \frac{X_2}{n_2} = \frac{100}{200} = 0.50$$

Now, the pooled or weighted proportion $\hat{p}_c$, is computed as follows:

$$\hat{p}_c = \frac{X_1 + X_2}{n_1 + n_2} = \frac{20 + 100}{100 + 200} = \frac{120}{300} = 0.40$$

Computation:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c \hat{q}_c \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

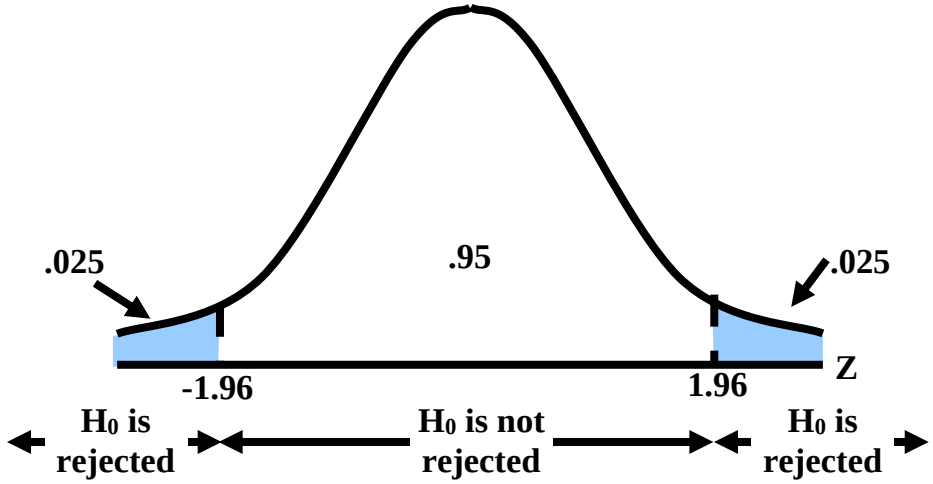
$$= \frac{0.20 - 0.50}{\sqrt{(0.40)(0.60)\left(\frac{1}{100} + \frac{1}{200}\right)}}$$

$$= \frac{-0.30}{0.06} = -5.00$$

Step-5:

Critical Region

Since H1 does not state any direction (such as p1 < p2), the test is two-tailed. Thus, the critical values for the .05 level are –1.96 and + 1.96. Two-Tailed Test, Areas of Rejection and Non-rejection, .05 Level of Significance:



Step-6:

CONCLUSION

The computed z of –5.00 is in the area of rejection, that is, to the left of –1.96. Therefore, the null hypothesis is rejected at the .05 level of significance.

In other words, we conclude that the proportion of young women in the population who prefer 'Fragrance' is not equal to the proportion of older women in the population who prefer 'Fragrance'.(The difference between the two sample proportion i.e. 0.30 is so large that it is highly unlikely that such a large difference could be due to chance (i.e. attributable to sampling fluctuations).)

In fact, the value z = -5.00 is even larger than -2.58, the critical value lying on the left tail of the sampling distribution if $\alpha = 0.01$. As such, we can say that our statistic is highly significant.(In such a situation, the statistic is said to be highly significant because of the fact that we are allowing as small a risk of committing Type-I error as 1%.)

Now, consider another situation:

Suppose that the computed value of our test-statistic comes out to be such that it falls between -1.96 and -2.58. In such a situation, we will reject H0 at the 5% level of significance, but we cannot reject H0 at the 1% level. This means that, if we are willing to allow as much as 5% risk of committing type I error, then we say that we are going to reject H0. But if we are willing to allow only 1% risk of committing type I error, then we conclude that the sample does not provide sufficient evidence to reject H0.Going back to the example of the perfume, obviously, the company would be interested in determining, which category of women prefers this perfume in greater numbers than the other?

The data clearly indicates that the proportion of women who prefer this particular perfume is higher in the population of older women. (This is the reason why the computed value of our test-statistic has come out to be negative.) Let us consolidate the above ideas by considering another example:

EXAMPLE

A candidate for mayor in a large city believes that he appeals to at least 10 per cent more of the educated voters than the uneducated voters. He hires the services of a poll-

taking organization, and they find that 62 of 100 educated voters interviewed support the candidate, and 69 of 150 uneducated voters support him at the 0.05 significance level.

Step-1:

The null and alternative hypothesis is
H0 : p1 – p2 > 0.10, and
H1 : p1 – p2 < 0.10, where p1 = proportion of educated voters, and p2 = proportion of uneducated voters.

Step-2:

Level of Significance:
$\alpha = 0.05$.

Step-3:

Test Statistic: $$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0.10}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$$

which for large sample sizes, is approximately standard normal.

Important Note
In this example, as the hypothesized value of p1 - p2 is not equal to zero, therefore are note estimating the same quantity, and, as such, we do not use in the formula of the test statistic.

Step-4:

Computation:

$$\text{Here} \quad \hat{p}_1 = \frac{62}{100} = 0.62, \text{ so that } \hat{q}_1 = 0.38,$$

$$\hat{p}_2 = \frac{69}{150} = 0.46, \text{ so that } \hat{q}_2 = 0.54.$$

$$\text{Thus} \quad z = \frac{(0.62 - 0.46) - 0.10}{\sqrt{\frac{(0.62)(0.38)}{100} + \frac{(0.46)(0.54)}{150}}}$$

$$= \frac{0.06}{\sqrt{0.002356 + 0.001656}} = \frac{0.06}{0.063} = 0.95.$$

Step-5:
Critical Region:
As this is a one-tailed test, therefore the critical region is given by
$Z < -z_{0.05} = -1.645$

Step-6:
Conclusion:
Since the calculated value z = 0.95 does not fall in the critical region, so we accept the null hypothesis

H0 : p1 – p2 > 0.10.The data seems to support the candidate's view.
Until now, we have discussed in considerable detail interval estimation and hypothesis-testing based on the standard normal distribution and the Z-statistic.
Next, we begin the discussion of interval estimation hypothesis-testing based on the t-distribution.

t-DISTRIBUTION

We begin by presenting the formal definition of the t-distribution and stating some of its main properties:

The Student's t-Distribution:

The mathematical equation of the t-distribution is as follows:

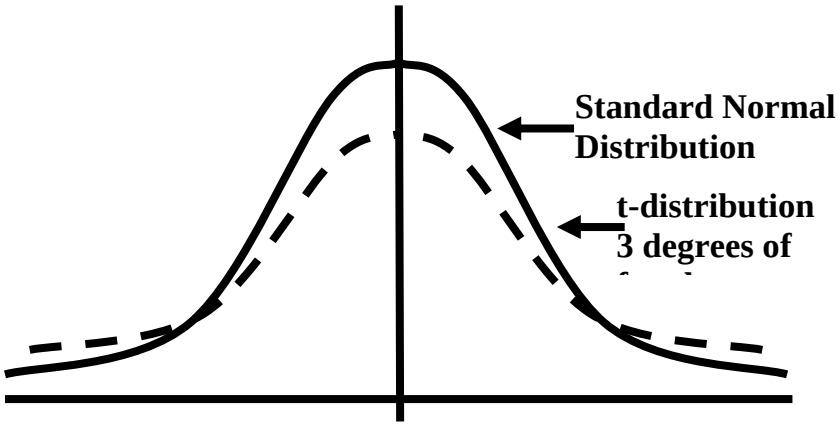
$$f(x)=\frac{1}{\sqrt{\nu}\,\beta\left(\frac{1}{2},\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2} \quad , -\infty < x < \infty$$

This distribution has only one parameter ν, which is known as the degrees of freedom of the t-distribution

PROPERTIES OF STUDENT'S t-DISTRIBUTION

The t-distribution has the following properties:

i) The t-distribution is bell-shaped and symmetric about the value t = 0, ranging from $-\infty$ to ∞.
ii) The number of degrees of freedom determines the shape of the t-distribution.
Thus there is a different t-distribution for each number of degrees of freedom.
As such, it is a whole family of distributions.
The t-distribution, for small values of ν, is flatter than the standard normal distribution which means that the t-distribution is more spread out in the tails than is the standard normal distribution.



As the degrees of freedom increase, the t-distribution becomes narrower and narrower, until, as n tends to infinity, it tends to coincide with the standard normal distribution.
(The t-distribution can never become narrower than the standard normal distribution.)
iii) The t-distribution has a mean of zero, when $\nu \geq 2$. (The mean does not exist when ν = 1.)

iv) The median of the t-distribution is also equal to zero.
v) The t-distribution is unimodal.The density of the distribution reaches its maximum at t = 0 and thus the mode of the t-distribution is t = 0.
(The students will recall that, for any hump-shaped symmetric distribution, the mean, median and mode are equal.)

vi) The variance of the t-distribution is given by $\sigma^2 = \frac{\nu}{\nu - 2}$ for $\nu > 2$.

It is always greater than1, the variance of the standard normal distribution. (This indicates that the t-distribution is more spread out than the standard normal distribution.) For $\nu \leq 2$, the variance does not exist. Next, we discuss the application of the t-distribution in statistical inference --- those situations where we need to carry out interval estimation and hypothesis - testing on the basis of the t-distribution.
(Situations where the t-distribution is the appropriate sampling distribution)
With reference to interval estimation and hypothesis-testing about μ, it has been mathematically proved that, if the population from which the sample has been drawn is normally distributed, the population variance is unknown, and the sample size is small (less than 30), then the statistic

$$t = \frac{\overline{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$(\text{where } s = \sqrt{\frac{\sum (X - \overline{X})^2}{n-1}})$$

Follows the t-distribution having n-1 degrees of freedom. First, we discuss the construction of a Confidence Interval for μ based on the t-distribution with the help of an example:

EXAMPLE

The masses, in grams, of thirteen ball bearings seen at random from a batch are
21.4, 23.1, 25.9, 24.7, 23.4, 24.5, 25.0, 22.5, 26.9, 26.4, 25.8, 23.2, 21.9
Calculate a 95% confidence interval for the mean mass of the population, supposed normal, from which these masses were drawn.

SOLUTION

The 95% confidence interval for the mean mass of the population μ, is given by

$$\overline{X} + t_{\alpha/2(n-1)} \frac{s}{\sqrt{n}}$$

(The derivation of the above confidence interval is very similar to that of the confidence interval for μ based on the Z-statistic.)
Now, in this problem, the sample mean $\overline{X}$ and s come out to be: $t_{\alpha/2(n-1)}$?

$$\overline{X} = \frac{\sum X}{n} = \frac{314.7}{13} = 24.21,$$

$$s = \sqrt{\frac{\sum (X - \overline{X})^2}{n-1}} = \sqrt{\frac{1}{n-1}\left[\sum X^2 - \frac{(\sum X)^2}{n}\right]}$$

$$= \sqrt{\frac{1}{12}[7655.59 - 7618.16]} = \sqrt{\frac{37.43}{12}} = \sqrt{3.12} = 1.77$$

The question is: 'How do we find

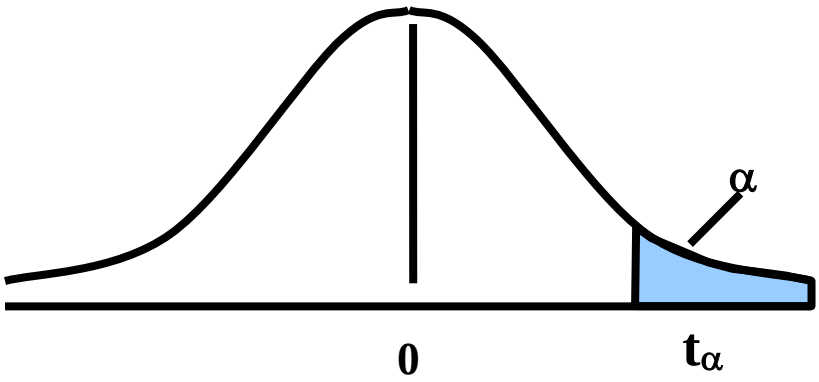
For this purpose, we will need to consult the table of areas under the t-distribution:

TABLE OF AREAS UNDER THE T-DISTRIBUTION

Upper Percentage Points of the t-Distribution							
ν \ α	0.25	0.10	0.05	0.025	0.01	0.005	0.001
1	1.000	3.078	6.314	12.706	31.821	63.657	318.310
2	0.816	1.886	2.920	4.303	6.965	9.925	22.327
3	0.765	1.838	2.353	3.182	4.541	5.841	10.214
4	0.741	1.533	2.132	2.776	3.747	4.604	7.173
5	0.727	1.476	2.015	2.571	3.365	4.032	5.893
6	0.718	1.440	1.943	2.447	3.143	3.707	5.208
7	0.711	1.415	1.895	2.365	2.998	3.499	4.785
8	0.706	1.397	1.860	2.306	2.896	3.355	4.501
9	0.703	1.383	1.833	2.262	2.821	3.250	4.297
10	0.700	1.372	1.812	2.228	2.764	3.169	4.144
11	0.697	1.363	1.796	2.201	2.718	3.106	4.025
12	0.695	1.356	1.782	2.179	2.681	3.055	3.930
13	0.694	1.350	1.771	2.160	2.650	3.012	3.852
14	0.692	1.345	1.761	2.145	2.624	2.977	3.787
15	0.691	1.341	1.753	2.131	2.602	2.947	3.733

Upper Percentage Points of the t-Distribution							
ν \ α	0.25	0.10	0.05	0.025	0.01	0.005	0.001
16	0.690	1.337	1.746	2.120	2.583	2.921	3.686
17	0.689	1.333	1.740	2.110	2.567	2.898	3.646
18	0.688	1.330	1.734	2.101	2.552	2.878	3.610
19	0.688	1.328	1.729	2.093	2.539	2.861	3.579
20	0.687	1.325	1.725	2.086	2.528	2.845	3.552
21	0.686	1.323	1.721	2.080	2.518	2.831	3.527
22	0.686	1.321	1.717	2.074	2.508	2.819	3.505
23	0.685	1.319	1.714	2.069	2.500	2.807	3.485
24	0.685	1.318	1.711	2.064	2.492	2.797	3.467
25	0.684	1.316	1.708	2.060	2.485	2.787	3.450
26	0.684	1.315	1.706	2.056	2.479	2.779	3.435
27	0.684	1.314	1.703	2.052	2.473	2.771	3.421
28	0.683	1.313	1.701	2.048	2.467	2.763	3.408
29	0.683	1.311	1.699	2.045	2.462	2.756	3.396
30	0.683	1.310	1.697	2.042	2.457	2.750	3.385
40	0.681	1.303	1.684	2.021	2.423	2.704	3.307
60	0.679	1.296	1.671	2.000	2.390	2.660	3.232
120	0.677	1.289	1.658	1.980	2.358	2.617	3.160
∞	0.674	1.282	1.645	1.960	2.326	2.576	3.090

The above table is an abridged version of the table by Fisher and Yates, and the entries in this table are values of $t_{\alpha,(\nu)}$ for which the area to their right under the t-distribution with ν degrees of freedom is equal to α, as shown below:



Now, in this problem, since n – 1 = 12, and the desired level of confidence is 95%, therefore, the right-tail area is 2½%, and, hence, (using the t-table) we obtain

$t_{0.025\ (12)} = 2.179$

Substituting these values, we obtain the 95% confidence interval for μ as follows:

$$24.21 \pm 2.179\left(\frac{1.77}{\sqrt{13}}\right)$$

or $24.21 \pm 2.179\ (0.49)$

or 24.21 ± 1.07 or 23.14 to 25.28

Hence, the 95% confidence interval for the mean mass of the ball bearings calculated from the given sample is (23.1, 25.3) grams.

LECTURE NO. 40

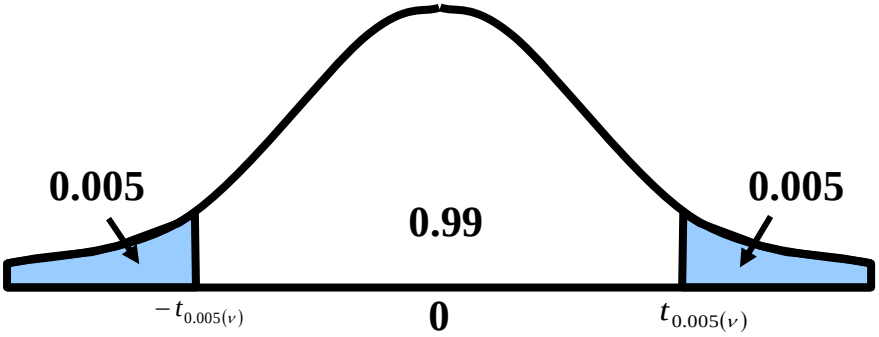
- Tests and Confidence Intervals based on the t-distribution

In the last lecture, we introduced the t-distribution, and began the discussion of statistical inference based on the t-distribution. In particular, we discussed the construction of the confidence interval for μ in that situation when we are drawing a small sample from a normal population having unknown variance $\sigma 2$. When the parent population is normal, the population variance is unknown, and the sample size n is small (less than 30), then the confidence interval for μ is given by

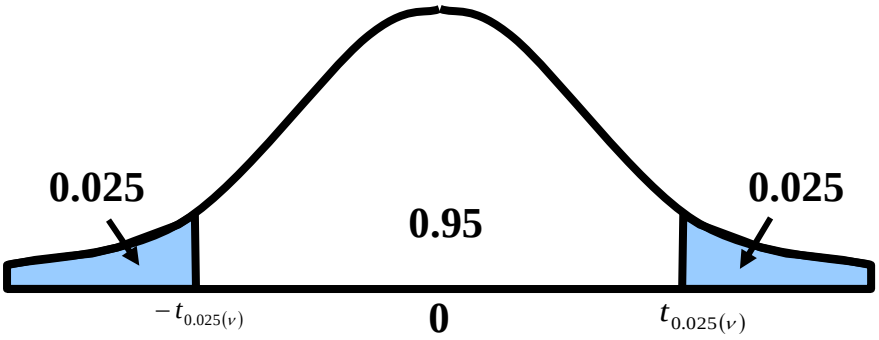
$$\bar{x} \pm t_{\alpha/2(n-1)} \frac{s}{\sqrt{n}}$$

where $\bar{x} = \frac{\sum x}{n}$ is the sample mean $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$ is the sample standard deviation n = sample size and $t_{(\alpha/2, \nu)}$ is found by looking in the t-table under the appropriate value of α against $\nu = n - 1$;

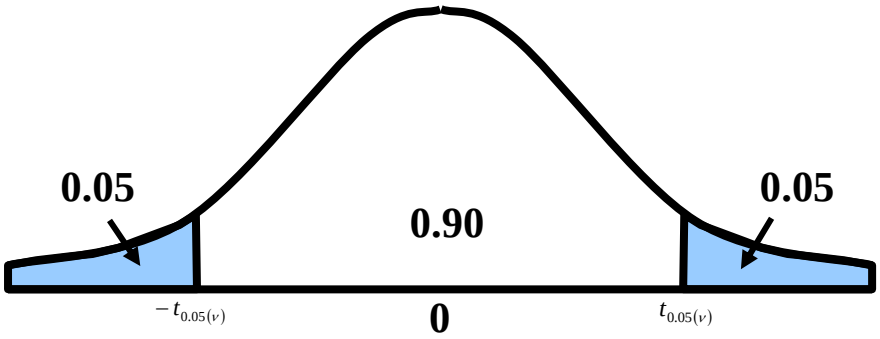
$\alpha/2$ = 0.005 if we desire 99% confidence:



$\alpha/2$ = 0.025 if we desire 95% confidence:



$\alpha/2$ = 0.05 if we desire 90% confidence:



Next, we discuss hypothesis - testing regarding the mean of a normally distributed population for which $\sigma 2$ is unknown and the sample size is small (n < 30).
This procedure is illustrated through the following example:

EXAMPLE-1

Just as human height is approximately normally distributed, we can expect the heights of animals of any particular species to be normally distributed. Suppose that, for the past five years, a zoologist has been involved in an extensive research-project regarding the animals of one particular species. Based on his research-experience, the zoologist believes that the average height of the animals of this particular species is 66 centimeters. He selects a random sample of ten animals of this particular species, and, upon measuring their heights, the following data is obtained.
63, 63, 66, 67, 68, 69, 70, 70, 71, 71
In the light of these data, test the hypothesis that the mean height of the animals of this particular species is 66 centimeters.

SOLUTION:

Hypothesis-Testing Procedure:

i) We state our null and alternative hypotheses as
H0 : $\mu = 66$ and H1 : $\mu \neq 66$.

ii) We set the significance level at $\alpha = 0.05$.
iii) Test Statistic:
The test-statistic to be used is

$$t = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$

which, if H0 is true, has the t-distribution with n – 1 = 9 degrees of freedom.

Important Note:
As indicated in the previous discussion, we always begin by assuming that H0 is true.(The entire mathematical logic of the hypothesis-testing procedure is based on the assumption that H0 is true.)
iv) CALCULATIONS

Individual No.	x_i	x_i^2
1	63	3969
2	63	3969
3	66	4356
4	67	4489
5	68	4624
6	69	4761
7	70	4900
8	70	4900
9	71	5041
10	71	5041
Total	678	46050

Now $\bar{x} = \frac{\sum x_i}{n} = \frac{678}{10} = 67.8$ inches,

and $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2 = \frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n}\right]$

$= \frac{1}{9}[46050 - 45968.4] = 9.0667,$

$s = \sqrt{9.0667} = 3.01$ inches.

$\therefore \quad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

$= \frac{67.8 - 66}{3.01/\sqrt{10}}$

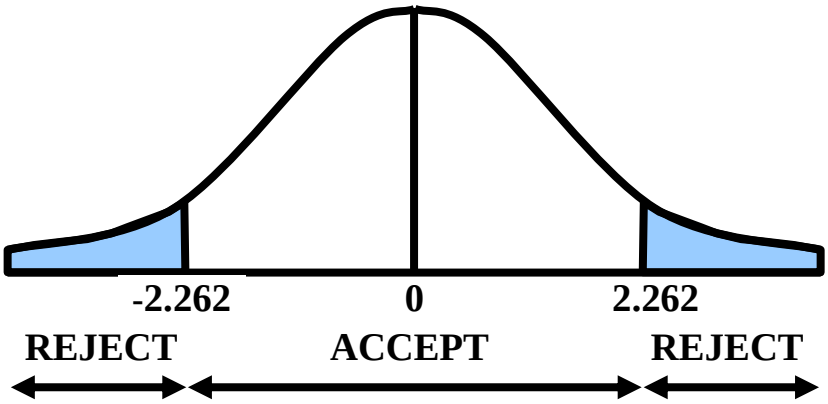
$= \frac{(1.8)(3.1623)}{3.01}$

$= 1.89$

V) Critical Region:

Since this is a two-tailed test, hence the critical region is given by

| t | > t0.025(9) = 2.262.



vi) Conclusion:
Since the computed value of t = 1.89 does not fall in the critical region, we therefore do not reject H0 and may conclude that the mean height of the animals of this particular species is 66 centimeters.

Next, we consider the construction of the confidence interval for $\mu 1-\mu 2$ in that situation when we are drawing small samples from two normally distributed populations having unknown but equal variances:
We illustrate this concept with the help of the following example:

EXAMPLE:
A record company executive is interested in estimating the difference in the average play-length of songs pertaining to pop music and semi-classical music. To do so, she randomly selects 10 semi-classical songs and 9 pop songs.

THE PLAY-LENGTHS (IN MINUTES) OF THE SELECTED SONGS ARE LISTED IN THE FOLLOWING TABLE

Semi-Classic Music	Pop Music
3.80	3.88
3.30	4.13
3.43	4.11
3.30	3.98
3.03	3.98
4.18	3.93
3.18	3.92
3.83	3.98
3.22	4.67
3.38	

Calculate a 99% confidence interval to estimate the difference in population means for these two types of recordings.

SOLUTION:

In this problem, we are dealing with a t-distribution with n1+n2 - 2 = 10 + 9 – 2 = 17 degrees of freedom. The table t-value for a 99% level of confidence and 17 degrees of freedom is t005.17 = 2.898.

Calculations:

Semi-Classical Music	Pop Music
$n_1 = 10$	$n_2 = 9$
$\overline{X}_1 = 3.465$	$\overline{X}_2 = 4.064$
$S_1 = .0.3575$	$S_2 = .0.2417$

Hence:

$$s_p = \sqrt{\frac{(.3575)^2(9)+(.2417)^2(8)}{10+9-2}}$$
$$= \sqrt{\frac{1.1503+0.4674}{17}}$$
$$= \sqrt{\frac{1.6177}{17}} \quad = \sqrt{0.0952}$$
$$= 0.31$$

The confidence interval is

$$(3.465-4.064)$$
$$\pm(2.898)(0.31)\sqrt{\frac{1}{10}+\frac{1}{9}} = -0.599 \pm 0.411$$

i.e. the C.I. is :

$$-1.010 \le \mu_1 - \mu_2 \le -.188$$

With 99% confidence, the record company executive can conclude that the true difference in population average length of play is between –1.01 minutes and –.188 minute. Zero is not in this interval, so she could conclude that there is a significant difference in the average length of play time between semi-classical music and pop music songs' recordings. Examination of the sample results indicates that pop music songs' recordings are longer. The result and conclusion obtained above can be used in the tactical and strategic planning for programming, marketing, and production of recordings.

EXAMPLE

From an area planted in one variety of guayule (a rubber producing plant), 54 plants were selected at random. Of these, 15 were off types and 12 were aberrant. Rubber percentages for these plants were:

Offtypes	6.21, 5.70, 6.04, 4.47, 5.22, 4.45, 4.84, 5.88, 5.82, 6.09, 6.06, 5.59, 6.74, 5.55
Aberrant	4.28, 7.71, 6.48, 7.71, 7.37, 7.20, 7.06, 6.40, 8.93, 5.91, 5.51, 6.36

Test the hypothesis that the mean rubber percentage of the Aberrants is at least 1 percent more than the mean rubber percentage of off types. Assume the populations of rubber percentages are approximately normal and have equal variances. Let subscript 1 stand for Aberrants, and let subscript 2 stand for off types. Then, we proceed as follows:

i) We formulate our null and alternative hypotheses as

H0 : $\mu 1 - \mu 2 > 1$,

and

H1 : $\mu 1 - \mu 2 < 1$

ii) We set the significance level at $\alpha = 0.05$.

iii) The test-statistic, if H_0 is true, is

$$t = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which has a Student's t-distribution with

$\nu = n_1 + n_2 - 2$, i.e. 25 degrees of freedom.

iv) Computations:

We have

$$\overline{x}_1 = \frac{\sum x_1}{n_1} = \frac{80.92}{12} = 6.74,$$

$$\overline{x}_2 = \frac{\sum x_2}{n_2} = \frac{84.25}{15} = 5.62,$$

And

$$\sum (x_1 - \overline{x}_1)^2 = \sum x_1^2 - \frac{(\sum x_1)^2}{n_1}$$

$$= 561.6402 - \frac{(80.92)^2}{12}$$

$$= 561.6402 - 545.6705$$

$$= 15.9697$$

$$\sum (x_2 - \overline{x}_2)^2 = \sum x_2^2 - \frac{(\sum x_2)^2}{n_2}$$

$$= 478.9779 - \frac{(84.25)^2}{15}$$

$$= 478.9779 - 473.2042$$

$$= 5.7737$$

$$Now\ s_p^2 = \frac{\sum (x_1 - \overline{x}_1)^2 + \sum (x_2 - \overline{x}_2)^2}{n_1 + n_2 - 2}$$

$$= \frac{5.9697 + 5.7737}{12 + 15 - 2}$$

= 0.8697,

so that $s_p = \sqrt{0.8697} = 0.93,$

Hence, the computed value of our test statistic comes out to be

$$\therefore t = \frac{(6.74 - 5.62) - 1}{0.93\sqrt{\frac{1}{12} + \frac{1}{15}}} = \frac{0.12}{0.36} = 0.33$$

v) Critical Region:

Since this is a left-tailed test, therefore the critical region is given by

t < -t0.05(25) i.e. t < -1.708

vi) Conclusion:

Since the computed value of t = 0.33 falls in the acceptance region, therefore we accept H0. We may conclude that the mean rubber percentage of the Aberrants is at least 1 percent more than the mean rubber percentage of Off types.

T-DISTRIBUTION IN THE CASE OF PAIRED OBSERVATIONS

In testing hypotheses about two means, until now we have used independent samples, but there are many situations in which the two samples are not independent. This happen when the observation are found in pairs such that the two observations of a pair are related to each other. Pairing occurs either naturally or by design. Natural pairing occurs whenever measurement is taken on the same unit or individual at two different times. For example, suppose ten young recruits are given a strenuous physical training programme by the Army. Their weights are recorded before they begin and after they complete the training. The two observations obtained for each recruit i.e. the before-and-after measurement constitute natural pairing. The above is natural pairing.

EXAMPLE:

Ten young recruits were put through a strenuous physical training programme by the Army. Their weights were recorded before and after the training with the following results:

Recruit	1	2	3	4	5	6	7	8	9	10
Weight before	125	195	160	171	140	201	170	176	195	139
Weight after	136	201	158	184	145	195	175	190	190	145

Using α = 0.05, would you say that the programme affects the average weight of recruits?

Assume the distribution of weights before and after to be approximately normal. When the observations from two samples are paired, we find the difference between the two observations of each pair, and the test-statistic in this situation is:

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

$$= \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

$$= \frac{\bar{d}}{s_d / \sqrt{n}}$$

LECTURE NO. 41

- Hypothesis-Testing regarding Two Population Means in the Case of Paired Observations (t-distribution)
- The Chi-square Distribution
- Hypothesis Testing and Interval Estimation Regarding a Population Variance (based on Chi-square Distribution)

In the last lecture, we began the discussion of hypothesis-testing regarding two population means in the case of paired observations. It was mentioned that, in many situations, pairing occurs naturally. Observations are also paired to eliminate effects in which there is no interest. For example, suppose we wish to test which of two types (A or B) of fertilizers is the better one. The two types of fertilizer are applied to a number of plots and the results are noted. Assuming that the two types are found significantly different, we may find that part of the difference may be due to the different types of soil or different weather conditions, etc. Thus the real difference between the fertilizers can be found only when the plots are paired according to the same types of soil or same weather conditions, etc.

We eliminate the undesirable sources of variation by taking the observations in pairs. This is pairing by design.

We illustrate the procedure of hypothesis-testing regarding the equality of two population means in the case of paired observations with the help of the same example that we quoted at the end of the last lecture:

EXAMPLE

Ten young recruits were put through a strenuous physical training programme by the Army. Their weights were recorded before and after the training with the following results:

Recruit	1	2	3	4	5	6	7	8	9	10
Weight before	125	195	160	171	140	201	170	176	195	139
Weight after	136	201	158	184	145	195	175	190	190	145

Using α = 0.05, would you say that the programme affects the average weight of recruits? Assume the distribution of weights before and after to be approximately normal.

SOLUTION

The pairing was natural here, since two observations are made on the same recruit at two different times. The sample consists of 10 recruits with two measurements on each. The test is carried out as below:

Hypothesis-Testing Procedure:

i) We state our null and alternative hypotheses as
H0 : $\mu d = 0$ and
H1 : $\mu d \neq 0$

ii) The significance level is set at α = 0.05.

iii) The test statistic under H0 is

$$t = \frac{\bar{d}}{s_d / \sqrt{n}},$$

which has a t-distribution with n – 1 degrees of freedom.

iv) Computations:

Recruit	Weight		Difference, d_i (after minus before)	d_1^2
	Before	After		
1	125	136	11	121
2	195	201	6	36
3	160	158	– 2	4
4	171	184	13	169
5	140	145	5	25
6	201	195	6	36
7	170	175	– 6	25
8	176	190	5	196
9	195	190	14	25
10	139	145	– 5	36
∑	1672	1719	6	673

$$\bar{d} = \frac{\sum d}{n} = \frac{47}{10} = 4.7,$$

$$s_{d^2} = \frac{\sum (d - \bar{d})^2}{n-1} = \frac{1}{n-1}\left[\sum d^2 - \frac{(\sum d)^2}{n}\right]$$

$$= \frac{1}{9}\left[673 - \frac{(47)^2}{10}\right] = \frac{673 - 220.9}{9} = 50.23,$$

so that

$$s_d = \sqrt{50.23} = 7.09.$$

Hence, the computed value of our test-statistic comes out to be :

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{4.7}{7.09/\sqrt{10}} = \frac{(4.7)(3.16)}{7.09} = 2.09.$$

v) The critical region is $|t| \geq t_{0.025}(9)$
= 2.262.

vi) Conclusion:

Since the calculated value of t = 2.09 does not fall in the critical region, so we accept H0 and may conclude that the data do not provide sufficient evidence to indicate that the programme affects average weight.

From the above example, it is clear that the hypothesis-testing procedure regarding the equality of means in the case of paired observations is very similar to the t-test that is applied for testing

H0 : $\mu = 0$.(The only difference is that when we are testing H0 : $\mu = 0$, our variable is X, whereas when we are testing H0 : $\mu d=0$, our variable is d.)

HYPOTHESIS-TESTING PROCEDURE REGARDING TWO POPULATIONS MEANS IN THE CASE OF PAIRED OBSERVATIONS

When the observations from two samples are paired either naturally or by design, we find the difference between the two observations of each pair. Treating the differences as a random sample from a normal population with mean $\mu d = \mu 1 - \mu 2$ and unknown standard deviation σd, we perform a one-sample t-test on them. This is called a paired difference t-test or a paired t-test.

Testing the hypothesis

H0 : $\mu 1 = \mu 2$ against
HA : $\mu 1 \neq \mu 2$ is equivalent to testing H0 : $\mu d = 0$ against
HA : $\mu d \neq 0$.
Let d = x1 – x2 denote the difference between the two samples observations in a pair. Then the sample mean and standard deviation of the differences are

$$\bar{d} = \frac{\sum d}{n} \text{ and } s_d = \frac{\sum (d - \bar{d})^2}{n-1}$$

where n represents the number of pairs.
Assuming that

1) $d_1, d_2, \ldots, d_n$ is a random sample of differences, and

2) the differences are normally distributed,

the test-statistic

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}} = \frac{\bar{d}}{s_d / \sqrt{n}}$$

follows a t-distribution with $\nu = n - 1$ degrees of freedom. The rest of the procedure for testing the null hypothesis H0 : $\mu d = 0$ is the same

EXAMPLE

The following data give paired yields of two varieties of wheat.

Variety I	45	32	58	57	60	38	47	51	42	38
Variety II	47	34	60	59	63	44	49	53	46	41

Each pair was planted in a different locality.

a) Test the hypothesis that, on the average, the yield of variety-1 is less than the mean yield of variety-2.
State the assumptions necessary to conduct this test.

b) How can the experimenter make a Type-I error?
What are the consequences of his doing so?

c) How can the experimenter make a Type-II error?

What are the consequences of his doing so?

d) Give 90 per cent confidence limits for the difference in mean yield.

Note: The pairing was by design here, as the yields are affected by many extraneous factors such as fertility of land, fertilizer applied, weather conditions and so forth.

SOLUTION:

a) In order to conduct this test, we make the following assumptions:

ASSUMPTIONS

- The differences in yields are a random sample from the population of differences,
- The population of differences is normally distributed.

i) We state our null and alternative hypotheses as

$$H_0 : \mu_d \geq 0 \ (\text{or} \ \ \mu_1 \geq \mu_2),$$

i.e. the mean yields are equal and

$$H_1 : \mu_d < 0 \ (\text{or} \ \ \mu_1 < \mu_2).$$

ii) We select the level of significance at $\alpha = 0.05$.

iii) The test statistic to be used is

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}} = \frac{\bar{d}}{s_d / \sqrt{n}}$$

where $\bar{d} = \bar{x}_1 - \bar{x}_2$ and sd2 is the variance of the differences di.

If the populations are normal, this statistic, when H0 is true, has a Student's t-distribution with (n – 1) d. f.

iv) Computations:

Let X1i and X2i represent the yields of Variety I and Variety II respectively. Then the necessary computations are given below:

X_{1i}	X_{2i}	$d_i = X_{1i} - X_{2i}$	d_i^2
45	47	–2	4
32	34	–2	4
58	60	–2	4
57	59	–2	4
60	63	–3	9
38	44	–6	36
47	49	–2	4
51	53	–2	4
42	46	–4	16
38	41	–3	9
Σ	—	–28	94

Now $\bar{d} = \frac{\sum d_i}{n} = \frac{-28}{10} = -2.8$, and

$$s_d^2 = \frac{1}{n-1}\left[\sum d_i^2 - \frac{(\sum d_i)^2}{n}\right] = \frac{1}{9}\left[94 - \frac{(-28)^2}{10}\right]$$

$$= \frac{15.6}{9} = 1.7333, \text{ so that } s_d = 1.32$$

$$\therefore \quad t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{-2.8}{1.32/\sqrt{10}} = \frac{(-2.8)(3.1623)}{1.32} = -6.71$$

v) As this is a one-tailed test therefore, the critical region is given by

$t < t_{0.05}(9) = -1.833$

vi) Conclusion

Since the calculated value of t = –6.71 falls in the critical region, we therefore reject H0. The data present sufficient evidence to conclude that the mean yield of variety-1 is less than the mean yield of variety-2.

b) The experimenter can make a Type-I error by rejecting a true null hypothesis.

In this case, the Type-I error is make by rejecting the null hypothesis when the mean yield of variety-1 is actually not different from the mean yield of variety-2.

In so doing, the consequences would be that we will be saying that variety-2 is better than variety-1 although in reality they are equally good.

c) The experimenter can make a Type-II error by accepting of false null hypothesis.

In this case, the Type-II error is made by accepting the null hypothesis when in reality the mean yield of variety-1 is less than the mean yield of variety-2 and the consequence of committing this error would be a loss of potential increased yield by the use of variety-2.

d) The 90% confidence limits for the difference in means $\mu 1 - \mu 2$ in case of paired observations, are given by

$$\bar{d} \pm t_{\alpha/2,(n-1)} \cdot \frac{s_d}{\sqrt{n}}$$

Substituting the values, we get

$$-2.8 \pm 1.833 \ \frac{1.32}{\sqrt{10}}$$

or -2.8 + 0.765

or -3.565 to -2.035

Hence the 90% confidence limits for the difference in mean yields, $\mu 1 - \mu 2$, are (-3.6, -2.0) .

Until now, we have discussed statistical inference regarding population means based on the Z-statistic as well as the t-statistic.

Also, we have discussed inference regarding the population proportion based on the Z-statistic.

In certain situations, we would be interested in drawing conclusions about the variability that exists in the population values, and for this purpose, we would like to carry out estimation or hypothesis-testing regarding the population variance $\sigma 2$.

Statistical Inference regarding the population variance is based on the chi-square distribution. We begin this topic by presenting the formal definition of the Chi-Square distribution and stating some of its main properties:

THE CHI-SQUARE ($\chi 2$) DISTRIBUTION

The mathematical equation of the Chi-Square distribution is as follows:

$$f(x)=\frac{1}{2^{\nu/2}\Gamma(\nu/2)}(x)^{(\nu/2)-1}.e^{-x/2},\ 0<x<\infty$$

This distribution has only one parameter ν, which is known as the degrees of freedom of the Chi-Square distribution.

PROPERTIES OF THE CHI-SQUARE DISTRIBUTION

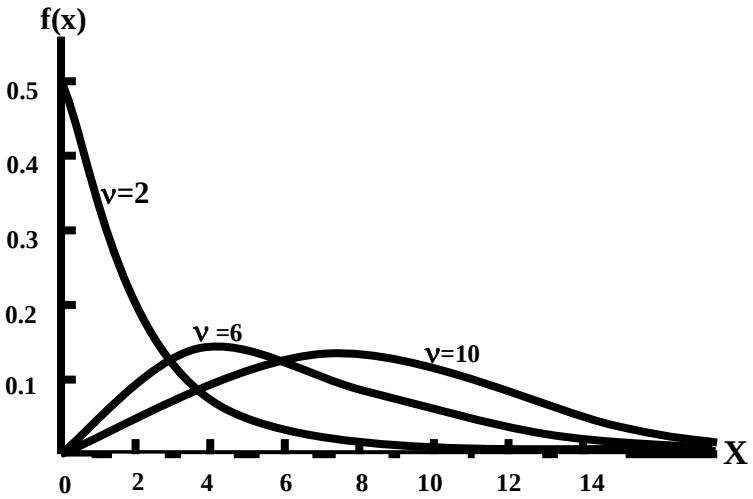
The Chi-Square ($\chi 2$) distribution has the following properties:

1. It is a continuous distribution ranging from 0 to $+\infty$.

The number of degrees of freedom determines the shape of the chi-square distribution. Thus there is a different chi-square distribution for each number of degrees of freedom. As such, it is a whole family of distributions.

2. The curve of a chi-square distribution is positively skewed.

The skewness decreases as ν increases.



$\chi 2$-distribution for various values of ν

As indicated by the above figures, the chi-square distribution tends to the normal distribution as the number of degrees of freedom approaches infinity.

3. The mean of a chi-square distribution is equal to ν, the number of degrees of freedom.
4. Its variance is equal to 2ν.
5. The moments about the origin are given by

$$\mu'_1 = \nu$$
$$\mu'_2 = \nu(\nu+2)$$
$$\mu'_3 = \nu(\nu+2)(\nu+4)$$
$$\mu'_4 = \nu(\nu+2)(\nu+4)(\nu+6)$$

As such, the moment-ratios come out to be

$$\beta_1 = \frac{8}{\nu}$$
$$\beta_1 = 3 + \frac{12}{\nu}$$

Having discussed the basic definition and properties of the chi-square distribution, we begin the discussion of its role in interval estimation and hypothesis-testing. We begin with interval estimation regarding the variance of a normally distributed population:

EXAMPLE:

Suppose that an aptitude test carrying a total of 20 marks is devised, and administered on a large population of students, and, upon doing so, it was found that the marks of the students were normally distributed. A random sample of size n = 8 is drawn from this population, and the sample values are 9, 14, 10, 12, 7, 13, 11, 12.
Find the 90 percent confidence interval for the population variance $\sigma 2$, representing the variability in the marks of the students.

SOLUTION:
The 90% confidence interval for $\sigma 2$ is given by

$$\frac{\sum(\mathrm{Xi}-\overline{\mathrm{X}})^2}{\chi^2_{0.05(\mathrm{n}-1)}} < \sigma^2 < \frac{\sum(\mathrm{X_i}-\overline{\mathrm{X}})^2}{\chi^2_{0.95(\mathrm{n}-1)}}$$

The above formula is linked with the fact that if we keep 90% area under the chi-square distribution in the middle, then we will have 5% area on the left-hand-side, and 5% area on the right-hand-side, as shown below:

$\chi 2$(N-1)-DISTRIBUTION

In order to apply the above formula, we first need to calculate the sample mean $\overline{\mathrm{X}}$, which is

$$\overline{\mathrm{X}} = \frac{\sum \mathrm{X}}{\mathrm{n}} = \frac{88}{8} = 11$$

Then, we obtain

$$\sum_{i=1}^{8} (X_i - \overline{X})^2 = (9-11)^2 + (14-11)^2 + \ldots + (12-11)^2 = 36$$

Next, we need to find :
1) the value of $\chi 2$ to the left of which the area under the chi-square distribution is 5%
2) the value of $\chi 2$ to the right of which the area under the chi-square distribution is 5%
For this purpose, we will need to consult the table of areas under the chi-square distribution.

THE CHI-SQUARE TABLE

The entries in this table are values of $x2\alpha(\nu)$, for which the area to their right under the chi-square distribution with ν degrees of freedom is equal to α.

Upper Percentage Points of the Chi-square Distribution

ν \ α	0.99	0.98	0.975	0.95	0.10	0.05	0.03	0.02	0.01
1	0.0002	0.001	0.001	0.004	2.71	3.84	5.02	5.41	6.64
2	0.020	0.040	0.051	0.103	4.61	5.99	7.38	7.82	9.21
3	0.115	0.185	0.216	0.352	6.25	7.82	9.35	9.84	11.34
4	0.297	0.429	0.484	0.711	7.78	9.49	11.14	11.67	13.28
5	0.554	0.752	0.831	1.145	9.24	11.07	12.83	13.39	15.09
6	0.87	1.13	1.24	1.64	10.64	12.59	14.45	15.03	16.81
7	1.24	1.56	1.69	2.17	12.02	14.07	16.01	16.62	18.48
8	1.65	2.03	2.18	2.73	13.36	15.51	17.54	18.17	20.09
9	2.09	2.53	2.70	3.32	14.68	16.92	19.02	19.68	21.67
10	2.56	3.06	3.25	3.94	15.99	18.31	20.48	21.16	23.21
11	3.05	3.61	3.82	4.58	17.28	19.68	21.92	22.62	24.72
12	3.57	4.18	4.40	5.23	18.55	21.03	23.34	24.05	26.22
13	4.11	4.76	5.01	5.89	19.81	22.36	24.74	25.47	27.69
14	4.66	5.37	5.63	6.57	21.06	23.68	26.12	26.87	29.14
15	5.23	5.98	6.26	7.26	22.31	25.00	27.49	28.26	30.58

Chi-Square Table (continued):

	α								
ν	**0.99**	**0.98**	**0.975**	**0.95**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**
16	5.81	6.61	6.91	7.96	23.54	26.30	28.84	29.63	32.00
17	6.41	7.26	7.56	8.67	24.77	27.59	30.19	31.00	33.41
18	7.02	7.91	8.23	9.39	25.99	28.87	31.53	32.35	34.81
19	7.63	8.57	8.91	10.12	27.20	30.14	32.85	33.69	36.19
20	8.26	9.24	9.59	10.85	28.41	31.41	34.17	35.02	37.57
21	8.90	9.92	10.28	11.59	29.62	32.67	35.48	36.34	38.93
22	9.54	10.60	10.98	12.34	30.81	33.92	36.78	37.66	40.29
23	10.20	11.29	11.69	13.09	32.01	35.17	38.08	38.97	41.64
24	10.86	11.99	12.40	13.85	33.00	36.42	39.36	40.27	42.92
25	11.52	12.70	13.12	14.61	34.38	37.65	40.65	41.57	44.31
26	12.20	13.41	13.84	15.38	35.56	38.88	41.92	42.86	45.64
27	12.88	14.12	14.57	16.15	36.74	40.11	43.19	44.14	46.96
28	13.56	14.85	15.31	16.93	37.92	41.34	44.46	45.42	48.28
29	14.26	15.57	16.05	17.71	39.09	42.56	45.72	46.69	49.59
30	14.95	16.31	16.79	18.49	40.26	43.77	46.98	47.96	50.89

From the $\chi 2$-table, we find that

$\chi^2_{0.05}(7) = 14.07$

and

$\chi^2_{0.95}(7) = 2.17$

Hence the 90 percent confidence interval for $\sigma 2$ is

$$\frac{\sum(Xi - \overline{X})^2}{\chi^2_{0.05,(7)}} < \sigma^2 < \frac{\sum(X_i - \overline{X})^2}{\chi^2_{0.95,(7)}}$$

or $$\frac{36}{14.07} < \sigma^2 < \frac{36}{2.17}$$

or $$2.56 < \sigma^2 < 16.61$$

Thus the 90% confidence interval for σ^2 is (2.56, 16.61).

If we take the square root of the lower limit as well as the upper limit of the above confidence interval, we obtain (1.6, 4.1).

So, we may conclude that, on the basis of 90% confidence, we can say that the standard deviation σ of our population lies between 1.6 and 4.1 .We can obtain a confidence interval for σ by taking the square root of the end points of the interval for $\sigma 2$, but experience has shown that σ cannot be estimated with much precision for small sample sizes.

The formula of the confidence interval for $\sigma 2$ that we have applied in the above example is based on the fact that:

If $\overline{X}$ and S2 are the mean and variance (respectively) of a random sample X1, X2, …, Xn of size n drawn from a normal population with variance $\sigma 2$, then the statistic

$$\frac{\sum(X_i - \overline{X})^2}{\sigma^2} = \frac{nS^2}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2}$$

follows a chi-square distribution with (n – 1) degrees of freedom.

Next, we consider hypothesis - testing regarding the population variance $\sigma 2$:
We illustrate this concept with the help of an example:

EXAMPLE

The variability in the tensile strength of a type of steel wire must be controlled carefully. A sample of the wire is subjected to test, and it is found that the sample variance is S2 = 31.5. The sample size was n = 16 observations.
Test the hypothesis that the population variance is 25 against the alternative that the variance is greater than 25. Use a 0.05 level of significance.

SOLUTION

a)i) We have to decide between the hypotheses

H0 : $\sigma 2 = 25$, and
H1 : $\sigma 2 > 25$

ii) The level of significance is $\alpha = 0.05$.

iii) The test statistic is $\chi^2 = \frac{nS^2}{\sigma_0^2}$, which under H_0, has a χ^2-distribution with (n–1) degrees of freedom, assuming that the population is normal.

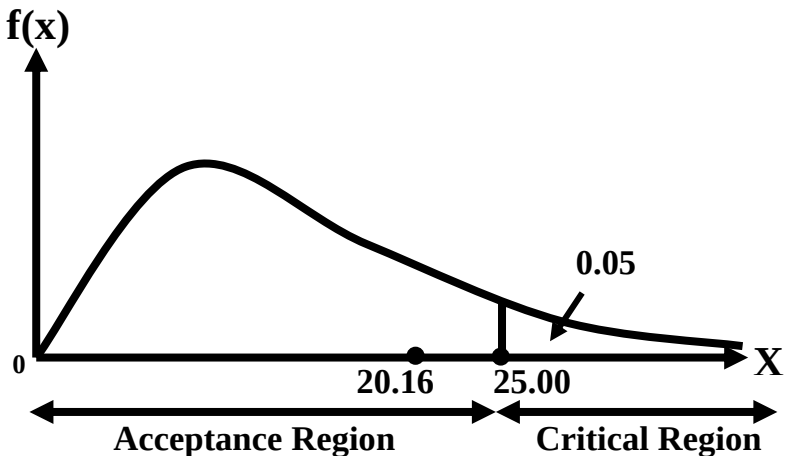
iv) We calculate the value of χ^2 from the sample data as

$$\chi^2 = \frac{nS^2}{\sigma_0^2} = \frac{16(31.5)}{25} = 20.16 .$$

v) The critical region is $\chi 2 > \chi^2_{0.05,(15)} = 25.00$ (one tailed test)

vi) Conclusion.

Since the calculated value of $\chi 2$ falls in the acceptance region, so we accept our null Hypothesis, i.e. we have reasonable evidence to conclude that $\sigma 2 = 25$.The Chi-Square Distribution with 15 degrees of Freedom:



The above example points to the following general procedure for testing a hypothesis regarding the population variance $\sigma 2$: Suppose we desire to test a null hypothesis H0 that the variance $\sigma 2$ of a normally distributed population has some specified value, say $\sigma 02$. To do this, we need to

draw a random sample $X_1, X_2, \ldots, X_n$ of size n from the normal population and compute the value of the sample variance S^2. If the null hypothesis $H_0 : \sigma^2 = \sigma^2_0$ is true, then the statistic $\chi^2 = \frac{nS^2}{\sigma_0^2}$ has a χ^2-distribution with (n–1) degrees of freedom.

LECTURE NO. 42

- The F-Distribution
- Hypothesis Testing and Interval Estimation in order to Compare the Variances of Two Normal Populations (based on F-Distribution)

Before we describe you statistical inference based on the F-distribution, let us consolidate the idea of hypothesis-testing regarding the population variance with the help of an example:

EXAMPLE

The manager of a bottling plant is anxious to reduce the variability in net weight of fruit bottled. Over a long period, the standard deviation has been 15.2 gm. A new machine is introduced and the net weights (in grams) in 10 randomly selected bottles (all of the same nominal weight) are 987, 966, 955, 977, 981, 967, 975, 980, 953, 972. Would you report to the manager that the new machine has a better performance?

SOLUTION

i) We have to decide between the hypotheses

H0 : $\sigma = 15.2$, i.e. the standard deviation is 15.2gm

H1 : $\sigma < 15.2$ i.e. the standard deviation has been reduced.

ii) We choose the significance level at $\alpha = 0.05$.

iii) The test-statistic is

$$\chi^2 = \frac{nS^2}{\sigma_0^2} = \frac{\sum(X_i - \overline{X})^2}{\sigma_0^2}$$

which under H_0, has a χ^2 -distribution with (n – 1) degrees of freedom, assuming that the weights are normally distributed.

iv) Computations.

n = 10, $\sum Xi = 9713$, $\sum X2i = 9435347$

v) The critical region is $\chi 2 < \chi 20.95\ (9) = 3.32$ (the lower 5% point)

NOW

$nS^2 = \sum(X_i - \overline{X})^2 = \sum X^2_i - (\sum X_i)^2/n$

$= 9435347 - (9713)^2/10 = 1110.1$

$$\therefore \quad \chi^2 = \frac{1110.1}{(15.2)^2} = \frac{1110.1}{231.04} = 4.81$$

vi) Conclusion:

Since the calculated value of $\chi^2 = 4.81$ does not fall in the critical region, we therefore cannot reject the null hypothesis that the standard deviation is 15.2 gm and hence we would not report to the manager that the new machine has a better performance.

The above example points to the fact that, if we wish to test a null hypothesis H0 that the variance $\sigma 2$ of a normally distributed population has some specified value, say $\sigma 02$, then, (having drawn a random sample X1, X2, …, Xn of size n from the normal population), we will compute the value of the sample variance S2.

The mathematics underlying this hypothesis-testing procedure states that:

If the null hypothesis $H_0 : \sigma^2 = \sigma_0{}^2$ is true, then the statistic $\chi^2 = \frac{nS^2}{\sigma_0^2}$ has a χ^2-distribution with (n–1) degrees of freedom.

A point to be noted is that, since the random variable X is distributed as chi-square, therefore we call it $\chi 2$.

If we do so, our equation of the chi-square distribution can be written as

$$f\left(\chi^2\right)=\frac{1}{2^{\nu/2}\Gamma\left(\nu/2\right)}\left(\chi^2\right)^{(\nu/2)-1}.e^{-\chi^2/2},\ 0<\chi^2<\infty$$

It should be obvious that the standard deviation of the normal population will be tested in the same way as the population variance is tested.

Next, we begin the discussion of statistical inference regarding the ratio of two population variances.

As this particular inference is based on the F-distribution, therefore we begin with the discussion of the mathematical definition and the main properties of the F-distribution.

THE F-DISTRIBUTION

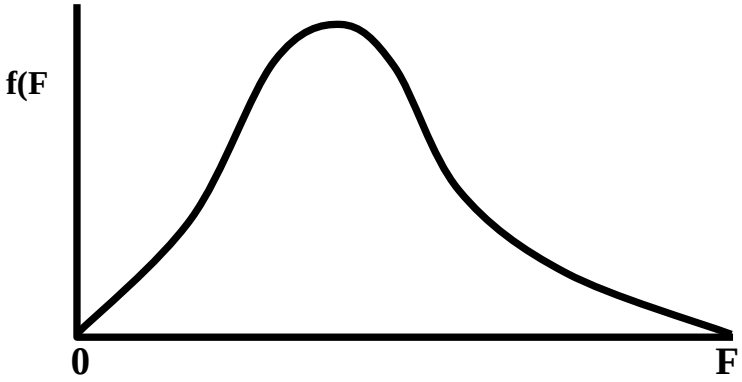
The mathematical equation of the F-distribution is as follows:

$$f\left(x\right)=\frac{\Gamma\left[\left(\nu_1+\nu_2\right)/2\right]\left(\nu_1/\nu_2\right)^{\nu_1/2}\ x^{(\nu_1/2)-1}}{\Gamma\left(\nu_1/2\right)\Gamma\left(\nu_2/2\right)\left[1+\nu_1\,x/\nu_2\right]^{(\nu_1+\nu_2)/2}},\quad 0<x<\infty$$

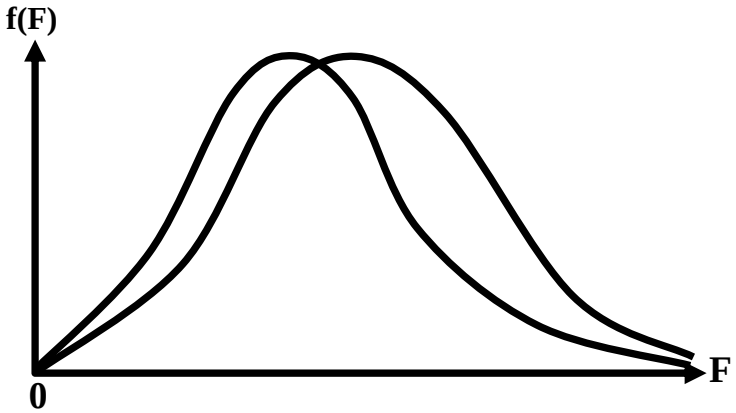
This distribution has two parameters $\nu 1$ and $\nu 2$, which are known as the degrees of freedom of the F-distribution.The F-distribution having the above equation have $\nu 1$ degrees of freedom in the numerator and $\nu 2$ degrees of freedom in the denominator. It is usually abbreviated as F($\nu 1$, $\nu 2$).

PROPERTIES OF F-DISTRIBUTION

1. The F-distribution is a continuous distribution ranging from zero to plus infinity.
2. The curve of the F-distribution is positively skewed.



But as the degrees of freedom $\nu 1$ and $\nu 2$ become large, the F-distribution approaches the normal distribution.



3. For $\nu 2 > 2$, the mean of the F-distribution is

$$\frac{\nu_2}{\nu_2 - 2}$$

which is greater than 1.

4. For $\nu 2 > 4$, the variance of the F-distribution is

$$\sigma^2 = \frac{2\nu_2^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 2)^2(\nu_2 - 4)}$$

5. The F-distribution for $\nu 1 > 2$, $\nu 2 > 2$ is unimodal, and the mode of the distribution with $\nu 1 > 1$ is at

$$\frac{\nu_2(\nu_1 - 2)}{\nu_1(\nu_2 + 2)}$$

which is always less than 1.

6. If F has an F-distribution with $\nu 1$ and $\nu 2$ degrees of freedom, then the reciprocal has an F-distribution with $\nu 2$ and $\nu 1$ degrees of freedom. Next, we consider the tables of the F-distribution. As the F-distribution involves two parameters, $\nu 1$ and $\nu 2$, hence separate tables have been constructed for 5%, 2½ % and 1% right-tail areas respectively, as shown below:

The F-table pertaining to 5% right-tail areas is as follows:

Upper 5 Percent Points of The F-Distribution i.e., $F_{0.05}(\nu_1, \nu_2)$

ν_2 \ ν_1	1	2	3	4	5	6	8	12	24	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	238.9	243.9	249.0	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.41	19.45	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.84	8.74	8.64	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.91	5.77	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.68	4.53	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.00	3.84	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.73	3.57	3.41	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.28	3.12	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.23	3.07	2.90	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.91	2.74	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	2.95	2.79	2.61	2.40
12	4.75	3.88	3.49	3.26	3.11	3.00	2.85	2.69	2.50	2.30
13	4.67	3.80	3.41	3.18	3.03	2.92	2.77	2.60	2.42	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.70	2.53	2.35	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.48	2.29	2.07

ν_2 \ ν_1	1	2	3	4	5	6	8	12	24	∞
16	4.49	3.63	3.24	3.01	2.85	2.74	2.59	2.42	2.24	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.55	2.38	2.19	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.51	2.34	2.15	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.48	2.31	2.11	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.28	2.08	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.42	2.25	2.05	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.40	2.23	2.03	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.38	2.20	2.00	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.36	2.18	1.98	1.73
25	4.24	3.38	2.99	2.76	2.60	2.49	2.34	2.16	1.96	1.71
26	4.22	3.37	2.98	2.74	2.59	2.47	2.32	2.15	1.95	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.30	2.13	1.93	1.67
28	4.20	3.34	2.95	2.71	2.56	2.44	2.29	2.12	1.91	1.65
29	4.18	3.33	2.93	2.70	2.54	2.43	2.28	2.10	1.90	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.09	1.89	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.00	1.79	1.51
60	4.00	3.15	2.76	2.52	2.37	2.25	2.10	1.92	1.70	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.02	1.83	1.61	1.25
∞	3.84	2.99	2.60	2.37	2.21	2.10	1.94	1.73	1.52	1.00

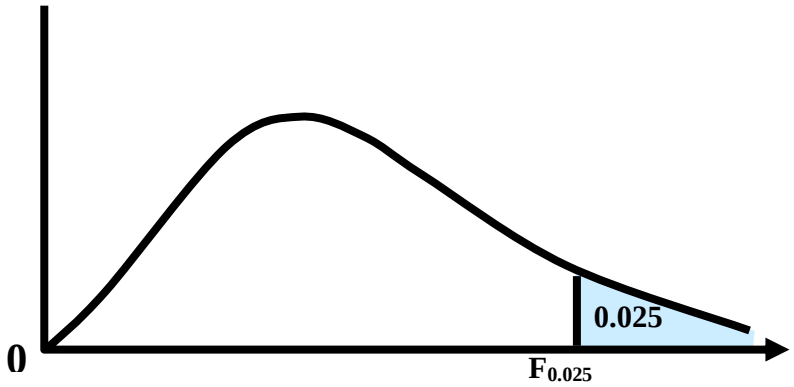
Similarly, the F-table pertaining to 2½% right-tail areas is as follows

Upper 2.5 Percent Points of the F-Distribution i.e. $F_{0.025}(\nu_1, \nu_2)$

ν_2 \ ν_1	1	2	3	4	5	6	8	12	24	∞
1	647.8	799.5	864.2	899.6	921.8	937.1	956.7	976.7	997.2	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.37	39.41	39.46	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.54	14.34	14.12	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	8.98	8.75	8.51	8.26
5	10.07	8.43	7.76	7.39	7.15	6.98	6.76	6.52	6.28	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.60	5.37	5.12	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.90	4.67	4.42	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.43	4.20	3.95	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.10	3.87	3.61	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.85	3.62	3.37	3.08
11	6.72	5.26	4.63	4.28	4.04	3.88	3.66	3.43	3.17	2.88
12	6.55	5.10	4.47	4.12	3.89	3.73	3.51	3.28	3.02	2.72
13	6.41	4.97	4.35	4.00	3.77	3.60	3.39	3.15	2.89	2.60
14	6.30	4.86	4.24	3.89	3.66	3.50	3.29	3.05	2.79	2.49
15	6.20	4.77	4.15	3.80	3.58	3.41	3.20	2.96	2.70	2.40

Upper 2.5 Percent Points of the F-Distribution i.e. $F_{0.025}(\nu_1, \nu_2)$ (Continued):

ν_2 \ ν_1	1	2	3	4	5	6	8	12	24	∞
16	6.12	4.69	4.08	3.73	3.50	3.34	3.12	2.89	2.63	2.32
17	6.04	4.62	4.01	3.66	3.44	3.28	3.06	2.82	2.56	2.25
18	5.98	4.56	3.95	3.61	3.38	3.22	3.01	2.77	2.50	2.19
19	5.92	4.51	3.90	3.56	3.33	3.17	2.96	2.72	2.45	2.13
20	5.87	4.46	3.86	3.51	3.29	3.13	2.91	2.68	2.41	2.09
21	5.83	4.42	3.82	3.48	3.25	3.09	2.87	2.64	2.37	2.04
22	5.79	4.38	3.78	3.44	3.22	3.05	2.84	2.60	2.33	2.00
23	5.75	4.35	3.75	3.41	3.18	3.02	2.81	2.57	2.30	1.97
24	5.72	4.32	3.72	3.38	3.15	2.99	2.78	2.54	2.27	1.94
25	5.69	4.29	3.69	3.35	3.13	2.97	2.75	2.51	2.24	1.91
26	5.66	4.27	3.67	3.33	3.10	2.94	2.73	2.49	2.22	1.88
27	5.63	4.24	3.65	3.31	3.08	2.92	2.71	2.47	2.19	1.85
28	5.61	4.22	3.63	3.29	3.06	2.90	2.69	2.45	2.17	1.83
29	5.59	4.20	3.61	3.27	3.04	2.88	2.67	2.43	2.15	1.81
30	5.57	4.18	3.59	3.25	3.06	2.87	2.65	2.41	2.14	1.79
40	5.42	4.05	3.46	3.13	2.90	2.74	2.53	2.29	2.01	1.64
60	5.49	3.93	3.34	3.01	2.79	2.63	2.41	2.17	1.88	1.48
120	5.15	3.80	3.23	2.89	2.67	2.52	2.30	2.05	1.76	1.31
∞	5.02	3.69	3.12	2.79	2.57	2.41	2.19	1.94	1.64	1.00



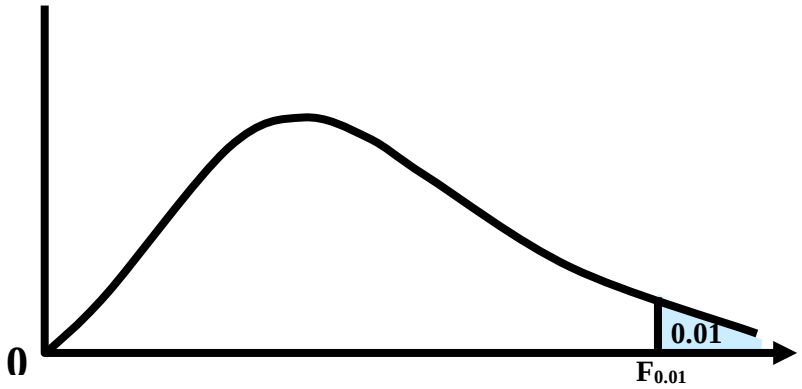
And, the F-table pertaining to 1% right-tail areas is as follows:

Upper 1 Percent Points of the F-Distribution i.e. $F_{0.01}(\nu_1, \nu_2)$

ν_2 \ ν_1	1	2	3	4	5	6	8	12	24	∞
1	4052	5000	5403	5625	5764	5859	5982	6106	6235	6366
2	98.50	99.00	99.17	99.25	99.30	99.33	99.37	99.42	99.46	99.50
3	34.12	30.82	29.46	28.71	28.24	27.91	27.49	27.05	26.60	26.12
4	21.20	18.00	10.69	15.98	15.52	15.21	14.80	14.37	13.93	13.46
5	16.26	13.27	12.06	11.39	10.97	10.67	10.29	9.89	9.47	9.02
6	13.75	10.92	9.78	9.15	8.75	8.47	8.10	7.72	7.31	6.88
7	12.25	9.55	8.45	7.85	7.46	7.19	6.84	6.47	6.07	5.65
8	11.26	8.65	7.59	7.01	6.63	6.37	6.03	5.67	5.28	4.86
9	10.56	8.02	6.99	6.42	6.06	5.80	5.47	5.11	4.73	4.31
10	10.04	7.56	6.55	5.99	5.64	5.39	5.06	4.71	4.33	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.74	4.40	4.02	3.61
12	9.33	6.93	5.95	5.41	5.06	4.82	4.50	4.16	3.78	3.36
13	9.07	6.70	5.74	5.20	4.86	4.62	4.30	3.96	3.59	3.17
14	8.86	6.51	5.56	5.03	4.69	4.46	4.14	3.80	3.43	3.00
15	8.68	6.36	5.42	4.89	4.56	4.32	4.00	3.67	3.29	2.87

Upper 1 Percent Points of the F-Distribution i.e. $F_{0.01}(\nu_1, \nu_2)$ (continued)

ν_2 \ ν_1	1	2	3	4	5	6	8	12	24	∞
16	8.53	6.23	5.29	4.77	4.44	4.20	3.89	3.55	3.18	2.75
17	8.40	6.11	5.18	4.67	4.34	4.10	3.79	3.45	3.08	2.65
18	8.28	6.01	5.09	4.58	4.25	4.01	3.71	3.37	3.03	2.57
19	8.18	5.93	5.01	4.50	4.17	3.94	3.63	3.30	2.92	2.49
20	8.10	5.85	4.94	4.43	4.10	3.87	3.56	3.23	2.86	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.51	3.17	2.80	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.45	3.12	2.75	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.41	3.07	2.70	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.36	3.03	2.66	2.21
25	7.77	5.57	4.68	4.18	3.86	3.63	3.32	2.99	2.62	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.29	2.96	2.58	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.26	2.93	2.55	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.23	2.90	2.52	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.20	2.87	2.49	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.17	2.84	2.47	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	2.99	2.66	2.29	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.82	2.50	2.12	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.66	2.34	1.94	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.51	2.18	1.79	1.00



Having discussed the basic definition and the main properties of the F-distribution, we now begin the discussion of the role of the F-distribution in statistical inference: First, we discuss interval estimation regarding the ratio of two population variances:

CONFIDENCE INTERVAL FOR THE VARIANCE RATIO σ12/σ22

Let two independent random samples of size n1 and n2 be taken from two normal population with variances σ12 and σ22 and let s12 and s22 be the unbiased estimators of σ12 and σ22.
Then, it can be mathematically proved that the quantity

$$F = \frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$$

has an F-distribution with (n1 – 1, n2 – 1) degrees of freedom.
The confidence interval for σ12/σ22 is given by

$$\left[\frac{s_1^2}{s_2^2} . \frac{1}{F_{\alpha/2}(n_1 - 1,\, n_2 - 1)},\ \frac{s_1^2}{s_2^2} . F_{\alpha/2}(n_2 - 1,\, n_1 - 1) \right]$$

We can also find a confidence interval for σ1/σ2 by taking the square root of the end points of the above interval.
We illustrate this concept with the help of the following example:

EXAMPLE

A random sample of 12 salt-water fish was taken, and the girth of the fish was measured. The standard deviation s1 came out to be 2.3 inches. Similarly, a random sample of 10 fresh-water fish was taken, and the girth of the fish was measured. The standard deviation of this sample i.e. s2 came out to be 1.5 inches. Find a 90% confidence interval for the ratio between the 2 population variances σ12/σ22. Assume that the populations of girth are normal.

SOLUTION

The 90% confidence interval for σ12/σ22 is given by

$$\left[\frac{s_1^2}{s_2^2} . \frac{1}{F_{0.05}(n_1 - 1,\, n_2 - 1)},\ \frac{s_1^2}{s_2^2} . F_{0.05}(n_2 - 1,\, n_1 - 1) \right]$$

Here $s_1^2 = (2.3)^2 = 5.29$,
$s_2^2 = (1.5)^2 = 2.25$,
$n_1 - 1 = 12 - 1 = 11$ and $n_2 - 1 = 10 - 1 = 9$

Hence,
$F_{0.05}(n_1 - 1, n_2 - 1) = F_{0.05}(11, 9) = 3.1$
and
$F_{0.05}(n_2 - 1, n_1 - 1) = F_{0.05}(9, 11) = 2.9$

With reference to the F-table, it should be noted that if it is an abridged table and the F-values are not available for all possible pairs of degrees of freedom, then the required F-values are obtained by the method of interpolation. In this example, for the lower limit of our confidence interval, we need the value of $F_{0.05}(11, 9)$, but in the above table pertaining to 5% right-tail area, values are available for $\nu_1 = 8$ and $\nu_1 = 12$, but not for $\nu_1 = 11$. Hence, we can find the F-value corresponding to $\nu_1 = 11$ by the method of interpolation: The F-value corresponding to $\nu_2 = 9$ and $\nu_1 = 8$ is 3.23 whereas the F-value corresponding to $\nu_2 = 9$ and $\nu_1 = 12$ is 3.07.If we wish to find the F-value corresponding to $\nu_2 = 9$ and $\nu_1 = 10$, we can find the arithmetic mean of 3.23 and 3.07 which is 3.15.If we wish to find the F-value corresponding to $\nu_2 = 9$ and $\nu_1 = 11$, we can find the arithmetic mean of 3.15 and 3.07 which is 3.11, which, upon rounding, is equal to 3.1.The above method of interpolation is based on the assumption that the F-values between any two successive F-values (printed in any row of the F-table) are equally spaced between the two given values.

If we do not wish to go through the rigorous procedure of interpolation, we can note that $\nu_1 = 11$ is close to $\nu_1 = 12$, and hence, we can consider that F-value which corresponds to $\nu_1 = 12$ (which in this case is 3.07 ~ 3.1 ----- exactly the same as what we obtained above (correct to one decimal place) by the method of interpolation). Going back to our example, the 90% confidence interval is

$$\left[\frac{5.29}{2.25}.\left(\frac{1}{3.1}\right), \frac{5.29}{2.25}(2.9)\right]$$

or (0.76, 6.81).

Taking the square root of the end points (0.76, 6.81), we obtain the 90% confidence interval for σ_1/σ_2 as (0.87, 2.61).

Next, we discuss hypothesis - testing regarding the equality of two population variances: Suppose that we have two independent random samples of size n1 and n2 from two normal populations with variances σ_1^2 and σ_2^2, we wish to test the hypothesis that the two variances are equal. The main steps of the hypothesis - testing procedure are similar to the ones that we have been discussing earlier. We illustrate this concept with the help of an example:

EXAMPLE

In two series of hauls to determine the number of plankton organisms inhabiting the waters of a lake, the following results were found:

Series I: 80, 96, 102, 77, 97, 110, 99, 88, 103, 1089

Series II: 74, 122, 92, 81, 104, 92, 92

In series I, the hauls were made in succession at the same place. In series II, they were made in different parts scattered over the lake. Does there appear to be a greater variability between different places than between different times at the same place?

SOLUTION

If X denotes the number of plankton organisms per haul, then for each of the two series, X can be assumed to be normally distributed.

Hypothesis-testing Procedure:

Step 1 :

H0 : $\sigma 12 \geq \sigma 22$ i.e. $\sigma 22 \leq \sigma 12$

HA : $\sigma 12 < \sigma 22$ i.e. $\sigma 22 > \sigma 12$

Step 2: Level of significance: $\alpha = 0.05$

Step 3: Test-statistic:

Since both the populations are normally distributed, hence, the statistic

$$F = \frac{s_2^2 / \sigma_2^2}{s_1^2 / \sigma_1^2}$$

will follow the F-distribution having (n2 - 1, n1 - 1) degrees of freedom.

Step 4 : Computations:

X_1	X_1^2
80	6400
96	9216
102	10404
77	5929
97	9409
110	12100
99	9801
88	7744
103	10609
108	11664
960	93276

X_2	X_2^2
74	5476
122	14884
92	8464
81	6561
104	10816
92	8464
92	8464
657	63129

Now
$$s_1^2 = \frac{1}{n_1 - 1}\left[\sum X_1^2 - \frac{\left(\sum X_1\right)^2}{n_1}\right]$$

and
$$s_2^2 = \frac{1}{n_2 - 1}\left[\sum X_2^2 - \frac{\left(\sum X_2\right)^2}{n_2}\right]$$

So
$$s_1^2 = \frac{1}{10 - 1}\left[9326 - \frac{(960)^2}{10}\right]$$

$$= \frac{1}{9}[93276 - 92160]$$

$$= \frac{1}{9}[1116] = 124$$

Similarly $s_2^2 = \frac{1}{n_2 - 1}\left[\sum X_2^2 - \frac{(\sum X_2)^2}{n_2}\right]$

$$= \frac{1}{7-1}\left[63129 - \frac{(657)^2}{7}\right]$$

$$= \frac{1}{6}[63129 - 61664.14]$$

$$= \frac{1}{6}[1464.86] = 244.14$$

Hence $F = \frac{s_2^2}{s_1^2} = \frac{244.14}{124} = 1.97$

Step 5 : Critical Region:

F > F0.05 (6, 9) = 3.37

Step 6: Conclusion:

Since 1.97 is less than 3.37, we do not reject H0; our data does not provide sufficient evidence to indicate that there is greater variability (in the number of plankton organisms per haul) between different places than between different times at the same place.

Let us consider another example:

EXAMPLE

Two methods of determining the moisture content of samples of canned corn have been proposed and both have been used to make determinations on proportions taken from each of 21 cans. Method I is easier to apply but appears to be more variable than Method II.

If the variability of Method I were not more than 25 per cent greater than that of Method II, then we would prefer Method I.

The sample results are as follows:

$$n_1 = n_2 = 21;\ \overline{X}_1 = 50;\ \overline{X}_2 = 53$$

$$\sum (X_1 - \overline{X}_1)^2 = 720; \qquad \sum (X_2 - \overline{X}_2)^2 = 340.$$

Based on the above sample results, which method would you recommend?

SOLUTION

In order to solve this problem, the first point to be noted is that, in this problem, our null and alternative hypotheses will be

H0: $\sigma 12 \leq 1.25\ \sigma 22$

and

H1: $\sigma 12 > 1.25\ \sigma 22$.

Null and Alternative Hypotheses:

In this problem, we need to test

H0 : $\sigma 12 \leq 1.25\ \sigma 22$

against

H1 : $\sigma 12 > 1.25\ \sigma 22$.

This is so, because 1.25 σ22 means 125% of σ22, and this means 25% greater than σ22. You are encouraged to work on this point on their own. The second point to be noted is that, in this problem, our test-statistic is not but is

$$F = \frac{s_1^{\,2}}{1.25\, s_2^{\,2}}.$$

Test Statistic:

$$\mathrm{F} = \frac{\mathrm{s}_1^{\,2}}{1.25\, \mathrm{s}_2^{\,2}}.$$

(Under the null hypothesis, s12 / 1.25 s22 has an F-distribution with ν1 = ν2 = 21-1 = 20 degrees of freedom.)
This is so because, (in accordance with the fact that has an F-distribution with (n1 – 1, n2 – 1) degrees of freedom),
it can be shown that:

$$F = \frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$$

If we have

H0: σ12/ σ22 = k

then

$$F = \frac{s_1^{\,2}}{s_2^{\,2}} \cdot \frac{1}{k}$$

has an F-distribution with (n1 - 1, n2 - 1) degrees of freedom. (In this problem, k = 1.25.)You are encouraged to work on this problem also on their own, and to carry out the rest of the steps of the hypothesis-testing procedure (which are the usual ones), and to decide whether to accept or to reject the null hypothesis.

LECTURE NO. 43

- Analysis of Variance
- Experimental Design

Earlier, we compared two-population means by using a two-sample t-test. However, we are often required to compare more than two population means simultaneously. We might be tempted to apply the two-sample t-test to all possible pairwise comparisons of means. For example, if we wish to compare 4 population means, there will be $\binom{4}{2} = 6$ separate pairs, and to test the null hypothesis that all four population means are equal, we would require six two-sample t-tests. Similarly, to test the null hypothesis that 10 population means are equal, we would need

$$\binom{10}{2} = 45$$

Separate two-sample t-tests. This procedure of running multiple two-sample t-tests for comparing means would obviously be tedious and time-consuming. Thus a series of two-sample t-tests is not an appropriate procedure to test the equality of several means simultaneously. Evidently, we require a simpler procedure for carrying out this kind of a test.
One such procedure is the Analysis of Variance, introduced by Sir R.A. Fisher (1890-1962) in 1923:

ANALYSIS OF VARIANCE (ANOVA)

It is a procedure which enables us to test the hypothesis of equality of several population means (i.e.

H0 : $\mu 1 = \mu 2 = \mu 3 = \ldots\ldots = \mu k$

against

HA: not all the means are equal)

The concept of Analysis of Variance is closely related with the concept of Experimental Design:

EXPERIMENTAL DESIGN

By an experimental design, we mean a plan used to collect the data relevant to the problem under study in such a way as to provide a basis for valid and objective inference about the stated problem. The plan usually includes:

- the selection of treatments whose effects are to be studied,
- the specification of the experimental layout, and
- the assignment of treatments to the experimental units.

All these steps are accomplished before any experiment is performed. Experimental Design is a very vast area. In this course, we will be presenting only a very basic introduction of this area. There are two types of designs:

SYSTEMATIC AND RANDOMIZED DESIGNS

In this course, we will be discussing only the randomized designs, and, in this regard, it should be noted that for the randomized designs, the analysis of the collected data is carried out through the technique known as Analysis of Variance.
Two of the very basic randomized designs are:

- The Completely Randomized (CR) Design,
- The Randomized Complete

- Block (RCB) Design

We will consider these one by one. We begin with the simplest design i.e. the Completely Randomized (CR) Design:

THE COMPLETELY RANDOMIZED DESIGN (CR DESIGN)

A completely randomized (CR) design, which is the simplest type of the basic designs, may be defined as a design in which the treatments are assigned to experimental units completely at random, i.e. the randomization is done without any restrictions. This design is applicable in that situation where the entire experimental material is homogeneous (i.e. all the experimental units can be regarded as being similar to each other). We illustrate the concept of the Completely Randomized (CR) Design (pertaining to the case when each treatment is repeated equal number of times) with the help of the following example.

EXAMPLE

An experiment was conducted to compare the yields of three varieties of potatoes. Each variety as assigned at random to equal-size plots, four times. The yields were as follow:

Variety		
A	B	C
23	18	16
26	28	25
20	17	12
17	21	14

Test the hypothesis that the three varieties of potatoes are not different in the yielding capabilities.

SOLUTION

The first thing to note is that this is an example of the Completely Randomized (CR) Design. We are assuming that all twelve of the plots (i.e. farms) available to us for this experiment are homogeneous (i.e. similar) with regard to the fertility of the soil, the weather conditions, etc., and hence, we are assigning the four varieties to the twelve plots totally at random. Now, in order to test the hypothesis that the mean yields of the three varieties of potato are equal, we carry out the six-step hypothesis-testing procedure, as given below:

Hypothesis-Testing Procedure:

i) H0 : $\mu A = \mu B = \mu C$
HA : Not all the three means are equal

ii) Level of Significance:
$\alpha = 0.05$

iii) Test Statistic:

$$F = \frac{MS\ Treatments}{MS\ Error}$$

which, if H0 is true, has an F distribution with $\upsilon 1 = k\text{-}1 = 3 - 1 = 2$ and $\upsilon 2 = n\text{-}k = 12 - 3 = 9$ degree of freedom

iv) Computations:

The computation of the test statistic presented above involves quite a few steps, including the formation of what is known as the ANOVA Table.
First of all, let us consider what is meant by the ANOVA Table (i.e. the Analysis of Variance Table).
In the case of the Completely Randomized (CR) Design, the ANOVA Table is a table of the type given below:

ANOVA TABLE IN THE CASE OF THE COMPLETELY RANDOMIZED (CR) DESIGN

Source of Variation	d.f.	Sum of Squares	Mean Square	F
Between treatments	k-1	SST	MST	MST/MSE
Within treatments (Error)	n-k	SSE	MSE	--
Total	n-1	TSS	--	--

Let us try to understand this table step by step:

The very first column is headed 'Source of Variation', and under this heading, we have three distinct sources of variation:

'Total' stands for the overall variation in the twelve values that we have in our data-set.

Variety		
A	B	C
23	18	16
26	28	25
20	17	12
17	21	14

As you can see, the values in our data-set are 23, 26, 20, 17, 18, 28, and so on. Evidently, there is a variation in these values, and the term 'Total' in the lowest row of the ANOVA Table stands for this overall variation.
The term 'Variation between Treatments' stands for the variability that exists between the three varieties of potato that we have sown in the plots.
(In this example, the term 'treatments' stands for the three varieties of potato that we are trying to compare)
(The term 'variation between treatments' points to the fact that:
It is possible that the three varieties or, at least two of the varieties are significantly different from each other with regard to their yielding capabilities. This variability between the varieties can be measured by measuring the differences between the mean yields of the three varieties.)

The third source of variation is 'variation within treatments'. This point to the fact that even if only one particular variety of potato is sown more than once, we do not get the same yield every time

Variety		
A	B	C
23	18	16
26	28	25
20	17	12
17	21	14

In this example, variety A was sown four times, and the yields were 23, 26, 20, and 17 --- all different from one another! Similar is the case for variety B as well as variety C. The variability in the yields of variety A can be called 'variation within variety A'.

Similarly, the variability in the yields of variety B can be called 'variation within variety B'. Also, the variability in the yields of variety C can be called 'variation within variety C'. We can say that the term 'variability within treatments' stands for the combined effect of the above-mentioned three variations. The 'variation within treatments' is also known as the 'error variation'. This is so because we can argue that if we are sowing the same variety in four plots which are very similar to each other, then we should have obtained the same yield from each plot!

If it is not coming out to be the same every time, we can regard this as some kind of an 'error'.

The second, third and fourth columns of the ANOVA Table are entitled 'degrees of freedom', 'Sum of Squares' and 'Mean Square'.

ANOVA TABLE IN THE CASE OF THE COMPLETELY RANDOMIZED (CR) DESIGN

Source of Variation	d.f.	Sum of Squares	Mean Square	F
Between treatments	k-1	SST	MST	MST/MSE
Within treatments (Error)	n-k	SSE	MSE	--
Total	n-1	TSS	--	--

The point to understand is that the sources of variation corresponding to treatments and error will be measured by computing quantities that are called Mean Squares, and 'Mean Square' can be defined as:

$$Mean\ Square = \frac{Sum\ of\ Squares}{Degrees\ of\ Freedom}$$

Corresponding to these two sources of variation, we have the following two equations:

$$1)\ 'MS\ Treatment' = \frac{'SS\ Treatment'}{d.f.}$$

AND

$$2)\ 'MS\ Error' = \frac{'SS\ Error'}{d.f.}$$

It has been mathematically proved that, with reference to Analysis of Variance pertaining to the Completely Randomized (CR) Design, the degrees of freedom corresponding to the Treatment Sum of Squares are k-1, and the degrees of freedom corresponding to the Error Sum of Squares are n-k. Hence, the above two equations can be written as:

$$1)\ 'MS\ Treatment' = \frac{'SS\ Treatment'}{k-1}$$

AND

$$2)\ 'MS\ Error' = \frac{'SS\ Error'}{n-k}$$

How do we compute the various sums of squares? The three sums of squares occurring in the third column of the above ANOVA Table are given by:

$$1)\ Total\ SS = TSS = \sum_i \sum_j X_{ij}^2 - C.F.$$

$$2)\ SS\ Treatment = SST = \frac{\sum_j T_{\cdot j}^2}{r} - C.F.$$

where C.F. stands for 'Correction Factor', and is given by

$$C.F. = \frac{T_{\cdot\cdot}^2}{n}$$

and r denotes the number of data-values per column (i.e. the number of rows). (It should be noted that this example pertains to that case of the Completely Randomized (CR) Design where each treatment is being repeated equal number of times, and the above formulae pertain to this particular situation. With reference to the CR Design, it should be noted that, in some situations, the various treatments are not repeated an equal number of times.

For example, with reference to the twelve plots (farms) that we have been considering above, we could have sown variety A in five of the plots, variety B in three plots, and variety C in four plots. Going back to the formulae of various sums of squares, the sum of squares for error is given by

$$3)\ SS\ Error = Total\ SS - SS\ Treatment$$

i.e.

$$SSE = TSS - SST$$

It is interesting to note that,
Total SS = SS Treatment + SS Error

In a similar way, we have the equation:
Total d.f. = d.f. for Treatment + d.f. for Error
It can be shown that the degrees of freedom pertaining to 'Total' are n - 1.
Now,

n-1 = (k-1) + (n-k)

i.e.
Total d.f. = d.f. for Treatment + d.f. for Error
The notations and terminology given in the above equations relate to the following table:

	Variety			Total	$\sum_j X_{ij}^2$
	A	B	C		
	23 (529) 26 (676) 20 (400) 17 (289)	18 (324) 28 (784) 17 (289) 21 (196)	16 (256) 25 (625) 12 (144) 14 (196)	-- -- -- --	1109 2085 833 926
$T._j$	86	84	67	237	4953
$T._{j^2}$	7396	7056	4489	18941	↑
$\sum_i X_{ij}^2$	1894	1838	1221	4953	Check ←

The entries in the body of the table i.e. 23, 26, 20, 17, and so on are the yields of the three varieties of potato that we had sown in the twelve farms. The entries written in brackets next to the above-mentioned data-values are the squares of those values.
For example:

529 is the square of 23,
676 is the square of 26,
400 is the square of 20,
and so on.

Adding all these squares, we obtain:

$$\sum_i \sum_j X_{ij}^2 = 4953$$

	Variety			Total	$\sum_j X_{ij}^2$
	A	B	C		
	23 (529) 26 (676) 20 (400) 17 (289)	18 (324) 28 (784) 17 (289) 21 (196)	16 (256) 25 (625) 12 (144) 14 (196)	-- -- -- --	1109 2085 833 926
$T._j$	86	84	67	237	4953
$T._{j^2}$	7396	7056	4489	18941	↑
$\sum_i X_{ij}^2$	1894	1838	1221	4953	Check ←

The notation T.j stands for the total of the jth column.(You must already be aware that, in general, the rows of a bivariate table are denoted by the letter 'i', whereas the columns of a bivariate table are denoted by the letter 'j'.

In other words, we talk about the 'ith row', and the 'jth column' of a bivariate table.)The 'dot' in the notation T.j indicates the fact that summation has been carried out over i (i.e. over the rows).

In this example, the total of the values in the first column is 86, the total of the values in the second column is 84, and the total of the values in the third column is 67.

	Variety			Total	$\sum_j X_{ij}^2$
	A	B	C		
	23 (529) 26 (676) 20 (400) 17 (289)	18 (324) 28 (784) 17 (289) 21 (196)	16 (256) 25 (625) 12 (144) 14 (196)	-- -- -- --	1109 2085 833 926
$T._j$	86	84	67	237	4953
$T._{j^2}$	7396	7056	4489	18941	↑ |
$\sum_i X_{ij}^2$	1894	1838	1221	4953	Check ←

Hence, ΣT.j is equal to 237.

ΣT.j is also denoted by T..

i.e.

T.. = ΣT.j The 'double dot' in the notation T.. indicates that summation has been carried out over i as well as over j.

The row below T.j is that of T.j2, and squaring the three values of T.j, we obtain the quantities 7396, 7056 and 4489.

Adding these, we obtain ΣT.j2 = 18941.

	Variety			Total	$\sum_j X_{ij}^2$
	A	B	C		
	23 (529) 26 (676) 20 (400) 17 (289)	18 (324) 28 (784) 17 (289) 21 (196)	16 (256) 25 (625) 12 (144) 14 (196)	-- -- -- --	1109 2085 833 926
$T_{\cdot j}$	86	84	67	237	4953
$T_{\cdot j^2}$	7396	7056	4489	18941	↑ | Check ←
$\sum_i X_{ij}^2$	1894	1838	1221	4953	

Now that we have obtained all the required quantities, we are ready to compute SS Total, SS Treatment, and SS Error:
We have

$$C.F. = \frac{T_{..}^2}{n} = \frac{(237)^2}{12} = 4680.75$$

Hence, the total sum of squares is given by

$$\begin{aligned} TSS &= \sum_i \sum_j X_{ij}^2 - C.F. \\ &= 4953 - 4680.75 \\ &= 272.25 \end{aligned}$$

Also, we have

$$\begin{aligned} SS\ Treatment = SST &= \frac{\sum_j T_{\cdot j}^2}{r} - C.F. \\ &= \frac{18941}{4} - 4680.75 \\ &= 54.50 \end{aligned}$$

And, hence:
SS Error = SSE = TSS - SST
= 272.25 - 54.50 = 217.75

In this example, we have n = 12, and k = 3, hence:

n-1=11,

k- 1 = 2, and n - k = 9.

Substituting the above sums of squares and degree of freedom in the ANOVA table, we obtain:

ANOVA TABLE

Source of Variation	d.f.	Sum of Squares	Mean Square	Computed F
Between treatments (i.e. Between varieties)	2	54.50		
Error	9	217.75		
Total	11	272.25		

Now, the mean squares for treatments and for error are very easily found by dividing the sums of squares by the corresponding degrees of freedom. Hence, we have

ANOVA TABLE

Source of Variation	d.f.	Sum of Squares	Mean Square	Computed F
Between treatments (i.e. Between varieties)	2	54.50	27.25	
Error	9	217.75	24.19	
Total	11	272.25	--	

As indicated earlier, the test-statistic appropriate for testing the null hypothesis

H0 : $\mu A = \mu B = \mu C$

versus

HA : Not all the three means are equal is:

$$F = \frac{MS\ Treatments}{MS\ Error}$$

which, if H0 is true, has an F distribution with $\upsilon 1 = k\text{-}1 = 3 - 1 = 2$ and $\upsilon 2 = n\text{-}k = 12 - 3 = 9$ degree of freedom Hence, it is obvious that F will be found by dividing the first entry of the fourth column of our ANOVA Table by the second entry of the same column i.e.

$$F = \frac{\text{MS Treatment}}{\text{MS Error}} = \frac{27.25}{24.19} = 1.13$$

We insert this computed value of F in the last column of our ANOVA table, and thus obtain:
ANOVA TABLE

Source of Variation	d.f.	Sum of Squares	Mean Square	Computed F
Between treatments (i.e. Between varieties)	2	54.50	27.25	1.13
Error	9	217.75	24.19	--
Total	11	272.25	--	--

The fifth step of the hypothesis - testing procedure is to determine the critical region. With reference to the Analysis of Variance procedure, it can be shown that it is appropriate to establish the critical region in such a way that our test is a right-tailed test. In other words, the critical region is given by:

Critical Region:

$F > F\alpha\ (k - 1, n - k)$

In this example:
The critical region is $F > F0.05\ (2,9) = 4.26$

vi) Conclusion:

Since the computed value of F = 1.13 does not fall in the critical region, so we accept our null hypothesis and may conclude that, on the average, there is no difference among the yielding capabilities of the three varieties of potatoes.

In this course, we will not be discussing the details of the mathematical points underlying One-Way Analysis of Variance that is applicable in the case of the Completely Randomized (CR) Design. One important point that the students should note is that the ANOVA technique being presented here is valid under the following assumptions:

- The k populations (whose means are to be compared) are normally distributed;
- All k populations have equal variances i.e. $\sigma 12 = \sigma 22 = \ldots = \sigma k2$. (This property is called homoscedasticity)
- The k samples have been drawn randomly and independently from the respective populations.

Next, we begin the discussion of the Randomized Complete Block (RCB) Design:

THE RANDOMIZED COMPLETE BLOCK DESIGN (RCB DESIGN)

A randomized complete block (RCB) design is the one in which

- The experimental material (which is not homogeneous overall) is divided into groups or blocks in such a manner that the experimental units within a particular block are relatively homogeneous.
- Each block contains a complete set of treatments, i.e., it constitutes a replication of treatments.
- The treatments are allocated at random to the experimental units within each block, which means the randomization is restricted.(A new randomization is made for every block.)The object of this type of arrangement is to bring the variability of the experimental material under control.

  In simple words, the situation is as follows:

We have experimental material which is not homogeneous overall. For example, with reference to the example that we have been considering above, suppose that the plots which are closer to a canal are the most fertile ones, the ones a little further away are a little less fertile, and the ones still further away are the least fertile.

In such a situation, we divide the experimental material into groups or blocks which are relatively homogeneous. The randomized complete block design is perhaps the most widely used experimental design. Two-way analysis of variance is applicable in case of the randomized complete block (RCB) design.

We illustrate this concept with the help of an example:

EXAMPLE

In a feeding experiment of some animals, four types of rations were given to the animals that were in five groups of four each. The following results were obtained

Groups	Rations			
	A	B	C	D
I	32.3	33.3	30.8	29.3
II	34.0	33.0	34.3	26.0
III	34.3	36.3	35.3	29.8
IV	35.0	36.8	32.3	28.0
V	36.5	34.5	35.8	28.8

The values in the above table represent the gains in weights in pounds. Perform an analysis of variance and state your conclusions. In the next lecture, we will discuss this example in detail, and will analyze the given data to carry out the following test:

H0 : $\mu A = \mu B = \mu C = \mu D$

HA : Not all the treatment-means are equal

LECTURE NO. 44

- Randomized Complete Block Design
- The Least Significant Difference (LSD) Test
- Chi-Square Test of Goodness of Fit

At the end of the last lecture, we introduced the concept of the *Randomized Complete Block (RCB) Design*, and we picked up an example to illustrate the concept. In this lecture, we begin with a detailed discussion of the same example:

EXAMPLE

In a feeding experiment of some animals, four types of rations were given to the animals that were in five groups of four each. The following results were obtained:

Groups	Rations			
	A	B	C	D
I	32.3	33.3	30.8	29.3
II	34.0	33.0	34.3	26.0
III	34.3	36.3	35.3	29.8
IV	35.0	36.8	32.3	28.0
V	36.5	34.5	35.8	28.8

The values in the above table represent the gains in weights in pounds. Perform an analysis of variance and state your conclusions.

SOLUTION

Hypothesis-Testing Procedure:

i a) Our primary interest is in testing:

H0 : $\mu A = \mu B = \mu C = \mu D$

HA : Not all the ration-means (*treatment-means*) are equal

i b) In addition, we can also test:

H'0 : $\mu I = \mu II = \mu III = \mu IV = \mu V$

H'A : Not all the group-means (*block-means*) are equal

ii) Level of significance

$\alpha = 0.05$

iii a) Test Statistic for testing H0 versus HA:

$$F = \frac{MS\ Treatment}{MS\ Error}$$

which, if H0 is true, has
an F-distribution with v1 = c-1= 4-1= 3 and v2 = (r-1)(c-1) =(5-1)(4-1) =4 × 3 = 12
degrees of freedom.

iii b) Test Statistic for testing
H ′0 versus H ′A:

$$F = \frac{\text{MS Block}}{\text{MS Error}}$$

which, if H0 is true, has
an F-distribution with v1 = r-1 = 5-1= 4 and
v2 = (r-1)(c-1) = (5-1)(4-1) =4 × 3 = 12 degrees of freedom.

Now, the given data leads to the following table:

iv) Computations:

Groups	Ration				$B_{i.}$	$B_{i.}^2$	$\sum_j X_{ij}^2$
	A	B	C	D			
I	32.3 (10.43.29)	33.3 (1108.89)	30.8 (948.64)	29.3 (858.49)	125.7	15800.49	3959.31
II	34.00 (1156.00)	33.0 (1089.00)	34.3 (1176.49)	26.0 (676.00)	127.3	16205.29	4097.49
III	34.3 (1176.49)	36.3 (1317.69)	35.3 (1246.09)	29.8 (888.04)	135.7	18414.49	4628.31
IV	35.0 (1225.00)	36.8 (1354.24)	32.3 (1043.29)	28.0 (784.00)	132.1	17450.41	4406.53
V	36.5 (1332.25)	34.5 (1190.25)	35.8 (1281.64)	28.8 (829.44)	135.6	18387.36	4633.58
$T_{.j}$	172.1	173.9	168.5	141.9	656.4	86258.04	21725.22
$T_{.j}^2$	29618.41	30241.21	28392.25	20135.61	108387.48		↑ | Check ←—
$\sum_i X_{ij}^2$	5933.03	6060.07	5696.15	4035.97	21725.22	←	

Hence, we have Total SS $= \sum\sum X_{ij}^2 - \frac{T_{..}^2}{n}$

$$= 21725.22 - \frac{(656.4)^2}{20}$$

$$= 21725.22 - 21543.05$$
$$= 182.17$$

Treatment SS $$= \frac{\sum_j T_{\cdot j}^2}{r} - \frac{T_{..}^2}{n}$$

$$= \frac{108387.48}{5} - \frac{(656.4)^2}{20}$$

= 21677.50 – 21543.05

= 134.45

Block SS $$= \frac{\sum_i B_{i.}^2}{c} - \frac{T_{..}^2}{n}$$

$$= \frac{86258.04}{4} - \frac{(656.4)^2}{20}$$
$$= 21564.51 - 21543.05$$

= 21.46

where c represents the number of observations per block (i.e. the number of *columns*)

And

Error SS= Total SS – (Treatment SS + Block SS)

= 182.17 – (134.45 + 21.46)

= 26.26

The degrees of freedom corresponding to the various sums of squares are as follows:

- Degrees of freedom for treatments: c - 1 (i.e. the number of treatments - 1)
- Degrees of freedom for blocks:
  r - 1 (i.e. the number of blocks - 1)
- Degrees of freedom for Total:
  rc - 1 (i.e. the total number of observations - 1)
- Degrees of freedom for error: degrees of freedom for Total minus degrees of freedom for treatments minus degrees of freedom for blocks

i.e. (rc-1) - (r-1) - (c-1)

= rc - r - c + 1

= (r-1) (c-1)

Hence the ANOVA-Table is:

ANOVA-TABLE

Source of Variation	d.f.	Sum of Squares	Mean Square	F
Between Treatments (i.e. Between Rations)	3	134.45	44.82	$F_1 = 20.47$
Between Blocks (i.e. Between Groups)	4	21.46	5.36	$F_2 = 2.45$
Error	12	26.26	2.19	--
Total	19	182.17	--	--

v a) Critical Region for Testing H0 against HA is given by
F > F0.05 (3, 12) = 3.49

v b) Critical Region for Testing H'0 against H'A is given by
F > F0.05 (4, 12) = 3.26

vi a) Conclusion Regarding Treatment Means

Since our computed value F1 = 20.47 exceeds the critical value F0.05 (3, 12) = 3.49, therefore we *reject* the null hypothesis, and conclude that there *is* a difference among the means of *at least two* of the treatments (i.e. the mean weight-gains corresponding to at least two of the rations are different).

vi b) Conclusion Regarding Block Means

Since our computed value F2 = 2.45 does not exceed the critical value F0.05(4, 12) = 3.26, therefore we accept the null hypothesis regarding the equality of block means and thus conclude that blocking (i.e. the *grouping* of animals) was actually *not required* in this experiment. As far as the conclusion regarding the block means is concerned, this information can be used when designing a similar experiment in the future.

[If blocking is actually not required, then a future experiment can be designed according to the Completely Randomized design, thus retaining more degrees of freedom for Error. (The more degrees of freedom we have for Error, the better, because an estimate of the error variation based on a greater number of degrees of freedom implies an estimate based on a greater amount of information (which is obviously good).)]

As far as the conclusion regarding the *treatment* means is concerned, the situation is as follows:

Now that we have concluded that there is a significant difference between the treatment means (i.e. we have concluded that the mean weight-gain is *not* the same for all four rations, then it is obvious that we would be interested in finding out, "Which of the four rations produces the greatest weight-gain?"

The answer to this question can be found by applying a technique known as the *Least Significant Difference (LSD) Test*.

THE LEAST SIGNIFICANT DIFFERENCE (LSD) TEST

According to this procedure, we compute the *smallest* difference that would be judged significant, and *compare* the absolute values of all differences of means with it. This smallest difference is called the least significant difference or LSD, and is given by:

LEAST SIGNIFICANT DIFFERENCE (LSD):

$$\mathrm{LSD} = t_{\alpha/2,(\nu)}\sqrt{\frac{2(\mathrm{MSE})}{r}},$$

where MSE is the Mean Square for Error, r is the size of equal samples, and $t\alpha/2\ (\nu)$ is the value of t at $\alpha/2$ level taken against the error degrees of freedom (ν).

The test-criterion that uses the least significant difference is called the LSD test.

Two sample-means are declared to have come from populations with significantly different means, when the absolute value of their difference *exceeds* the LSD.

It is customary to *arrange* the sample means in ascending order of magnitude, and to draw a *line* under any pair of adjacent means (or set of means) that are not significantly different.

The LSD test is applied *only* if the null hypotheses is rejected in the Analysis of Variance. We will not be going into the mathematical details of this procedure, but it is useful to note that this

procedure can be regarded as an alternative way of conducting the t-test for the equality of two population means.
If we were to apply the usual two-sample t-test, we would have had to *repeat* this procedure quite a few times!
(The six possible tests are:
H0 : $\mu A = \mu B$
H0 : $\mu A = \mu C$
H0 : $\mu A = \mu D$
H0 : $\mu B = \mu C$
H0 : $\mu B = \mu D$
H0 : $\mu C = \mu D$)
The LSD test is a procedure by which we can compare *all* the treatment means simultaneously.
We illustrate this procedure through the above example:
The Least Significant Difference is given by

$$\begin{aligned} LSD &= t_{\alpha/2,(\nu)}\sqrt{\frac{2(MSE)}{r}} \\ &= t_{0.025(12)}\sqrt{\frac{2(2.19)}{5}} \\ &= 2.179\sqrt{\frac{2(2.19)}{5}} \\ &= 2.179\times 0.936 \\ &= 2.04. \end{aligned}$$

Going back to the given data:

Groups	Rations			
	A	B	C	D
I	32.3	33.3	30.8	29.3
II	34.0	33.0	34.3	26.0
III	34.3	36.3	35.3	29.8
IV	35.0	36.8	32.3	28.0
V	36.5	34.5	35.8	28.8
Total	172.1	173.9	168.5	141.9
Mean	34.42	34.78	33.70	28.38

We find that the four treatment means are:

$$\overline{X}_A = 34.42$$
$$\overline{X}_B = 34.78$$
$$\overline{X}_C = 33.70$$
$$\overline{X}_D = 28.38$$

Arranging the above means in *ascending* order of magnitude, we obtain:

$\overline{X}_D$	$\overline{X}_C$	$\overline{X}_A$	$\overline{X}_B$
28.38	33.70	34.42	34.78

Drawing lines under pairs of adjacent means (or sets of means) that are not significantly different, we have:

$\overline{X}_D$	$\overline{X}_C$	$\overline{X}_A$	$\overline{X}_B$
28.38	33.70	34.42	34.78

From the above, it is obvious that rations C, A and B are *not* significantly different from each other with regard to weight-gain. The only ration which *is* significantly different from the others is ration D.

Interestingly, ration D has the *poorest* performance with regard to weight-gain. As such, if our primary objective is to *increase* the weights of the animals under study, then we may recommend *any* of the other three rations i.e. A, B and C to the farmers (depending upon availability, price, etc.), but we must *not* recommend ration D.

Next, we will consider two important tests based on the chi-square distribution. These are:

- The chi-square test of *goodness of fit*
- The chi-square test of *independence*

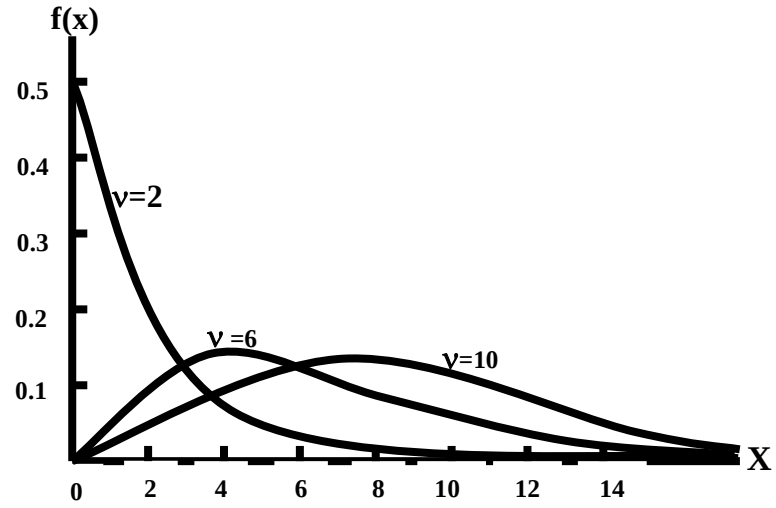
Before we begin the discussion of these tests, let us review the basic properties of the chi-square distribution:

PROPERTIES OF THE CHI-SQUARE DISTRIBUTION

The Chi-Square ($\chi 2$) distribution has the following properties:

1. It is a continuous distribution ranging from 0 to $+\infty$ .The number of degrees of freedom determines the *shape* of the chi-square distribution. (Thus, there is a different chi-square distribution for each number of degrees of freedom.
As such, it is a whole *family* of distributions.)

2. The curve of a chi-square distribution is *positively skewed*. The skewness *decreases as* ν *increases*.



χ2-distribution for various values of ν

As indicated by the above figures, the chi-square distribution tends to the normal distribution as the number of degrees of freedom approaches infinity. Having reviewed the basic properties of the chi-square distribution; we begin the discussion of the Chi-Square Test of Goodness of Fit:

CHI-SQUARE TEST OF GOODNESS OF FIT

The chi-square test of goodness-of-fit is a test of hypothesis concerned with the comparison of observed frequencies of a sample, and the corresponding expected frequencies based on a theoretical distribution. We illustrate this concept with the help of the same example that we considered in Lecture No. 28 --- the one pertaining to the fitting of a binomial distribution to real data:

EXAMPLE:

The following data has been obtained by tossing a *LOADED* die 5 times, and noting the number of times that we obtained a *six*. Fit a binomial distribution to this data.

No. of Sixes (x)	0	1	2	3	4	5	Total
Frequency (f)	12	56	74	39	18	1	200

SOLUTION

To fit a binomial distribution, we need to find n and p.

Here n = 5, the largest x-value.

To find p, we use the relationship $\bar{x} = np$.

We have:

No. of Sixes (x)	0	1	2	3	4	5	Total
Frequency (f)	12	56	74	39	18	1	200
fx	0	56	148	117	72	5	398

Therefore:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$= \frac{0 + 56 + 148 + 117 + 72 + 5}{200}$$

$$= \frac{398}{200} = 1.99$$

Using the relationship $\bar{x} = np$,
we obtain

$p = 1.99$ or $p = 0.398$.

Letting the random variable X represent the number of sixes, the above calculations yield the fitted binomial distribution as

$$b(x;5,\ 0.398) = \binom{5}{x} (0.398)^x\ (0.602)^{5-x}$$

Hence the *probabilities* and *expected frequencies* are calculated as below:

No. of Sixes (x)	Probability f(x)		Expected frequency
0	$\binom{5}{0} q^5 = (0.602)^5$	**= 0.07907**	**15.8**
1	$\binom{5}{1} q^5 p = 5.(0.602)^4\ (0.398)$	**= 0.26136**	**52.5**
2	$\binom{5}{2} q^3 p^2 = 10.(0.602)^3\ (0.398)^2$	**= 0.34559**	**69.1**
3	$\binom{5}{3} q^2 p^3 = 10.(0.602)(0.398)^3$	**= 0.22847**	**45.7**
4	$\binom{5}{4} qp^4 = (0.602)(0.398)^4$	**= 0.07553**	**15.1**
5	$\binom{5}{5} p^5 = (0.398)^5$	**= 0.00998**	**2.0**
Total		**= 1.00000**	**200.0**

Comparing the observed frequencies with the expected frequencies, we obtain:

No. of Sixes x	Observed Frequency o_i	Expected Frequency e_i
0	12	15.8
1	56	52.5
2	74	69.1
3	39	45.7
4	18	15.1
5	1	2.0
Total	200	200.0

The above table seems to indicate that there is not much discrepancy between the observed and the expected frequencies. Hence, in Lecture No.28, we concluded that it was a reasonably *good* fit. But, it was indicated that *proper* comparison of the expected frequencies with the observed frequencies can be accomplished by applying the *chi-square test of goodness of fit*. The Chi-Square Test of Goodness of Fit enables us to determine in a *mathematical* manner whether or not the theoretical distribution fits the observed distribution reasonably well.

The procedure of the chi-square of goodness of fit is very *similar* to the *general* hypothesis-testing procedure:

HYPOTHESIS-TESTING PROCEDURE

Step-1:

H_0 : The fit is good

H_A : The fit is not good

Step-2:

Level of Significance: $\alpha = 0.05$

Step-3: Test-Statistic:

$$\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i}$$

which, if H0 is true, follows the chi-square distribution having k - 1 - r degrees of freedom(where k = No. of x-values (*after* having carried out the necessary mergers), and r = number of parameters that we estimate from the sample data).

Step-4: Computations:

No. of Sixes x	Observed Frequency o_i	Expected Frequency e_i	$o_i - e_i$	$(o_i - e_i)^2$	$(o_i - e_i)^2/e_i$
0	12	15.8	-3.8	14.44	0.91
1	56	52.5	3.5	12.25	0.23
2	74	69.1	4.9	24.01	0.35
3	39	45.7	-6.7	44.89	0.98
4	18 } **19**	15.1 } **17.1**	1.9	3.61	0.21
5	1 }	2.0 }			
Total	200	200.0			2.69

IMPORTANT NOTE

In the above table, the category x = 4 has been merged with the category x = 5 because of the fact that the expected frequency corresponding to x = 5 was less than 5.

[It is one of the basic requirements of the chi-square test of goodness of fit that the expected frequency of any x-value (or any combination of x-values) should not be less than 5.]Since we have combined the category x = 4 with the category x = 5, hence k = 5. Also, since we have estimated one parameter of the binomial distribution (i.e. p) from the sample data, hence r = 1. (The other parameter n is already known.)

As such, our statistic follows the chi-distribution having k - 1 - r = 5 - 1 - 1 = 3 degrees of freedom. Going back to the above calculations, the computed value of our test-statistic comes out to be $\chi 2 = 2.69$.

Step-5: Critical Region:

Since $\alpha = 0.05$, hence, from the Chi-Square Tables, it is evident that the critical region is:$\chi 2 \geq \chi 20.05\ (3) = 7.82$

Step-6:

Conclusion:

Since the computed value of $\chi 2$ i.e. 2.69 is less than the critical value 7.82, hence we accept H0 and conclude that the fit is good.(In other words, with only 5% risk of committing Type-I error, we conclude that the distribution of our random variable X can be regarded as a binomial distribution with n = 5 and p = 0.398.)

LECTURE NO. 45

- Chi-Square Test of Goodness of Fit (in continuation of the last lecture)
- Chi-Square Test of Independence
- The Concept of Degrees of Freedom
- p-value
- Relationship Between Confidence; Interval and Tests of Hypothesis

An Overview of the Science of Statistics in Today's World (including Latest Definition of Statistics)

The students will recall that, towards the end of the last lecture, we discussed the chi-square test of goodness of fit. We applied the test to the example where we had fitted a binomial distribution to real data, and, since the computed value of our test statistic turned out to be insignificant, therefore we concluded that the fit was good.

Let us consider another example:

EXAMPLE

The platform manager of an airline's terminal ticket counter wants to determine whether customer arrivals can be modelled by using a Poisson distribution. The manager is especially interested in late-night traffic.

Accordingly, data for the time period of interest have been collected, as follows:

Number of Arrivals Per Minute	Frequency
0	84
1	114
2	70
3	60
4	32
5	16
6	15
7	4
8	5
	400

Is the distribution Poisson?

SOLUTION:

First of all, we fit a Poisson distribution to the given data Because a mean is not specified, it must be estimated from the sample data. The mean of the frequency distribution can be found by using the formula

$$\bar{x} = \frac{\sum fx}{n}$$

where $n = \Sigma f$.

Thus we have the following calculations:

Number of Arrivals x	Frequency f	fx
0	84	0
1	114	114
2	70	140
3	60	180
4	32	128
5	16	80
6	15	90
7	4	28
8	5	40
	400	800

Hence:

$$Mean = \bar{x} = \frac{\sum fx}{n} = \frac{800}{400} = 2$$

Replacing μ by , the formula for the Poisson probabilities is

$$f(x) = \frac{e^{-\bar{x}}\bar{x}^{x}}{x!} = \frac{e^{-2}2^{x}}{x!}$$

Hence, we obtain:

Number of Customer Arrivals	Observed Frequencies	Poisson Probabilities f(x)	Expected Frequencies 400 f(x)
0	84	0.1353	54.12
1	114	0.2707	108.28
2	70	0.2707	108.28
3	60	0.1804	72.16
4	32	0.0902	36.08
5	16	0.0361	14.44
6	15	0.0120	4.80
7	4	0.0034	1.36
8	5	0.0009	0.36
9 or more	0	0.0002	0.08
	400	1	400

Next, we apply the chi-square test of goodness of fit according to the following procedure:

HYPOTHESIS-TESTING PROCEDURE

Step-1:

H0 : Arrivals are Poisson-distributed.

H1 : The distribution is not Poisson.

Step-2:

Level of Significance: $\alpha = 0.05$

Step-3: Test-Statistic: $\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i}$

which, if H0 is true, follows the chi-square distribution having k - 1 - r degrees of freedom; (where k = No. of x-values (after having carried out the necessary mergers), and r = number of parameters that we estimate from the sample data)

Step-4: Computations:

The necessary calculations are shown in the following table:

Number of Customer Arrivals	Observed Frequency o_i		Expected Frequency e_i		(0 – e)	$(0-e)^2$	$(0-e)^2/e$
0	84		54.12		29.88	892.81	16.50
1	114		108.28		5.72	32.72	0.30
2	70		108.28		-38.28	1465.36	13.53
3	60		72.16		-12.16	147.87	2.05
4	32		36.08		-4.08	16.65	0.46
5	16		14.44		1.56	2.43	0.17
6	15	24	4.80	6.60	17.40	302.76	45.87
7	4		1.36				
8	5		0.36				
9 or more	0		0.08				
	400		400				χ^2=78.88

With reference to the above, it should be noted that, since some of the expected frequencies are less than the required minimum of 5, it became necessary to combine some of those classes. Combination is best accomplished working from the bottom up.

In order that we obtain a number greater than 5, the last four expected frequencies had to be combined.

Hence, the effective number of categories becomes 7.

Step-5:

Determination of the Critical Region:

Since the effective number of categories becomes 7

Therefore k = 7.

Also, since the one lone parameter of the Poisson distribution has been estimated from the sample data, hence r = 1.

Hence: Our statistic follows the chi-square distribution having

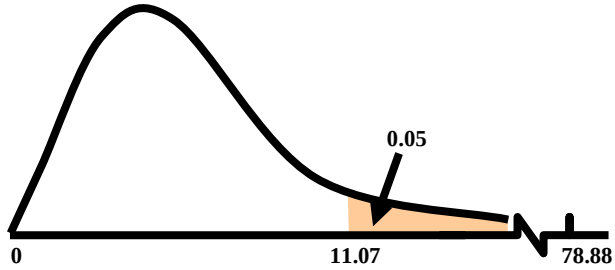
k - 1 - r = 7 - 1 - 1 = 5

degrees of freedom.

The critical region is given by

$\chi 2 \geq \chi 20.05\ (5) = 11.07$

CRITICAL REGION:



Step-6:

Conclusion:

Since the computed value of our test statistic i.e. 78.88 is much larger than the critical value 11.07, therefore, we reject H0 and conclude that the distribution is probably not a Poisson distribution with parameter 2.

(With only 5% risk of committing Type-1 error, we conclude that the fit is not good.)

In fact, the computed value of our test statistic i.e. 78.88 is so large that it is possible that if we had set the level of significance at 1%, even then it would have exceeded the critical value. The students are encouraged to check this up themselves. If the computed value does fall in the critical region corresponding to 1% level of significance, then our result is highly significant

RATIONALE OF THE CHI-SQUARE TEST OF GOODNESS OF FIT

It is clear that $\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i}$ will be a small quantity when all the oi's are close to the corresponding ei's.

(In fact, if the observed frequencies are exactly equal to the expected ones, then $\chi 2$ will be exactly equal to zero.)

The $\chi 2$ - statistic will become larger when the differences between the oi's and ei have become larger. Thus, $\chi 2$ measure the amount of deviation (or discrepancy) between the observed and the expected results.

ASSUMPTIONS OF THE CHI-SQUARE TEST OF GOODNESS OF FIT

While applying the chi-square test of goodness of fit, certain requirements must be satisfied, three of which are as follows:

1. The total number of observations (i.e. the sample size) should be at least 50.
2. The expected number ei in any of the categories should not be less than 5. (So, when the expected frequency ei in any category is less than 5, we may combine this category with one or more of the other categories to get ei $\geq$ 5.)
3. The observations in the sample or the frequencies of the categories should be independent.

Next, we begin the discussion of the Chi-Square Test of Independence:

In this regard, it is interesting to note that, (since the formula of chi-square in this particular situation is very similar to the formula that we have just discussed), therefore, the chi-square test of independence can also be regarded as a kind of chi-square test of goodness of fit. We illustrate this concept with the help of an example:

EXAMPLE

A random sample of 250 men and 250 women were polled as to their desire concerning the ownership of personal computers. The following data resulted:

	Men	Women	Total
Want PC	120	80	200
Don't Want PC	130	170	300
Total	250	250	500

Test the hypothesis that desire to own a personal computer is independent of sex at the 0.05 level of significance.

SOLUTION

i) H0 : The two variables of classification (i.e. gender and desire for PC) are independent, and

H1 : The two variables of classification are not independent.

ii) The significance level is set at $\alpha = 0.05$.

iii) The test-statistic to be used is $\chi^2 = \sum_i \sum_j (0_{ij} - e_{ij})^2 / e_{ij}$

This statistic, if H0 is true, has an approximate chi-square distribution with (r - 1) (c - 1) = (2 - 1) (2 - 1) = 1 degrees of freedom.

iv) Computations:

In order to determine the value of $\chi 2$, we carry out the following computations:

The first step is to compute the expected frequencies. The expected frequency of any cell is obtained by multiplying the marginal total to the right of that cell by the marginal total directly below that cell, and dividing this product by the grand total.

In this example,

$$e_{11} = \frac{(200)(250)}{500} = 100,$$

$$e_{12} = \frac{(200)(250)}{500} = 100,$$

$$e_{21} = \frac{(300)(250)}{500} = 150,$$

and

$$e_{22} = \frac{(300)(250)}{500} = 150.$$

Hence, we have:

Expected Frequencies:

	Men	Women	Total
Want PC	100	100	200
Don't Want PC	150	150	300
Total	250	250	500

Next, we construct the columns of oij - eij, (oij - eij)2 and (oij - eij)2 eij , as shown below:

Observed Frequency o_{ij}	Expected Frequency e_{ii}	$o_{ij} - e_{ij}$	$(o_{ij} - e_{ij})^2$	$(o_{ij} - e_{ij})^2/e_{ii}$
120	100	20	400	4.00
130	150	-20	400	2.67
80	100	-20	400	4.00
170	150	20	400	2.67
				$\chi^2 = 13.33$

$$\chi^2 = 13.33.$$

Hence, the computed value of our test-statistic comes out to be

v) Critical Region:

$\chi 2 \geq \chi 20.05(1) = 3.84$

vi)

Conclusion:

Since 13.33 is bigger than 3.84, we reject H0 and conclude that desire to own a personal computer set and sex are associated. Now that we have concluded that gender and desire for PC are associated, the natural question is, "Which gender is it where the proportion of persons wanting a PC is higher?" We have:

	Men	Women	Total
Want PC	120	80	200
Don't Want PC	130	170	300
Total	250	250	500

A close look at the given data indicates clearly that the proportion of persons who are desirous of owning a personal computer is higher among men than among women.

And, (since our test statistic has come out to be significant), therefore we can say that the proportion of men wanting a PC is significantly higher than the proportion of women wanting to own a PC.

Let us consider another example:

EXAMPLE

A national survey was conducted in a country to obtain information regarding the smoking patterns of the adults males by marital status. A random sample of 1772 citizens, 18 years old and over, yielded the following data :

MARITAL STATUS	SMOKING PATTERN			Total
	Total Abstinence	Only at times	Regular Smoker	
Single	67	213	74	**354**
Married	411	63	129	**1173**
Widowed	85	51	7	**143**
Divorced	27	60	15	**102**
Total	**590**	**957**	**225**	**1772**

Use this data to decide whether there is an association between marital status and smoking patterns. The students are encouraged to work on this problem on their own, and to decide for themselves whether to accept or reject the null hypothesis.(In this problem, the null and the alternative hypotheses will be:

H0: Marital status and smoking patterns are statistically independent.

HA : Marital status and smoking patterns are not statistically independent.)

This brings us to the end of the series of topics that were to be discussed in some detail for this course on Statistics and Probability. For the remaining part of today's lecture, we will be discussing some interesting and important concepts.

First and foremost, let us consider the concept of

DEGREES OF FREEDOM

As you will recall, when discussing the t-distribution, the chi-square distribution, and the F-distribution, it was conveyed to you that the parameters that exists in the equations of those distributions are known as degrees of freedom.

But the question is, 'Why these parameters are called degrees of freedom?' Let us try to obtain an answer to this question by considering the following:

Consider the two-dimensional plane, and consider a straight line segment in the plane. If one edge of the line segment is fixed at some point (x0, y0), the line segment can be rotated in the plane such that the fixed edge stays in its place. In other words, we can say that the line segment is free to move in the plane with one restriction. Hence, if we fix one end-point of the line segment, then we are left with one degree of freedom for its movement. Next, consider the case when we fix both end-points of the line segment in the plane. In this case, both degrees of freedom are lost, and therefore the line can no longer move in the plane. But, if we view the above situation with reference to the three-dimensional space --- the one that we live in --- we note that the whole plane (containing the fixed line segment) can move in three dimensions, and hence, we have one degree of freedom for its movement. Let us try to understand this concept in another way: Suppose we have a sample of size n = 6, and suppose that the sum of the sample values is 20. That is, we have the following situation: Our Sample:

Sr. No.	Value
1	
2	
3	
4	
5	
6	
Total	20

Now, the point is that, given this total of 20, if we choose the first 5 values freely, we are not free to choose the sixth value. Hence, one degree of freedom is lost. This point can also be explained in the following alternative way. Given that the sum of the six values is 20, if we have knowledge of the first five values, but the sixth value is missing, then we can re-generate the sixth value. This can also be expressed as follows.

If there are six observations and you find their sum; next, you throw away one of the six observations; then, you can re-generate that observation (because of the fact that you have already computed the sum).

Since, the number of values that can be re-generated is one, hence, the degrees of freedom are n minus one.

(The one which can be re-generated is not the one that we can choose freely.)

Going back to sampling distributions such as the t-distribution, the chi-square distribution and the F-distribution, 'degrees of freedom' can be defined as the number of observations in the sample minus the number of population parameters that are estimated from the sample data (from those observations). For example, in lecture number 39, we noted that the statistic follows the t-distribution having n-1 degrees of freedom.

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Here n denotes the number of observations in our sample, and since we are estimating one population parameter i.e. σ from the sample data, hence the number of degrees of freedom is n-1.

$$\frac{s_1^2}{s_2^2}$$

Similarly, referring to lecture number 42, the students will recall that it was stated that the statistic

Follows the F-distribution having (n1-1, n2-1) degrees of freedom

Here n1 denotes the number of observations in the first sample, and since we are estimating one parameter of the first population i.e. $\sigma 12$ from the sample data, hence the number of degrees of freedom for the numerator of our statistic is n1 minus one. Similarly, n2 denotes the number of observations in the second sample, and since we are estimating one parameter of the second population i.e. $\sigma 22$ from the sample data, hence the number of degrees of freedom for the denominator of our statistic is n2 minus one. In addition, in today's lecture, you learnt that the statistic

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(0_{ij} - e_{ij}\right)^2}{e_{ij}},$$

follows the chi-square distribution having (r-1)(c-1) degrees of freedom. Let us try to understand this point: Consider the 2 × 2 contingency table ,similar to the one that we had in the example

regarding the desire for ownership of a personal computer. In this regard, suppose that we have two variables of classification, A and B, and the situation is as follows:

	A_1	A_2	Total
B_1			200
B_2			300
Total	250	250	500

The point is that, given the marginal totals and the grand total, if we choose the frequencies of the first cell of the first row freely, we are not free to choose the frequency of the second cell of the first row. Also, given the frequency of the above-mentioned first cell, we are not even free to choose the frequency of the second cell of the first column.

Not only this, it is interesting to note that, given the above, we are not even free to choose the frequency of the second cell of the second row or the second column !Hence, given the marginal and grand totals, we have only degree of freedom (i.e. $1 = 1 \times 1 = (2-1)(2-1)$ degrees of freedom).A similar situation holds in the case of a 2 x 3 contingency table. The students are encouraged to work on this point on their own, and to realize for themselves that, in the case of a 2 x 3 contingency table, there exist (2 - 1) (3 - 1) = 2 degrees of freedom . Next, let us consider the concept of p-value:

You will recall that, with reference to the concept of hypothesis-testing, we compared the computed value of our test statistic with a critical value. For example, in case of a right-tailed test, we rejected the null hypothesis if our computed value exceeded the critical value, and we accepted the null hypothesis if our computed value turned out to be smaller than the critical A hypothesis can also be tested by means of what is known as the p-value.

P-VALUE

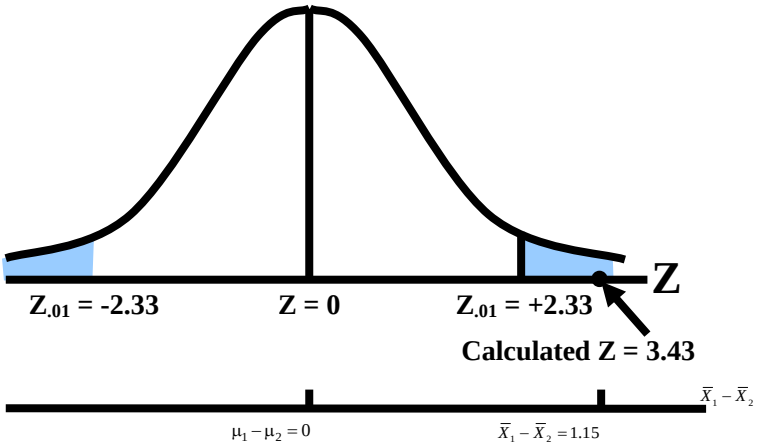
The probability of observing a sample value as extreme as, or more extreme than, the value observed, given that the null hypothesis is true. We illustrate this concept with the help of the example concerning the hourly wages of computer analysts and registered nurses that we discussed in an earlier lecture. The students will recall that the example was as follows:

EXAMPLE

A survey conducted by a market-research organization five years ago showed that the estimated hourly wage for temporary computer analysts was essentially the same as the hourly wage for registered nurses. This year, a random sample of 32 temporary computer analysts from across the country is taken. The analysts are contacted by telephone and asked what rates they are currently able to obtain in the market-place. A similar random sample of 34 registered nurses is taken. The resulting wage figures are listed in the following table.

Computer Analysts			Registered Nurses		
$ 24.10	$25.00	$24.25	$20.75	$23.30	$22.75
23.75	22.70	21.75	23.80	24.00	23.00
24.25	21.30	22.00	22.00	21.75	21.25
22.00	22.55	18.00	21.85	21.50	20.00
23.50	23.25	23.50	24.16	20.40	21.75
22.80	22.10	22.70	21.10	23.25	20.50
24.00	24.25	21.50	23.75	19.50	22.60
23.85	23.50	23.80	22.50	21.75	21.70
24.20	22.75	25.60	25.00	20.80	20.75
22.90	23.80	24.10	22.70	20.25	22.50
23.20			23.25	22.45	
23.55			21.90	19.10	

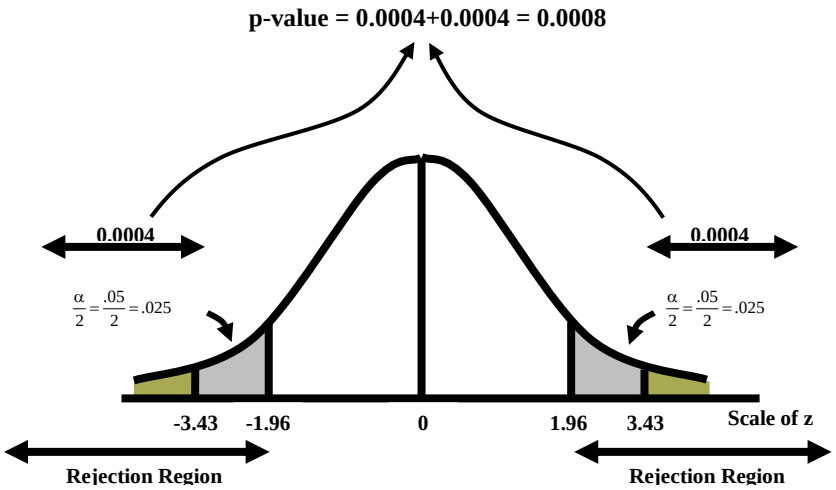
Conduct a hypothesis test at the 2% level of significance to determine whether the hourly wages of the computer analysts are still the same as those of registered nurses. In order to carry out this test, the Null and Alternative Hypotheses were set up as follows:
Null and Alternative Hypotheses:
H0 : $\mu 1 - \mu 2 = 0$
HA : $\mu 1 - \mu 2 \neq 0$
(Two-tailed test)
The computed value of our test statistic came out to be 3.43, whereas, at the 5% level of significance, the critical value was 2.33, hence, we rejected H0.



Hence, we concluded that there was a significant difference between the average hourly wage of a temporary computer analyst and the average hourly wage of a temporary registered nurse. This conclusion could also have been reached by
using the

P-VALUE METHOD

I. Looking up the probability of Z > 3.43 in the area table of the standard normal distribution yields an area of .5000 – .4996 = .0004.
II. To compute the p-value, we need to be concerned with the region less than –3.43 as well as the region greater than 3.43 (because the rejection region is in both tails).



The p-value is 0.0004 + 0.0004 = 0.0008. Since this value is very small, it means that the result that we have obtained in this example is highly improbable if, in fact, the null hypothesis is true. Hence, with such a small p-value, we decide to reject the null hypothesis.

The above example shows that: The p-value is a property of the data, and it indicates "how improbable" the obtained result really is. A simple rule is that if our p-value is less than the level of significanceα, then we should reject H0, whereas if our p-value is greater than the level of significanceα, then we should accept H0. (In the above example, α = 0.02 whereas the p-value is equal to 0.0008, hence we reject H0.)

RELATIONSHIP BETWEEN CONFIDENCE INTERVAL AND TESTS OF HYPOTHESIS

Some of the students may already have an idea that there exists some kind of a relationship between the confidence interval for a population parameter θ and a test of hypothesis aboutθ. (After all: When deriving the confidence interval for μ, the area that was kept in the middle of the sampling distribution of $\bar{X}$ was equal to 1-α so that the area in each of the right and left tails was equal to $\alpha/2$. And, when testing the hypothesis H0 : $\mu = \mu 0$ versus HA : $\mu \neq \mu 0$ at level of significance α, the area in each of the right and left tails was again equal to $\alpha/2$.)Hence, consider the following proposition: Let [L, U] be a 100(1 - α)% confidence interval for the parameter θ.Then we will accept the null hypothesis H0 : $\theta = \theta 0$ against H1 : $\theta \neq \theta 0$ at a level of significance α if $\theta 0$ falls inside the confidence interval, but if $\theta 0$ falls outside the interval [L, U], we will reject H0.In the language of hypothesis testing, the (1 - α) 100% confidence interval is known as the acceptance region and the region outside the confidence interval is called the rejection or critical region. The critical values are the end points of the confidence interval. The students are encouraged to work on this point on their own. As we approach the end of this course, we present an Overview of the Science of Statistics in Today's World: Statistics is a vast discipline! In this course, we have discussed the very basic and fundamental concepts of statistics and probability. But, there are numerous other topics that could have been discussed if we had the time. We could have talked about the Latin Square Design, we could have considered Inference Regarding Regression and Correlation Coefficients, we could have discussed Non-Parametric Statistics, and so on, and so forth.

The students are encouraged to study some of these concepts on their own --- as and when time permits --- in order to develop a better understanding and appreciation of the importance of the science of Statistics. In this course, numerous examples were discussed and many numerical problems were presented.

The solutions of these problems were presented in detail, and the various steps were worked out. In doing so, the purpose was to develop in the students a better understanding of the core concepts of the various techniques that were applied. But, it is interesting and useful to

note that, a lot many of these numerical problems can be solved within seconds by using the wide variety of statistical packages that are available. These include **SPSS, SAS, Statistica, Statgraph, Minitab, Stata, S-Plus, etc.** (The students are welcome to try out some of these packages on their own.)Towards the end of this course, we present one of the latest definitions of Statistics:

LATEST STATISTICAL DEFINITION

Statistics is a science of decision making for governing the state affairs. It collects analyzes, manages, monitors, interprets, evaluates and validates information. Statistics is Information Science and Information Science is Statistics. It is an applicable science as its tools are applied to all sciences including humanities and social sciences.

- THE END –